VISION AND SENSOR-BASED SIGNER-INDEPENDENT FRAMEWORK FOR ARABIC SIGN LANGUAGE RECOGNITION

AHMAD SAMI ABD ALKAREEM AL-SHAMAYLEH

FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
UNIVERSITY OF MALAYA
KUALA LUMPUR

2020

Universiti Malaya

VISION AND SENSOR-BASED SIGNER-INDEPENDENT FRAMEWORK FOR ARABIC SIGN LANGUAGE RECOGNITION

AHMAD SAMI ABD ALKAREEM AL-SHAMAYLEH

THESIS SUBMITTED IN FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
UNIVERSITY OF MALAYA
KUALA LUMPUR

2020

Universiti Malaya

UNIVERSITY OF MALAYA
ORIGINAL LITERARY WORK DECLARATION

Name of Candidate: AHMAD SAMI ABD ALKAREEM AL-SHAMAYLEH
Registration/Matric No: 17020074/1 (WHA140039)

Name of Degree: DOCTOR OF PHILOSOPHY

Title of Project Paper/Research Report/Dissertation/Thesis ("this Work"):
VISION AND SENSOR-BASED SIGNER-INDEPENDENT FRAMEWORK FOR ARABIC SIGN LANGUAGE RECOGNITION

Field of Study: SOFTWARE ENGINEERING (COMPUTER SCIENCE)

I do solemnly and sincerely declare that:

(1) I am the sole author/writer of this Work;
(2) This Work is original;
(3) Any use of any work in which copyright exists was done by way of fair dealing and for permitted purposes and any excerpt or extract from, or reference to or reproduction of any copyright work has been disclosed expressly and sufficiently and the title of the Work and its authorship have been acknowledged in this Work;
(4) I do not have any actual knowledge nor do I ought reasonably to know that the making of this work constitutes an infringement of any copyright work;
(5) I hereby assign all and every rights in the copyright to this Work to the University of Malaya ("UM"), who henceforth shall be owner of the copyright in this Work and that any reproduction or use in any form or by any means whatsoever is prohibited without the written consent of UM having been first had and obtained;
(6) I am fully aware that if in the course of making this Work I have infringed any copyright whether intentionally or otherwise, I may be subject to legal action or any other action as may be determined by UM.

Candidate's Signature Date:

Subscribed and solemnly declared before,

Witness's Signature Date:
Name:
Designation:

Witness's Signature Date:
Name:
Designation:

Witness's Signature Date:
Name:
Designation:

Universiti Malaya

VISION AND SENSOR-BASED SIGNER-INDEPENDENT FRAMEWORK FOR ARABIC SIGN LANGUAGE RECOGNITION

ABSTRACT

Hearing and speech-impairment disability is widespread throughout the world. At present, 15 million people have this disability in the Arab world, and about 86% of them come from low- and middle-income countries. Meanwhile, sign language (SL) can be classified into standard Arabic sign language (ArSL) and local Arabic sign language (LArSL). ArSL is the formal standard and is the more acceptable SL in the Arab world; it is also considered as the medium of instructions for schools and universities as well as television news, shows and programmes. With the absence of usable ArSL recognition (ArSLR) platforms, hearing- and speech-impaired people tend to be isolated and face serious difficulties in communication and interaction.

The focus of this thesis is to design and create a new ArSL and an ArSLR framework, which cover all ArSLR approaches and ArSL forms based on the standard criteria and consensus of SL experts. Essentially, this thesis proposes two frameworks for modelling and developing ArSLR. The first framework represents a usable static backhand and a signer-independent approach for the ArSLR of letters and number signs based on the Vision-Based Recognition (VBR) approach using a smartphone camera. The second framework represents a usable hybrid VBR and sensor-based recognition (SBR) approach for signer-independent continuous ArSLR development and evaluation using Microsoft Kinect and Smart Data Gloves.

To accomplish the research objective, an extensive systematic literature review has been conducted to create research taxonomies and to identify the research gaps on ArSLR approaches and ArSL forms of signs.

Universiti Malaya

For the first framework, the input signs to the ArSLR framework were first split into open and closed hand signs. Then, the ArSLR framework identified suitable approaches of recognising open and closed hand signs, such as the discrete wavelet transform, which was integrated with 1D-signature signals in closed hand signs. The open hand signs were differentiated through the distribution of quantised area levels, which were generated from the Run-Length Matrix. Statistical analysis approaches were employed to compute the feature descriptors. In our modelled framework, the multi-classification approach was implemented in the recognition phase. The presented framework has the ability to recognise ArSL static backhand finger numbers and letters with a recognition accuracy of 95.63%.

The second framework shows how to jointly exploit the VBR and SBR for accurate ArSLR made by the signer-independent. Firstly, a dedicated solution for the combined calibration of the two different devices was proposed. Then, a set of novel feature descriptors was used both for the smart gloves and for Microsoft Kinect. Finally, the proposed feature sets were fed to the multi-class support vector machines. Experimental results show that the obtained accuracies for the Arabic alphabet, Arabic numbers, isolated words, and continuous sentences datasets of this ArSL database are 99.38%, 99.42%, 99.15% and 99.55%, respectively. In conclusion, the created database and the proposed frameworks are able to increase the performance of the ArSLR.

Keywords: Arabic Sign Lanuage, Vision-based recognition, Sensor-based recognition; Signer-Independent.

Universiti Malaya

Rangkakerja Penanda-Bebas Berdasarkan penglihatan dan sensor untuk Pengesanan bagi Bahasa Isyarat Arab

ABSTRAK

Kecacatan pendengaran dan pertuturan adalah semakin berleluasa. Ada hamper 15 juta orang yang mempunyai kecacatan ini di dunia Arab di mana 86% daripadanya datang dari negara berpendapatan rendah ke sederhana. Bahasa isyarat boleh dikelaskan kepada Bahasa Isyarat Arab Piawai (ArSL) dan ArSL tempatan (LArSL). ArSL adalah bahasa piawai formal dan lebih diterima di dunia Arab yang merupakan medium arahan bagi sekolah, universiti dan bagi berita, pertunjukan dan program television. Tanpa platform pengecaman ArSL yang mudah digunakan, mereka yang mempunyai kecacatan pendengaran dan pertuturan lebih terpinggir dan menghadapi masalah serius dalam komunikasi dan interaksi.

Fokus tesis ialah merekabentuk dan mencipta suatu pangkalan dada ArSL dan rangkakerja ArSLR yang dapat meningkatkan prestasi pendekatan sedia ada berdasarkan kirteria dan consensus pakar Bahasa isyarat. Pada dasarnya, tesis ini mencadangkan dua rangkakerja untuk pemodelan dan pembangunan ArSLR. Rangkakerja pertama mewakili pendekatan tangan belakagn statik yang boleh digunakan dan pendekatan penandatangan bebas untuk huruf dan nombor ArSLR berdasarkan pengecaman berasaskan penglihatan (*Vision-Based Recognition - VBR*) menggunakan kamera telefon pintar. Seterusnya, rangkakerja kedua mewakili pendekatan boleh guna VBR hibrid dan Pendekatan Pengecaman Berasaskan Sensor (SBR) automatik bagi pembangunan ArSLR yang menggunakan penandatangan bebas, dinamik dan. Penilaian rangkakerja menggunakan Microsoft Kinect dan sarung tangan data pintar.

Untuk mencapai objektif tesis, kajian sastera yang sistematik dan luas telah dijalankan untuk membangunkan taksonomi kajian dan mengenalpasti jurang kajian bagi pendekatan ArSLR dan isyarat ArSL.

Universiti Malaya

Untuk rangkakerja pertama, isyarat input kepada rangkakerja ArSLR di pecahkan kepada isyarat tangan terbuka dan isyarat tangan tertutup. Kemudiannya, rangkakerja ArSLR mengenalpasti pendekatan yang sesuai bagi pengecaman isyarat tangan terbuka dan tertutup, seperti transformasi wavelet diskret yang diintegrasi dengan isyarat 1-D tandatangan dalam isyarat tangan tertutup manakala dalam isyarat tangan terbuka ia dikenakan melaui pertaburan tahap kawasan terkuantisasi, di mana ia di janakan daeipada matrik jangka-panjang. Pendekatan analisis statistik digunakan untuk mengira ciri deskriptor. Dalam rangkakerja ini, pendekatan multi-klasifikasi dilaksanakan dalam fasa pengecaman. Rangkakerja ini mempunyai kebolehan mengenalpasti tangan belakang statik jari nombor ArSL dan huruf dengan ketepatan pengecaman 95.63%.

Rangkakerja kedua menunjukkan bagaimana menggabungkan dua jenis berlainan VBR dan SBR untuk ArSLR yang tepat dibuat oleh penandatangan bebas. Pertamanya, kita bentangkan penyelesaian khusus untuk gabungan kalibarasi bagi dua alatan yang berlainan. Kemudian, satu set ciri diskriptor dibentangkan untuk sarung tangan pintar dan Microsoft Kinect. Akhirnya, set ciri yang dicadangkan ini di gunakan untuk multi-kelas *Support Vector Machines*. Hasil eksperimen, menunjukkan ketepatan pengecaman untuk dataset huruf Arab, nombor Arab, kata kata terpencil dan kata kata berterusan bagi pangkalan data ArSL ialah 99.38%, 99.42%, 99.15% dan 99.55% masing-masing. Kesimpulannya, pangkalan data dan rangkakerja yang dibangunkan ini berupaya meningkatkan prestasi bagi ArSLR .

Keywords: Bahasa Isyarat Arab, pengecaman berasaskan penglihatan, Pengecaman Berasaskan Sensor; penandatangan bebas.

Universiti Malaya

ACKNOWLEDGEMENTS

First and foremost ‘Alhamdulillah’, All praise is for Allah for the strengths He has given me and the generous blessings He has endowed me to successfully complete this study.

I express my deepest gratitude to my principal supervisors, Associate Prof. Dr. Rodina Ahmad, Dr. Nazean Jumhari, and Assoc. Prof. Dr. Mohammad A. M. Abushariah. Thank you for believing in me, and for your invaluable instructions, and for your unlimited support, encouragement, patience and understanding. Throughout this journey, you have been a great source of inspiration and strength to me.

I further thank my Father ‘Sami’, who has always provided me with his love and encouraged me to progress and succeed. May God bless him. My Mother ‘Abla’, the virtuous educator, who has always believed in me and lit my road with a special light by surrounding me with her prayers, love and compassion. May God Bless her. My Aunts and Sisters ‘Khloud, Dr. Sumaia, Eslam, Esra’a, Duaa’, Safa’ and Qamar, with whom I have always felt the warmth of brotherhood and who have always been good examples for me to follow so as to succeed and excel. May God protect them. Finally, I wish to thank all my family members and friends, who have always supported me with their encouragement and feelings to do my best, excel and go beyond.

Universiti Malaya

TABLE OF CONTENTS

ABSTRACT..... iii

ABSTRAK v

ACKNOWLEDGEMENTS vii

TABLE OF CONTENTS viii

LIST OF FIGURES xii

LIST OF TABLES xv

LIST OF ABBREVIATIONS xvii

CHAPTER 1: INTRODUCTION **01**
1.0 General Introduction 01
1.1 Research Background 02
1.2 Research Motivation 06
1.3 Problem Statement and Open Issues 07
1.4 Research Aim and Objectives 09
1.5 Research Questions 10
1.6 Research Scope 11
1.7 Research Methodology 11
1.8 Research Contributions 14
1.9 Thesis Outline 15
1.10 Summary 17

CHAPTER 2: LITERATURE REVIEW **18**
2.1 General Introduction 18
2.2 Hearing and Speech-Impaired 19
2.3 Gesture Definition 20
2.4 Gesture Recognition 20
2.5 Architecture of Gesture Recognition 22
2.5.1 Data Acquisition Phase 23
2.5.2 Pre-processing and Segmentation phase 24
2.5.3 Feature Extraction Phase 25
2.5.4 Classification Phase 27
2.6 Vision-based Gesture Recognition Devices 28
2.7 Sensor-based Gesture Recognition Devices 32
2.8 Automated Sign Languages Recognition (ASLR) 35
2.9 Background on Arabic Language and Arabic Sign Language 35
2.9.1 Difficulties in Arabic Language 35
2.9.2 Difficulties in Arabic Sign Language Recognition 37
2.9.3 Unified Arabic Sign Language Dictionary (UASLD) 42
2.9.4 Related works for ArSLR Approaches. 43
2.9.3.1 Alphabet Sign Language Recognition 43
2.9.3.2 Isolated Words Sign Language Recognition 44
2.9.3.3 Continuous Sign Language Recognition 44
2.10 Open Issues and Gaps for Arabic Sign Language Recognition 56
2.11 Summary 58

Universiti Malaya

CHAPTER 3: RESEARCH METHODOLOGY 59
3.1 Introduction 59
3.2 Literature Review Phase 60
3.3 Development Phase 65
3.3.1 Creating ArSL Database 65
3.3.1.1 Qualitative Method 67
3.3.1.2 Quantitative Method 68
3.3.2 Design and Model ArSL Frameworks 69
3.4 Evaluation Phase 77
3.5 Summary 81

CHAPTER 4: SYSTEMATIC LITERATURE REVIEW 82
4.1 Introduction 82
4.2 Research Design 83
4.2.1 Systematic Literature Review Research Questions 83
4.2.2 Search Process 84
4.2.3 Search Terms 85
4.3 Review Process 86
4.3.1 Inclusion and Exclusion Criteria 87
4.3.2 Study Selection 87
4.3.3 Data Extraction Form 88
4.4 Synthesis 88
4.5 Systematic literature review Results and Discussion 91
4.6 Summary and Discussion 122

CHAPTER 5: ARABIC SIGN LANGUAGE DATABASE 123
5.1 Introduction 123
5.2 Database Description 124
5.3 Different Condition Considerations 124
5.3.1 Lighting and background conditions for vision-based dataset 124
5.3.2 Data Organization 125
5.3.3 Signer Independence 125
5.4 Static backhand VBR database 128
5.5 Hybrid approach ArSL database 129
5.5.1 Smart Data Gloves 129
5.5.2 Microsoft Kinect 134
5.6 Data Collection and Tagging of Gestures 134
5.6.1 Data Collection 134
5.6.2 Data Tagging 135
4.6.2.1 Microsoft Kinect Tagging 136
4.6.2.2 Smart Data Gloves Tagging 136
5.7 Summary 139

CHAPTER 6: FRAMEWORK DESIGN AND IMPLEMENTATION 140
6.1 Introduction 140
6.2 Framework 1: Static backhand Arabic Sign Language Recognition 140
6.2.1 Pre-Processing Phase 142
6.2.1.1 Image Segmentation and Detection 142
6.2.1.2 Signs status categories 143
6.2.1.3 Shape Factor and Axis Ratio Method 149
6.2.2 Feature Extraction Phase 155

Universiti Malaya

6.2.2.1 Feature Extraction of Closed Hand Signs 155
6.2.2.2 Feature Extraction from Open Hand Signs 160
6.2.3 Recognition and Classification Phase 168
6.2.3.1 Naïve Bayes Classifier (NBC) 169
6.2.3.2 Multi-Layer Perceptron Neural Networks (MLP) 170
6.2.4 Multi-Classifier Combination 173

6.3 Framework 2: Hybrid Approach for ArSLR Based on Data Combining 174
6.3.1 Pre-processing and feature extraction phase from the smart glove 176
6.3.1.1 Pre-processing Phase to smart data gloves device 176
6.3.1.2 Feature Extraction from smart data gloves device 182
6.3.2 Hand Segmentation and Feature Extraction from the Microsoft Kinect 188
6.3.2.1 Hand Segmentation to Microsoft Kinect device 188
6.3.2.2 Extraction the relevant Features for Kinect data 198
6.3.3 Combining Features for Recognition and Classification Phase 209
6.4 Summary 211

CHAPTER 7: EVALUATION RESULTS AND DISCUSSION **212**
7.1 Introduction 212
7.2 Results of Framework 1: Static Backhand Arabic Sign Language Recognition ... 212
7.2.1 System Experimental Setups 212
7.2.1.1 Training and Testing Phases 213
7.2.1.2 Lighting Condition Test 213
7.2.1.3 Selection of the optimal number of hidden layers for MLP classifier... 214
7.2.1.4 Selection of the optimal number of features 215
7.2.2 Performance Metrics of Static Backhand Closed Signs 217
7.2.3 Performance Metrics of Static Backhand Open Sings 218
7.3 Results of Framework 2: Hybrid Approach for ArSLR 220
7.3.1 System Experimental Setups 220
7.3.1.1 Training and Testing Phases 220
7.3.1.2 Optimized parameters for pre-processing and feature extraction 221
7.3.1.3 Selecting the optimal extracted features set 222
7.3.2 Performance metrics results for hybrid Approach 227
7.4 Result Discussion 235
7.4.1 Experimental Results and Discussions for Framework 1 235
7.4.1.1 Recorded Accuracy and Evaluation Metrics for Applied Classifiers ... 235
7.4.1.2 Confusion matrices for Closed Hand Sign 236
7.4.1.3 Confusion matrices for Open Hand Sign 237
7.4.1.4 Error Analysis for open and closed hand signs cases 241
7.4.1.5 Comparison with related works 243
7.4.2 Experimental Results and Discussions for Framework 2 244
7.4.2.1 Confusion matrices for different input datasets 244
7.4.2.2 Comparison with previous works 259
7.5 Summary and Discussion 262

CHAPTER 8: CONCLUSION AND FUTURE RESEARCH WORK **264**
8.1 Overview 264
8.2 Responses to Research Objectives 265
8.3 Future Work 271

REFERENCES **272**

University of Malaya

LIST OF PUBLICATIONS AND PAPERS PRESENTED ……... 287

APPENDIX A ……. 288

Universiti Malaya

LIST OF FIGURES

Figure 1.1: Mapping the problem statement, research questions and objectives..... 10
Figure 1.2: Overview of Research Methodology Phases 12
Figure 2.1: Outlines the structure of chapter 2 18
Figure 2.2: Difference in Usage of Various Body Parts 22
Figure 2.3: Summarized the phases in architecture of gesture recogntion 23
Figure 2.4: Image Segmentation Techniques..... 25
Figure 2.5: Microsoft Kinect Sensor 29
Figure 2.6: Creative Senz3D Interactive Gesture Camera 30
Figure 2.7: Depthsense 325 30
Figure 2.8: Leap Motion Controller 31
Figure 2.9: P5 Glove 33
Figure 2.10: "Hand Talk" Data Glove Architecture 34
Figure 2.11: Hardware Specification of Smart Glove 34
Figure 2.12: ArSL Alphabet 38
Figure 2.13: Summary of the number of conducted ArSL research 58
Figure 3.1: Outlines the structure of chapter 3..... 59
Figure 3.2: Different Stages Based On the Following Methodology 61
Figure 3.3: Design of the proposed framework 1 70
Figure 3.4: Design of the proposed framework 2 74
Figure 3.5: Recognition and Classification Phase 76
Figure 3.6: Accuray metric..... 77
Figure 3.7: Sensitivity metric..... 78
Figure 3.8: Specificity metric..... 79
Figure 3.9: Precision metric..... 79
Figure 3.10: FPR metric..... 80
Figure 3.11: FNR metric..... 80
Figure 3.12: F1-Score metric..... 80
Figure 4.1: Flowchart of systematic literature review process 83
Figure 4.2: Systematic literature review activity 85
Figure 4.3: Systematic Review: Selection Procedure For Publications 86
Figure 4.4: Study Selection Process 88
Figure 4.5: Proportion of selected studies 89
Figure 4.6: References by category specifically dedicated to gesture recognition 90
Figure 4.7: Chronological distribution of studies over the years 90
Figure 4.8: Statistical analysis of the usage of sensor-based and vision-based 94
Figure 4.9: Taxonomy of hand gesture recognition 98
Figure 4.10: Several types of smart gloves 98
Figure 4.11: Classification of Hand Gestures Recognition 99
Figure 4.12: Hand Gesture Recognition Taxonomies 101
Figure 4.13: Example for static and dynamic gestures 103
Figure 4.14: A Representation of HGR 106
Figure 4.15: Statistical analysis of the usage of 3D model & appearance-based 108
Figure 4.16: Techniques being involved in VB hand gesture recognition 109
Figure 4.17: K-means when K=3 111
Figure 4.18: K-NN when k=3 112

Universiti Malaya

Figure 4.19: Mean Shift Algorithm Procedure for a Data Point xi ……. 113
Figure 4.20: SVM example ……..……...….. 114
Figure 4.21: Representation of HMM With 3 Hidden States and 3 Output States ….. 114
Figure 4.22: Aligning T and B, and Optimal Warping Path in Solid Squares ……..… 115
Figure 4.23: An Example of One-Layer Time-Delay Neural Network …… 116
Figure 4.24: Finite-State Machine Example ……..… 117
Figure 4.25: Statistical analysis of the main scopes and applications …….. 117
Figure 5.1: Outlines the structure of chapter 5…….. 123
Figure 5.2: The image acquisition environment …….. 129
Figure 5.3: Smart Data Gloves Package …….. 130
Figure 5.4: Smart Data Gloves. ……... 131
Figure 5.5: Smart Data Gloves Sensors ……... 131
Figure 5.6: Pitch and Roll Angles ……... 132
Figure 5.7: Axes (x, y and z) Reference ……..…… 133
Figure 5.8: Data Collection and Tagging Flow ……... 135
Figure 5.9: Data Tagging flowchart …… 138
Figure 6.1: The Proposed Framework for Static Backhand ArSLR ……..… 141
Figure 6.2: Image Segmentation Using an Iteration Thresholding …… .143
Figure 6.3: Closed Static Backhand ArSL Letters and Numbers …… 144
Figure 6.4: Open Static Backhand ArSL Letters and Numbers …….. 145
Figure 6.5: Input Image Example ……..……... 146
Figure 6.6: Hand Configuration Analysis …… 147
Figure 6.7: Left-Backhand Fingers State ……. 147
Figure 6.8: Fingertips in Extension and Flexion ……. 147
Figure 6.9: Hand Configuration Analysis …… 148
Figure 6.10: Region of Interest of the Close Hand Signs …… 149
Figure 6.11: Circularity Shape Factor for Closed Backhand Signs ……. 150
Figure 6.12: Circularity Shape Factor for Open Backhand Signs ……... 150
Figure 6.13: Flowchart for Proposed Sign Category Algorithm ……. 150
Figure 6.14: Ratio of the Major Axis to the Minor Axis ……. 151
Figure 6.15: Polarization Ellipse ……. 154
Figure 6.16: Major and Minor Axes of Open and Closed Hand Signs ……... 154
Figure 6.17: Finger Area Window ……...…. 156
Figure 6.18: Laplacian Filter In X Direction and Y Directions …….. 157
Figure 6.19: Example of Flexing and Extending Thumb …… 159
Figure 6.20: Edge Detection for Closed Hand Signs …….. 160
Figure 6.21: Adjacent and Spaced Fingers ……. 161
Figure 6.22: Hand Finger Directions …….. 161
Figure 6.23: Open Backhand signs with two fingers out …… 162
Figure 6.24: Finger Locations and Directions in 5*5 & 10*10 Block Masks …… 164
Figure 6.25: Examples of Coding Patterns Based On Occupied Blocks ……. 165
Figure 6.26: Run-Length Matrix Computation …… 166
Figure 6.27: Area Level Distribution ……..…… 166
Figure 6.28: Structure of a MLP classifier ……. 170
Figure 6.29: Example of MLP with Five Inputs …… 171
Figure 6.30: Framework of Combining Multi-Classifiers ……... 173

Universiti Malaya

Figure 6.31: Main phases for proposed hybrid approaches for automatic ArSLR ….. 175
Figure 6.32: Architecture of the proposed smart gloves sign language recognition ... 176
Figure 6.33: Original and resampling feature vectors of a signal for sentence 5 …… 178
Figure 6.34: Original and resampling feature vectors of a signal for sentence 7 …… 178
Figure 6.35: Original and resampling feature vectors of a signal for sentence 25 ….. 179
Figure 6.36: Normalization Effect for Sentence 5 ……... 180
Figure 6.37: Normalization Effect for Sentence 7 ……... 180
Figure 6.38: Normalization Effect for Sentence 25 ……. 181
Figure 6.39: An Example of Statistical Mean for Sentence 5 ……. 185
Figure 6.40: An Example of Standard Deviation for Sentence 5 …… 185
Figure 6.41: An Example of Statistical Mean for Sentence 7 ……. 186
Figure 6.42: An Example of Statistical Mean for Sentence 7 ……. 186
Figure 6.43: An Example of Statistical Mean for Sentence 25 ……..……. 187
Figure 6.44: An Example of Statistical Mean for Sentence 25 ……..…. 187
Figure 6.45: Architecture of the Proposed Microsoft Kinect for ArSLR …… 188
Figure 6.46: Acquired Color Image ……. 188
Figure 6.47: Acquired Microsft Kinect Depth Image …….. 189
Figure 6.48: Extracted Hand Samples from Microsoft Kinect …… 191
Figure 6.49: Output of the Gaussian Filter Applied on the Mask Corresponding ….. 192
Figure 6.50: Circle Fitted on the Hand with the Point $\mathbf{C_p}$ in Blue …….. 194
Figure 6.51: The Regions Hand Subdivision ……... 194
Figure 6.52: Angular sectors used for the computation of the ellipse ……. 195
Figure 6.53: The Fitting of the Ellipse Over the Palm …… 196
Figure 6.54: Reference System ……..… 197
Figure 6.55: Finger Edges F …… 199
Figure 6.56: Associated Histogram $L(\theta)$ with the Regions Corresponding …… 200
To the Different Features
Figure 6.57: Samples of aligned distance histogram $L_g\ (\theta)$ ……..……. 202
Figure 6.58: Samples of Curvature Descriptors for Three Sample Frames …….. 206
Figure 6.59: Regions Corresponding To the Various Area Features Shown ……..… 207
Over a Sample Gesture
Figure 6.60: Cross-validation techniques ……. 210
Figure 7.1: Optimal Number of Hidden Layers in MLP ……. 215
Figure 7.2: Recorded Accuracy Percentages Related To Each Data Type in Terms .. 226
Of Extracted Features Set
Figure 7.3: Accuracy of Classification Techniques …….. 236
Figure 7.4: Recognition Rates of Closed Backhand Signs ……. 237
Figure 7.5: Recognition Rates of Open Hand Signs …… 238
Figure 8.1: Outlines the structure of chapter 8…… 264

Universiti Malaya

LIST OF TABLES

Table 2.1: Syntax in Arabic Sign Language …….. 39
Table 2.2: Syntax variation between Arabic language and ArSL ……. 40
Table 2.3: Example of Syntax in Arabic Sign Language ……... 41
Table 2.4: Differences between Negative Syntax in Arabic Language and ArSL …….. 41
Table 2.5: Performance Comparison of Alphabet Sign Language for ArSL …….. 45
Research Efforts Using Vision-Based
Table 2.6: Performance Comparison of Isolated Words Sign Language for ArSL …… 46
Research Efforts Using Vision-Based
Table 2.7: Performance Comparison of Isolated Words Sign Language for ArSL …… 48
Research Efforts Using Sensor-Based
Table 2.8: Performance Comparison of Continuous Sign Language for ArSL ……..… 49
Research Efforts Using Vision-Based
Table 2.9: Performance Comparison of Continuous Sign for ArSL Research ……..… 50
Efforts Using Sensor-Based
Table 2.10: Performance Comparison of VB Static Sign Recognition for ……. 50
Other Languages
Table 2.11: Performance Comparison of VB dynamic Sign Recognition for ……. 51
Other Languages
Table 2.12: Performance Comparison of Active techniques and SB Sign …….. 52
Recognition for Other Languages
Table 2.13: Summarize the advantages and disadvantges for the ArSL related work... 54
Table 3.1: The Relation between Problem Statement, RO, RQ and Outcome …… 62
Table 4.1: Inclusion and exclusion criteria involved in the search process ……..…….. 87
Table 4.2: Data Extraction ……. 88
Table 4.3: Number of paper in final selection ……. 88
Table 4.4: Synthesis of hand gesture analysis approaches ……. 93
Table 4.5: Comparison between sensor-based and vision-based gesture recognition … 93
Table 4.6: Synthesis of representations techniques of VB gesture recognition …… 107
Table 4.7: Comparison of core computing applications …… 119
Table 4.8: Comparison of different VB gesture recognition devices and software ….. 121
Table 5.1: Arabic Letters Dataset ……. 122
Table 5.2: Arabic Numbers Dataset ……..… 126
Table 5.3: Arabic Isolated-Words Dataset …….. 126
Table 5.4: Arabic Continuous Dataset 1 …….. 127
Table 5.5: Arabic Continuous Dataset 2 ……... 128
Table 7.1: Accuracy of Lighting Conditions ……...…… 214
Table 7.2: Features Number for Closed Backhand Signs …… 216
Table 7.3: Features Number for Open Backhand Signs …….. 216
Table 7.4: Evaluation Matrices Values For Closed Backhand Signs …….. 217
Table 7.5: Evaluation Matrices Values For Open Backhand Signs …….…… 218
Table 7.6: Optimized Parameters and Recognition Rates (%) …….…… 222
Table 7.7: Accuracy of Alphabet Gestures in Terms of Extracted Features Set …..... 223
Table 7.8: Accuracy of Numbers Gestures in Terms of Extracted Features Set.....… 223

University of Malaya

Table 7.9: Accuracy of Isolated Words in Terms of Extracted Features Set …… 224
Table 7.10: Accuracy of Continuous Dataset 1 in Terms of Extracted Features Set... 224
Table 7.11: Accuracy of Continuous Dataset 2 in Terms of Extracted Features Set... 225
Table 7.12: Computed Evaluation Performance Metrics for Arabic Letters …… 227
Table 7.13: Computed Evaluation Performance Metrics for Arabic Numbers …… 229
Table 7.14: Computed Evaluation Performance Metrics for Arabic Isolated-Words... 230
Table 7.15: Computed Evaluation Performance Metrics for Continuous Dataset 1 … 231
Table 7.16: Computed Evaluation Performance Metrics for Continuous Dataset 2 … 233
Table 7.17: Accuracy of classification techniques based on all dataset size …… 236
Table 7.18: Confusion Matrix of Closed Backhand Signs …….. 237
Table 7.19: Confusion Matrix of Open Backhand Signs ……..… 228
Table 7.20: Misclassified Letter Signs By The Proposed System ……... 241
Table 7.21: Comparison among our Framework and other ArSL frameworks …… 242
Table 7.22: Comparison among our framework and related works on other languages 243
Table 7.23: Confusion matrix for Arabic Alphabet letters.…….. 245
Table 7.24: Confusion matrix for Arabic Numbers ……. 248
Table 7.25: Confusion matrix for Arabic Isolated-Words …… 250
Table 7.26: Confusion matrix for Arabic Continuous Dataset 1 …….. 254
Table 7.27: Confusion matrix for Arabic Continuous Dataset 2 …….. 257
Table 7.28: Comparison between Isolated words framework and other frameworks … 259
Table 7.29: Comparison between Continuous signs framework and other frameworks 260
Table 7.30: Comparison among our framework and related works on other languages 261

Universiti Malaya

LIST OF ABBREVIATIONS

ANFIS	: Adaptive Neuro-Fuzzy Inference System
ANNs	: Artificial Neural Networks
AR	: Accuracy Rate
ArSLR	: Arabic Sign Language Recognition
ArSLRT	: Arabic Sign Language Recognition Technique
ASLR	: Automatic Sign Language Recognition
CV	: Computer Vision
DTW	: Dynamic Time Warping
FN	: False Negative
FNR	: False Negative Rate
FP	: False Positive
FPR	: False Positive Rate
FSM	: Finite-State Machine
GR	: Gesture Recognition
HCI	: Human-Computer Interaction
HGR	: Hand Gesture Recognition
HGRT	: Hand Gesture Recognition Technique
HMM	: Hidden Markov Model
HSV	: Hue, Saturation, Value
HTK	: Hidden Markov Model Toolkit
IPA	: International Phonetic Alphabet
KNN	: K-Nearest Neighbor
MLP	: Multilayer Perceptron
NB	: Naïve Bayes

Universiti Malaya

NLP : Natural Language Processing

NNT : Nearest-Neighbor Technique

PCA : Principal Component Analysis

SBR : Sensor-Based Recognition

SL : Sign Language

SVM : Support Vector Machine

TDNN : Time delay neural network

TN : True Negative

TP : True Positive

UASLD : Unified Arabic Sign Language Dictionary

VBR : Vision-Based Recognition

VR : Virtual Reality

Universiti Malaya

CHAPTER 1: INTRODUCTION

1.0 General Introduction

Human-Computer Interaction (HCI) is a branch of computer science that is concerned with the design, implementation, and evaluation of interactive computing systems for human use. It focuses on the interfaces between human and computers, and involves several different fields including computer science, cognitive psychology, assistive technology, and communication. The HCI field has evolved over the last two decades to include Gesture Recognition (GR), text-based interaction systems, graphical user interfaces and voice user interfaces for speech recognition and speech synthesis (Saffiong, 2018).

Being among the main sub-fields in HCI, GR is one of the intuitive methods for interaction that plays a crucial role in Human-Robot Interaction (HRI), Automatic Sign Language (ASL), smart environments, spontaneous services, and many more (Mitra, Member, Acharya, & Member, 2007). In addition, GR is one of the central research fields of computer vision (CV), one of the core components in the thriving research field of HCI, and one of the central problems in the rapidly growing fields of HCI and human-robot interaction. GR is among the approaches that can possibly contribute to enhanced developments in HCI (Wachs et al., 2011; Mitra & Acharya, 2007). However, effective GR is still a challenging task until today.

GR is among the main considerations for deaf and hearing-impaired people, because they use Sign Language (SL) to communicate with each other and with normal people. The way in which people interact with each other relies on their sensory skills, gestures, postures, and body language. The ability to understand body gesture comes handy in case

Universiti Malaya

a person needs to communicate with another who is a deaf, hearing and speech-impaired, or does not know how to employ signs or body language that could facilitate communication and interaction (Al-Shamayleh, Ahmad, Abushariah, Alam, & Jomhari, 2018). In general, normal people have difficulties with SL, therefore; they need an interpreter that supports communication. Hence, Automatic Sign Language Recognition (ASLR) is needed to help hearing and speech-impaired people get integrated with social community (Huong et al., 2015). ASLR systems are used to convert SL into speech or text to enable communication with normal people who do not know these gestures. ASLR is an assistive technology that supports communication and interaction, which can be employed by various applications, and be used to assist in situations where a human interpreter may not be available. In addition, it can help to bridge the gap between people with and without hearing and speech-impairment.

This chapter presents the research background in Section 1.1. The research motivation is elaborated in Section 1.2. The problem description is stated in Section 1.3. In addition, Section 1.4 lists the research aim and objectives, followed by the research scope and research questions in Section 1.5 and Section 1.6, respectively. Section 1.7 provides the research methodology, whereas Section 1.8 presents the research contributions. The thesis outline is finally presented in Section 1.9.

1.1 Research Background

HCI is getting increasingly more important as computers has more influence in our lives. It is becoming more significant with the present advancement in computers and technology. Existing HCI devices such as mouse and keyboard are no longer able to satisfy the increasing demands anymore. Software developers are trying to make HCI faster, easier, and more natural by employing human-to-human interaction techniques such as GR.

Universiti Malaya

GR is an important way for people to express their feelings and thoughts, and also supports the knowledge delivered in their daily discussions. It has increasingly gained popularity due to its wide range of potential applications in the field of HCI, such as sign language recognition, multimedia application control, video games, and medical systems. Research efforts on GR are classified into communicative gestures, manipulative gestures, controlling gestures, and conversational gestures (Wu & Huang, 1999).

The importance of using Hand Gestures Recognition (HGR) for communication becomes clearer when sign language (SL) is considered. SL is a collection of gestures, movements, postures, and facial expressions corresponding to letters and words in natural languages (El-Bendary et al., 2010). SL assists a normal person to communicate with hearing and speech-impaired people (Antonakos, Roussos, & Zafeiriou, 2015; Al-Shamayleh et al., 2018). SL in general is a well-constructed code gesture, where each gesture has a particular significance. Gesture based communication is the salient mean of correspondence for hearing and speech-impaired people. With the progression of science and innovation, numerous systems have been created to minimize the issues of hard of hearing individuals (Antonakos et al., 2015).

For hearing and speech-impaired people to indulge in social activities and opportunities, SL is considered as the most natural, expressive, and useful way for daily communication. It includes the use of different parts of the human body namely; fingers, hands, arms, and head, to deliver ideas and knowledge. Majority of SL include upper parts of the body from waist level upwards (Cheok, Omar, & Jaward, 2019). Unfortunately, SL is a common language only among the hearing and speech-impaired community, not among normal people since very few are able to understand it. Thus, this

Universiti Malaya

limitation forms an obstacle when communication and interaction occurs between hearing and speech-impaired community and normal people.

Communication capabilities and social integration for hearing and speech-impaired have improved due to the advancement in ASLR, which is part of communicative gestures (Bhuyan, Ghoah, & Bora, 2006; Al-Shamayleh et al., 2018). Real time ASLR is a multidisciplinary research area involving image segmentation, pattern recognition, Computer Vision (CV), and Natural Language Processing (NLP). ASLR is a global issue due to the complication of the shapes of control. It requires an understanding of hands position, shape, motion and orientation (Kuznetsova, Leal-Taixe, & Rosenhahn, 2013; Molina et al., 2013; Lu et al., 2014). Functional ASLR systems can be utilized to create speech and text making the hearing and speech-impaired citizens more autonomous. The most difficult part of any ASLR system is to detect, analyze and recognize the simplest and easiest hand gestures that must be recognized in the picture (Kang et al., 2012).

ASLR is increasingly getting attention from researchers due to its applicability to various domains such as GR systems for hearing and speech-impaired people, Virtual Reality (VR), machine control in the industrial domain, and many others. ASLR research efforts can be divided into two main groups, namely; 1) Static sign language, and 2) Dynamic sign language.

ASLR is adopted in many sign languages such as America sign language (ASL) (Pansare, Gawande, & Ingle, 2012; Abhishek, Qubeley, & Ho, 2016), Indian sign language (Jayshree et al., 2013), Dutch sign language (Lichtenauer, Hendriks, & Reinders, 2008), Korean sign language (Kim, Jang, & Bien, 1996) and many others. As far as Arabic sign language (ArSL) is concerned, ArSL resources and research efforts are

Universiti Malaya

still limited and insufficient. Therefore, it is important to explore and investigate the capabilities of ASLR and apply them to ArSL in order to contribute to the state-of-the-art of ArSLR research in particular, and to the adaptation and integration of technology in terms of computerization and applications, and research for Arabic language in general.

It is important to highlight that Arabic language is the largest Semitic language, which is still in existence and one of the six official languages of the United Nations (UN). The number of first language speakers of Arabic exceeds 250 million, whereas its number of second language speakers can reach four times the number of first language speakers. It is the official language in 22 countries situated in Levant, Gulf, and Africa, Arabic language is ranked as fourth after Mandarin, Spanish and English in terms of the number of first language speakers (Abushariah et al., 2012; Elons, Abull-Ela, & Tolba, 2013; Abushariah, 2017). In spite of its importance, its research is lacking in many aspects, and research effort on ArSLR still requires more emphasis worldwide.

Unified Arabic Sign Language Dictionary (UArSLD) is the major form of ArSL. UArSLD includes both the Standard Arabic sign language (ArSL) and the Local Arabic sign language (LArSL). The ArSL is the formal standard of Arabic sign language, which is widely used across the Arab world. ArSL is the standard form of SL that is normally taught in schools and universities, and integrated with some television's news, shows, and programs. Tubaiz, Shanableh, & Assaleh (2015) stated that ArSL is the acceptable form of Arabic sign language for all hearing and speech-impaired people. LArSL varies from one country to another and includes the daily gesture Arabic sign language that deviates from the ArSL.

Universiti Malaya

Research efforts on ArSLR can be divided into Vision-Based Recognition (VBR) and Sensor-Based Recognition (SBR) approaches. The classification of VBR and SBR approaches form our primary research taxonomy in this thesis. Based on this classification, the research community has been focusing on recognizing hand configurations including movements, orientation, and position using VBR and SBR approaches. ArSL has three forms of signs including alphabet (letters), isolated signs, and continuous sentences, which form our secondary research taxonomy in this thesis. VBR and SBR approaches are discussed in detail in Chapter 2. Therefore, the primary and secondary taxonomies are discussed with sufficient detail in Chapter 2.

1.2 Research Motivation

Hearing and speech-impairment is a form of disability that affects more than 15 million people in the Arabic world (Aujeszky & Eid, 2015). To these people, communication is a substantially harder task compared to people not suffering from its consequences. These people often face two problems which are lack of convenient means of real-time digital sign language based communication, and low level of knowledge of sign language among people without hearing-impairment. Since more than 86% of hearing-impaired people come from low and middle income countries (Aujeszky & Eid, 2015), it is important to develop affordable ASLR systems, in order to maximize the potential reach of the system, which can be treated as one of the motivational factors for this research.

The fact that ArSL is the official language of 22 countries located in different regions of Levant, Gulf and Africa (Elhadj, Zemirli, & Ayyadi, 2013), and the growing demands and interests to ArSLR the research community has witnessed, it acts as important motivations for this research. Arabic language still needs more research to achieve matured ArSLR technology and highly performing applications. Therefore, selecting the

Universiti Malaya

ArSL for the purpose of this research is another important motivational factor due to its importance, complexity, limited research attempts, and acceptability for all hearing and speech-impaired Arab people.

Based on our literature investigation, there is an urgent need for language resources for ArSL within the Arab world and other countries around the world. This high demand for ArSL resources is also an essential motivation to conduct this research work.

1.3 Problem Statement and Open Issues

In order to address issues and problems pertaining to automatic ArSLR research, it is important to discuss them from technology and language perspectives (Database ArSL).

From technological perspective, major issues pertaining to automatic ArSLR can be summarised into signer-dependent versus signer-independent approaches, forehand versus backhand static sings, forms of signs ranging from alphabet signs, isolated word signs and continuous sentence signing as well as the use of VBR or SBR approaches individually versus the use of the VBR and SBR hybrid approach.

Based on literature investigation of the existing research attempts for designing and developing ArSLR systems for ArSL, there is a noticeable attention for signer-dependent approach using forehand static signs based on VBR approach. In addition, less attention and sometimes no attention for signer-independent approach using backhand static signs based on VBR approach. In addition, the main problem still remains in continuous ArSLR under signer-independent framework. Furthermore, VBR and SBR approaches have been used individually in most research attempts with less emphasis on fusing VBR and SBR approaches to produce a new hybrid approach to improve the recognition performance rate, and to overcome the existing difficulties with less recognition errors than other existing approaches, which can be produced by maximizing the advantages and minimizing the disadvantages of each approach. Finally, the absence of a unique

Universiti Malaya

framework that is capable of recognizing the three issues mentioned, forms of signs remains an issue for automatic ArSLR. It is noticed that the related research works normally develop a separate framework for each form of signs, which makes it difficult to hearing and speech-impaired people to use multiple frameworks to interact and communicate with normal people. Therefore, a broad and comprehensive framework is necessary for the hearing and speech-impaired community (Tuffaha, Shanableh, & Assaleh, 2015; Kong & Ranganath, 2014; Shanableh & Assaleh, 2011; AL-Rousan, Assaleh, & Tala'a, 2009; Shanableh et al., 2007).

From language perspective, major issues pertaining to automatic ArSLR can be summarized into ArSL resources collection, availability of ArSL datasets, availability of benchmark datasets to be used with VBR and SBR approaches, and adherence to ArSL standards. Based on ~~our~~ literature investigation of the related research works for designing and developing ArSLR systems, there is a remarkable emphasis on websites and personal writings for the collection of ArSL resources in a random manner. In addition, it is also recognized that the available ArSL datasets are scarce and very small in terms of size, and there exists no standard database for ArSL. Finally, and unfortunately, ArSL standards and measures as set by experts are mostly neglected, and there is no apparent adherence to these standards by researchers and developers of automatic ArSLR. The role and participation of ArSL experts is not clear from many related works in neither the design, preparation, and collection of ArSL resources and datasets, nor the design, development, and evaluation of automatic ArSLR tools, applications, and systems (Cheok, Omar, & Jaward, 2019; Al-Shamayleh et al., 2018; Amin et al., 2016; Mohamed Elhadj et al., 2013; Tuffaha, Shanableh, & Assaleh, 2015; Kong & Ranganath, 2014; Shanableh & Assaleh, 2011; AL-Rousan, Assaleh, & Tala'a, 2009; Shanableh et al., 2007).

Universiti Malaya

1.4 Research Aim and Objectives

This thesis aims to prepare and create a new SL database for ArSL along with two main frameworks, which can be used to design, develop and evaluate a hybrid approach, the signer-independent ArSLR. The first framework refers to a static backhand and signer-independent approach for ArSLR based on the VBR approach using a smartphone camera, which translates signs pertaining to letters and numbers into their corresponding texts. In addition, the second framework refers to a hybrid approach for the signer-independent continuous ArSLR development and evaluation, which integrates the VBR approach using Microsoft Kinect and the SBR approach with the Smart Data Gloves. Specifically, the research objectives are presented below.

- ✓ **RO1:** To create research taxonomies based on a comprehensive literature investigation that summarize the research efforts on ArSLR approaches and ArSL forms of signs.
- ✓ **RO2:** To create a comprehensive SL database for ArSL that covers vision and sensor-based approaches and forms of signs for ArSLR development and evaluation based on SL experts standard criteria and consensus.
- ✓ **RO3:** To model a static backhand signer-independent ArSLR framework based on VBR approach that translates signs of letters and numbers into their corresponding text using a smartphone camera.
- ✓ **RO4:** To model a hybrid approach, signer-independent continuous ArSLR framework that integrates the VBR approach using Microsoft Kinect and the SBR approach using Smart Data Gloves.
- ✓ **RO5:** To evaluate the performance of the modeled frameworks in terms of accuracy, sensitivity, specificity, precision, False Positive Rate (FPR), False Negative Rate (FNR), and F1-score, and compare the accuracy performance with the related works.

Universiti Malaya

1.5 Research Questions

Stated below are the research questions, which need to be dealt with in conducting this research and meeting the research aim and objectives:

- ✓ **RQ1:** What are the main taxonomies of ArSLR approaches and ArSL forms of signs?
- ✓ **RQ2:** What are the standard criteria used by SL experts for creating ArSL database?
- ✓ **RQ3:** How to model a static backhand and signer-independent ArSLR framework for ArSL letters and numbers?
- ✓ **RQ4:** How to model a hybrid approach for signer-independent continuous ArSLR framework?
- ✓ **RQ5:** What are the performance measures used to evaluate and compare ArSLR research works?

The relationship between the problem statement, research questions, and research objectives of this thesis and their mapping is shown in Figure 1.1.

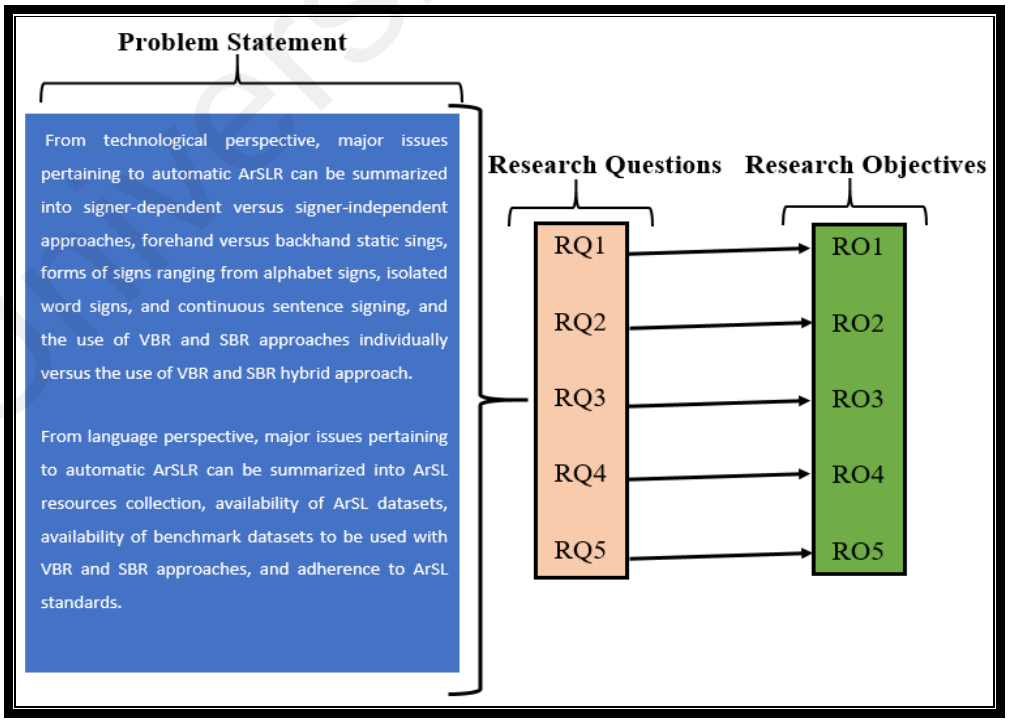


Figure 1.1: Mapping the problem statement, research questions and research objectives

Universiti Malaya

1.6 Research Scope

This research focuses on Standard ArSL as the main form of ArSL for creating the SL resources and database, which can be used for modeling, developing, and evaluating the two automatic ArSLR frameworks as formulated in this thesis, in which the first framework is based on the VBR approach using IPhone X camera and the second framework is based on a hybrid approach that integrates VBR and SBR approaches using Microsoft Kinect and Smart data gloves. It is important to highlight that the hybrid ArSLR framework is able to receive ArSL forms of signs with different data types, and transforms them into their corresponding text or voice. In addition, this thesis focuses on all required components including pre-processing and segmentation, features extraction, and classification and recognition, in order to model, develop, and evaluate a hybrid approach, signer-independent continuous ArSLR.

1.7 Research Methodology

This research is performed under three main phases which are literature review phase, development phase, and evaluation phase. In literature review phase, a systematic literature review method (Kitchenham et al., 2010) is used to investigate ArSLR scope and to create a research taxonomies and to identify the research gaps on ArSLR. In development phase, a development method (Construction of techniques) is used to design and model the ArSL framework. In addition, an empirical research methods (qualitative and qualitative method) are used to create and prepare the ArSL database. While, in evaluation phase, we used an experimental methods (Evaluation method) to evaluate our framework and compare to other related work. Figure 1.2 shows the main phases and provide brief explanation. More details in chapter three.

Universiti Malaya

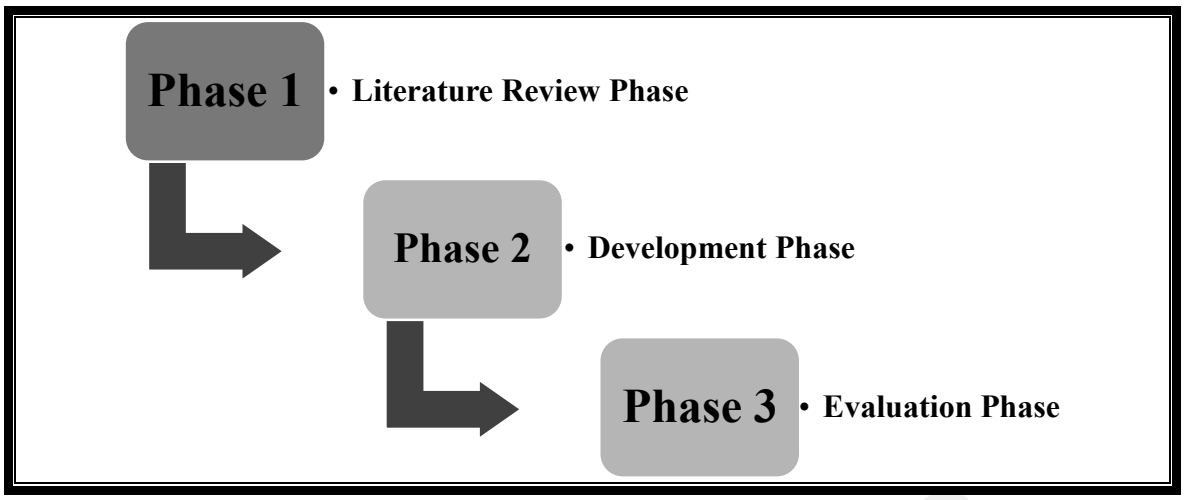


Figure 1.2: Overview of Research Methodology Phases

1. Conduct a comprehensive literature review on ArSL in order to identify open issues, research gaps, related theory, dominating techniques, and suitable tools and devices.
2. Formulat the primary taxonomy that summarizes ArSLR approaches, and the secondary taxonomy that summarizes the related works pertaining to ArSL forms of signs.
3. Identify research objectives and questions.
4. Form the research scope for research focus.
5. Propose and model a static backhand signer-independent ArSLR framework, and a hybrid approach, signer-independent continuous ArSLR framework, which can bridge the research gap for ArSLR research and development.
6. Prepare the questionnaire and interview qualitative methods to elicit and analyze the needs and requirements of SL experts and hearing and speech-impaired users that are related to the standard criteria for collecting the ArSL database.
7. Distribute the questionnaire and conduct the interview sessions with ArSL experts and practitioners who work at Prince Ali Bin Al-Hussein Club for the Deaf – Alzarqaa – Jordan, Al-Amal Primary Mixed School for the Deaf – Alzarqaa –

Universiti Malaya

Jordan, and the Support Division for Students with Special Needs at the Deanship of Student Affairs – The University of Jordan – Amman – Jordan.

8. Collect the necessary Arabic words and sentences from the SL experts and practitioners based on their feedback from the questionnaire and interview in accordance to the two proposed frameworks, which form the Arabic text database for further action.
9. Evaluate the Arabic text database by the SL experts and practitioners and conduct necessary pre-processing activities such as removing redundant words and sentences in order to produce a ready to use Arabic text database for ArSL purposes.
10. Perform the Arabic text database by SL experts and practioners, and conduct pre-processing activities such as checking the correctness of the recorded words and sentences in order to produce a ready to use ArSL database for ArSLR purposes.
11. Develop all required components including pre-processing and segmentation, features extraction, and classification and recognition, for a static backhand signer-independent ArSLR framework using VBR approach for signs of Arabic letters and numbers as performed on smartphone camera.
12. Develop all required components including pre-processing and segmentation, features extraction, and classification and recognition, for a hybrid approach, signer-independent continuous ArSLR framework using VBR and SBR approaches as performed on Microsoft Kinect and Smart Data Gloves.
13. Evaluate the performance of the ArSLR frameworks using relevant performance measures such as the Accuracy Rate, Sensitivity, Specificity, Precision, False Positive Rate (FPR), False Negative Rate (FNR), and F1-score.
14. Optimize and tune the training and testing parameters in order to improve the performance of the ArSLR frameworks.

Universiti Malaya

15. Identify the metrics and compare the performance of the ArSLR frameworks in terms of accuracy rate with that produced by other related research works.

1.8 Research Contributions

Outcome of this research has contributed in several ways as follows:

- ✓ Primary and secondary research taxonomies are formulated based on a comprehensive literature investigation spanning from 1996 until 2019, where the primary research taxonomy summarizes the research efforts on ArSLR approaches and the secondary research taxonomy summarizes the research efforts on ArSL forms of signs. These research taxonomies provide comprehensive information to the research community and furnish them with sufficient related works and background on GR in general and ArSLR in specific.

- ✓ A comprehensive ArSL database is successfully created, which meets SL experts' standard criteria and bridges the gap between currently available ArSL resources and the research community expectations and recommendations.

- ✓ The static backhand and signer-independent ArSLR framework based on VBR approach is modeled, which intends to provide a usable platform and support to hearing and speech-impaired users pertaining letters and numbers using their smartphone camera.

- ✓ The hybrid approach, signer-independent continuous ArSLR framework is modeled, which intends to provide a usable platform and support to hearing and speech-impaired users pertaining all ArSL forms of signs with different data types that are received using Microsoft Kinect and Smart Data Gloves. Based on our literature investigation and to the best of our knowledge, this framework is

Universiti Malaya

amongst the earliest research efforts that integrates VBR and SBR approaches for ArSLR.

- ✓ The development of highly accurate signer-independent ArSLR frameworks based on the newly created standard ArSL database is considered the final wholesome contribution of this thesis.

1.9 Thesis Outline

The thesis is documented with the following thesis organization:

- ✓ Chapter 1: General Introduction

  Chapter 1 is an introductory chapter that presents the research motivation and background reflecting the significance of study, problem description, research aim and objectives, research scope, research questions, and research methodology conducted.

- ✓ Chapter 2: Literature Review

  This chapter describes human gestures, and the definition of gesture, types, and the concept of gesture recognition with its applications. In addition, this chapter provides a comprehensive literature review on ArSLR. Arabic language forms and available database resources are highlighted. Taxonomies of ArSLR research, techniques, and tools used are also investigated with detailed comparison of systems performance.

- ✓ Chapter 3: Research Methodology

  Chapter 3 presents an overview of the general stages of the proposed frameworks, which includes research methodology, background information about the data collection, image processing, and their extraction in general.

Universiti Malaya

- ✓ Chapter 4: Systematic Literature Review

  Chapter 4 explains a systematic literature review that has been conducted. It review relevant literature in ASLR and ArSLR using a highly rigorous and systematic process.

- ✓ Chapter 5: Arabic Sign Language Database

  Chapter 5 discusses the procedure used to create our database in detail. The process of creating the sign language database is also described. In addition, informative analysis is performed on the collected data, which are also presented in this chapter.

- ✓ Chapter 6: Research Design And Implementation

  Chapter 6 presents the proposed ArSLR frameworks, which contain two frameworks of ArSLR.

- ✓ Chapter 7: Evaluation Results and Discussion

  Chapter 7 contains experimental results of the four development phases of the experimental work of this research based on default and modified parameters. It also analyzes and discusses the testing results and reports on the evaluation.

- ✓ Chapter 8: Conclusion and Future Research Work

  Chapter 8 summarizes the work accomplished and research findings, and discusses possibilities and recommendations for future direction of research.

- ✓ Appendix:

  Appendix A presents the Questionnaires was used to collect the ArSL database.

Universiti Malaya

1.10 Summary

This chapter presented the introduction of this thesis as well as the overview and the background of the problem to provide preliminary ground about the research. This chapter includes a specific statement of problem, defined research questions and research objectives to be achieved over the course of the research. In the next chapter, summary of the existing literature on gesture recognition and automatic Arabic sign language, will be presented.

Universiti Malaya

CHAPTER 2: LITERATURE REVIEW

This chapter introduces the general introduction for GR and ArSLR and summarises the architecture of GR. It also presents previous related works that address issues on ArSLR. The research gaps are identified and analysed to justify this thesis work. The subsequent introductory discussions on ASLR and ArSLR, which are helpful in identifying the current study research gaps, are also presented in this chapter. Figure 2.1 outlines the structure of this chapter.

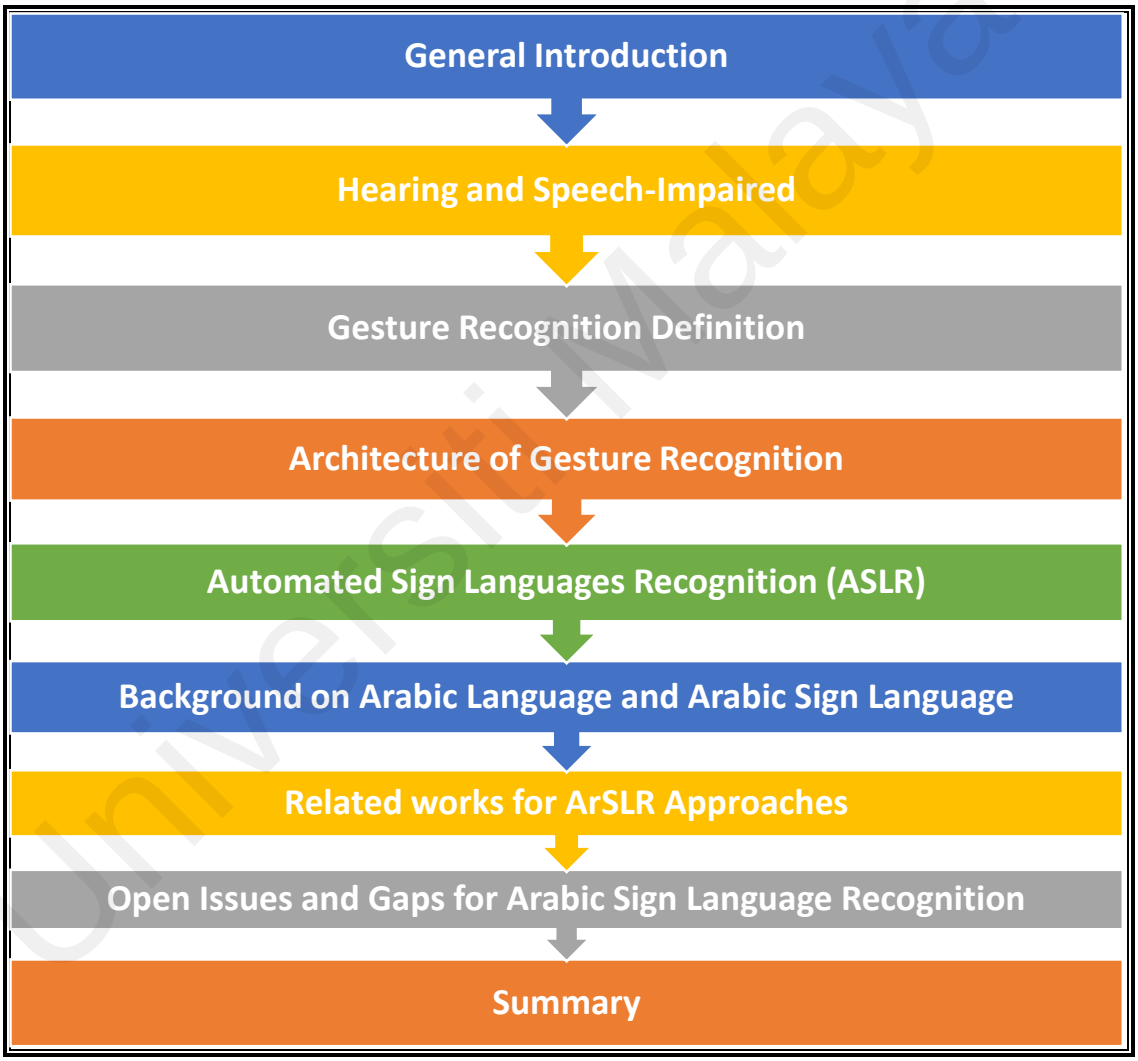


Figure 2.1: Outlines the structure of chapter 2

2.1 General Introduction

Due to the increased concurrent development in the interactive computing domain, the need for implementing an effective human computer interaction continues to grow. Gesture recognition (GR) is one of the approaches that could contribute to the enhanced

University Malaya

development in that domain. This process functions by releasing some gestures that could be recognized by the receiver (Al-Shamayleh et al., 2018).

Different researches have been conducted on gesture recognition techniques in the last two decades. These tools have been employed in various applications such as in web development, playing games and other tools that have been integrated to enhance the virtual environment for users. However, there is a difference between each tool that is used within each of these applications. Moreover, the constant decrease in the cost of laptops has create a major role in our everyday life now and mostly in the future. Thereby, the efficient use of computers requires the integration of more recognition interactions. Therefore, the Human Computer Interaction (HCI) technology has been receiving global attention in the past few years (Just, 2006).

The way in which people interact with each other relies on their sensation skills, as well as gesture, posture and body language abilities. The ability of understanding body gesture can be useful in case a person needs to communicate with another person who is deaf or speech-impaired and that person does not know how to unveil some sign or body language that could facilitate communication and interaction. The perception of gestures by a speech disabled individual should be recognized easily without the need of using sensors or wired marks on human body. Presently, the development of technology advanced devices has enhanced the process of gestures recognition.

2.2 Hearing and Speech-Impaired

The term "Hearing and Speech-Impaired" is a technically accurate description of someone who is having difficulty in hearing or who has no hearing (*Hearing Impaired Definition*, 2012). Hearing impaired individuals face a lot of difficulty in communicating with ordinary people. This should not be a reason for them to be separated from the community or set aside from practicing religion.

Universiti Malaya

Early studies of the intelligence of the hearing impaired suggested that as a group, these individuals averaged 10 to 15 IQ points below normal. Later studies indicated that if appropriate nonverbal or performance measures were used, the average scored were comparable with those of the hearing population (Domino & Domino, 2006). This research revealed that it is important to have proper way of teaching deaf students in order to increase their performance. However, another research also found out that an estimated 40% of children with hearing loss have additional impairments that can affect education and adaptive development (Schum, 2004).

2.3 Gesture Definition

The following definition of gesture can be found in the Oxford Advanced Learner's Dictionary: “Gesture - a movement of part of the body, especially, a hand or the head to express an idea or meaning” (Gesture). The word gesture is used for various phenomena that involve human movement, especially of the hands and arms. However, some of these gestures are interactive or communicative (Nehaniv et al., 2005). Gestures differ from pure functional movements, which can be achieved with other actions. Movements that show or symbolize something and contain a message are called gestures. For example, steering a car is a pure functional movement without information. However, gestures that describe the size of a round object by circling the hand contains information about the size of the object. Writing down words on a sheet of paper is a pure functional movement. Words contain the information, not the movement of the hand. Writing can be replaced by typing. Therefore, writing is not a gesture (Cadoz & Wanderley, 2000).

2.4 Gesture Recognition

Gestures are actions that can be used to interact and communicate with others. The recognition of gestures means that a person has the ability to understand what another

Universiti Malaya

person intends to deliver. Computers' ability to recognize human gestures is known as "gesture recognition." Gestures can be expressed through the physical movement of one's fingers, arms, hands or body to deliver a message (Lantz, 2011).

In computer science, gesture recognition has been employed to interpret and analyze human gestures through the use of mathematical algorithms. Gestures are performed through body movements but commonly originate from the face or hands. According to a study performed by Mitra and Acharya (Mitra et al., 2007), "Gestures are expressive, meaningful body motions involving physical movements of the fingers, hands, arms, head, face, or body with the intent of: 1) conveying meaningful information or 2) interacting with the environment." In order to determine and perceive body gestures, it is necessary to integrate the theoretical aspects involved in a particular domain along with the powerful use of hardware devices.

Human perception of body gestures is different and more spontaneous compared to a machine's ability to recognize information. Humans have a better ability to recognize gestures spontaneously through the use of senses, such as hearing and vision.

The universal use of a combination of gestures by integrating object screen devices that can help perceive gestures was developed by Ivan Sutherland in Sketchpad in 1963 (Myers, 1998). On the other hand, modern advancements of devices and equipment have facilitated the process of gesture recognition through the implementation of algorithm structures.

The hand is the body part that is most frequently involved in gesturing in contrast to other body parts. Thereby, the hand is the most suitable communication part and best suited for integration in HCI. Figure 2.2 illustrates the different body parts or objects used for gesturing (Al-Shamayleh et al., 2018).

Universiti Malaya

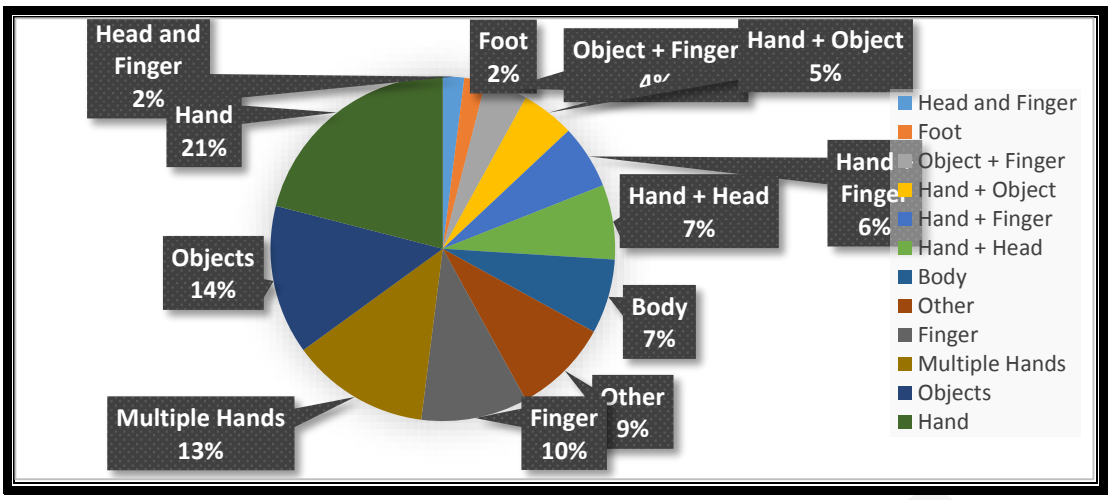


Figure 2.2: Difference in usage of various body parts or objects involved in gesturing

Humans naturally use gesture to communicate. Young children can readily learn to communicate with machines before they learn to talk (Kjeldsen, 1997). Gestures are frequently used as a means of communication. Gestures are used for everything, from pointing at a person to getting their attention to conveying information about space and characteristics. Gestures are not merely used as qualifiers for oral communication but actually form part of the main language generation process (Shet, Prasad, Elgammal, Yacoob, & Davis, 2004).

Gestures in modern societies are everything, from a smile to hand-and-arm movements. Most people add meaning to their words by drawing pictures with their hands, which is done subconsciously and is therefore hard to suppress. They do it even when they speak on the phone or when they talk to themselves. Deaf or mute people might use sign language as the sole means to communicate. Instructors in diverse fields such as military or aerobics use arm signals to give commands (Winnemöller, 1999).

2.5 Architecture of Gesture Recognition

ASLR and GR comprises of four major phases in order to identify the correct gestures, namely; 1) Data Acquisition, 2) Pre-Processing and Segmentation, 3) Feature Extraction and finally 4) Classification. Figure 2.3 is summarized the phases in architecture of gesture recogntion.

Universiti Malaya

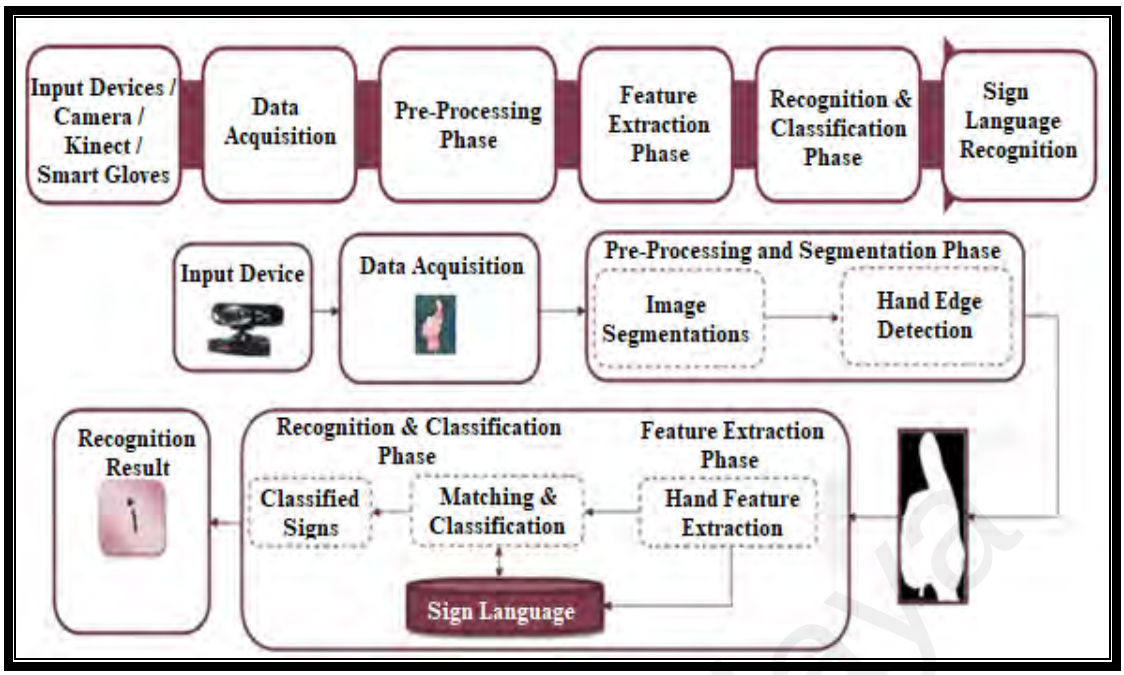


Figure 2.3: Summarized the phases in architecture of gesture recogntion.

Once the hand image is captured using a suitable input device, the image is then segmented in order to find the location of the hand from the (cluttered) background and any other parts of the body and the image is then processed in order to remove noises, detect edges/contours, and normalize for generating the desired model.

The features such as shape, position, orientation, movements and location of the hand are then extracted for classification purposes from the segmented and pre-processed images and gestures. Finally, input images are recognized as meaningful gestures based on gesture modeling and analysis (Aujeszky & Eid, 2015; Surve Pranjali, 2015).

2.5.1 Data Acquisition Phase

The more perfect the data acquisition is, the better and more efficient hand gesture recognition can be (Surve Pranjali, 2015; Cooper, Holt, & Bowden, 2011). Frame of images is normally acquired as input, which is collected using webcam, digital camera, stereo and thermal camera. Leap Motion Controller and Microsoft Kinect can also be used as advanced active techniques. It is important to note that 3D cameras such as stereo

Universiti Malaya

cameras, Microsoft Kinect and LMC can collect depth information (Cheok et al., 2019). This can be done through selecting suitable input device to be used in data acquisition. Many input devices are available to be used to acquire data; including but not limited to the hand images that are acquired from drawings, Kinect 3D sensor, marker, data gloves, stereo camera and webcam (Surve Pranjali, 2015; Hasan & Mishra, 2012; Chen et al., 2017). ASLR can acquire data by using two different approaches, namely; 1) vision-based 2) Sensor-based approaches.

2.5.2 Pre-processing and Segmentation phase (Gesture Modeling Phase)

This phase is vital as to conduct successful gesture recognition. The various data collected from such devices should be modeled in a proper way through appropriate application type. The respective modeling shall be carried out via different steps, namely; 1) pre-processing and segmentation, 2) noise/filter removal, 3) contour/edge detection, and lastly 4) normalization (Wu & Huang, 2001).

Image pre-processing phase is applied on image or video inputs for system performance improvement and noise reduction using median and Gaussian filters followed by morphological operations in the acquired images or videos. In some studies, the captured gestures are downsized into a smaller resolution prior to subsequent stages, which is reduces the resolution of the input image and is able to improve the computational efficiency. For enhancing the contrast of the input images taken under different environment to uniform the brightness and illumination of the images, histogram equalization is used (Cheok et al., 2019).

Segmentation phase is conducted to partition images into multiple distinct parts, where the Region of Interest (ROI) is segmented from the remaining of the image. Segmentation methods can either be contextual or non-contextual, where contextual segmentation considers the spatial relationship between features such as edge detection

Universiti Malaya

techniques, and a non-contextual segmentation groups pixels based on global attributes (Cheok et al., 2019). In Vision-based Arabic Sign Language Recognition (ArSLR) system, hand segmentation is most important and challenging step towards gesture recognition. Figure 2.4 is presents the main classification of image segmentation.

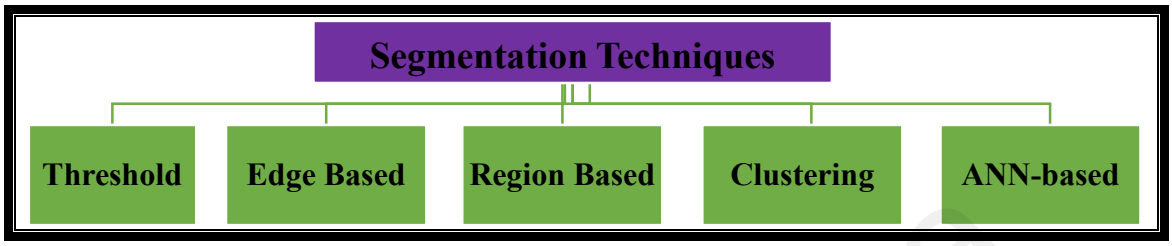


Figure 2.4: Image segmentation techniques

2.5.3 Feature Extraction Phase

Perfect segmentation process leads to perfect features extraction process (Bairagi & Kapse, 2018; Zaman Khan, 2012). The features are considered basic elements in order to establish hand gesture recognition. In gesture recognition context, the features extracted should contain relevant information from the hand gestures input and represented in a compact version which serves as an identity of the gesture to be classified apart from other gestures.

We can use huge number of features extraction; i.e. shaping, distance, motion, contour, textures, centre of gravity, orientation, velocity and etc. For hand gesture recognition, we can identify hand gesture by using geometric features such as finger detections, hand contour and fingertips. However, these features are not available at all times, nor reliable, due to illuminations and occlusions. On the other hand, there are some of non-geometric features; i.e. Texture, silhouette, and colour that are also available for recognition (Surve Pranjali, 2015; Murthy & Jadon, 2009). The important feature extraction techniques using in sign language, Shift-invariant feature transform (SIFT), Principal component analysis (PCA), Linear discriminant analysis (LDA), Speeded up robust feature (SURF), Convexity defects and K‑curvature (Cheok et al., 2017; Bay, Tuytelaars, & Van, 2006; Rekha et al., 2011).

Universiti Malaya

1. **Shift-invariant feature transform (SIFT)** describes an image by its interest points, where detection requires multi-scale approach. At each level of the pyramid, the image is rescaled and smoothed using Gaussian function (Cheok et al., 2019). SIFT is a scale and rotation invariant feature extraction technique introduced by Lowe (Lowe, 2004).

2. **Principal component analysis (PCA):** is a mathematical operation which utilizes orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of uncorrelated variables called principal components (G. Kumar & Bhatia, 2014).

3. **Linear discriminant analysis (LDA):** Both LDA and PCA approaches find the linear combination of features that best describe the data. LDA maximizes class separability by finding linear combination of features that best discriminate classes of objects (Cheok et al., 2019) (Delac, Grgic, & Grgic, 2005). PCA only finds the direction of maximal variance among features and does not consider the difference in classes (Suriya, Sathyapriya, Srinithi, & Yesodha, 2016). LDA can be applied as a linear classifier and a dimensionality reduction method (Cheok et al., 2019).

4. **Speeded up robust feature (SURF):** is developed based on SIFT, where SIFT constructs scale pyramid, convolving the upper and lower scales of the image with difference-of-Gaussian (DoG) operator and searching the local extreme in scale space. Meanwhile, SURF scales filter up instead of iteratively reduce the image size. In SIFT, Laplacian of Gaussian (LoG) is approximated with DoG for finding scale-space (Cheok et al., 2019).

5. **K- curvature and Convexity defects:** involve finding convex hull, convex and defects, center of the palm, angle between fingertips and palm center. Several researches use global features with convexity defects to identify the gestures (Cheok et al., 2019).

Universiti Malaya

6. **Features extraction in frequency domain:** involves transformation of time domain input data into frequency domain. This includes Cosine Transform, Fourier Transform and Wavelet Transform (Cheok et al., 2019).

2.5.4 Classification Phase

The last phase of recognition system is represented by classification of hand gestures. We have to consider making efficient classifier and recognition method available herein which shall represent a very important role in any gesture recognition system. This phase goes forward in line with the machine learning and pattern recognition field. Hand gestures can be classified using rule-based and machine learning-based approaches (Surve Pranjali, 2015; Murthy & Jadon, 2009), which are further discussed in the following sections.

A. **Rule-based Approach:** In this approach, a number of manually encoded relationships or so called rules between feature inputs are developed. Therefore, features are extracted from the input gesture, and compared against the encoded rules. The matched rule is finally resulted as the gesture. This approach lacks the human ability for encoding the rules and limits the recognition process success (Murthy & Jadon, 2009).

B. **Machine Learning-based Approach:** Due to the shortcoming of rule-based approach in gesture recognition as stated in the previous section, many researchers have turned into machine learning approach in order to find mappings between high-dimensional feature sets and gestures. Machine learning-based approach considers gesture as the result of stochastic processes.

Conditional random fields (CRF) (Lafferty, McCallum, & Pereira, 2001), Naïve Bayes Classifier (NBC) (Duda, Hart, & Stork, 2012), K-means (Kanungo et al., 2002), K-

Universiti Malaya

nearest neighbor (K-NN) (Thirumuruganathan, 2010), Mean-shift clustering model (Derpanis, 2005), Support Vector Machine (SVM) (Burges, 1998), Hidden Markov Model (HMM) (Ramage, 2007), Dynamic Time Warping (DTW) (Senin, 2008), Time-Delay Neural Network (TDNN) (Wohler & Anlauf, 1999), Finite-State Machine (FSM) (Holzmann, 1990), and Artificial Neural Networks (ANNs) are used in sign language and gesture recognition. Other researchers have also employed Gaussian mixture distribution to achieve gestures classification and Euclidian distance measure (Murthy & Jadon, 2009; Chaudhary et al., 2011; Choi & Kim, 2017; Al-Shamayleh et al., 2018).

2.6 Vision-based Gesture Recognition Devices

In general, gesture recognition algorithms to work efficiently we need to capture the gestures using proper instruments such as cameras, imaging and tracking devices. However, gesture recognition is commonly and widely used within the gaming community to ubiquitously interact with the games.

A- Microsoft Kinect

Kinect was originally released as a motion sensing input device by Microsoft for the Xbox 360 video game console. Later Microsoft released a version for Windows PCs which turned out as a device to be used for purposes other than developing games. (Tang, 2011) proposed and implemented a novel method for recognizing hand gestures using RGB (Red, Green and Blue color model) and depth data from Microsoft's Kinect sensor. Figure 2.5 shows the Microsoft Kinect.

Universiti Malaya

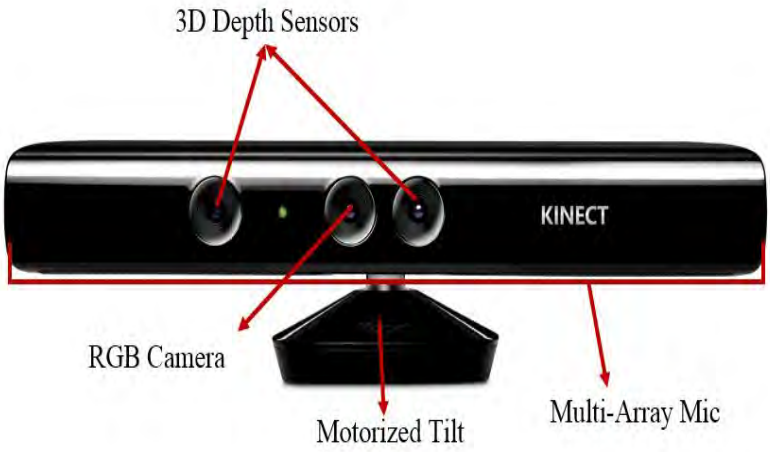


Figure 2.5: Microsoft Kinect Sensor

Although Microsoft Kinect is among the most popular vision-based device available on the market, it also has its own drawbacks. When it comes to detection of finger movements, Kinect is not a good choice. This is due to the low-resolution of the Kinect depth map which works well when tracking a large object such as the human body; it is however difficult to track fingers of human hand (Ren, Yuan, & Zhang, 2011). Due to these limitations, this study cannot entirely rely on the results from the camera.

B- Creative Senz3D Interactive Gesture Camera by Intel

The Senz3D camera by Intel detects hand gestures and head movement, which enables developers to create fun interactive games and interesting ways to interact with the computer.

In comparison to Microsoft Kinect, the Senz3D is made for only one user at a time, and it can get confused when multiple people or hands appear in front of it at the same time. It also has a relatively short maximum range of just 1 meter. The camera can also capture right down to the contours of face, creating many new applications and game development opportunities for professional and amateur developers.

Universiti Malaya



Figure 2.6: Creative Senz3D Interactive Gesture Camera

In addition to finger tracking, the camera has the ability to recognize and differentiate between static hand poses, and moving hand gestures. It also boasts facial detection and analysis and voice recognition. The SDK available for developers is called Perceptual Computing SDK. On the official website of Intel it was stated that "with the Intel® Perceptual Computing SDK 2013, developers can create exciting new applications that take advantage of the SDK's core capabilities: speech recognition, close-range hand and finger tracking, face analysis, augmented reality, and background subtraction.

C- Depthsense 325 Camera by SoftKinetic

Depthsense 325 is a 3D camera developed by SoftKinetic, a Belgian company which develops gesture recognition hardware and software for real-time range imaging (3D) cameras (such as time-of-flight cameras). SoftKinetic is a company that provides gesture recognition solutions based on its technology to the interactive digital entertainment, consumer electronics, health & fitness, and serious game industries. SoftKinetic technology has been applied to interactive digital signage and interactive television.



Figure 2.7: Depthsense 325

Universiti Malaya

In the official website of SoftKinetic, it was stated that their camera will "interact with devices at short range, using your hands and fingers, but without touching a screen, keyboard, trackball or mouse. The pocket-sized DS325 camera sees in 3D (depth) and high-definition 2D (color) enabling one to build precise finger and hand tracking into a wide range of applications including laptop, desktop and automotive applications. One of the disadvantages in using this camera as a developer is because of the limitation in the free SDK.

D- Leap Motion Controller

The Leap Motion Controller tracks all 10 fingers up to 1/100th of a millimeter but it can only detect the movements of hand within a very close range. It has a superwide 150° field of view and a Z-axis for depth. This means it can detect hands in 3D, just like we do in the real world.



Figure 2.8: Leap motion controller

One of the advantages of Leap motion controller is the availability of SDK for Microsoft Windows, Macintosh OSX and Linux. This will enable more developers to utilize this SDK and can contribute to the developer's community. Leap motion controller detects the movements of fingers using two monochromatic IR cameras and three infrared LEDs, the device observes a roughly hemispherical area, to a distance of about 1 meter (3 feet). The LEDs generate a 3D pattern of dots of IR light and the cameras generate almost 300 frames per second of reflected data. This data is then sent through a USB cable to the host computer, where it is analyzed by the Leap Motion controller software using a complex algorithm.

Universiti Malaya

2.7 Sensor-based Gesture Recognition Devices

Sensor-based approach can be used to overcome the limitations of Kinect camera by accurately tracking the movements and states (i.e. open and closed) of fingers. Wired-gloves or data-gloves provide very accurate measurements of hand shape and movements of the fingers. In Sensor-based systems, the common technique for hand pose tracking is to instrument the hand with a glove which is equipped with a number of sensors to provide input to the computer about hand position, orientation, and flex of the fingers using magnetic or inertial tracking devices. There are various types of gloves which can be used to track the movements of hand and fingers.

One of the first data glove was designed by Ryan Patterson in the year 2002 (Braddock, Rizzolo, Thompson, & Bell, 2004). He used a golf glove and used flex sensors and accelerometers to detect the hand and finger movements. He developed a simple sign language translator using the information received from the glove. The glove is integrated with a small microcontroller on the back of the hand which converts the change in the electrical current into digital signals and transmits them wireless to a computer. The computer then reads the numerical values and converts them into the alphabetical letters which appear on the screen.

Some of the famous wired or data-gloves are P5 glove (see Figure 2.9), PEREGRINE Wearable and 5DT Data Glove 5 Ultra. These peripheral devices, based upon proprietary bend sensors and remote tracking technologies that provides users total intuitive interaction with 3D and virtual environments. With its accuracy in detecting the movement of fingers and orientation of the hand, wired gloves will be a good choice to combine the vision-based gesture recognition to increase the accuracy and efficiency.

A- P5 Data Glove

The P5 Data Glove is a 3D input device capturing finger-bend and relative hand-position that enables intuitive interaction with 3D environments. The Essential Reality P5

Universiti Malaya

glove fits over the hand and senses all its movements in three dimensions, becoming the interface to a PC. The P5 uses Essential Reality's proprietary bend sensor and tracking technologies to enable full interaction with a 3D environment. It is also fully USB compliant, the first generation P5 is designed for compatibility with the Microsoft Windows 98, Windows ME and MS 2000 operating systems.



Figure 2.9: P5 Glove

B- "Hand Talk" data glove

A group of engineering students at Carnegie Mellon University invented a data glove named "Hand Talk" which is a sensor equipped glove that translates finger and hand gestures into spoken words. This was recognized as the first demonstrator model to show the functionality based on a limited vocabulary of 32 words. Sensors in the glove pick up gestures and transmit the data wirelessly via Bluetooth to a cell phone which runs Text to Speech software. The sensor data are converted first into text and then to voice output. A person not knowledgeable in Sign language can listen via the cell phone what the other person is saying in Sign language form. The main advantage with this design was its simplicity and the cheap components these students used to create this amazing and truly interactive glove that could help to improve greatly the communication barrier between deaf persons and people (Gujrati, Singh, Soral, & Ambikapathy, 2013).

Universiti Malaya

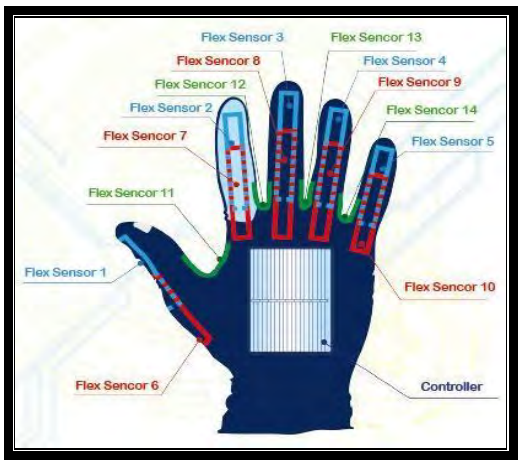


Figure 2.10: "Hand Talk" data glove architecture

C- Smart Glove

"Smart Glove" is also a similar data glove which was engineered by Muhammad Hisyam Bin Idris (Idris, 2012), an engineering student in *Universiti Teknologi Malaysia* for his undergraduate final year project. The glove incorporates with 5 flex sensors for 5 fingers, an accelerometer and a Bluetooth sender for transmitting sensor data as shown in Figure 2.11.

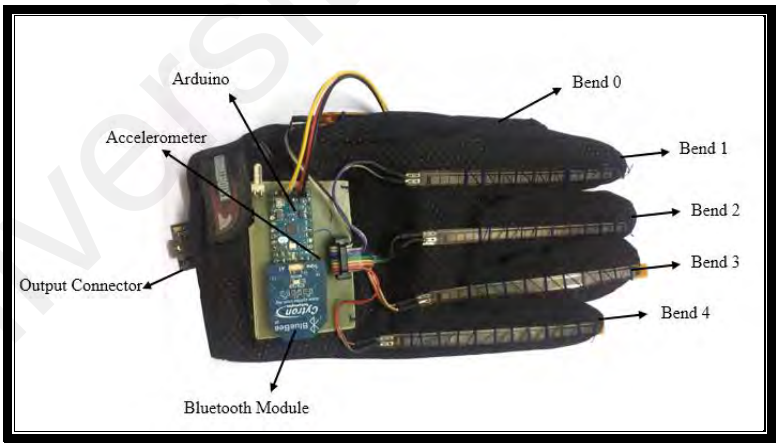


Figure 2.11: Hardware specification of Smart Glove

According to (Idris, 2012), the main controller chosen for the project was the Arduino Leonardo, which is a microcontroller board based on the ATmega32u4. This microcontroller has 20 digital input output pins including 7 pins that can be used as PWM outputs and 12 pins as analog inputs (Idris, 2012).

Universiti Malaya

2.8 Automated Sign Languages Recognition (ASLR)

ASLR is a visible gesture language which incorporates face, hands and arms to pass on considerations and implications. It is produced in deaf community to integrate deaf and hearing impaired citizens with their peers, friends, relatives, and the society at large. They utilize their hands, face, arms, and body for communication. It is important to note that there is no international Sign Language (SL) (M. Abhishek & Divakar, 2017; Raees & Ullah, 2014; Avraam, 2014). In fact, every country or region has its own SL, which is discussed further in this chapter. These SLs can be different from one country to another in terms of syntax and grammar. The SL that is used in Australia for instance is known as Australia Sign Language (Auslan), whereas the SL used in Japan is known as Japanese Sign Language (JSL), and the SL used in the Arab world is known as Arabic Sign Language (ArSL), and so on so forth (Shanableh & Assaleh, 2011; Sami M Halawani et al., 2013; Wei et al., 2016). The SL is used among the deaf and hearing impaired citizens, including friends and relatives who have the same impairment, as well as the interpreters. In addition, the SLs are normally not familiar out of the scope of these communities; hence communication constraints emerge between deaf and hearing impaired citizens with those outside the scope of their SLs.

2.9 Background on Arabic Language and Arabic Sign Language

This section provides thorough background on the difficulties of using Arabic language as well as the difficulties in Arabic sign language. A taxonomy of ArSLR approaches is also provided.

2.9.1 Difficulties in Arabic Language

Being a Semitic language, and syntactically and morphologically rich language, vocabularies in Arabic language are expected to be very huge and larger than many other languages (Kirchhoff et al., 2003) (Habash, 2010). Arabic language consists of 28 letters,

Universiti Malaya

and 6 short and long vowels. Arabic text is written from right to left cursively, which is the most noticeable feature of Arabic writing (Althobaiti & Lu, 2017).

Being a highly inflected agglutinative and morphologically rich and productive language, words in Arabic language can be concatenated using certain conjunctions, prepositions, articles, and pronouns through inserting prefixes and suffixes to the word stem, which leads to a huge list of potential word forms and increases the out-of-vocabulary rate for applications such as Automatic Speech Recognition (ASR) (Kirchhoff et al., 2003; Alghamdi, Elshafei, & Al-Muhtaseb, 2007; Abushariah, 2017).

Arabic language is highly demanded by the NLP community due to socio-political importance and NLP challenges, which are presented by its dialect differences, diglossia, complex morphology, and non-transparent orthography (Soudi, Neumann, & Van den Bosch, 2007). Therefore, Arabic language is one of the most complex natural languages especially when processed using machine learning. The complexity of Arabic language is due to the followings:

1. Arabic language consists of three main forms namely (1) Classical Arabic (CA), (2) Modern Standard Arabic (MSA), and (3) Colloquial or Dialectal Arabic (DA). Each form has some distinct characteristics (Abushariah et al., 2012; Elmahdy, Gruhn, Minker, & Abdennadher, 2009).
2. Diacritical marks is MSA text are mostly omitted and readers must infer missing diacritical marks from the context (Elmahdy et al., 2009). However, mistakes in identifying the right diacritics may result in a different meaning of the same word.
3. The lack of dedicated letters to represent short vowels can change the form of the letter depending on its position in the word (Farghaly & Shaalan, 2009).
4. Every Arabic letter can be written in various ways that can result in two or four forms according to its position in a word, which could be at the beginning of a word, middle, end, or stand-alone (Althobaiti & Lu, 2017).

Universiti Malaya

5. The absence of capitalization and the inconsistency and irregular use of punctuation marks cause serious challenges to many NLP tasks such as tokenization, part of speech tagging, name entity recognition, parsing, and many others (Soudi et al., 2007; Farghaly & Shaalan, 2009).
6. The NLP tools that were initially developed for other languages are not easily adaptable to Arabic language, which is due to the specific features of Arabic language itself (Farghaly & Shaalan, 2009).
7. Typographically, Arabic character set is different from the Latin character set. In order to view Arabic fonts and process Arabic scripts correctly, computers need to be Arabic enabled (Attia & Somers, 2008).
8. Arabic word formation involves attaching prefixes, suffixes, and infixes to stems with many morphophonemic processes. This makes root recovering and internal analyzing structure of Arabic words a real challenge for computer processing and linguistic theory and description (Soudi et al., 2007).

2.9.2 Difficulties in Arabic Sign Language Recognition

Sign language may vary according to the country or even to the region, because it is not considered as a universal language. There are many efforts to establish and standardize the sign language in the Arab World such as Gulf States, Jordan, Tunisia, Algeria, Iraq, Palestine, Syria, Sudan and Djibouti, in order to spread it among members of the deaf community. Consequently, many sign languages are produced as many as the Arabic-speaking countries using the same sign alphabets (El-Bendary et al., 2010; S.M. Halawani, 2008). Figure 2.12 shows the gestures used in Arabic sign language alphabets.

Arabic sign language contains more than nine thousand signs and gestures, and also uses twenty six (26) static hand gestures and five (5) dynamic hand gestures to represent

Universiti Malaya

the Arabic language alphabets (Aujeszky & Eid, 2016) (El-Bendary, Zawbaa, Daoud, Hassanien, & Nakamatsu, 2010b) (M. Tolba, Samir, & Abul-Ela, 2012).

In Arab countries, assisting deaf students to identify suitable environments is the main aim of hard-of-hearing schools. One gesture (sign) is produced through combining specific and clear hand shapes, palm orientations and movements. However, sign language is considered as universal native language for hearing impaired and deaf. ArSL is considered as the official language for deaf people in the Arab countries.

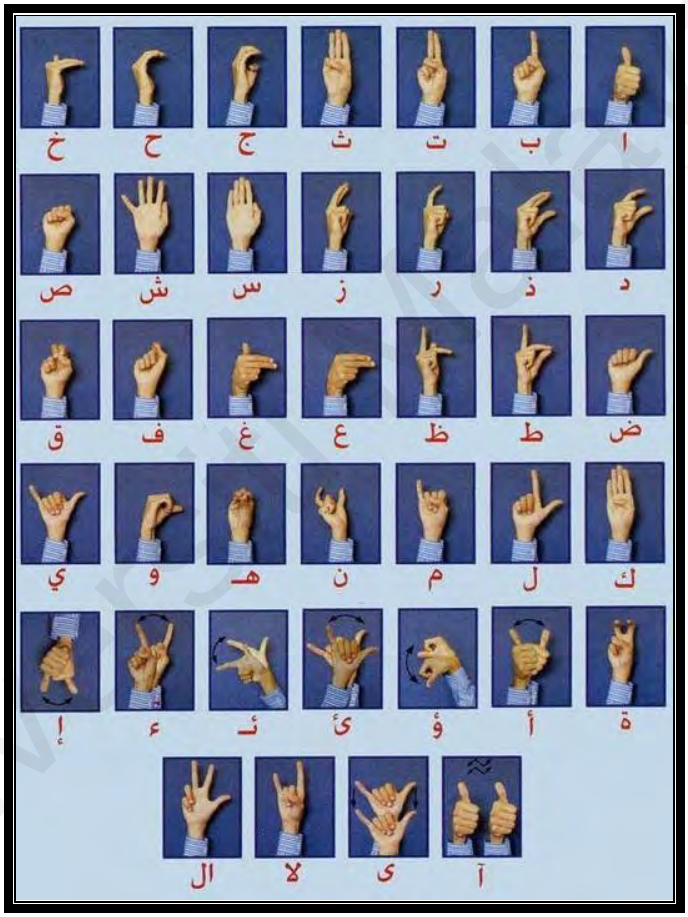


Figure 2.12: ArSL alphabet ("The Arabic Dictionary of Gestures for the Deaf.," 2007)

Stated below are examples of the various problems that normally face ArSL researchers when translating into Arabic sign language (Abdel-Fattah, 2005) (Almohimeed, Wald, & Damper, 2011):

1. Lack of syntax, linguistic and morphology studies on Arabic sign language.

2. Huge size of the translation corpus while generating and creating an Arabic sign language translation technique.
3. Complicated to represent Arabic sign language translation output.
4. Difficult to find method to evaluate Arabic sign language translation output.

Arabic sign language is similar with other sign languages in that they are gestural and spatial languages. However, Arabic sign language differs in terms of syntax, morphology and language structure from Arabic spoken language, which becomes hard to compare the sign language with the corresponding spoken form similar to other languages, whereby many concepts in spoken languages are described inadequately for sign languages (Jemina, Rachel, & Goswell, 2010).

In Arabic sign language, signed and gestured sentences do not have plural, dual, and singular agreements as shown in Table 2.1. There are many countable nouns in Arabic language, but they do not exist in Arabic sign language. For example, the Arabic word /فُنْجَانَيْنْ/ /***fundʒaa:najn***/ which means ***two cups*** in English language is represented in the Arabic sign language by two consecutive words as follows: the sign/gesture for the word /فُنْجَانْ/ / ***fundʒaa:n*** / which means ***cup*** in English language, followed by the sign/gesture of the Arabic number /إِثْنَيْنْ/ /***ʔiθnajn***/ which means ***two*** in English language, in order to indicate the dual.

Table 2.1: Syntax in Arabic sign language

No	Arabic Language Syntax	Arabic Sign Language Syntax
1	Singular	Singular
2	Dual	Singular + Two
3	Plural	Singular + Corresponding Number

On the other hand, ArSL does not benefit from grammatical tenses similar to those that exist in written and spoken forms. Tenses in ArSL are easily and practically applied.

Universiti Malaya

Future, present and past tenses are specified and indicated at the beginning (starting points) of a discussion and a conversation, and also switch when there is a need to indicate another tense. In addition, interrogatives and negatives can be expressed in more than one way. Facial expressions, longer signing time, dramatization and repetition are used for emphasis, whereas adverbs can be explained manually by one hand's position in relation to the other. Furthermore, features such as declension and passivation are nonexistent, whereas nominal features of facial expressions and context are used to achieve conditional expressions, sentence boundaries, and turn taking (Abdel-Fattah, 2005). Table 2.2 presents the syntax variation between Arabic language and Arabic sign language.

Table 2.2: Syntax variation between Arabic language and Arabic sign language (Altun & Albayrak, 2011)

No	Arabic Language Syntax	Arabic Sign Language Syntax
1	Subject + Verb	Subject + Verb
2	Verb + Subject	Subject + Verb
3	Subject + Predicate	Subject + Predicate
4	Subject + Verb + Object	Subject + Object + Verb
5	Subject + Verb + Object (Adjective, Adverb)	Subject + Object + Verb (Adjective, Adverb)
6	Subject + Predicate + (Adjective, Adverb)	Subject + Predicate + (Adjective , Adverb)
7	Subject + Verb + Pronoun	Subject + Verb
8	Verb + Object	Object + Verb

Syntax in sign language is synchronous with a parallel spatial and temporal arrangement but the syntax of sentences in oral language (spoken language) is linear; word followed by another word. The syntax of Arabic Language sentences are partitioned into two categories of sentences.

First category includes Subject, Verb, Predicate and Object. Table 4 presented a syntax variation between Arabic language and Arabic sign Language. Second category

Universiti Malaya

sometimes starts with Subject and sometimes with Verb (Abdel-Fattah, 2005). However, it is better to start the Arabic sign language sentence with Subject. Table 2.3 shows an example of an Arabic language sentence and its representation in the Arabic sign language.

Table 2.3: Example of Syntax in Arabic sign language

Criteria	Arabic Language Representation	ArSL Representation
Sentence	قَامَ الطَبِيْبُ بِإجرَاءِ عَمَلِيةٍ فِيْ القَلْبْ	الطَبِيْبُ عَمَلِيةٍ القَلْبْ
English Translation	The Doctor conducted a surgery in the heart	Doctor surgery heart
IPA Representation	qaa:ma a:ltˤabijbu biʔdʒraa:ʔi ʕamalijti: fij a:lqalb	/a:ltˤabijbu ʕamalijti: a:lqalb/

In Arabic sign language, ordering of negative sentences is not similar to the sentences in Arabic language and spoken languages. Furthermore, the adjectives can be translated from Arabic language to Arabic sign language by two ways, whereby the first way is using the adjectives sign immediately and directly, whereas the second way is using negation of an equivalent negative verb to the adjective. For example, the word /حُبْ/ /***ħub***/ which means ***love*** in English language does not necessarily be the word /كُرْه/ /***kurh***/ which means ***hatred*** in English language as shown in Table 2.4.

Table 2.4: Differences between negative syntax in Arabic language and Arabic sign language (Altun & Albayrak, 2011)

No	Arabic Language Syntax	Arabic Sign Language Syntax
1	Negative + Verb	Verb + Negative
2	Subject + Negative + Verb + Object	Object + Subject + Verb + Negative
3	Subject + Negative + Verb	Subject + Verb + Negative
4	Negative + (Adjective, Adverb)	(Adjective , Adverb) + Negative
5	Adjective	Verb + Negative

Universiti Malaya

2.9.3 Unified Arabic Sign Language Dictionary (UASLD)

Based on the above, it is clear that Arabic sign language has many varieties for different Arab countries. After investigating the history of Arabic sign languages, they started to be documented back in 1972 for the Egyptian dictionary, which was published by the Egyptian Association for Deaf and Hearing Impaired (Smreen, 2004). Consequently, other Arab countries made several attempts to standardize and spread the local Arabic sign language among their deaf community. As a result, official sign languages appeared in many Arabic countries such as Qatar (in 2010), Sudan (in 2009), Yemen (in 2009), Kuwait (in 2007), Tunisia (in 2007), Jordan (in 1993), Libya (in 1992), Morocco (in 1987), Iraq, UAE, Palestine, and Saudi Arabia (Mohamed Elhadj et al., 2013).

It is important to note the similarity of cultural values and gestural repertoire among the Arab countries, which have created the belief that ArSLs are almost identical for all Arab countries. It is crucial to produce a standardized dictionary for Arabic sign language of all Arabic countries. Therefore, 18 Arab countries launch the Unified Arabic Sign Language Dictionary (UASLD) for the deaf in 2007, which was an initiative in 1999 from the League of Arab States (LAS) and the Arab League Educational, Cultural and Scientific Organization (ALECSO). This UASLD was divided into two parts, whereby the first part was published in Tunisia in 2001 and contains more than 1600 words classified in categories: family, home, food, etc., and the second part contains more signs, better items organization and was published with a strong collaboration between the Supreme Council for Family Affairs in Qatar, ALECSO, LAS, and the Arab Union for Deaf (AUD), with support of the Arab deaf in 2006. A DVD version with video-clips of gestures of the two parts appeared in 2007 (Mohamed Elhadj et al., 2013).

Universiti Malaya

2.9.4 Related works for ArSLR Approaches

The research efforts on ArSLR have witnessed a significant increase in last few years. The researchers used various methods, frameworks, and techniques in this domain. Based on our literature investigation, the primary research taxonomy of research efforts on ArSLR is categorized into two main types namely 1) VBR, and 2) SBR. While the secondary research taxonomy for ArSLR research efforts using VBR and SBR approaches range from developing systems that can recognize small units and segments such as alphabets, moving to larger yet small units such as isolated words, and finally to the largest and most difficult units such as full sentences. The larger the unit to be recognized, the harder is the ASLR task. For simplicity of the secondary research taxonomy, this work recommends classifying Arabic ASLR research efforts into three broad categories, which include 1) alphabet sign language recognition, 2) isolated words sign language recognition, and 3) continuous sign language recognition. The following sections provide a related literature investigation and comparison between various research efforts for each category.

2.9.4.1 Alphabet Sign Language Recognition

Based on our related literature investigation, there various research attempts and efforts for Arabic alphabet sign language recognition using VBR approach as shown in Table 2.5, which depends on performing each alphabet independently. Generally, alphabets are performed by a static position or posture and the vocabularies database is limited in size. Although there are 28 alphabets for Arabic language, ArSL utilizes 39 signs as previously mentioned in Figure 2.12 (“The Arabic Dictionary of Gestures for the Deaf.,” 2007). Each Arabic alphabet is represented by one sign and the 11 remainders represent other orthographical representation of the Arabic alphabets including (أ/إ/ئ/ؤ/ء/ئ/آ/ى/ة/لا/ال).

Universiti Malaya

In addition, the case of (لا and ال), they represent 2 (ل and ا) Arabic alphabets but they have special case when used in sign language recognition and they are represented as one sign too. To our knowledge, we were unable to find any research publication for Arabic alphabet sign language recognition using SBR approach and backhand ArSLR.

2.9.4.2 Isolated Words Sign Language Recognition

Based on our literature investigation, there various research attempts and efforts for Arabic isolated words sign language recognition using both VBR approach as shown in Table 2.6 and SBR approach as shown in Table 2.7. The isolated words sign language recognition depends on performing and analyzing each isolated word separately by analyzing a set of images to perform one sign only.

2.9.4.3 Continuous Sign Language Recognition

Based on our literature investigation, there were very limited research attempts and efforts for Arabic continuous sign language recognition using VBR approach as shown in Table 2.8 and SBR approach as shown in Table 2.9. We were able to find only two research attempts using SBR approach, which were recently published in 2015 (Tuffaha et al., 2015; Tubaiz et al., 2015). This shows lack of research attempts and efforts for Arabic continuous sign language recognition using VBR and SBR approaches, which opens room for further research in this gap.

The continuous sign language recognition depends on performing and analyzing full sentences, which is more challenging than the two previous types (alphabet and isolated words). Continuous sign language recognition is considered more efficient and appropriate in real life. Continuous ASLR systems are more suitable for impaired hearing and deaf Community. An ideal continuous ArSLR should have a reliable and available in real-time with a high recognition rate. Detecting the additional movements from the transition between a set of signs, recognizing and modeling are considered as major challenges in continuous sign language recognition.

Universiti Malaya

Table 2.5: Performance comparison of alphabet sign language for ArSL research efforts using vision-based

Ref.	Year	Main Task	Techniques		Tools	Sign language Data		Signer Dependency	Recognition Rate (%)
			Features Extraction	Features Classification		Training	Testing		
(Mohandes et al., 2014)	2014	ArSLR using the LMC (28 alphabet)	Hus moments	NB, MLP	MATLAB	2800	2800	N/A	98% with the NB, 99% using the MLP
(Samir Elons et al., 2013)	2013	Arabic alphabet sign language recognition (30 alphabet)	N/A	PCNN	N/A	60	240	Signer-Dependent	90%
(Naoum et al., 2012)	2012	Arabic alphabet sign language recognition (28 alphabet)	Finding histograms	KNN	N/A	N/A	N/A	N/A	White gloves: 80%, a black gloves: 65%, a red gloves: 75%, Naked hand: 50%.
(Hemayed et al., 2011)	2011	Arabic sign language Alphabets Recognition (converts signs into voice)	YCbCr space, PCA, Prewitt edge	KNN	N/A	150	N/A	N/A	97%
(El-Bendary, Zawbaa, Daoud, Hassanien, & Nakamatsu, 2010c)	2010	Arabic alphabet sign language recognition (30 alphabet)	Edge detection stage, feature vector creation stage	NNT	N/A	N/A	N/A	N/A	91.3%
(Maraqa et al., 2008)	2008	Arabic alphabet sign language recognition using one hand (28 alphabet)	color layers expanded	Fully recurrent network, Elman network algorithms	Matlab, C	900	900	2 Signer-Dependent	Recurrent Networks: 95.11%, Elman Network: 89.66%

Universiti Malaya

(Al-Jarrah et al., 2007)	2007	Arabic alphabet sign language recognition (30 gesture)	boundary information and region information	ANFIS	N/A	1200	600	N/A	97.5%
(Assaleh et al., 2005)	2005	Arabic alphabet sign language recognition (30 alphabet)	segmented color regions , geometric measures	polynomial classifier	N/A	1625	698	signer-Dependent	93.41%
(Al-Rousan et al., 2001)	2001	Arabic alphabet sign language recognition (28 alphabet)	Hand region	ANFIS	N/A	N/A	N/A	N/A	95.5%
(Al-Jarrah et al., 2001)	2001	Arabic alphabet sign language recognition (30 gesture)	Border Information	ANFIS	Matlab	1200	1800	N/A	93.55%
(Mohandes, 2001)	2001	Arabic alphabet sign language recognition	Hu moments	SVMs	N/A	N/A	N/A	N/A	87%

Table 2.6: Performance comparison of isolated words sign language for ArSL research efforts using vision-based

Source	Year	Main Task	Techniques		Tools	Sign language Data		Signer Dependency	Recognition Rate (%)
			Features Extraction	Features Classification		Training	Testing		
(Amin et al., 2016)	2016	Arabic isolated word sign language recognition (40 isolated gesture)	Go-Stop Detector	HMM	Microsoft Kinect	2400	1200	Signer-Dependent, Signer-Independent	95.125% for user-Dependent, (92.5% for signer-independent
(Ahmed et al., 2014)	2014	Arabic isolated word sign language recognition (23 isolated words)	Appearance-based , LBP and PCA	HMM	N/A	2415	1035	3 Signer-Dependent	99.97%
(Elons et al., 2013)	2013	Arabic isolated word sign language recognition (50 isolated words)	Fitness function	Hybrid PCNN, graph matching approach	N/A	N/A	N/A	Signer-Independent	96% for Pose-Invariant constraints with indulgence of up to 90º

Universiti Malaya

(Fares et al., 2013)	2013	Arabic isolated word sign language recognition (40 isolated words)	GMM	TDNN	Matlab	N/A	N/A	Signer-Dependent	77.43%
(Mohandes et al., 2012)	2012	Arabic isolated word sign language recognition (300 isolated words)	Gaussian skin color, simple region growing, Two colored gloves	HMM	Matlab	1000	500	Signer-Independent	50 signs was 98%, 300 signs was 95%
(Albelwi et al., 2012)	2012	Arabic isolated word sign language recognition (180 isolated words)	Fourier descriptors (FDs)	KNN	Visual c#, OpenCV, LifeCam VX-5500	N/A	N/A	N/A	90.55 %.
(Shanableh et al., 2011)	2011	Arabic isolated word sign language recognition (23 isolated gesture)	Zonal coding, Discrete cosine transform (DCT)	KNN	N/A	2300	1150	3 Signer-Independent	87%
(Youssif et al., 2011)	2011	Arabic isolated word sign language recognition (20 isolated gesture)	N/A	HMM	N/A	900	810	Signer-Independent	82.22%
(Al-Rousan et al., 2010)	2010	Arabic isolated word sign language recognition (40 isolated gesture)	spatial domain analysis	HMM	N/A	1920	1280	Signer-Independent, Signer-Dependent	70.5% user independent mode, 92.5% user-dependent mode
(AL-Rousan et al., 2009)	2009	Arabic isolated word sign language recognition (30 isolated gesture)	discrete cosine transform (DCT)	HMM	camera	2730	1315	Signer-Independent, Signer-Dependent	94.2% & 90.6% user-independent for offline & online respectively.97.4% & 93.8% user-dependent for offline & online respectively.
(Shanableh et al., 2007)	2007	Arabic isolated word sign language recognition (23 isolated gesture)	Zonal coding, Discrete cosine transform (DCT)	HMM	N/A	805	345	Signer-Independent	95%

Universiti Malaya

Table 2.7: Performance comparison of isolated words sign language for ArSL research efforts using sensor-based

Ref.	Year	Main Task	Techniques		Tools	Sign language Data		Signer Dependency	Recognition Rate (%)
			Features Extraction	Features Classification		Training	Testing		
(Mohandes et al., 2013)	2013	Arabic isolated word sign language recognition (20 isolated gesture)	Linear Discriminant Analysis (LDA)	Minimum Distant (MD), Dempster-Shafer theory of evidence (DS)	Two CyberGlove (GLV), two Flock-of-Birds (FOB)	1000	1000	N/A	Hand tracking: 84.7%, Cyber Glove: 91.3%. Feature based fusion: 96.2%. Fusion at the decision level: 98.1%
(Khaled Assaleh, Shanableh, & Zourob, 2012)	2012	Arabic isolated word sign language recognition (10 isolated gesture)	accumulated differences (ADs), a regression technique	KNN with Manhattan distance measure.	DG5-VHand data glove	500	500	Signer-Independent, Signer-Dependent	92.5% in the signer independent mode, 95.3% in the signer dependent.
(Ritchings et al., 2012)	2012	Developed Intelligent Computer-Based System for learning ArSL	N/A	N/A	DataGlove, Bend sensors, push button	N/A	N/A	4 Signer-Independent	93%
(Mohandes, 2004)	2004	Arabic isolated word sign language recognition (42 isolated gesture)	Statistical features	SVMs	PowerGlove	720	120	N/A	90%

Universiti Malaya

Table 2.8: Performance comparison of continuous sign language recognition for ArSL research efforts using vision-based

Ref.	Year	Main Task	Techniques		Tools	Sign language Data		Signer Dependency	Recognition Rate (%)
			Features Extraction	Features Classification		Training	Testing		
(Tolba et al., 2013)	2013	Arabic continuous sign language recognition (3 to 4 words) (30 sentences)	PCNN, Graph matching	MLP	N/A	N/A	N/A	N/A	70%
(Assaleh et al., 2010)	2010	Arabic continuous sign language recognition (40 sentences)	DCT, Spatio-temporal	HMM	HTK	532	228	Signer-Dependent	73.3%
(Assaleh et al., 2008)	2008	Arabic continuous sign language recognition (40 sentences)	DCT	HMM	HTK Toolkits, digital video camera	N/A	N/A	Signer-Dependent	75%

Table 2.9: Performance comparison of continuous sign language recognition for ArSL research efforts using sensor-based

Ref.	Year	Main Task	Techniques		Tools	Sign language Data		Signer Dependency	Recognition Rate (%)
			Features Extraction	Features Classification		Training	Testing		
(Tuffaha et al., 2015)	2015	Arabic continuous sign language recognition using Sensor-Based (40 sentences)	Window-based approach	Modified the polynomial classifier to suitable with sequential data	Two DG5-VHand data gloves	280	120	N/A	85%
(Tubaiz et al., 2015)	2015	Arabic continuous sign language recognition using Sensor-Based (40 sentences)	Window-based approach	MKNN	Two DG5-VHand data gloves	70% of dataset	30% of dataset	Signer-Dependent	98.9%

Universiti Malaya

Table 2.10: Performance comparison of VBR static sign recognition for other languages

Ref.	Year	Segmentation	Feature extraction	Classification	Accuracy/sample size	Scope
(Pan et al., 2016)	2016	Skin color with Gaussian Mixture Model	LDA, and PCA, Hu-moments and SIFT	SVM	94% - 36 × 2425 samples	ASL
(Gupta et al., 2016)	2016	-	SIFT & Histograms of oriented gradients	k-NN	26 × 30 samples	ISL
(Bastos et al., 2015)	2015	MP-ANN on skin color	Histograms of oriented gradients	-	96.8% - 40 × 240 samples	Libras
(Huong et al., 2015)	2015	Skin color	PCA	Template matching by Mahalanobis distance	91.5% - 25 × 10 samples	VSL
(Jasim et al., 2014)	2014	Skin color	LDA	K-NN	88.6% - 15 × 7 samples	Bangla SL
(Jasim et al., 2014)	2014	-	LDA & LBP	K-NN	92.4%	CSL
(Hartanto et al., 2014)	2014	Skin color	SURF	Euclidian distance & Template matching	63% - 24 × 12 samples	Indonesian SL
(Jayshree et al., 2013)	2013	Skin color (YIQ) Thinning method	Fourier transform, Edge Oriented Histogram and DCT	Euclidian distance & Template matching	10 × 2 samples	ISL
(Rajathi et al., 2013)	2013	Thresholding	Eccentricity and orientation, minor axis length, major axis length, perimeter, Solidity, and convex area	Proximal SVM	91% - 20 samples	TSL and face
(Sethi et al., 2012)	2012	Region growing &Skin color	SIFT	Correlation matching	-	ASL
(Jayashree et al., 2012)	2012	Sobel edge detection	Area and centroid of hand edges	Euclidian distance & Template matching	90.19% - 26 × 100 samples	ASL
(Rekha et al., 2011)	2011	Skin color (RGB) & K-means clustering	SURF, SIFT, Hu moment, and SURF	K-NN & SVM	15 × 80 samples	ASL
(Kurdyumov et al., 2011)	2011	Background subtraction & Normalization	Fourier transform and Relative area	SVM, Linear kernel & K-NN	93% - 25 × 100 samples	ESL

Universiti Malaya

(Karami et al., 2011)	2011	Cropping image	DWT	MLP-NN	94.06% - 32 × 20 samples	PSL
(Rokade et al., 2009)	2009	Skin color, Histogram matching & Thinning method	vertical distance, horizontal distance & Angle	-	92.13% - 10 × 60 samples	ASL
(Akmeliawati et al., 2009)	2009	CAM Shift	PCA & LPP	ANN	92.02% - 13 samples	NZSL
(Binh et al., 2005)	2005	Skin color (HSV) & Kalman filter	-	GRFNN	92.19% - 36 × 200 samples	ASL

Table 2.11: Performance comparison of VBR dynamic sign recognition for other languages

Ref.	Year	Segmentation	Feature Extraction	Classification	Dataset Size - Accuracy	Scope
(Baranwal et al., 2017)	2017	Otsu thresholding Edge Detection	Mel-Frequency Cepstral Coefficients (MFCC)	K-NN & SVM	19 × 8 dynamic	ISL
(Koller et al., 2015)	2015	Simple colour segmentation	Histogram of oriented gradients	HMM & CMLLR	1081 signs, 9 signers 47.0%	DGS
(Fagiani et al., 2015)	2015	-	Canny Filter	HMM	147 signs, 10 signers 48.06%	LIS
(Hrúz et al., 2012)	2012	Skin Color	Geometric Moments, Local Binary Pattern	HMM	40 signs, 11 signers 57.54%	Czech SL
(Cooper et al., 2012)	2012	Skin Color	Appearance-based	HMM & Sequential Pattern Boosting	40 signs, 15 signers - 76%	DGS
(Chanda et al., 2012)	2012	-	Upright speed-upobust feature (U-SURF)	Fuzzy C-means (FCM)	42 signs, 17 signers 87.71%	Thai SL
(Zaki et al., 2011)	2011	Skin Color	Motion Chain Code & PCA	HMM	RWTH-Boston 50 -89.1%	ASL
(Rekha et al., 2011)	2011	Skin Color with Single Gaussian Model	Convexity Defect	K-NN, DTW & SVM	3×22 sample - 77.2%	ISL Two hands

Universiti Malaya

(Agris et al., 2008)	2008	skin color	Statistical Modeling	HMM & MLLR & MAP	153 signs, 25 signers 78.6%	GGS
(Lichtenauer et al., 2008)	2008	-	Discriminative Feature (DF)	DFFM, CDFD & SDTW	120 signs & 75 signers	DSL
(Rybach, 2006)	2006	Motion Tracking & Score Function	LDA & PCA	HMM	RWTH-Boston 104	ASL & Body

Table 2.12: Performance comparison of active techniques and SBR for other languages

Ref.	Year	Device Type	Feature Extraction	Classification	Dataset Size - Accuracy	Scope
(Abhishek et al., 2016)	2016	Capacitive touch sensor	Electrode signals	Finger posture recognition	26 static - 92%	ASL
(Abreu et al., 2016)	2016	Myo Armband EMG and IMU	EMG signals	SVM	20 static - 89.6%	Libras
(Bukhari et al., 2015)	2015	Data Glove	PCA	Template Matching by Euclidian Distance	26 × 250 samples - 92%	ASL
(Funasaka et al., 2015)	2015	LMC	Hand position, velocity & movement	Decision Tree	24 static samples 82.71%	ASL
(Chuan et al., 2014)	2014	LMC	Seven features from LMC	SVM & K-NN	26 × 4 samples - 79.83%	ASL
(Marin et al., 2014)	2014	LMC and Kinect	(LMC) Angle, distance, orientation, (Kinect) Color and depth	SVM	10 × 140 samples - 91.3%	ASL
(Kong et al., 2014)	2014	CyberGloves	-	CRF & SVM & BN	107 signs, 8 signers 86.6%	CSL
(Chai et al., 2013)	2013	Kinect	3D trajectory of hand	Template matching by Euclidian distance	239 × 5 sentences 96.32%	CSL

Universiti Malaya

(Zhang et al., 2011)	2011	Accelerometer and EMG	3-axis ACC & 5-channel EMG signals	Fuzzy K-means clustering	72 × 2 sentences - 93.1%	CSL
(Fang et al., 2002)	2002	Data Glove	Self-organizing feature maps(SOFM)	HMM+FDT	208 signs, 7 signers 83.7%	CSL
(Liang et al., 1998)	1998	Data Glove	10-finger flexion, position, angle, motion trajectory	HMM	50 static - 80.4% 30 dynamic - 92%	TSL
(Kim et al., 1996)	1996	VPL Data Glove	x and y axis movement 16 raw angle, orientation, position data	Fuzzy NN	25 words - 85%	KSL Two hand

Universiti Malaya

Table 2.13: Summary of the advantages and disadvantges of ArSLR

Ref.	Advantages	Disadvantages
Related works for alphabet sign language		
(Mohandes et al., 2014)	- Multi-classifier - Using LMC	- Hus moments not suitable in all sign cases. - limited scope (forehand) - Singer-Dependent
(Samir Elons et al., 2013)	- Recognition rate is Satisfying	- Need large number of training - Training and testing set are not Satisfying - limited scope (forehand) - Singer-Dependent
(Naoum et al., 2012)	- Feature extraction and Features - Classification are simple	- Training and testing set is missed - Recognition rate is not Satisfying - limited scope (forehand) - Singer-Dependent
(Hemayed et al., 2011)	- Recognition Rate is Satisfying	- Need large number of training - Training and testing set are not Satisfying - limited scope - Singer-Dependent
(El-Bendary et al., 2010c)	- Feature extraction and Features Classification are easy to implement	- Not suitable for in case have many edges - Training and testing set are not Satisfying - limited scope (forehand) - Singer-Dependent
(Maraqa et al., 2008)	- No need of previous information, simplest method	- Complex calculation of gradients - Failed in some cases - limited scope (forehand) - Singer-Dependent
(Al-Jarrah et al., 2007)	- Recognition Rate is Satisfying	- More computational complexity - limited scope (forehand) - Singer-Dependent
(Assaleh et al., 2005)	- Recognition Rate is Satisfying - Feature extraction and classification are simple	- Singer Dependent framework - Failed in some cases - Feature extraction is difficult to compute - limited scope (forehand) - Singer-Dependent
(Al-Rousan et al., 2001)	- The solution of nonlinear problems and robust	- More wastage of time in training - must be trained for each problem - Training and testing set is missed - limited scope (forehand) - Singer-Dependent
(Al-Jarrah et al., 2001)	- Feature extraction is easy to implement	- More wastage of time in training - must be trained for each problem - limited scope (forehand)

Universiti Malaya

(Mohandes, 2001)	- Classification are easy to implement and robust	- Singer-Dependent - Hus moments not suitable in all sign cases - limited scope (forehand) - Singer-Dependent
Related works for isolated words sign language		
(Amin et al., 2016)	- Good for signs having better contrast between objects.	- Complicated feature extraction - More wastage of time in training - limited scope (isolated words only)
(Ahmed et al., 2014)	- The solution is robust - Recognition Rate is Satisfying	- Complicated feature extraction - More wastage of time in training - limited scope (isolated words only) - Singer-Dependent
(Elons et al., 2013)	- The solution is robust - Recognition Rate is Satisfying	- The system is insensitive to signer position, view angle and background effects - limited scope (isolated words only) - Training and testing set is missed
(Fares et al., 2013)	- Classification are easy to implement and robust	- Fixed of finite order - Training time is very large - Training and testing set is missed - limited scope (isolated words only) - Recognition Rate is not satisfying - Singer-Dependent
(Mohandes et al., 2012)	- Recognition Rate is Satisfying - Feature extraction and Features - Classification are simple	- Have a large number of unstructured parameters - limited scope (isolated words only)
(Albelwi et al., 2012)	- Simple to implement - Can learn non-linear boundary	- Time complexity - Need large number of training - Training and testing set is missed - limited scope (isolated words only) - Singer-Dependent
(Shanableh et al., 2011)	Training and testing set are Satisfying	- Time complexity - Need large number of training - Recognition Rate is not satisfying - limited scope (isolated words only)
(Youssif et al., 2011)	- Proposed solution is signer independent	- Have a large number of unstructured parameters - Training and testing set is small and limited - Recognition Rate is not satisfying
(Mohandes et al., 2013)	- Multi-classifier framework - The solution is robust	- Expensive devices - dataset is limited - limited scope (isolated words only) - Singer-Dependent
(Ritchings et al., 2012)	- Proposed solution is signer independent - Recognition Rate is Satisfying	- Training and testing set is missed - limited scope (isolated words only) - Training and testing set is small and limited

Universiti Malaya

(Mohandes, 2004)	- No need of previous information, simplest method	- Not suitable for large data sets and not perform very well, when the data set has more noise - Recognition Rate is not satisfying - Singer-Dependent - limited scope (isolated words only)
Related works for continuous sign language		
(Tolba et al., 2013)	- Very good algorithm to use for the regression and mapping. - Good to model with nonlinear data with large number of inputs	- It is computationally very expensive and time consuming to train with traditional CPUs. - Depend a lot on training data. - limited scope - Singer-Dependent - limited scope
(Assaleh et al., 2010)	- Training and testing set are Satisfying - The solution is robust	- Recognition rate is not Satisfying - Singer-Dependent - limited scope
(Tubaiz et al., 2015)	- Recognition rate is Satisfying - The solution is robust	- Not work well with large dataset - Sensitive to noisy data - Singer-Dependent - Very expensive devices - limited scope

2.10 Open Issues and Gaps for Arabic Sign Language Recognition

By reviewing the relevant related literature in previous sections, it has become evident that ArSLR poses several issues into two perspective: language resources and technology.

A. Language Perspective:

A critical issue in existing ArSL research is that the available databases are limited, small in terms of size, and have no clear commitment to ArSL standards. Due to the limited size of the database, such research works suffer applicability and usage unless larger datasets are used. The size of the database plays an essential role in the overall system accuracy and performance regardless of the framework type. Finally, all the related works use their own databases; thus, no standard database in all ArSL forms and approaches has been conceptualized and created.

Universiti Malaya

B. Technology Perspective:

The existing research attempts focused on the modelling and development of ArSLR. There is a noticeable attention for the signer-dependent approach but less attention for signer-independent. In addition, the related works investigated the same scope and limited areas of study. For example, almost approaches focused on forehand signs or one form of signs (i.e. alphabet, isolated or continuous). To the best of our knowledge, no comprehensive approach has been proposed to cover all the sign forms under the signer-independent approach. Furthermore, the VBR and SBR approaches have been used individually in most research attempts, with less emphasis on fusing both approaches to produce a new hybrid approach, which can be produced by maximising the advantages and minimising the disadvantages of each approach.

In conclusion, ArSLR using static forehand under the VBR approach has achieved a satisfactory recognition rate accuracy. However, all previous research attempts and reviews on SLR for ArSL used forehand signs. This shows that the backhand signs have not been considered for ArSL tasks, thus creating another important research gap to be filled up. In addition, no study on SLR for the Arabic alphabet has used the SBR approach.

On the basis of the literature investigation for the continuous SLR conducted in the current study, research using the VBR approach has achieved a recognition rate of 75% (Assaleh et al., 2008), whereas other attempts using the SBR approach has achieved a recognition rate of 98.90% for signer-dependent testing (Tubaiz et al., 2015). However, all previous research attempts and reviews on SLR for ArSL used one approach (vision- or sensor-based), and no study has used the hybrid approach for ArSL. The ArSLR studies in this category are still inadequate, and the performance is not yet satisfactory compared to other languages. Hence, more research attempts should use the hybrid approach (combining vision- and sensor-based approaches), especially for the signer-independent framework, in order to achieve better performance. Figure 2.13 presents a summary of the recently conducted ArSL research.

Universiti Malaya

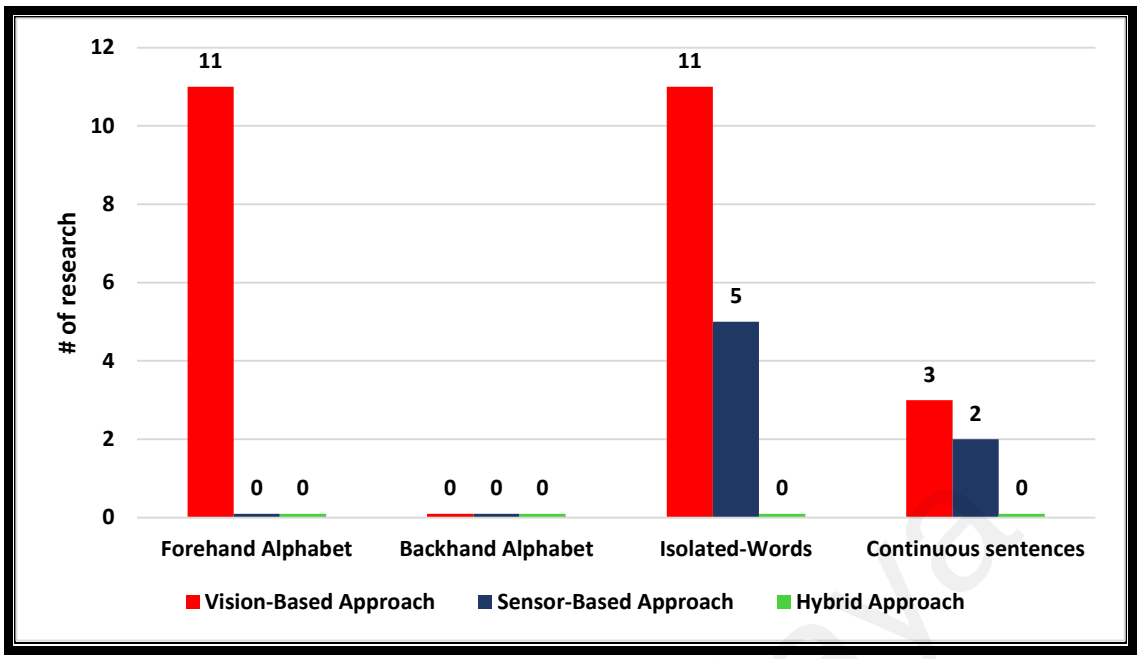


Figure 2.13 Summary of the number of conducted ArSL research

2.11 Summary

This chapter presented a comprehensive review of literature on the following categories: Hearing and Speech-Impaired people, gesture recognition and gesture recognition Taxonomies, enabling technologies for hand gesture recognition, algorithms and methods used in gesture recognition, vision and sensor-based gesture recognition devices, automated SLR, background on Arabic language and ArSL, difficulties in Arabic language and ArSLR, unified ArSL dictionary and taxonomy of ArSL approaches.

This chapter also highlighted the research gaps in the existing literature on ArSLR, which could be classified into two main perspectives, namely, the language and technology perspectives. A comprehensive ArSL database and framework solution for ArSLR was proposed based on the identified research gaps. The next chapter shall present the overall research methodology of the current research.

Universiti Malaya

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the research methodology that has been applied in the current research. It aims to describe the selected research methods used in order to achieve the defined research objective. This chapter precisely identifies and outlines the various methods that will be utilised in accordance with the aims and objectives of each phase. Figure 3.1 outlines the structure of this chapter.



Figure 3.1: Outlines the structure of chapter 3

Generally, three main phases of the research are illustrated, namely, the literature review phase, the development phase and the evaluation phase. Figure 3.2 presents the research methodologies used corresponding to the different phases. Table 3.1 presents the relationships between the problem statement, research objectives, research questions, methodology and outcomes.

This chapter carries out the modelled ArSLR frameworks and all related tasks, namely, identifying the research approach and research tools, determining the overall component framework, identifying the dataset used for ArSLR and defining the evaluation criteria. The current chapter also describes the framework design, the database creation, the comprehensive theoretical background of the methodology utilised and the various methods adopted.

Universiti Malaya

3.2 Literature Review Phase

The literature review phase aims to reveal all aspects of the current related works in order to precisely identify the research gaps and research scope as well as to formulate the main objectives.

In order to achieve the first research objective, this thesis employed a systematic literature review method. This is one of the methods of identifying, evaluating and interpreting all available previous studies relevant to a particular research question (Kitchenham et al., 2010), topic area or phenomenon of interest. The main purpose of performing a systematic literature review is that it enables the researcher to develop cumulative knowledge based on the research questions, thus ensuring that the research work is relevant, and to build a knowledge base for the research scope. The knowledge base can help establish the research direction, assist in gap analysis and provide a strong platform that can guide the research direction in order to address any gaps and support the hypothesis. A systematic literature review aims to find as many relevant studies on a particular research question as possible and to use explicit methods to identify what can be reliably said on the basis of these studies. The methods should not only be explicit but systematic, with the aim of producing varied and reliable results. In this way, systematic review reduces the bias that tend to occur in other approaches in reviewing research evidence.

Universiti Malaya

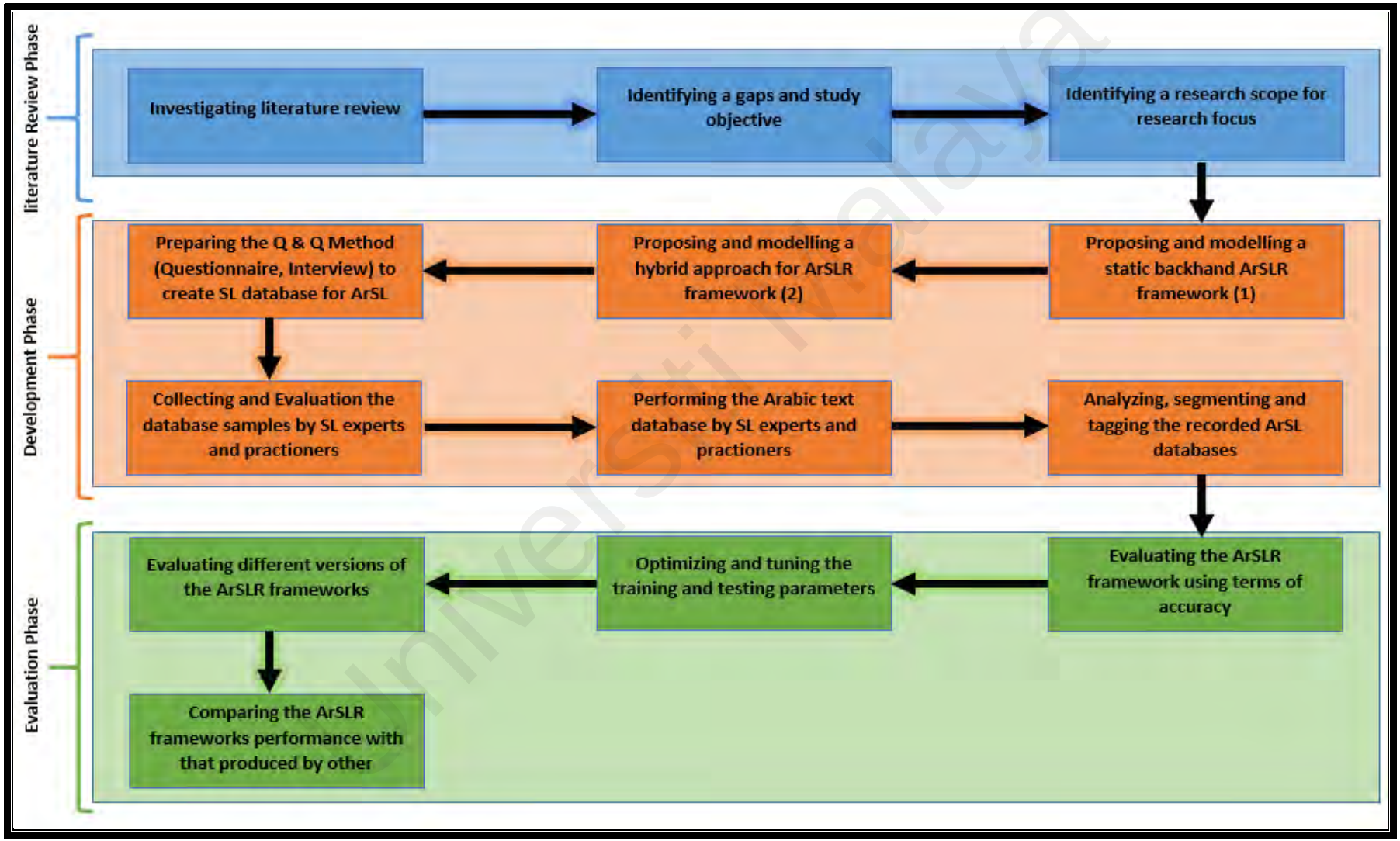


Figure 3.2: Different stages based on the following methodology

Table 3.1: The Relation between problem statement, RO, RQ, methodology, and outcome

Problem Statement	Research Objective	Research Question	Methodology	Outcome
Technological Perspective & Language Perspective:	**RO1:** To create research taxonomies based on a comprehensive literature investigation that summarize the research efforts on ArSLR approaches and ArSL forms of signs.	**RQ1:** What are the main taxonomies of ArSLR approaches and ArSL forms of signs?	**Literature Review Method:** Conducting a comprehensive literature review on ArSL pertaining the related works in order to identify open issues, research gaps, related theory, dominating techniques, and suitable tools and devices. Formulating the primary taxonomy that summarizes the related works pertaining ArSLR approaches, and the secondary taxonomy that summarizes the related works pertaining ArSL forms of signs. Identifying the research objectives and questions. Forming the research scope for research focus.	**Finding for RQ1:** Primary and secondary research taxonomies are formulated based on a comprehensive literature investigation spanning from 1996 until 2019, where the primary research taxonomy summarizes the research efforts on ArSLR approaches and the secondary research taxonomy summarizes the research efforts on ArSL forms of signs. These research taxonomies provide comprehensive information to the research community and furnish them with sufficient related works and background on GR in general and ArSLR in specific.
Language Perspective:	**RO2:** To create a comprehensive SL database for ArSL that covers vision and sensor-based approaches and forms of signs for ArSLR development and evaluation based on SL experts standard criteria and consensus.	**RQ2:** What are the standard criteria used by SL experts for creating ArSL database?	**Empirical methods (Q & Q Method):** Preparing the questionnaire and interview qualitative methods to elicit and analyze the needs and requirements of SL experts and hearing and speech-impaired users that are related to the standard criteria for collecting the ArSL database. Distributing the questionnaire and conducting the interview sessions with ArSL experts and practitioners. Collecting the necessary Arabic words and sentences. Evaluating the Arabic text database by the SL experts and conducting necessary pre-processing activities. Performing the Arabic text database by SL experts and practioners.	**Finding for RQ2:** A state-of-the-art ArSL database is successfully created, which meets SL experts' standard criteria and bridges the gap between currently available ArSL resources and the research community expectations and recommendations.

Universiti Malaya

Cont.

Problem Statement	Research Objective	Research Question	Methodology	Outcome
Technological Perspective	**RO3:** To model a static backhand signer-independent ArSLR framework based on VBR approach that translates signs of letters and numbers into their corresponding text using a smartphone camera. **RO4:** To model a hybrid approach, signer-independent continuous ArSLR framework that integrates the VBR approach using Microsoft Kinect and the SBR approach using Smart Data Gloves.	**RQ3:** How to model a static backhand and signer-independent ArSLR framework for ArSL letters and numbers? **RQ4:** How to model a hybrid approach for signer-independent continuous ArSLR framework?	**Development Method (Construction of Techniques) :** Proposing and modelling a static backhand and signer-independent ArSLR framework, and a hybrid approach, signer-independent continuous ArSLR framework, which can bridge the research gap for ArSLR research and development. Developing all required components including pre-processing and segmentation, features extraction, and classification and recognition, for a usable static backhand and signer-independent ArSLR framework using VBR approach for signs of Arabic letters and numbers as performed on smartphone camera. Developing all required components including pre-processing and segmentation, features extraction, and classification and recognition, for a usable hybrid approach, signer-independent continuous ArSLR framework using VBR and SBR approaches as performed on Microsoft Kinect and Smart Data Gloves. The database was splitted in 30% for testing and 70% for training.	**Finding for RQ3, RQ4:** The static backhand and signer-independent ArSLR framework based on VBR approach is modeled, which intends to provide a usable platform and support to hearing and speech-impaired users pertaining letters and numbers using their smartphone camera. The hybrid approach, signer-independent continuous ArSLR framework is modeled, which intends to provide a usable platform and support to hearing and speech-impaired users pertaining all ArSL forms of signs with different data types that are received using Microsoft Kinect and Smart Data Gloves. Based on our literature investigation and to the best of our knowledge, this framework is amongst the earliest research efforts that integrates VBR and SBR approaches for ArSLR.

Universiti Malaya

Cont.

Problem Statement	Research Objective	Research Question	Methodology	Outcome
Technological Perspective & Language perspective	**RO5:** To evaluate the performance of the modeled frameworks in terms of accuracy, sensitivity, specificity, precision, False Positive Rate (FPR), False Negative Rate (FNR), and F1-score, and compare the accuracy performance with the related works.	**RQ5:** What are the performance measures used to evaluate and compare ArSLR research works?	**Experimental methods (Evaluation method):** Evaluating the performance of the ArSLR frameworks using relevant performance measures such as the Accuracy Rate, Sensitivity, Specificity, Precision, False Positive Rate (FPR), False Negative Rate (FNR), and F1-score. Optimizing and tuning the training and testing parameters in order to improve the performance of the ArSLR frameworks. Comparing the performance of the ArSLR frameworks in terms of accuracy rate with that produced by other related research works. Evaluating different versions of the ArSLR system in order to examine the effects of different training and testing parameters, and user's characteristics.	**Finding for RQ5:** Evaluate the proposed framework based on Accuracy Rate, Sensitivity, Specificity, Precision, False Positive Rate (FPR), False Negative Rate (FNR), and F1-score. Evaluate the proposed framework based on SL experts and hearing and speech-impaired. Our newly developed ArSL database and the proposed frameworks are able to have positive impact on the performance of the ArSLR.

Universiti Malaya

3.3 Development Phase

The development phase is one of the standard processes in the methodology flow, which aims to identify all the essential procedures and steps to construct the design and to analyse, train, test and evaluate framework. The framework development life cycle is defined as a process of building, modifying and manipulating the system setting in order to achieve the desired output or performance, that is, the development phase based on particular methodologies. In this thesis, the development phase structure based on the research gaps reveal and define the requirements, which include various steps and procedures, such as those listed below.

- Preparing the empirical research methods (qualitative and qualitative methods) to elicit and analyse the needs and requirements of SL experts as well as hearing and speech-impaired users who are related to the standard criteria for collecting the ArSL database.
- The research used a development method (construction of techniques) to design and model a static backhand framework based on a signer-independent ArSLR and a hybrid approach, namely, the signer-independent continuous ArSLR framework, which can bridge the research gap for ArSLR research and development.

3.3.1 Creating ArSL Database

In the context of targeted language, the main challenge and complexity faced by the researchers is that there is no common standard database for ArSL in the domain. This can be attributed to the complexity of the Arabic language and the divergence in dialects among countries that are different from the standard Arabic language. Consequently, the gestures of the Arabic language also diverge and vary. Thus, the created ArSL database has to contain the necessary Arabic letters and numbers, isolated-words and sentences.

Universiti Malaya

In this regard, empirical research methods such as quantitative and qualitative methods were used to observe and collect data to create ArSL database. One of the common methods in software engineering is proposed in (Easterbrook, Singer, Storey, & Damian, 2008). The advantages of this method that it have robust way to prepare and collect the data from questionnaires and interviews, in addition, it contributes to existing knowledge, it adds high value to the research. Moreover, it flexibles and saves a lot of time. The mixture of these two research methods is also known as triangulation method. These two types of methods have different complementary strengths and when used together can lead to a more comprehensive understanding of a phenomenon.

SL expert, practitioners, and hearing and speech-impaired will be involved in database collection, experimentation, and evaluation of the proposed framework. The mixed method has been performed as follows:

- ✓ Apply the quantitative method using the questionnaire for data collection, which aim to identify the complexity of the Arabic signs from singers' and interpreters' perspective, which will reduce the time consumption in solving simple aspects such as the hand position detections, or detection of the sequentially demonstrated gesture, and more significance and complex aspect such as detection of non-gesture movements.
- ✓ Applying the qualitative method using interview tools in conducting the sign language interpreters in order to explore the difficulty and importance aspect that may not be covered by the quantitative method, and obtain in-depth understanding and widen the relayed perspectives especially the differences among singers.
- ✓ Then a quantitative approach is re-utilized for testing and refinement purposes, which were considered the most appropriate in fulfilling the aim in evaluating the performance, which cover a large number of feedback and portion of the population.

Universiti Malaya

The more details of preparing the Qualitative Method (Questionnaire Format, Interviews) to build an ArSL database to be used for training and testing purposes are revealed in the following sections.

3.3.1.1 Qualitative Method

The purpose of a questionnaire is to identify the signs or gestures that are difficult to sign from a signer's viewpoint; human interpreters cannot easily recognise those signs, and it is not easy to identify how a signer expects the interpreter to recognise his/her signs (whether they show a special attention for the hand movements or hand-signs or both) (Easterbrook et al., 2008).

Questionnaire Participants: SL experts, practitioners and hearing- and speech-impaired individuals. The hearing- and speech-impaired individuals included males and females aged between 15–40 years. They possessed knowledge about ArSL ranging from the basic, intermediate and advance levels. The SL expert and practitioners were females and males with more than five years interpreting experience and their knowledge about ArSL ranged from the basic, intermediate and advance levels.

Questionnaire Structure: Two questionnaire structures were created, 'Questionnaire A' was distributed to the hearing- and speech-impaired individuals, and 'Questionnaire B' was distributed to the SL experts and practitioners. The questionnaires used in this thesis for collecting the ArSL database are shown in Appendix A. The following are the key points covered in both questionnaires.

The key points of the Questionnaire A.

- ✓ Knowledge about ArSLs forms.
- ✓ Knowledge in identifying letters that have the same hand movements but different hand sign.

Universiti Malaya

- ✓ Duration in studying about ArSL.
- ✓ The difficulty faced in signing gestures, which have similar hand movements but different hand signs.
- ✓ Identifying the importance of their perspective in showing the correct hand signs and movements.
- ✓ Signer's opinion in ignoring the hand movements when showing the gesture.
- ✓ To rate the difficulty level of the gestures from a given list.
- ✓ To write down a list of easiest and most difficult signs to interpret.

The key points of the Questionnaire B.

- ✓ Work experience as an SL expert.
- ✓ Duration of interpreting and teaching SL used in ArSL method.
- ✓ Difficulty in distinguishing characters having similar hand movements but different hand-sign.
- ✓ The difficulty they face in interpreting gestures, which have similar hand movements but different hand signs.
- ✓ Interpreter's opinion when signer ignores the hand movements but shows only the hand-sign.
- ✓ To rate the difficulty level of interpreting gestures from a given list.
- ✓ To write down a list of most difficult and easiest signs to interpret.

3.3.1.2 Quantitative Method

Interviews were conducted among the SL experts only as they could express their difficulty and experience easily compared to the deaf students. They also tend to have a broader viewpoint of the whole problem, which could help improve the system

Universiti Malaya

significantly, compared to the limited experience gained by the researcher throughout the research. The main points covered during the interview are as follows:

- ✓ Difficulties that they encountered when the gestures are ambiguous;
- ✓ Differences among signers (SL experts and hearing- and speech-impaired individuals); and
- ✓ Difficulties in interpretation due to the variations in the speed of hand movements when they sign.

3.3.2 Design and Model ArSL Frameworks

In this thesis, two frameworks are modelled: one using a VBR approach only (smartphone camera) and another using a hybrid approach (VBR and SBR) for the first and the second frameworks, respectively. Two taxonomies are illustrated in this thesis in which the first taxonomies are based on the approach (VBR, SBR and hybrid VBR and SBR), whereas the second taxonomies are based on ArSLR in the form of Arabic letters, numbers, isolated words and continuous sentences. The VBR framework was used for static backhand signer-independent ArSLR framework, which translated the signs of letters and numbers into their corresponding texts. The second framework is a hybrid approach signer-independent continuous ArSLR framework for ArSL forms (letters, numbers, isolated words and continuous sentences).

Framework 1: Static Backhand Arabic Sign Language Recognition

This framework is proposed to recognise the static backhand and signer-independent ArSLR framework based on the VBR approach, which translates the signs of letters and numbers into their corresponding text using a smartphone camera. Figure 3.3 shows the general framework 1 phases. As shown in Figure 3.3, the framework consists of three main phases, which are described below.

Universiti Malaya

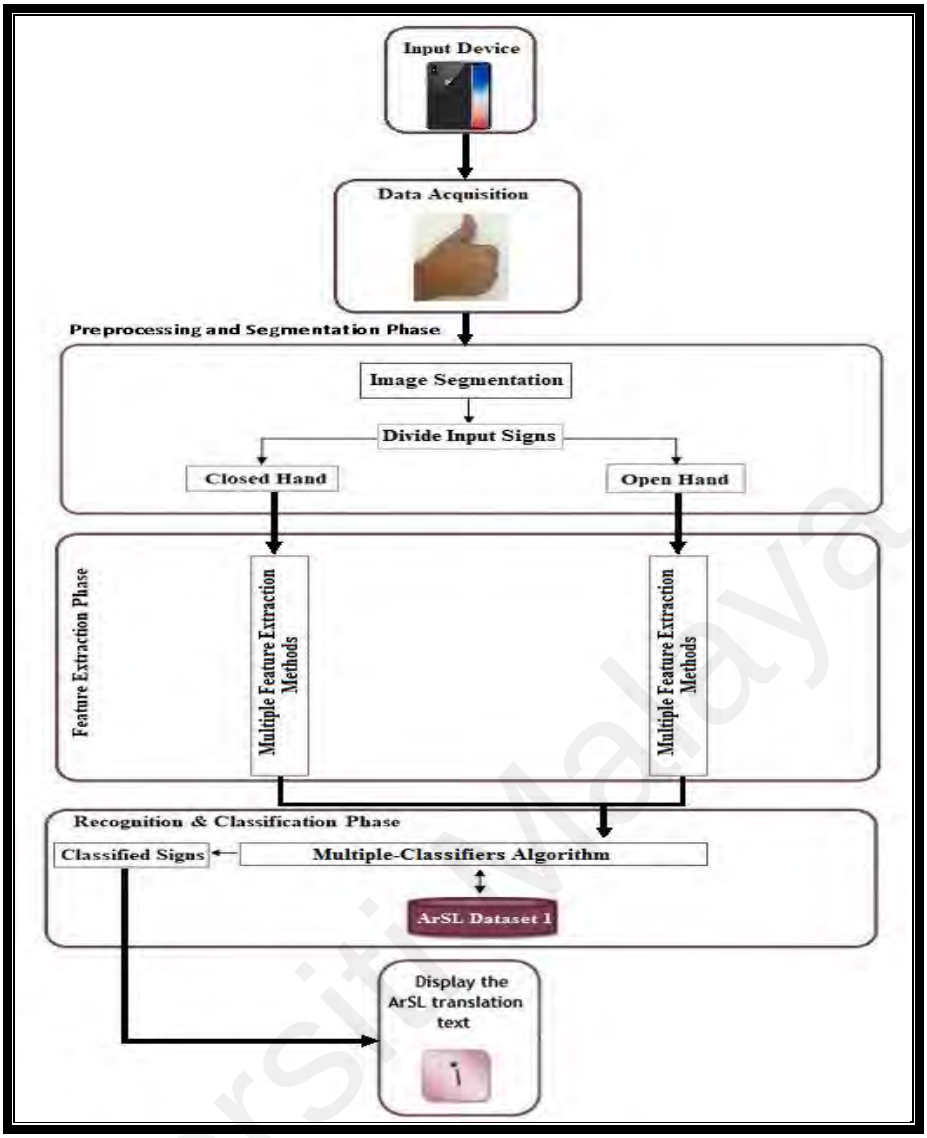


Figure 3.3: Design of the proposed framework 1

As shown in Figure 3.3, the framework consists of three main phases, which are:

1- The pre-processing and segmentation phase: This phase is further subdivided into two steps. The first step is the segmentation for the detection of hand localisation in the image through the translation of the obtained image from RGB format to white and black images, whereas the second function is the determination of the hand status (whether open or close), which is important to know as some characters are totally performed via closed hand or open hand. Therefore, the shape of each status, and the related features shall differ accordingly, and the detection of the hand status is helpful for the next phase of feature-related decision making.

Universiti Malaya

2- Feature extraction phase: This is the second phase of the ArSLR framework in which the extraction of the raw element for recognition approaches and techniques is applied in the next phases. The feature types vary according to the variation of hand status. The following features are detected: the spread degree and status of fingers (extension or flexion), shape factors and axis ratio. These shall be discussed in detail in Chapter 5. The palm and finger recognition is detected and classified into two regions (the palm region and the finger region) based on both open hand and closed hand status, and each area is represented with a distinct symbol, as shown in the Figure 3.3.

Next, the closed hand feature extraction utilises edge detection, signal construction and discrete wavelet transform, whereas the open hand status uses the coding pattern block mask, length matrix and extracting consecutive pixels. Both are described statistically by using statistical descriptors. The aforementioned techniques for both closed and open hand status are described in detail in Chapter 6.

3- The recognition and classification phase: Different classifiers, such as MLP, SVM, NBC, ANN, KNN and many more, are often used for the recognition and classification of data. However, choosing an appropriate classifier is not straightforward, because each classifier has its own strengths and weaknesses, and each dataset has its own characteristics. Thus, researchers can avoid the dilemma of choosing an individual classifier out of many to achieve an optimised classification result by integrating many classifiers together. In this phase, two classifiers are applied and integrated, namely, the NBC and MLP.

The Naïve Bayes (NB) classifier is a mathematical model classifier that classifies based on likelihood probability. In the same context, the multi-layer perceptron is a complicated mathematical model compared to the NBC, which is based on various differential equations. NBC is known as the simplest and highly scalable classification algorithm. It does not require a large training data set and can converge quicker compared

Universiti Malaya

to the discriminative model. Thus, it is suitable for both discrete and continuous data and can be adopted for binary or multi-classification issues if they are configured as conditional independence.

Multilayer Perceptron (MLP) contains multiple-layer types that consist of an input layer, hidden layer and output layer, which are stimulated by the human brain's neural system. The number of hidden layers varies among systems and may increase or decrease based on the system requirement and accuracy obtained. The MLP has many advantages: it can handle complicated or imprecise data and the recognition of complex noticed trends, it has an adaptive learning property, and it does not make assumptions based on any probability density functions.

The result of each classifier and the combination of the multi-classifier is obtained separately for comparison purposes, with the expectation of an increase of the discriminatory power of the system based on the difference between the likelihood probabilities obtained.

Universiti Malaya

Framework 2: Hybrid Approach for ArSLR Based on Data Combining

Many kinds of SL, such as ArSL, have ambiguous hand sign problems if dynamic hand or fingers are not shown, not clear, or ignored. To overcome this problem, a technology to detect the hand signs and fingers with the hand movements is required to accurately recognise the gesture. If a single device cannot perform this functionality, a hybrid approach could be used by integrating and combining the capability of two approaches (VBR and SBR) to detect signs efficiently and accurately.

This framework is modelled with a hybrid approach for an automatic continuous ArSLR framework based on signer-independent that integrates the VBR approach using Microsoft Kinect and the SBR approach that uses Smart Data Gloves. Figure 3.4 shows the general phase for the hybrid continuous ArSLR framework.

Universiti Malaya

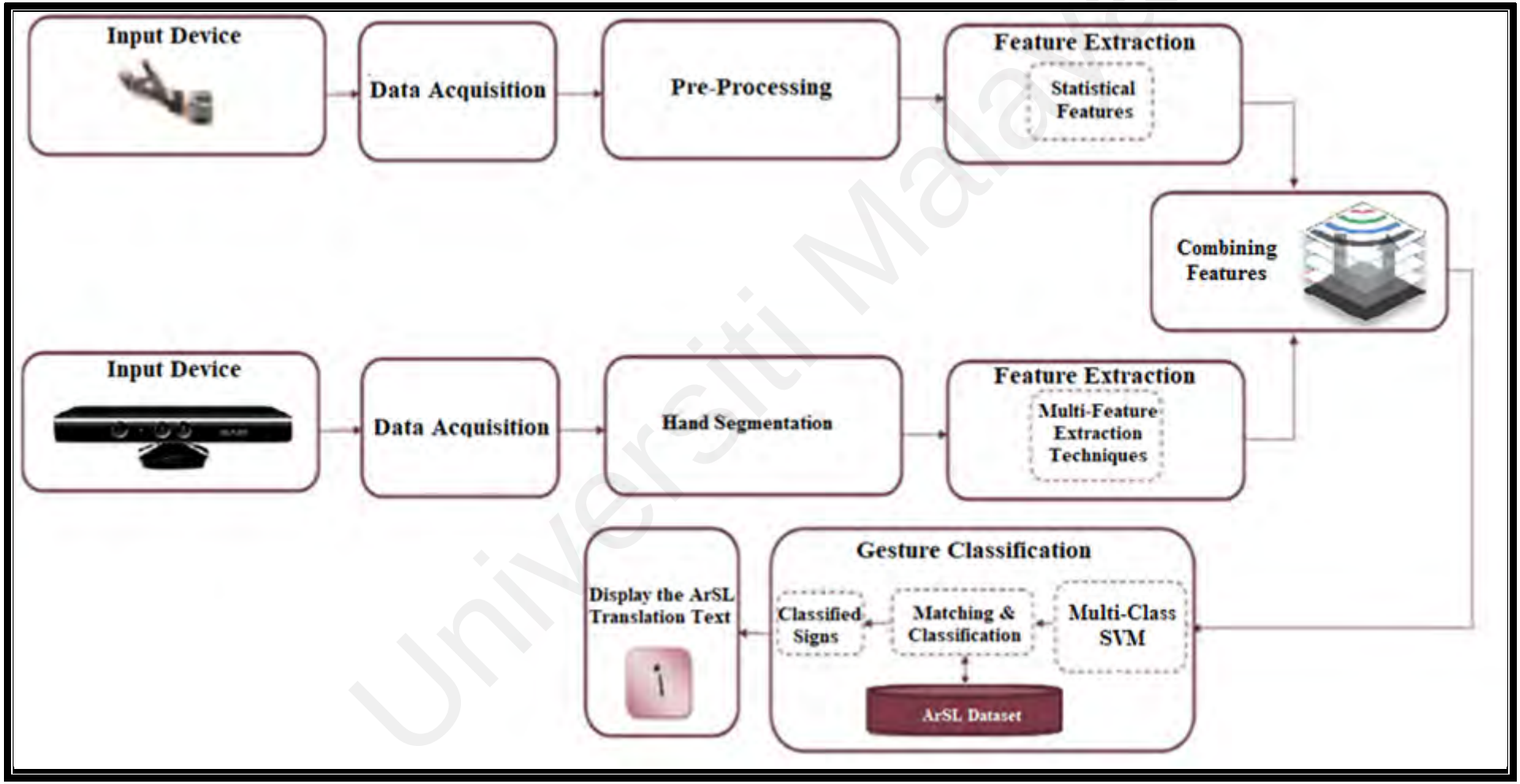


Figure 3.4: Design of the proposed framework 2

Universiti Malaya

1- Pre-Processing and Segmentation Phase:

- Smart Data Glove: Given that the input of smart gloves data is the sensor reading, then the pre-processing aim is to eliminate the distortion suppressed with the input signal. Accordingly, multiple techniques are used, such as resampling for reducing redundant feature vectors and normalisation for unifying each feature vector to the common range, signal denoising and removing unnecessary data points. Each method is discussed in detail in Chapter 6.
- Microsoft Kinect data: According to the raw depth data obtained from the Kinect controller, which does not have a regular frequency, hand segmentation is needed for pre-processing. This is described in detail in Chapter 5. However, it is worth mentioning that the hand segmentation in this approach is based on the colour and depth of images.

2- Feature Extraction Phase:

- Smart Data Glove: The features set extracted from the smart glove data conform with the sensor readings of the amount of bend in the fingers, hand acceleration and hand orientation. Accordingly, the extracted features are the flexion measurement, instantaneous acceleration measurements and hand orientation angles, respectively. The feature set details, such as ranges and measurement scales, are further discussed in Chapter 6. Multiple statistical analysis approaches are used to increase the representation of features.

- Microsoft Kinect: The features are extracted from the segmentation regions in the images. A set of the feature is extracted from Microsoft Kinect depth data defined in distance, elevation, curvature and palm area feature.

Universiti Malaya

3. Recognition and Classification Phase

It is worth mentioning that one classification approach is used for the classification, namely, multi-class SVM.

- Multi-class Support Vector Machine: SVM, which considers a suitable approach to use with the binary data classification (the multi-class SVM), has various forms. In the current research, the one-against-one approach is adopted because it has the advantage of requiring a small number of training samples, thus reducing the training time and the nonlinearity causes. The SVM also has advantages, including flexibility arising from the kernel function and threshold capabilities, the non-assumption of the functional form, high robustness based on the good out-of-sample organisation and the ability to deliver unique solutions. Figure 3.5 shows the recognition and classification phase.

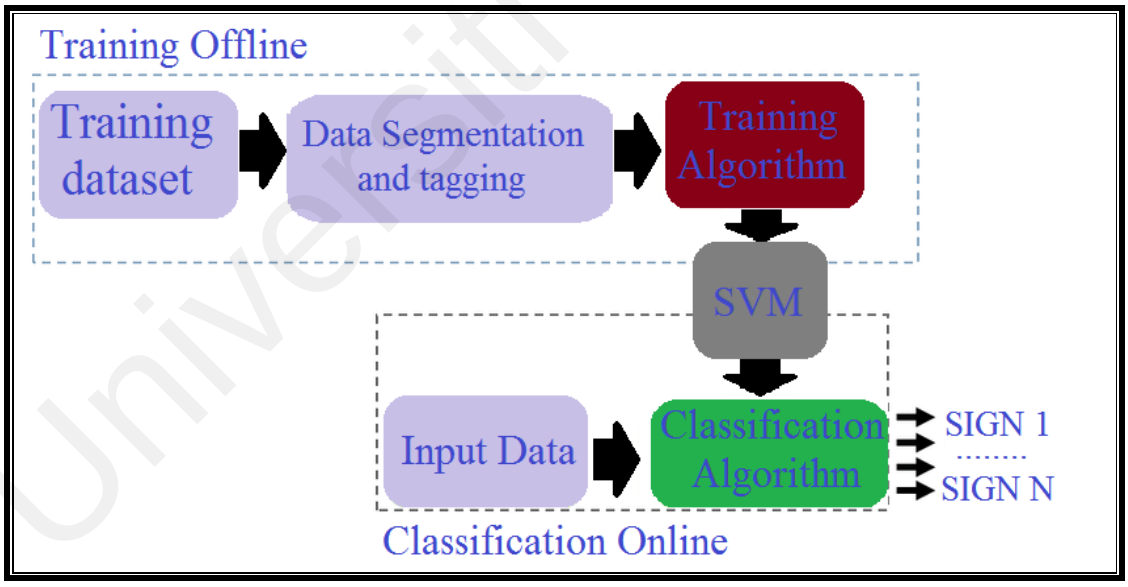


Figure 3.5: Recognition and Classification Phase

Universiti Malaya

3.4 Evaluation Phase

In the evaluation phase, an experimental evaluation is used to measure and evaluate our framework and compare the ArSLR system's performance result with the related work, which served as the performance evidence of the proposed framework. In addition, it provides a fault notation caused by the wrong recognition or classification compared to the real or actual selected case. However, for achieving the evaluating purposes, it evaluates different ArSLR frameworks in order to examine the effect of different training and testing parameters and user characteristics.

The performance of the proposed framework is evaluated based on its capability to recognise signs correctly corresponding with the input signs. The performance measure that is used to accomplish this task is called the recognition rate. The recognition rate is defined as the ratio of the number of correctly recognised gestures to the total number of input gesture samples, as shown in Equation 3.1.

Accuracy is calculated as the number of all correct predictions divided by the total number of the dataset. The best accuracy is 1.0, whereas the worst is 0.0. It can also be calculated by 1 – ERR.

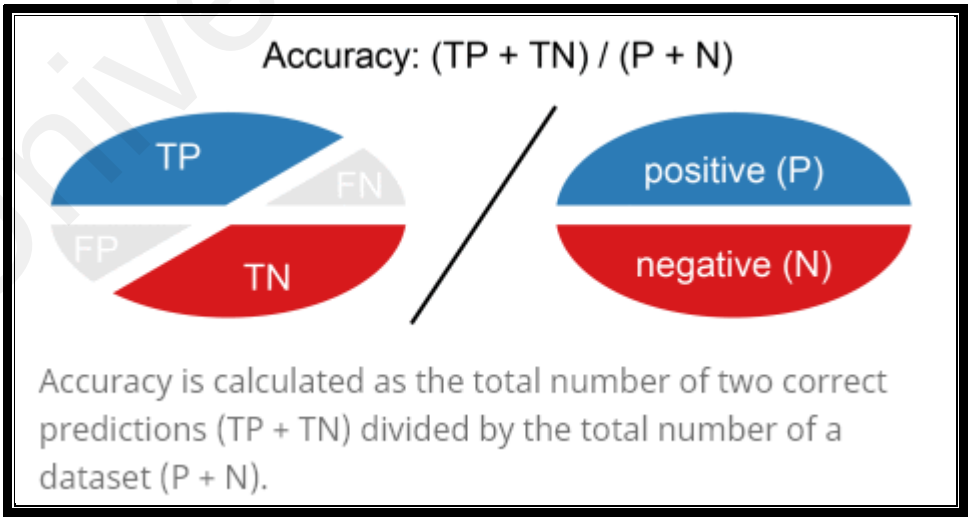


Figure 3.6: Accuray metric

$$\textbf{Recognition rate} = \frac{\textbf{Number of correctly recognized signs}}{\textbf{Total number of signs}} \times 100\% \qquad (3.1)$$

Universiti Malaya

In addition to the recognition rate, Sensitivity, Specificity, Precision, False Positive Rate (FPR), False Negative Rate (FNR), and F1-score are also considered for each sign.

Practically, the classified hand signs by each classifier may or may not be the actual sign status. Therefore, four cases are defined:

- True Positive (TP): correctly classified signs.
- False Positive (FP): incorrectly classified signs.
- True Negative (TN): correctly misclassified signs.
- False Negative (FN): incorrectly misclassified signs.

Based on these cases, the following evaluation metrics are defined:

➢ **Sensitivity (recall or true positive rate):** is calculated as the number of correct positive predictions divided by the total number of positives. It is also called recall (REC) or true positive rate (TPR). The best sensitivity is 1.0, whereas the worst is 0.0. It measures the classification accuracy of true cases as follows in equation 3.2:

$$\boldsymbol{Sensitivity} = \frac{\boldsymbol{TP}}{\boldsymbol{TP+FN}} \quad \textbf{(3.2)}$$

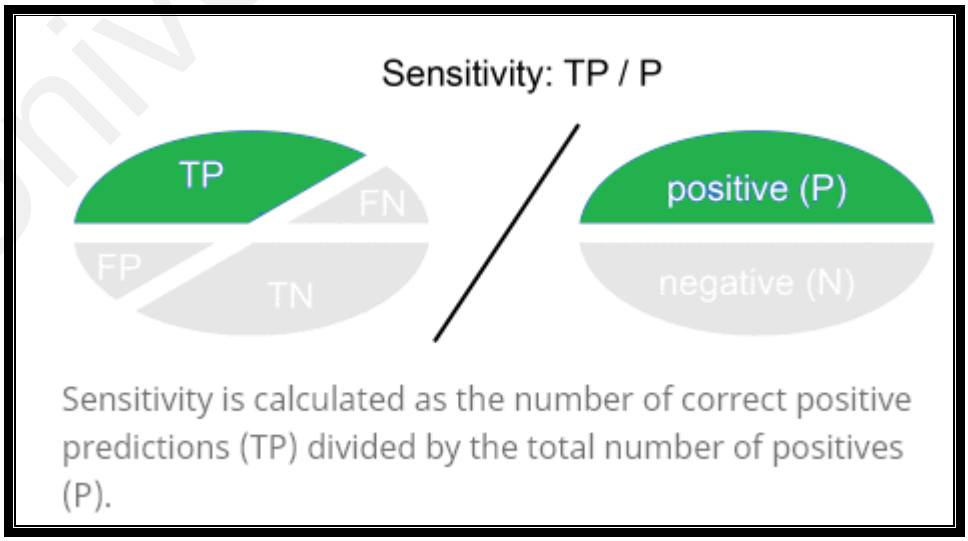


Figure 3.7: Sensitivity metric

Universiti Malaya

- **Specificity (true negative rate):** Specificity (SP) is calculated as the number of correct negative predictions divided by the total number of negatives. It is also called true negative rate (TNR). The best specificity is 1.0, whereas the worst is 0.0.

It measures the classification accuracy of false cases as follows in equation 3.3:

$$\boldsymbol{Specificity} = \frac{\boldsymbol{TN}}{\boldsymbol{TN+FP}} \quad \textbf{(3.3)}$$

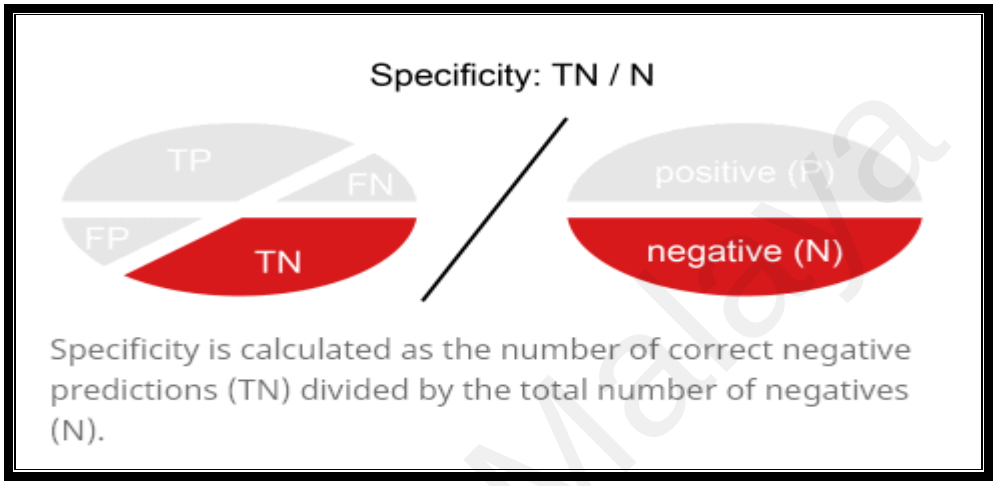


Figure 3.8: Specificity metric

- **Precision:** is calculated as the number of correct positive predictions divided by the total number of positive predictions. It is also called positive predictive value (PPV). The best precision is 1.0, whereas the worst is 0.0.

It describes random errors and measures the statistical variability.

$$\boldsymbol{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP+FP}} \quad \textbf{(3.4)}$$

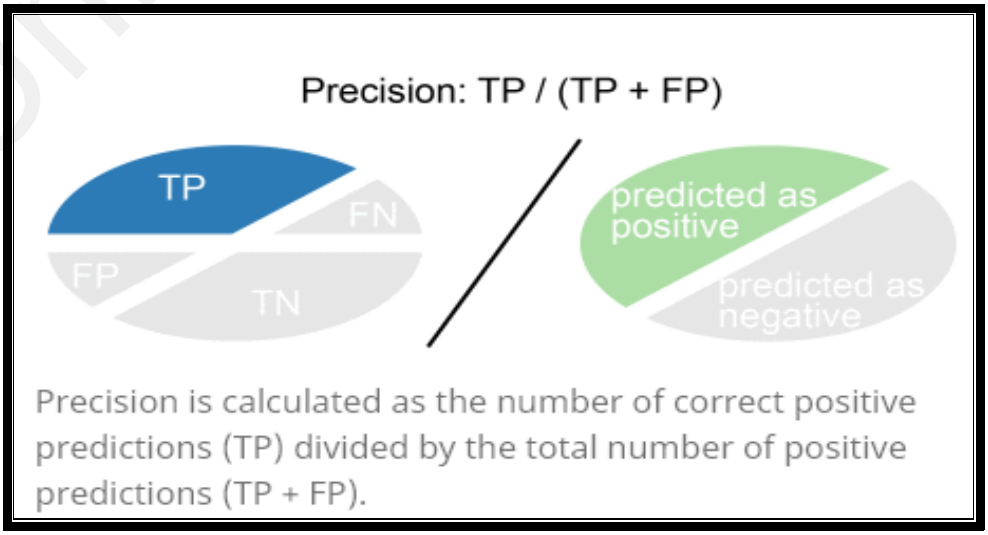


Figure 3.9: Precision metric

Universiti Malaya

➢ **False Positive Rate (FPR):** is calculated as the number of incorrect positive predictions divided by the total number of negatives. The best false positive rate is 0.0 whereas the worst is 1.0. It can also be calculated as 1 – specificity.

$$\mathbf{FPR} = \frac{\mathbf{FP}}{\mathbf{FP+TN}} = \mathbf{1 - Specificity} \quad \textbf{(3.5)}$$

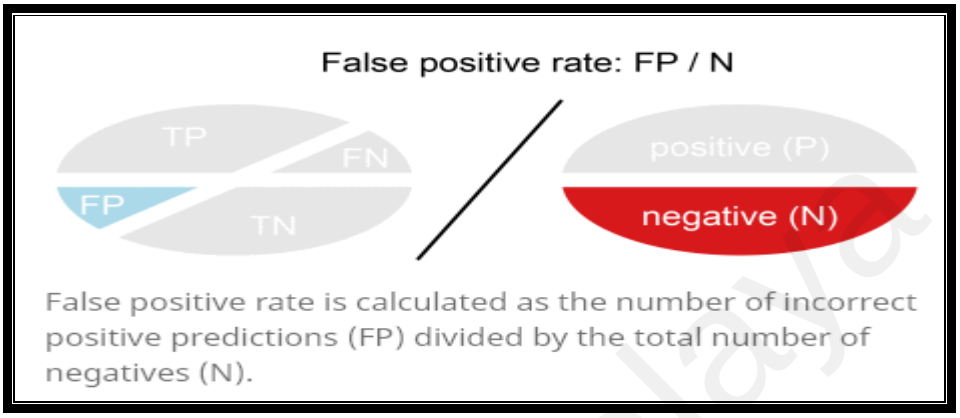


Figure 3.10: FPR metric

➢ **False Negative Rate (FNR):** it is the percentage of positives that offered negative results

$$\mathbf{FNR} = \frac{\mathbf{FN}}{\mathbf{TP+FN}} = \mathbf{1 - Sensitivity} \quad \textbf{(3.6)}$$

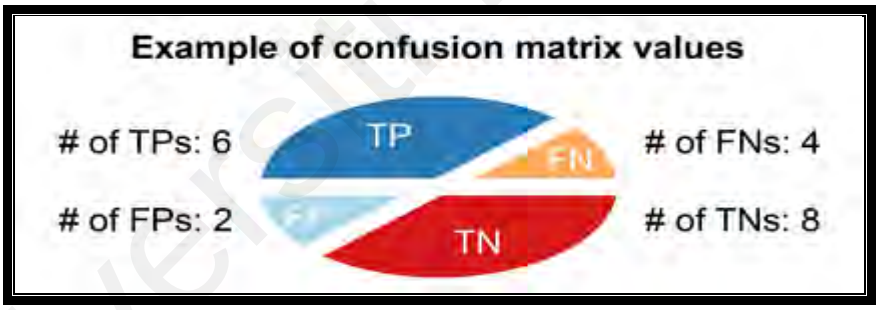


Figure 3.11: FNR metric

➢ **F1-Score (F-measure):** It measures the balance among the sensitivity and precision, in which its best value is 1.

$$\mathbf{F1 - Score} = 2 * \left(\frac{\mathbf{Precision * sensitivity}}{\mathbf{Precision + sensitivity}}\right) \quad \textbf{(3.7)}$$

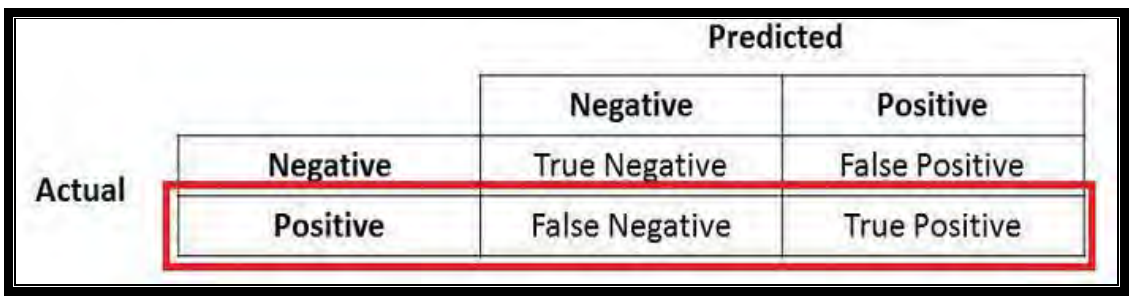


Figure 3.12: F1-Score metric

Universiti Malaya

3.5 Summary

This chapter described the roadmap used in this thesis to achieve the stated research objectives. Quantitative and qualitative data were both used to create a new SL database. In addition, a systematic literature review was conducted to identify the gaps and issues in the ArSLR domain. The above research methods were selected to achieve the research objectives of this thesis. In addition, this chapter described the performance measures used to evaluate the frameworks. The next chapter shall describe the systematic literature review method used in this research.

Universiti Malaya

CHAPTER 4: SYSTEMATIC LITERATURE REVIEW

4.1 Introduction

This chapter focuses on a systematic literature review to explore the relevant literature in ASLR and ArSLR through a highly rigorous and systematic process, thus helping us identify the main gaps and issues in this scope. A systematic literature review covers not only the content found in the literature but the methods used to find the literature, the search strategies used and how and where the search is made. A systematic literature review also importantly focuses on the inclusion or exclusion criteria used to evaluate the literature found. The systematic literature review can help identify the research gap and direction.

A well-planned search plan is essential to ensure that relevant evaluation models and criteria may appear in the search results. A systematic literature review aims at identifying, evaluating and interpreting available research related to a specific field of interest. A systematic literature review needs to follow the rigorous, fair and unbiased search plan, which can ensure the completeness of research for assessment. No systematic study at this point has provided a rigorous review and analysis of existing research in this particular domain. The current study aims to fill this gap by conducting a systematic literature review using the guidelines provided in a past work (Kitchenham & Charters, 2007; Kitchenham et al., 2010). This review process comprises numerous steps that need to be carried out in a systematic and disciplined way, as illustrated in Figure 4.1. These steps include the development of a systematic review protocol, conduct of a review according to a defined protocol, analysis, reporting and visualisation of results and, finally, the discussion of the major findings.

Universiti Malaya

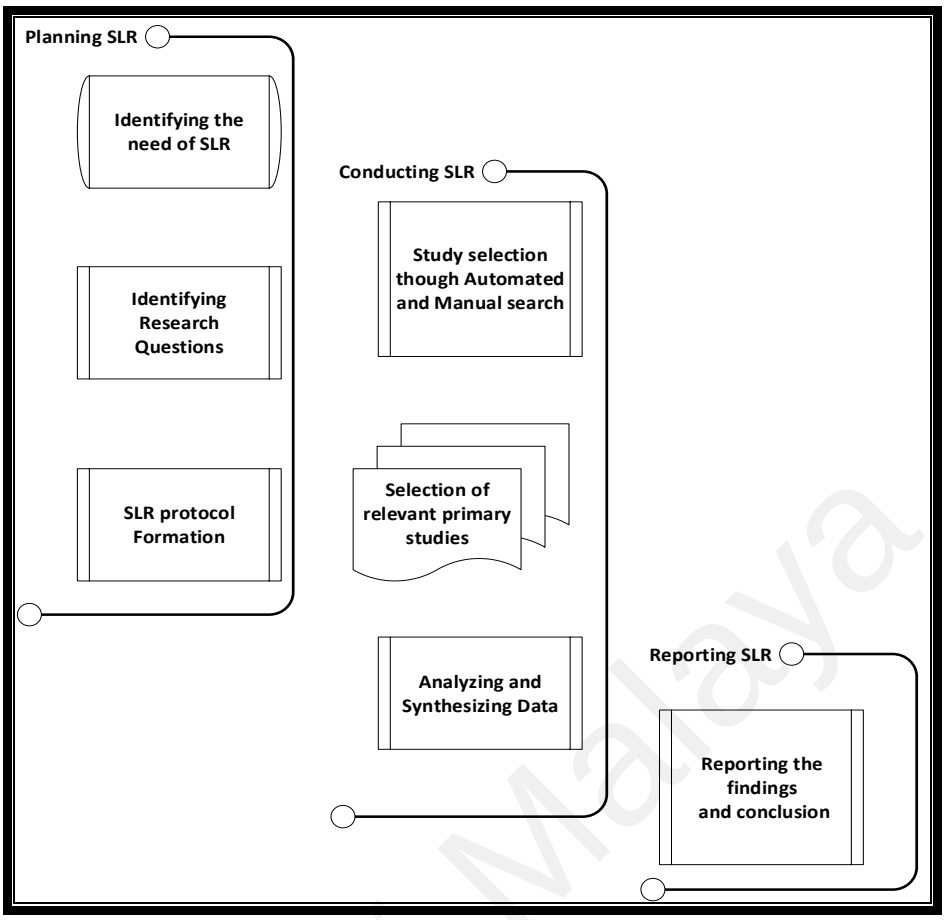


Figure 4.1: Flowchart of systematic literature review process

4.2 Research Design

This section of the study illustrates the current research requirements by identifying the research questions and keywords related to the research questions.

4.2.1 Systematic Literature Review Research Questions

This research is intended to identify the current issues with GR and ASLR, existing solutions, techniques, and technologies that affect desired with ASLR. For purpose of identifies the problem and technical gaps, the research utilized the descriptive evidence-based systematic literature review methodology.

Universiti Malaya

The first stage in a systematic literature review is about framing the main question of the review, which in our case was:

1- "What are the existing solutions and enabling technologies for GR and ASLR?"
2- "What are the available ASLR taxonomies and representation techniques used in VBR and SBR?"
3- "What are the main VBR and SBR techniques and the algorithms used for recognition specifically?"
4- "What are the application domains, commercial software, and devices that use ASLR for interaction in different environments?"

4.2.2 Search Process

In this stage, it is identifying the relevant publication and research. This systematic literature review concentrates on searching several digital database libraries and electronic sources, as it is assumed that the main research results from books and reports are also usually described or referenced in scientific papers. The selected electronic databases are including IEEE Xplore, Springer Link, Science Direct, ACM, Emerald Insight, Wiley, and Google Scholar.

The relevant studies was chosen, were the relevance was the inclusion and exclusion criteria for selection relevant work, then a 100 scientific publications were chosen, published in last two decades (1996-2018). Systematic literature review activates were presented in the Figure 4.2. The selection procedures were presented in the Figure 4.2.

Universiti Malaya

4.2.3 Search Terms

The following search keywords were used to find relevant studies from titles, keywords, and abstracts.

The main keywords used in the research were:

"Automatic sign language recognition" OR "sign language recognition" OR "gesture recognition" OR "hand gesture recognition" OR "hand gesture approach" OR "gesture recognition technologies" OR "sensor based" OR "vision based" OR "Sensor based" OR "recognition techniques" OR "gesture taxonomy" OR "gesture representation" OR "automatic sign language recognition" AND "gesture recognition" OR "hand gesture recognition" OR "sign language" AND "vision based recognition" OR "sensor based recognition" OR "Arabic sign language recognition" OR "sign language recognition" AND "hand gesture recognition" OR detection OR tracking OR recognition OR "vision based recognition" OR "sensor based recognition" OR "human computer interaction" OR "gesture recognition" AND "sign language recognition" OR "application of sign language" OR application of hand gesture recognition" OR "natural interfaces".

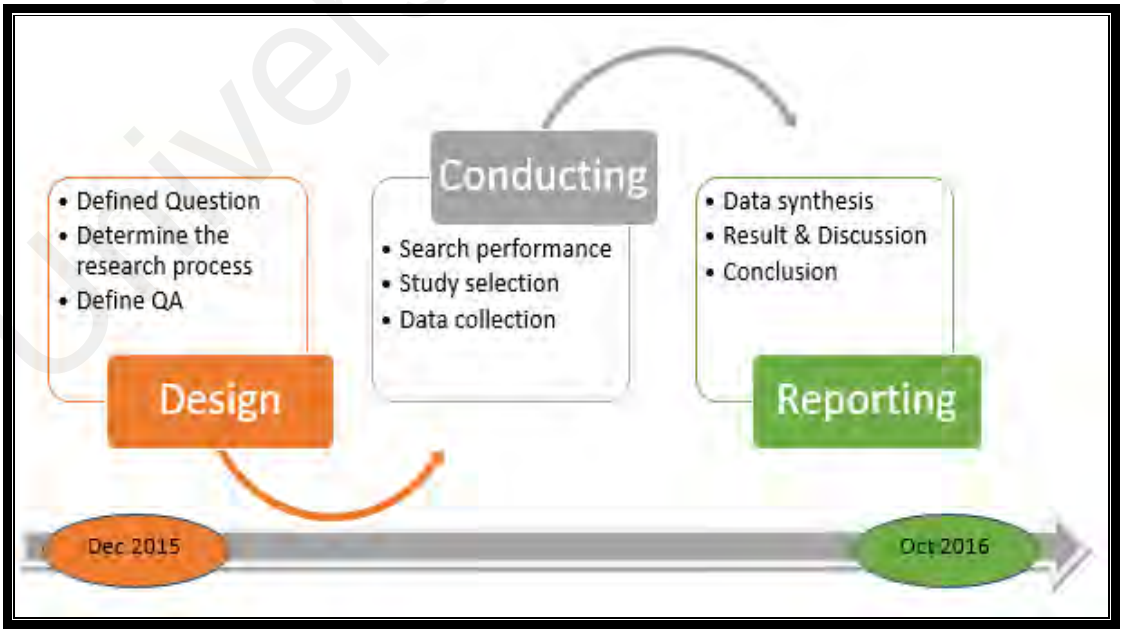


Figure 4.2: Systematic literature review activity

Universiti Malaya

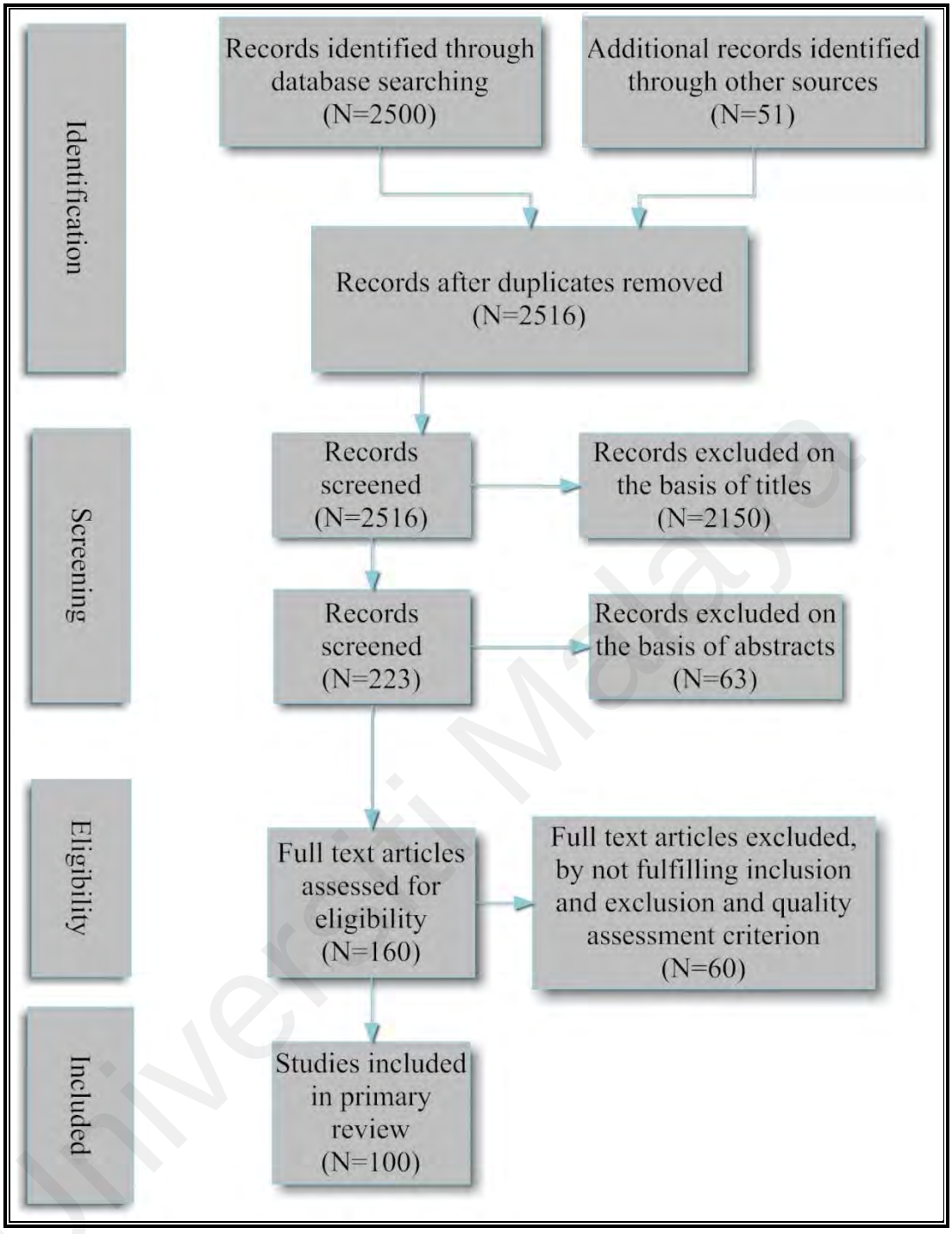


Figure 4.3: Systematic literature review: selection procedure for publications

4.3 Review Process

This section defines the review protocol for conducting the Systematic literature review. The review protocol of this thesis is defined based on the structure and rules of conducting reviews.

Universiti Malaya

4.3.1 Inclusion and Exclusion Criteria

Table 4.1 lists the inclusion and exclusion criteria for the conducted Systematic literature review. The intentions for inclusion and exclusion criteria are to select for potentially relevant studies for review from data sources to answer research questions and avoid irrelevant studies base on Kitchenham (Kitchenham et al., 2010).

Table 4.1: Inclusion and exclusion criteria involved in the search process

Inclusion Criteria
1. Articles that are peer reviewed original articles
2. Articles that have a proposed approach for gesture recognition and Sign language
3. Articles that utilize hand gesture recognition and Automatic sign language
4. Articles that use sensor-based/vision-based techniques
5. Most recent articles in the case of multiple studies with the same purpose
Exclusion Criteria
1. Studies written in languages other than English
2. Studies with no validation of the proposed techniques and algorithms
3. Articles providing unclear results or findings
4. Short papers, posters, short communications and patents.
5. Duplicated studies (by title or content)

4.3.2 Study Selection

The study selection process contains 5 main parts as shown in Figure 4.4. To prevent bias, another researcher was allocated to check and review the primary studies selected.

Universiti Malaya

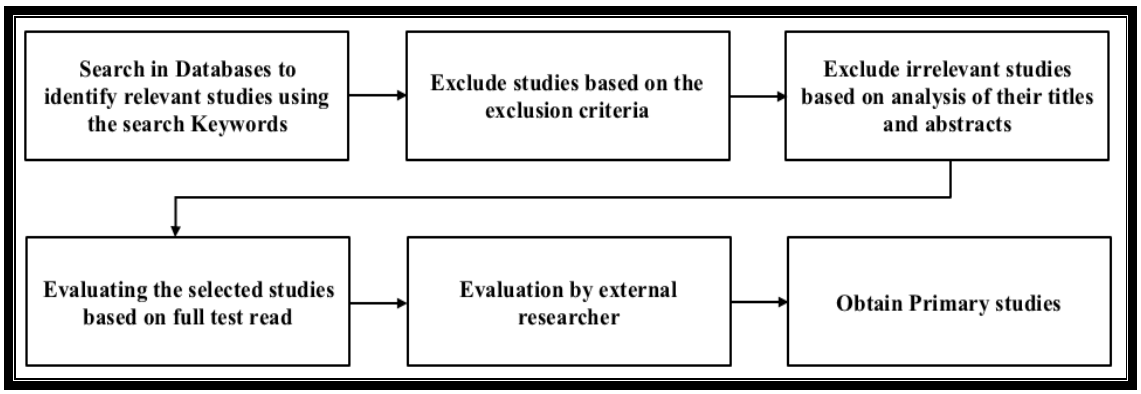


Figure 4.4: Study selection process

4.3.3 Data Extraction Form

The desired information was extracted from the selected papers based on the paper selection form shown in Table 4.2. Data extraction allows gathering desired information from selected papers to answer the defined search questions. The following table presents the data extraction employed for the primary studies in order to carry out an in-depth analysis.

Table 4.2: Data Extraction

Data Extraction Form	
(1) Identify of study	(2) Publication venue
(3) Bibliographic reference	(4) Type of study
(5) Empirical research method (experiment, case study, survey, other)	(6) Contribution / Finding (method, model, framework, tool, other)
(7) Research questions	(8) Classification (according to predefined classification schema)
(9) Focus of study	(10) Current issues

4.4 Synthesis

Table 4.3 and Figure 4.5 shows the final number of papers found per source based on the keyword searches in the selected databases. The numbers of papers are those selected from each source after the inclusion process. The significant gap between studies found

Universiti Malaya

and candidate studies from Google Scholar constitutes papers that were duplicates, short, irrelevant or inappropriate.

Table 4.3: Number of paper in final selection

Database Num.	Electronic Database	Final selected studies
Database 1	ACM	10
Database 2	Wiley	5
Database 3	IEEE Xplore	49
Database 4	Science Direct	16
Database 5	Emerald Insight	4
Database 6	Springer Link	9
Database 7	Google Scholar	7
		100

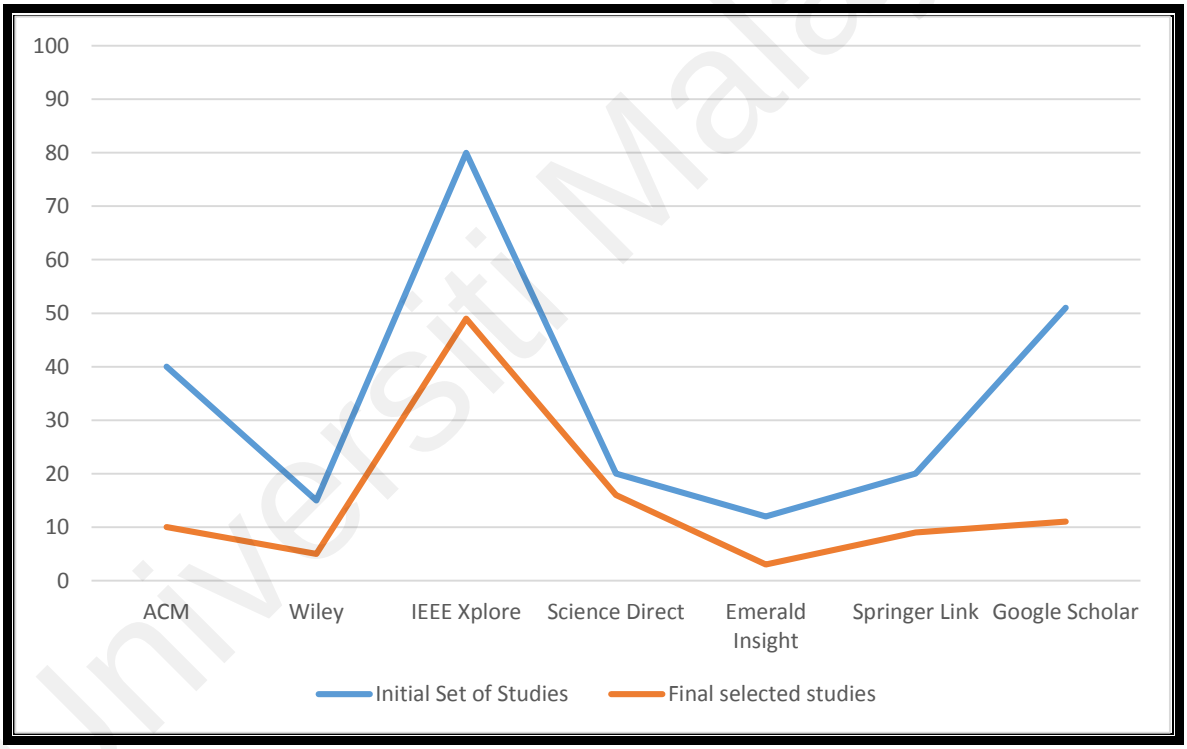


Figure 4.5: Proportion of selected studies

All publications evaluated were split into four categories, as shown in Figure 4.6. For this research, the proportion of scientific journals perused was 60% of all references employed. In addition, the proportion of conference proceedings was 36%. Finally, workshop papers and theses comprised 2% of all references used. Figure 4.7 presents a chronological distribution of studies over the years.

Universiti Malaya

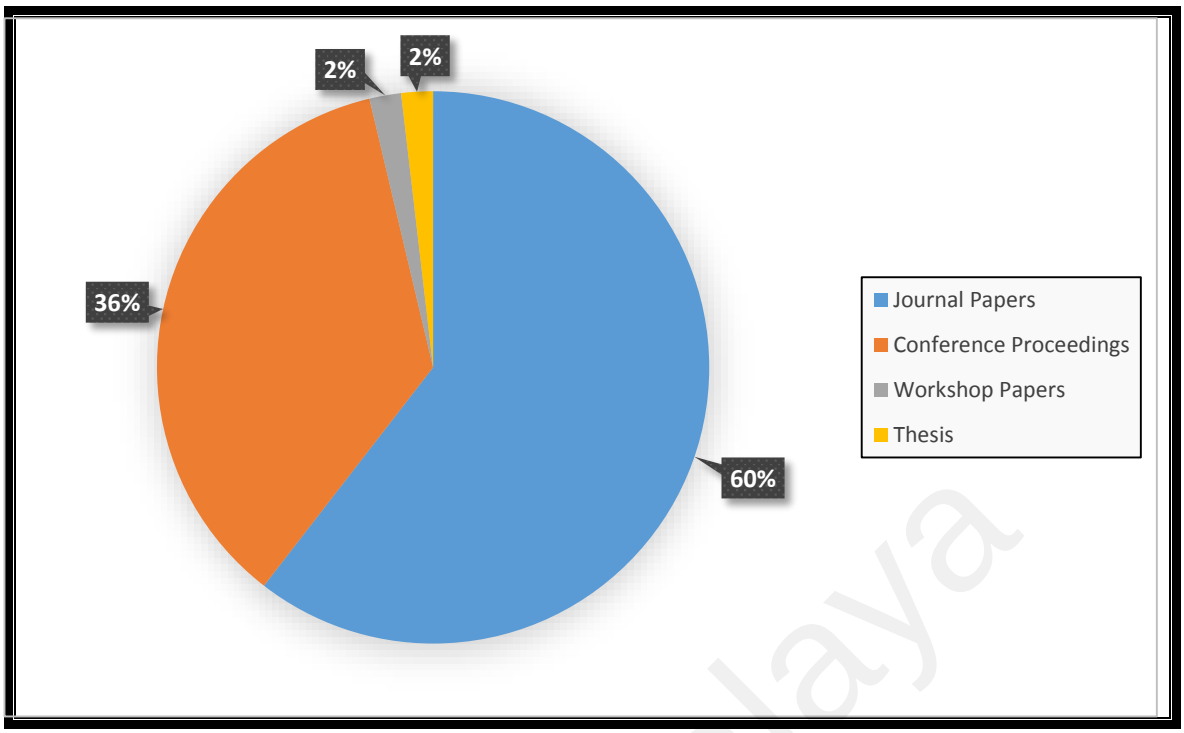


Figure 4.6: References by category specifically dedicated to gesture recognition from over 100 scientific documents evaluated.

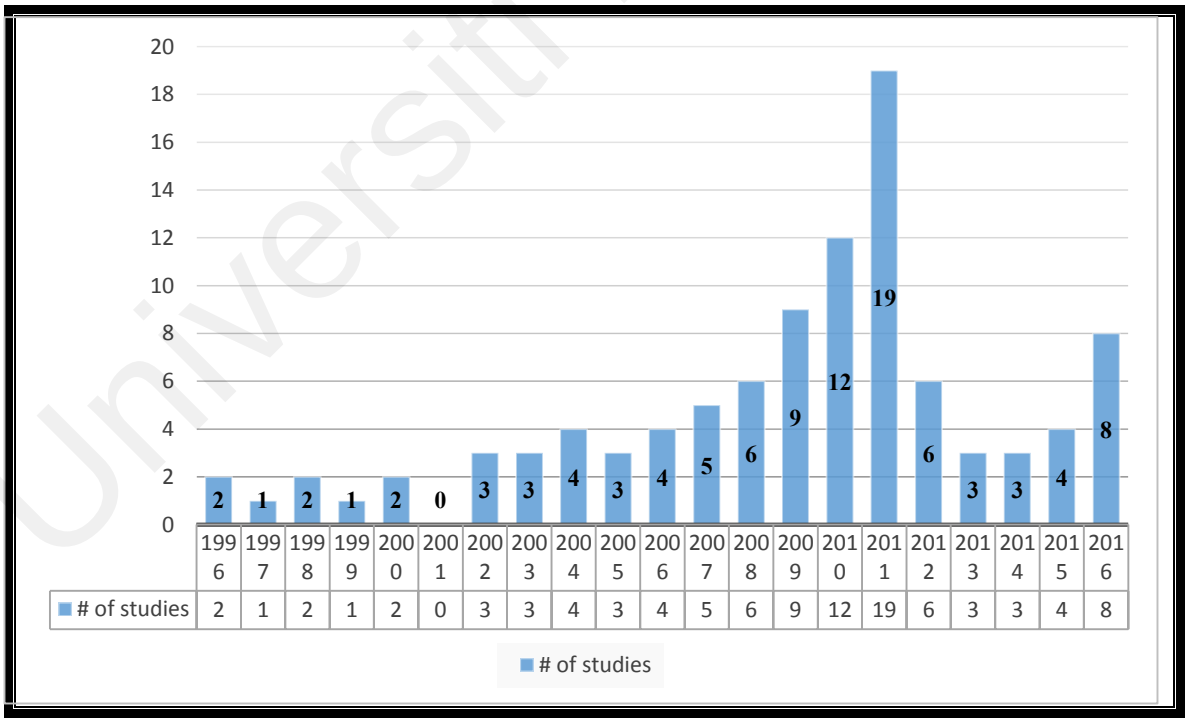


Figure 4.7: Chronological distribution of studies over the years

Universiti Malaya

4.5 Systematic literature review Results and Discussion

Once the primary studies are identified, data can be extracted based on the extraction form defined earlier. The purpose of extracting data from the primary studies selected is to answer the formulated research questions. Thus, in accordance with the meaning of each question and the extraction form, the findings of this Systematic literature review are obtained. This section represents the findings and a discussion of this review in order to answer the defined Systematic literature review research questions. In addition, a general discussion on the primary studies selected based on the defined data extraction form is expressed. As a result, this section contains findings and a discussion on the Systematic literature review research questions.

4.5.1 What are the existing solutions and enabling technologies for ASLR?

4.5.1.1 Analysis Solutions Approaches for ASLR

Generally, there are two types of ASLR analysis approaches that are used for perceiving hand gestures: 1) sensor-based analysis and 2) vision-based analysis (Pavlovic, Sharma, & Huang, 1997; Ionescu, Coquin, Lambert, & Buzuloiu, 2005).

The first type of approaches requires the integration of optical or mechanical sensors in a glove that contains electrical signals implemented in the finger flexors to perform hand gestures. The hand position can be identified by using an extra sensor. An acoustic or magnetic sensor is then connected to a special glove that can receive signals. This glove is connected to a software toolkit to convey data that is relevant to hand posture recognition.

The software toolkit and glove are provided together to obtain some data about glove applications involved in hand posture recognition. This approach is used to identify and understand American Sign Language (Parvini & Shahabi, 2007), with a 75% hand gesture recognition rate determined. However, this approach requires utilizing a cumbersome device and a large number of wires connected to a computer, which are among the

Universiti Malaya

limitations of its use (Pavlovic et al., 1997). Another hand gesture model was proposed, which uses a scale of 0 to 10, where each number on the scale represents particular hand gestures (Swapna, Pravin, & Dharaskar Rajiv, 2011). This system involves various stages of image capturing and other recognition applications. The recognition rate of this system is reportedly 89%, but this system functions only in accordance with some criteria that may include the use of colored hand gloves and a black background.

The second type of approaches known as Vision-based Analysis depend on how humans recognize signals pertaining to their surrounding environment (Ionescu et al., 2005). With this method, various extraction techniques are used to extract some features derived from particular images. Techniques that can be used to extract certain hand gesture include the Orientation Histogram (Freeman & Roth, 1995), Wavelet Transform (Triesch & von der Malsburg, 1996), Fourier Coefficients of Shape (Licsár & Szirányi, 2002), Zernic Moment (C. C. Chang, Chen, Tai, & Han, 2006), Gabor filter (A. Amin & Hong, 2007)(Huang, Hu, & Chang, 2009)(S. Gupta, Jaafar, & Ahmad, 2012), Vector Quantization (Hu, Shen, & Zhao, 2014), Edge Codes (Meng & Liu, 2003), Hu Moment (Yun & Peng, 2009), and Finger-Earth Mover's Distance (FEMD) (Ren, Yuan, Meng, & Zhang, 2013).

The majority of ASLR techniques integrate mechanical sensing, which is normally used to control signals revealed by the environment and surroundings. Mechanical sensing is also used in communications that involve only symbols. On the other hand, the use of mechanical sensing for hand gesture recognition is linked with some issues pertaining to the accuracy and reliability of the data obtained as well as noise that might be produced by electromagnetic devices. Visual sensing can facilitate the process involved in gesture interaction, but sometimes it produces sophisticated problems linked to machine errors (Symeonidis, 1996). Table 4.4 presents a selected synthesis of studies on hand gesture analysis approaches that are classified in two dimensions. The first

Universiti Malaya

dimension highlights sensor-based analysis while the second dimension highlights vision-based analysis. Table 4.5 presents a comparison between sensor-based and vision-based gesture recognition. Figure 4.8 shows a statistical analysis of the usage of sensor-based and vision-based techniques in vision-based gesture recognition.

Table 4.4: Synthesis of hand gesture analysis approaches

Analysis approaches	Studies
Sensor-based analysis	S2, S18, S19, S25, S28, S29, S32, S36, S37, S42, S47, S48, S49, S50, S59, S70, S74, S75, S85, S86, S87, S88, S89.
Vision-based analysis	S1, S3, S4, S5, S6, S7, S8, S9, S10, S11, S12, S13, S14, S15, S16, S17, S20, S21, S22, S24, S25, S26, S27, S29, S30, S33, S34, S35, S38, S39, S40, S41, S43, S44, S45, S46, S51, S52, S53, S54, S55, S56, S57, S58, S60, S61, S62, S63, S64, S65, S66, S67, S68, S69, S71, S72, S73, S76, S77, S78, S79, S80, S81, S82, S83, S84, S88, S90, S91, S92, S93, S94, S95, S96, S97, S98, S99, S100.

Table 4.5: Comparison between sensor-based and vision-based gesture recognition

Concept	Sensor-Based Gesture Recognition	Vision-Based Gesture Recognition
Feature Extraction	Relatively easier	Challenging
User Experience	Inconvenient	Good
Cost	High	Low
User Dependency	Less prone	Highly prone
Calibration	Required but stable	Environment-dependent
Accuracy	High	Low
Speed	High	Low
Open Issues	Low	High

Universiti Malaya

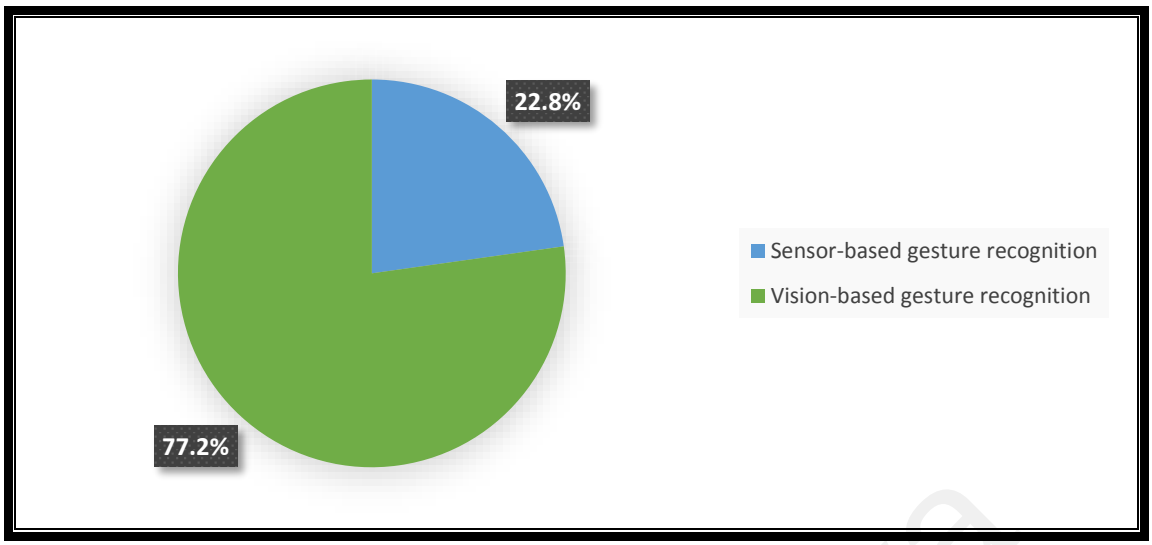


Figure 4.8: Statistical analysis of the usage of sensor-based and vision-based techniques

Based on Figure 4.8, it is found that the research community tends to use vision-based over sensor-based techniques in gesture recognition, as 77.2% of the research articles address vision-based methods whereas 22.8% address sensor-based methods.

4.5.1.2 Enabling Technologies for Hand Gesture Recognition

Human Computer Interaction design involves the use of various human body parts and is of interest to decision-makers. The complex procedure involved in the interaction process between humans and machines is not as clear as the process involved in human interaction itself. The processes differ according to the various factors involved in the usability, functionality and financial aspects of the machines employed (Karam, 2006).

A gesture is defined as the movement of a certain body part, such as an arm, hand or face to convey or deliver a message (Mitra et al., 2007). Karam (2006) reported in their study that the hand is the body part most commonly used for communication between humans as compared with other body parts. Therefore, the hand may be the most suitable part for use in computerized human interaction technologies. Indeed, the integration of hand gestures in this field is considered as one of the most interesting research topics in gesture taxonomy. Moreover, the process involved in analyzing and interpreting hand

Universiti Malaya

gestures into useful commands is known as Gesture Recognition. Thereby, the main goal of using hand gesture recognition is to build a system that can be used to recognize human hand gestures as input and that can subsequently use these gestures to interpret and analyze data in order to produce output data according to certain criteria provided in the input. ASLR technologies are divided in two domains: 1-Sensor-based and 2-Vision-based as shown in Figure 4.9.

A. Vision-based Recognition Approach (VBR):

VBR approach depends on providing a group of static and dynamic images (such as video) (Badi, 2016). Generally, the signers are requested to stop shortly between the signs so that the produced signs are of good quality and to ease the signs segmentation process. The major benefit of vision-based approach in ArSL is that the users are not required to wear the ponderous DataGlove. On the other hand, there are various challenges posed by using the vision-based approach in ArSL such as images background, hand segmentation, face segmentation, lighting conditions. In addition, the segmentation of lips, facial expressions and hands gesture is computationally expensive. Nowadays, the techniques and algorithms have ability to implement and perform the segmentation in real time. However, VBR approach is still limited and needs more to be developed (Al-Shamayleh et al., 2018; Cheok et al., 2017; Mitra et al., 2007):

- **Traditional monocular cameras (Single Camera):** These are the most commonly used cameras owing to their affordable cost. Specific variants can be implemented, such as smartphone camera, webcam, video camera, fisheye cameras to provide a clear vision, and time-of-flight cameras to measure information with thorough details.

Universiti Malaya

- **Active techniques (Stereo camera):** use the projection of structured light and using multiple monocular cameras to provide depth information such as Kinect and LMC.
- **PTZ cameras:** Pan-Tilt-Zoom cameras allow the vision system to particularly focus on details in the captured scene to identify the nature of the scene.
- **Hand markers:** Vision-based system that employs hand markers, which are responsible for detecting human gestures and hand motions. The hand markers are categorized as reflective markers of unknown nature. However, LED light is considered more active and it releases flashes occasionally. The camera used within such system has a specialized marker to identify positions in 2D and it shines when it passes through normal light. Throughout the pre-processing step, data are extracted for analyzing and identifying positions in a 3D frame.
- **Infrared cameras:** Normally used for night vision, infrared cameras release a brittle view of the human silhouette.

B. Sensor-Based Recognition Approach (SBR):

SBR approach processes data obtained from Smart Gloves, whereby such gloves depend on sensors. The Power-Glove, Data-Glove, and Cyber-Glove have been generally used for GR as shown in Figure 2.8. Using the data obtained from the smart gloves, huge number of features can be extracted such as finger bending, orientation of the hand, movement, rotation, and position. In addition, these features can be used by the classifier to detect and recognize the optimal sign. These devices can be classified into five categories (Kaaniche, 2009; Cheok et al., 2017):

- **Mechanical:** ImmersionC proposed "Cyber-Glove IIR," a wireless instrumented glove that can be used for gesture recognition. Animazoo suggested a body suit known as "IGS-190R" to capture body gestures. This type of device is usually

Universiti Malaya

used along with other types of devices. For example in (Ranganath, Ghosh, & Kevin, 2004), a method was proposed that integrates trajectory modelling in gesture recognition along with magnetic trackers and cyber-gloves. Similarly, the IGS-190 body suit comes with eighteen inertial devices (gyroscopes) that allow for motion detection.

- **Inertial:** These are a type of device that can be used to measure magnetic field variations in order to detect the earth's motions. Two kinds of such device are accelerometers, e.g. Wii-moteR, and gyroscopes, e.g. IGS-190R. A model was proposed to identify gestures along with a Wii-controller that is independent of the target system used in HMM (Schlömer et al., 2008). The user can recognize personalized gestures that are implemented in multimodal intuitive media browsing. The authors in (Noury et al., 2003) and (Bourke, O'Brien, & Lyons, 2007) proposed detection techniques for normal gestures through the use of accelerometers.
- **Haptics:** Multi-touch screens are becoming more popular in everyday life, such as tablets, PCs, Applec and iPhoneR. The authors in (Webel, Keil, & Zoellner, 2008) proposed a system to recognize multi-touch gestural interactions by using HMM.
- **Magnetic:** These devices require the integration of a tool that measures the variations of an artificial magnetic field to detect motions. Unlike inertial devices, magnetic devices present some health issues associated with artificial electro-magnetism (Kaaniche, 2009).
- **Ultra-sonic**: These are motion trackers that consist of three types of devices: (1) sonic emitters that release ultrasound, (2) sonic discs that reflect ultrasound (wired by the person) and (3) multiple sensors that measure the time it takes a pulse to return (Kaaniche, 2009).

Universiti Malaya

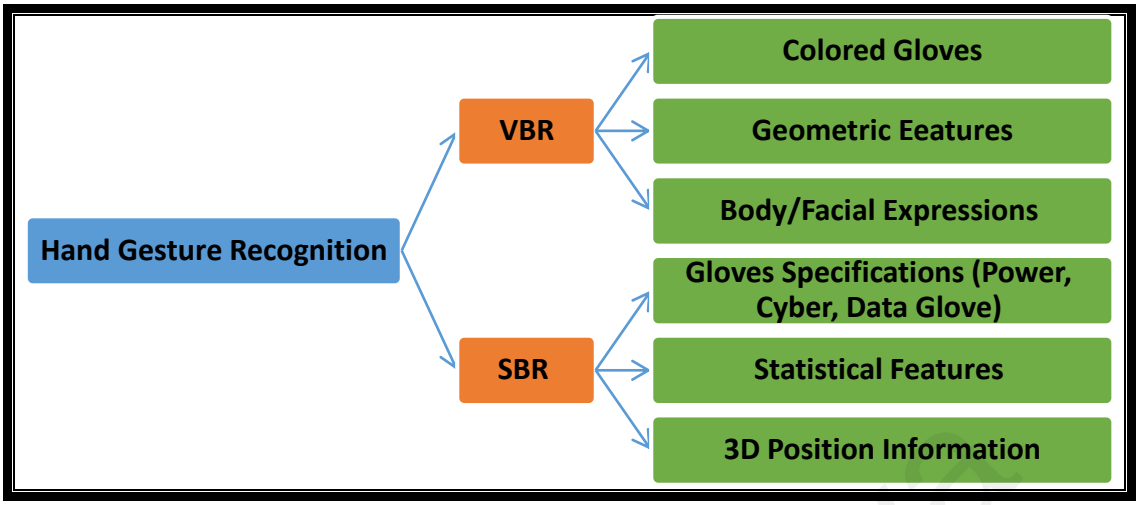


Figure 4.9: Taxonomy of hand gesture recognition



Figure 4.10: Several types of smart gloves: Power Glove (left), DT-Data Glove (Center), CyberGlove (right).

4.5.2 What are the available ASLR taxonomies and representation techniques used in VBR and SBR?"

4.5.2.1 Hand Gesture Recognition Taxonomies:

Signs are a technique of transferring non-vocal communication. Sometimes it is combined with verbal communication to deliver commands that hold meaningful information. Signs are conjugated with various parts of the body and an essential part of human communication and are considered an essential aspect of hand gesture recognition. However, meanings are linked with various types of gestures that differ according to cultural background. Gestures based on cultural background do not usually have universal meanings in some instances. Thereby, gesture interpretation is usually correlated with a particular cultural background.

Universiti Malaya

In theory, signs are classified in two types as shown in Figure 4.11, namely static and dynamic gestures. Static gestures are correlated with hand positions as well as the orientation in a given space and a certain period of time without being involved in any kind of movement. However, gestures that are described as dynamic are related to hand orientation with a certain type of movement. Dynamic gestures may include movements of some parts of the hand, such as waving. However, static gestures may include a single orientation of the hand without performing any kind of movement. Figure 4.12 presents a taxonomy of hand gesture categories. According to a study (Kaaniche, 2009), around 35% of communication among humans involves verbal conversations, whereas 65% is based on the use of nonverbal communication. Gestures are divided into five types, including regulators, emblems, illustrators and adapters, and displays. Emblematic and emblem gestures are based on the direct transformation of short verbal communication, which may include nodding for assurance and hand waving.

Quotable gestures are based on cultural aspects. Gestures that convey inner intention or sensation are known as affect displays. Generally speaking, affect displays are not associated with culture.

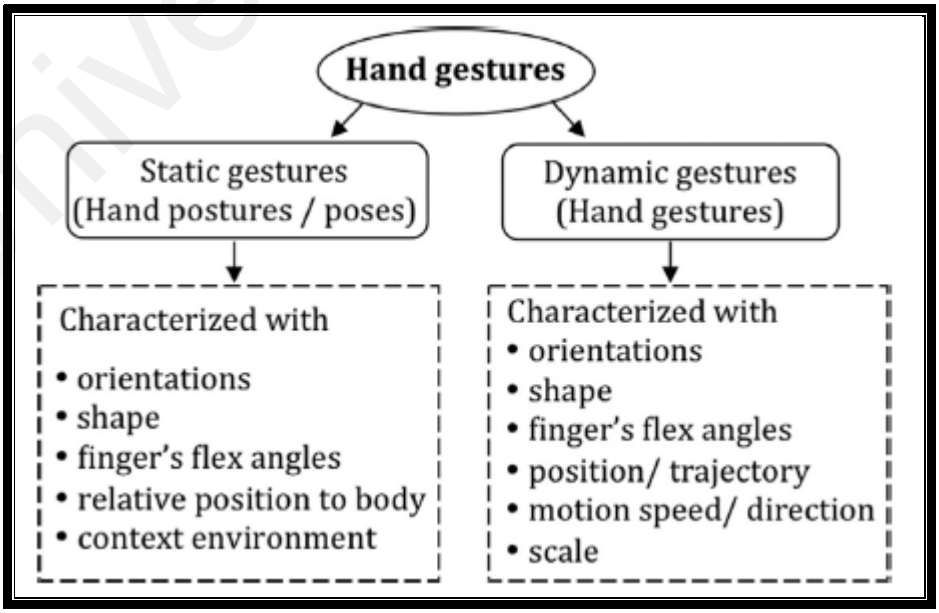


Figure 4.11: Classification of hand gestures recognition (Pisharady & Saerbeck, 2015)

Universiti Malaya

Gestures that are used to define interactions are known as regulators. Other gestures used for head shaking and foot movements to relieve tension are known as adapters, which are not intended to be used in communication. Illustrator gestures emphasize important points in speech and are thereby based on the speaker's speech and thought process. Illustrators are divided in five classes: beats, deictic gestures, iconic gestures, metaphoric gestures, and cohesive gestures.

- Beats Gestures are quick, short, and concurrent.
- Deictic Gestures involve the process of identifying a person or an object at a particular time.
- Iconic Gestures involve the process of moving the hand to represent or illustrate some figures.
- Metaphoric Gestures involve revealing abstractions.
- Cohesive Gestures are related to each other only thematically but are occasionally irrelevant. The separation of thematically related gestures is the result of a misunderstanding between the communicator and any other communicator.

Universiti Malaya

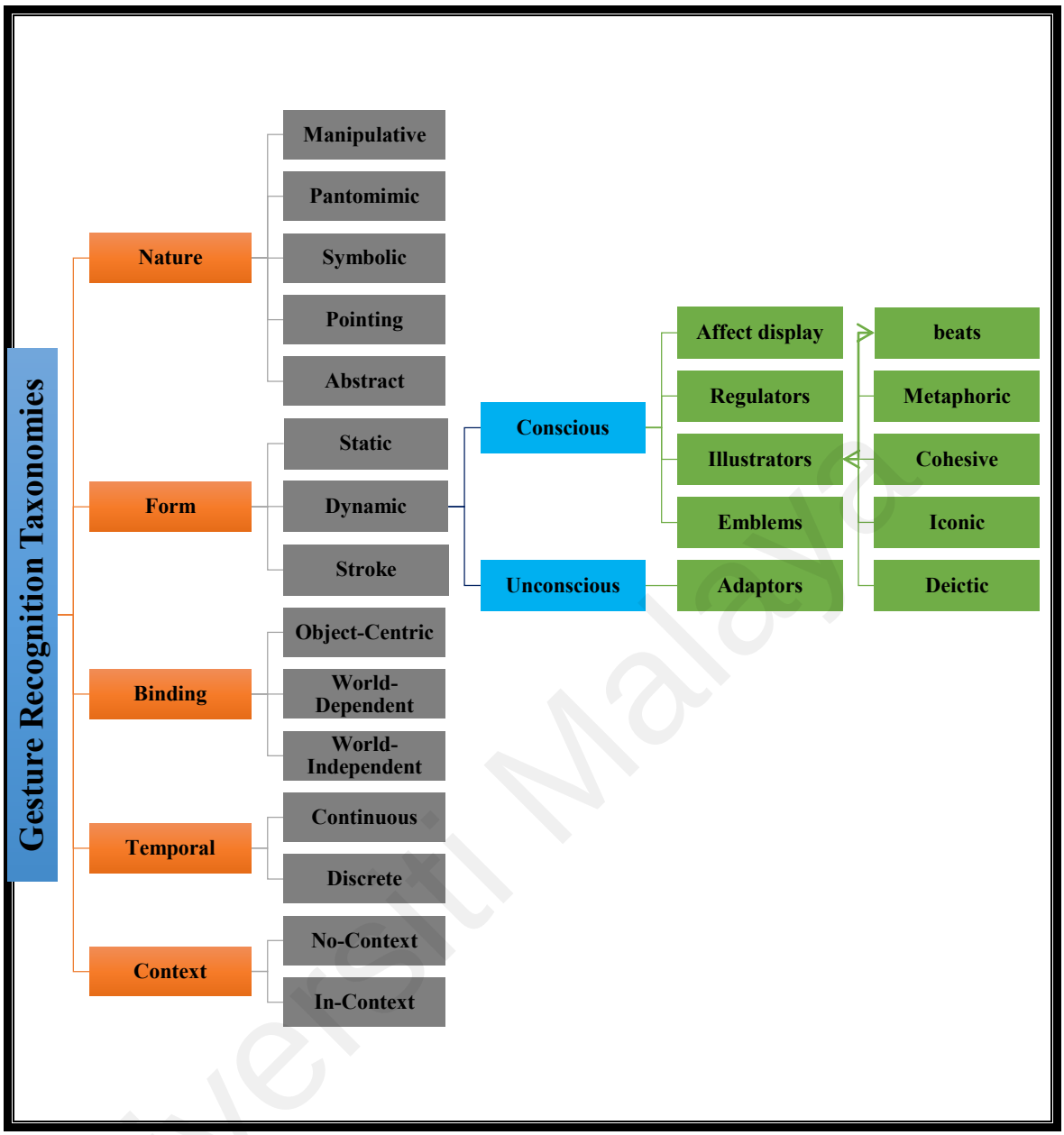


Figure 4.12: Hand Gesture Recognition Taxonomies

Static Gestures, Liang (Lamar, Bhuiyan, & Iwata, 2000) provided the following definition of static gesture or hand posture: *"Posture is a specific combination of hand position, orientation, and flexion observed at some time instance."*

Posture of static gestures is not a time-varying signal. Thus, they can be completely analyzed using a single image or a set of images of the hand taken at a specific time, the hand signs for "OK," or the "STOP" sign , in a simple picture are examples of hand postures since they convey enough meaning for complete understanding.

Universiti Malaya

Dynamic Gestures, Liang (Lamar et al., 2000) provided the following definition of "gesture" to describe dynamic gestures: *"Gesture is a sequence of postures connected by motion over a short time span."*

A gesture can be regarded as a sequence of postures. The individual frames in a video signal define the postures, whereas the video sequence defines the gesture. The head movements for "No" and "Yes" and the hand "goodbye" or "come here" gestures that can only be recognized by taking the temporal context information, are good examples of dynamic gestures.

Static gestures including hand postures and hand poses are normally correlated with the hand position and orientation in a given space and time without making any kind of movement. On the other hand, dynamic gestures (Hand gesture) normally include movements of some parts of the hand such as waving, which are also related to hand orientation with a certain type of movement. In addition, static gestures may include a single orientation of the hand without performing any movement (Al-Shamayleh et al., 2018).

Static gesture takes single frames of signs as input, while dynamic gesture takes continuous frames of signs in a video as input. Static gestures are those in which the hand position does not change during the gesturing period. Static gestures mainly rely on the shape and flexure angles of the fingers. In dynamic hand gestures, the hand position changes continuously with respect to time. Dynamic gestures generally have three motion phases: preparation, stroke, and retraction. The message in a dynamic gesture is mainly contained in the temporal sequence in the stroke phase.

According to (Kausar & Javed, 2011), static gestures are characterized with orientations, shape, finger's flex angles, relative position to body, and context environment, while dynamic gestures are characterized with orientations, shape, finger's

Universiti Malaya

flex angles, hand trajectories and orientations, motion speed and direction, and scale. Figure 4.13 is present an example for static and dynamic gesture.

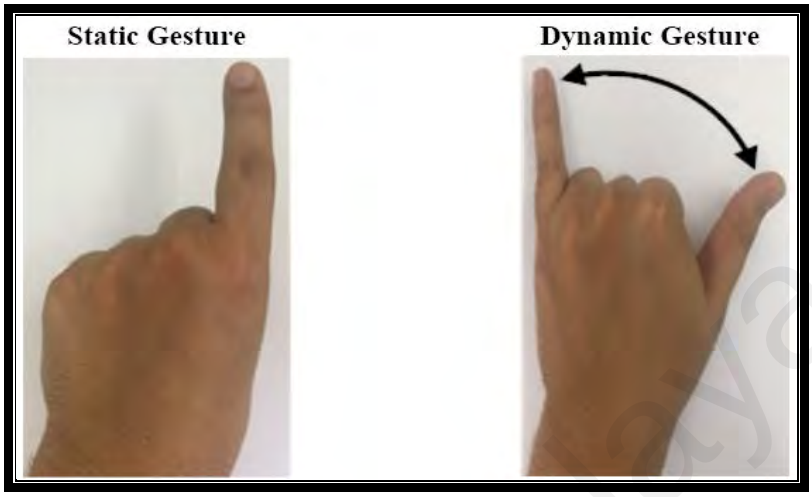


Figure 4.13: Example for static and dynamic gestures

4.5.2.2 The Representations Techniques of Vision-Based Gesture

There are different models and representations of gestures that describe the movements of various human body gestures. The two main approaches used to represent gestures are the 3D Model-based and Appearance-based approaches, as shown in Figure 4.14. Selected synthesized studies on representation techniques of vision-based gestures are illustrated in Table 4.6. The first dimension highlights the 3D Model-based while the second dimension highlights the Appearance-based approaches. The recognition method involved in 3D Model-based recognition necessitates various techniques involved in gesture representation. These techniques include the 3D Volumetric Model, 3D Geometric Model and 3D Skeleton Model. However, the representation of gestures according to the Appearance Models may include the Color-based Model, Silhouette Geometry Model, Deformable Gabarit Model and Motion-based model. The 3D representation in Model-based approaches is the human hand in a 3D spatial structure while revealing the temporal structure through an automation process. Through this process, the temporal representations of a gesture are classified in three stages: the pre-

Universiti Malaya

stroke phase that occurs before the stroke, the stroke phase that occurs during the stroke, and the post-stroke phase, which involves the reaction to a stroke. Each of these stages is related to single or various transitions of the 3D spatial states involved in the human model. Thereby, different numbers of cameras can be used to find the target and work with various parameters to identify the most appropriate match to the target, subsequently monitoring its movement throughout the recognition process in a 3D model. Therefore, the advantage of a 3D model is related to the advantage of updating the model parameters and checking for consistency of the transition level in the model parameters, thereby achieving a higher rate of gesture recognition. However, this process becomes more complex on account of the need to have dedicated hardware. There are several techniques that combine silhouette extraction along with 3D model projection fitting by directing it to a particular target orientation. To achieve this purpose, three models can be implemented to indicate the location of a target. First is the 3D volumetric model that provides in-depth details about the backbone body parts and further details regarding the skin surface, However, the 3D volumetric models are considered more accurate than the 3D geometric models in terms of information related to skin surface, whereas 3D geometric models consist of fundamental details pertaining to the human body backbone structure. Gestures based on appearance can be divided in two categories, which are the 2D static and motion models. Each of these categories may include various parameters. The most widely employed 2D model is based on color and involves the use of certain markers to monitor the movements of some human body parts. Boulay (Boulay, 2007) revealed that HCI technique use requires the integration of multi-scale hierarchal models, color features, and particle filtering. Silhouette geometry-based models require the integration of several spatial silhouette properties that may include the surface, perimeter, centroid, elongation, orientation, convexity, bounding box, or ellipse and rectangularity.

Universiti Malaya

Geometry features involving the skin surface were employed to identify various gestures that require using the hand (Bretzner, Laptev, & Lindeberg, 2002).

The Deformable Gabarit model depends on the formation of active deformable contouring. However, Birdal and Hassanpour (Birdal & Hassanpour, 2008) employed snakes in order to monitor movements and other features used for interpreting hand gesture properties in technical talks and potentially related to video indexing. Moreover, models based on motion were employed to identify a certain entity and its movement to capture its sequence. Another model that is known as a local motion histogram involves the use of an Ad boost model, which was introduced by Ju et al. (Ju, Black, Minneman, & Kimber, 1997) for action model learning.

Three types of volumetric models are usually employed. The 1-3D volumetric models consist of in-depth detailed information that is relevant to the skin surface and human backbone structure. The 2-3D geometric models are not as accurate as 3D volumetric models in terms of information related to the skin surface but also consist of information related to the human skeleton structure.

Gesture models that are based on appearance can be divided in two main classifications, which are the 2D static and motion approaches. Each of these subcategories have several variables. The most widely used 2D models may consist of the following:

- Color-based models that integrate body markers to track the movements of certain body parts. Bretzner et al. (Bretzner et al., 2002) suggested a hand gesture recognition system that involves the use of the properties of certain colors, particle filtering and hierarchical models.
- Silhouette geometry models encompass the integration of various geometrical features regarding silhouette, which may include rectangularity, perimeter, elongation, orientation, surface, centroid bounding box/ellipse, and convexity.

Universiti Malaya

The geometric features of the skin surface can be employed to identify various hand gestures (Birdal & Hassanpour, 2008).

- Models that are based on deformable Gabarit are linked with active deformable contouring, such as parameterized snakes, which are integrated along with movement and other variables. Ju et al. (Ju et al., 1997) integrated the use of snakes to analyze human body gestures derived from conversations or actions.
- Models based on motion or movement can be employed to identify the movements of an entity within the sequences of an image. A model that is based on motion analysis was introduced by Luo et al. (Luo, Kong, Zeng, & Fan, 2010), which is used for analyzing learned actions.

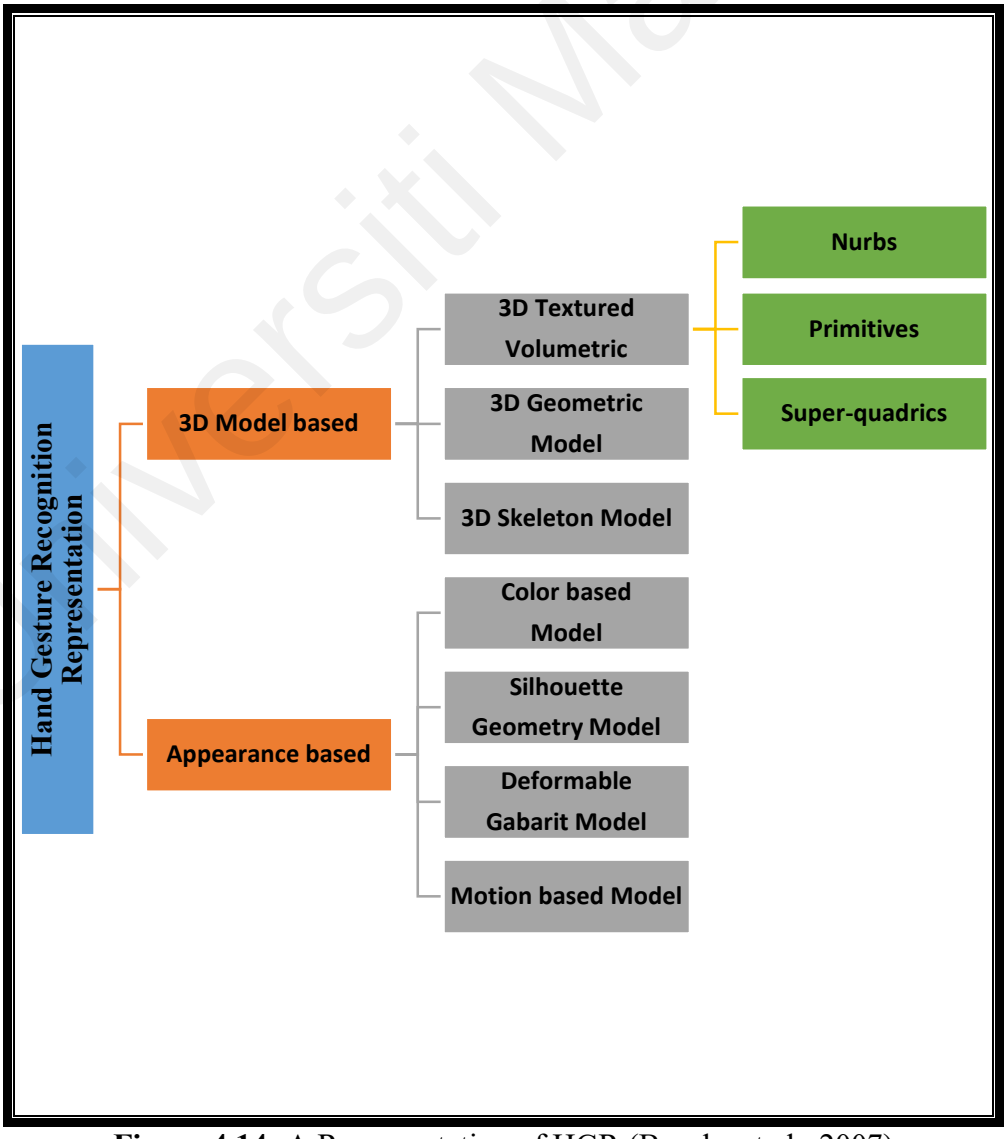


Figure 4.14: A Representation of HGR (Bourke et al., 2007)

Universiti Malaya

Table 4.6: Synthesis of representations techniques of vision-based gesture recognition

Classification	Techniques	Description	Advantages/Disadvantages	Studies
3D Model-Based	3-D Volumetric Model	These models show highly precise human details in terms of the skeleton and skin surface.	**Advantages:** The most effective methods The most accurate descriptions More flexible **- Disadvantage:** More complex (require very good segmentation)	S1, S14, S16, S19, S20, S24, S25, S26, S27, S29, S45, S46, S47, S56, S57, S58, S60, S65, S66, S67, S70, S71, S72, S73, S79, S82, S88, S89, S90, S91, S96, S100.
	3-D Geometric Model	These models show basic skeleton information and less precise information in terms of skin surface.		
	3-D Skeleton Model	These models have simpler features and higher adaptability, and are thus quite common. In terms of skeleton information, only the articulations and their 3 degrees of freedom (DoF) are provided.		
Appearance-Based	Color-Based Model	These models utilize basic body markers for body motion tracking.	**- Advantages:** The most efficient methods The most Simplicity methods More easier to implement (for motion based model) **- Disadvantages:** More ambiguity	S3, S4, S5, S6, S7, S8, S9, S10, S11, S12, S13, S15, S17, S21, S22, S25, S29, S30, S33, S34, S35, S38, S39, S40, S41, S43, S44, S51, S52, S53, S54, S55, S61, S62, S63, S64, S68, S69, S76, S77, S78, S80, S81, S83, S84, S92, S93, S94, S95, S97, S98, S99.
	Silhouette Geometry Model	These models use distinguished geometric feature of the silhouette, such as perimeter, convexity, surface, bounding box, ellipse, elongation, rectangularity, centroid, and orientation.		
	Deformable Gabarit Model	These models observe deformities of active contours, such as a variety of snake parameters through motions.		
	Motion-Based Model	This models are divided in two variants: global motion descriptors and local motion descriptors.		

Universiti Malaya

Figure 4.15 shows a statistical analysis of 3D model-based and appearance-based technique usage in vision-based gesture recognition.

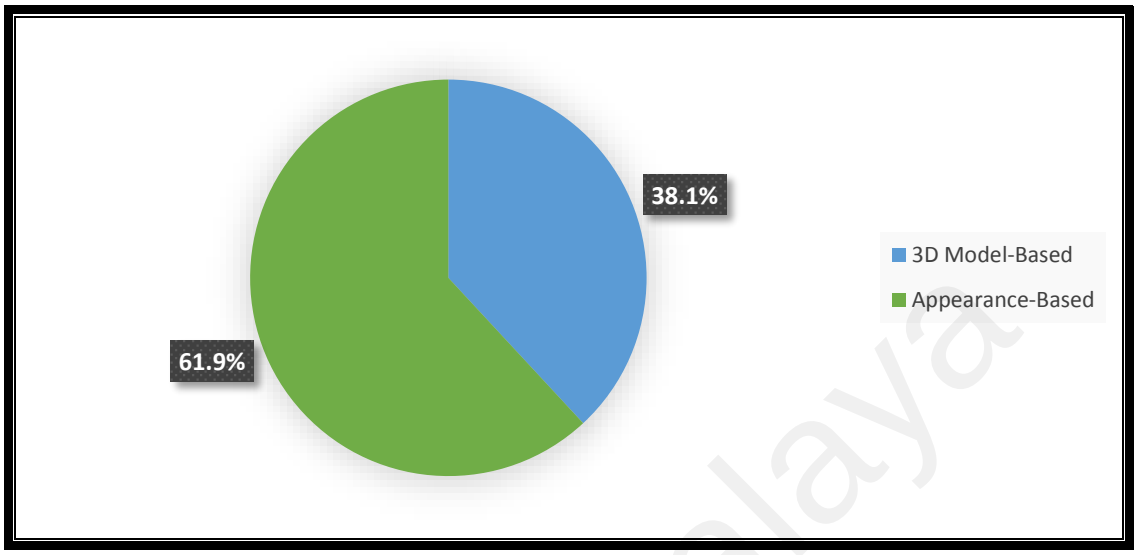


Figure 4.15: Statistical analysis of the usage of 3D model-based and appearance-based

Based on Figure 4.15, it seems the research community tends to prefer appearance-based techniques over 3D model-based and vision-based gesture recognition. Of the research articles, 61.9% addressed appearance-based, whereas 38.1% addressed 3D model-based approaches.

4.5.3 What are the main VBR and SBR techniques and the algorithms used for recognition specifically?

The majority of interactive hand mechanism techniques involve the use of building blocks that depend on a vision-based hand gesture recognition system. These consist of three essential techniques, which are the process of detecting a suitable target, tracking the movement of certain body parts, and recognition. This section sheds light on the promising developments in hand gesture recognition techniques. These techniques are employed by most researchers and are classified into three vertical representations of the three essential stages of detection, tracking and recognition, as illustrated in Figure 4.16.

Universiti Malaya

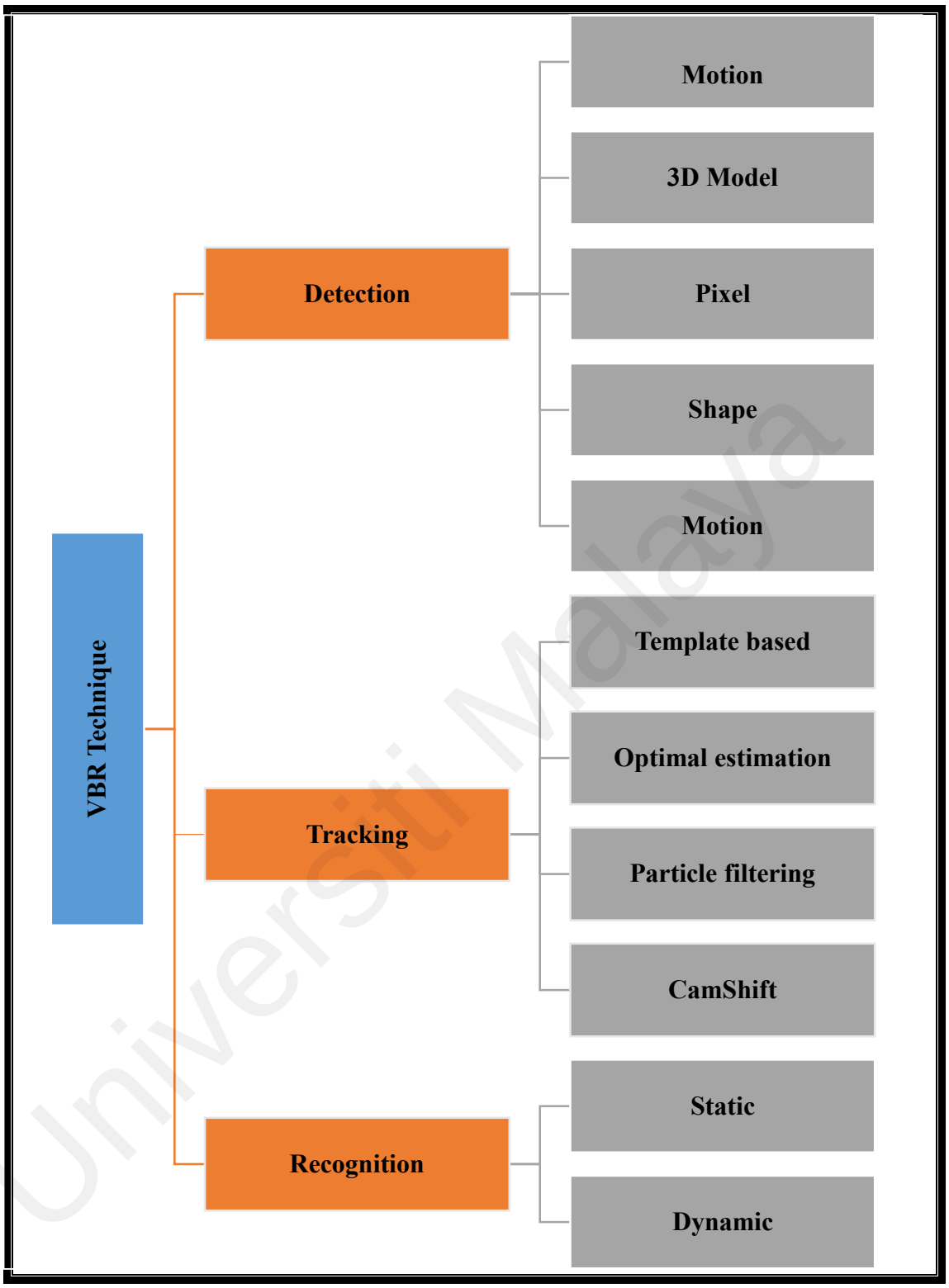


Figure 4.16: Techniques being involved in vision based hand gesture recognition

A- Detection Techniques

The first and foremost task in the hand gesture recognition domain is the identification of hands and the division process in the relevant image space. The division of relevant

Universiti Malaya

tasks from the image background is essential before moving on to subsequent stages. Different types of techniques have been suggested in the literature, which require integrating different types of visual features or combinations of features. Some of the features that can be detected may include shape, motion, skin color, and other tangible models of hands. Cote, Payeur, & Comeau (2006) conducted a comparative study to identify the performance of some techniques for hand segmentation.

B- Tracking Techniques

In case the detection task works at a fast pace to find a suitable frame, the image tracking process functions at a suitable rate. Nonetheless, the hand tracking process is considered difficult since the hands can move quickly and are more likely to change in appearance (Little, 2006). Tracking is known as the Frame-to-Frame correspondence of a hand in a particular region or feature for the purpose of understanding hand movement. There are two important phases in robust tracking. First, tracking can provide the inter-frame that links to the appearance of the hand or finger, taking into account the trajectories of some features within a certain period of time. These trajectories carry essential information relating to gestures and can also be used in raw form. For instance, by using control applications such as virtual drawing, the tracked hand can provide guidance to the drawing operation as well as the recognition of particular types of hand gesture recognition. Moreover, model-based techniques provide a means of tracking some model variables and also other characteristics that cannot be detected in a certain period of time.

C- Recognition Techniques

The main objective of hand gesture recognition is the morphological analysis of gestures, postures or locations of hand transfers. The recognition of vision-based hand gesture techniques can also be classified into dynamic and static gesture recognition. To find static gestures, a MLP or NBC can be used. In contrast, detecting dynamic hand

Universiti Malaya

gestures requires the integration of techniques such as SVM, Random Forests. Classifying learning algorithms takes into account the data outcome. Thereby, it is important to differentiate between supervised learning (i.e. finding samples to label), unsupervised learning (i.e. sample clusters with labels), semi-supervised learning (i.e. a mixture of un-labelled and labelled data), and transduction (i.e. supervised learning with prediction, Li and Wechsler (Li & Wechsler, 2005), and learning to learn (Baxter, 2000)). Choosing a learning algorithm depends on the hand gesture representation.

K-means (Kanungo et al., 2002): This type of classification is based on a statistical mean for identifying similar groups within an expanded area. The system can be initiated in an unordinary manner by identifying the k clusters in a given space. Figure 4.17 shows a K-means example when K=3, and Equation (4.1) represents a K-means model as a simple representation.

$$\boldsymbol{arg_s min} = \sum_{i=1}^{k} \sum_{x \epsilon s_i} \|x - \mu_i\|^2 = \boldsymbol{arg_s min} = \sum_{i=1}^{k} |S_i| \, Var \, S_i$$

(4.1)

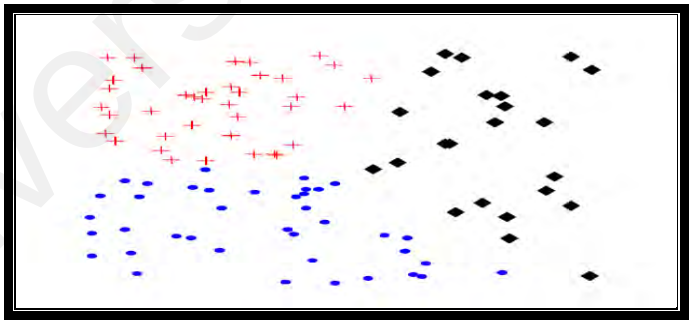


Figure 4.17: K-means when K=3

K-nearest neighbors (K-NN) (Thirumuruganathan, 2010): This technique can be used to separate objects related to their nearest training models in a particular place. K-NN is based on a spontaneous learning model, which requires integrating an approximate function and deferring all other computations until the classification process begins. Figure 4.18 shows a K-NN example when K=3, Equation (4.2) represents a Euclidean

Universiti Malaya

equation, Equation (4.3) represents a Manhattan equation, and Equation (4.4) represents a Minkowski equation.

$$\sqrt{\sum_{i=1}^{k} (x_i - y_i)^2} \quad \ldots. \textit{ Euclidean equation} \tag{4.2}$$

$$\sum_{i-1}^{K} |x_i - y_i| \quad \ldots. \textit{ Manhattan equation} \tag{4.3}$$

$$\left(\sqrt{\sum_{i-1}^{k} (|x_i - y_i|)^q}\right)^{1/q} \ldots. \textit{Minkowski equation} \tag{4.4}$$

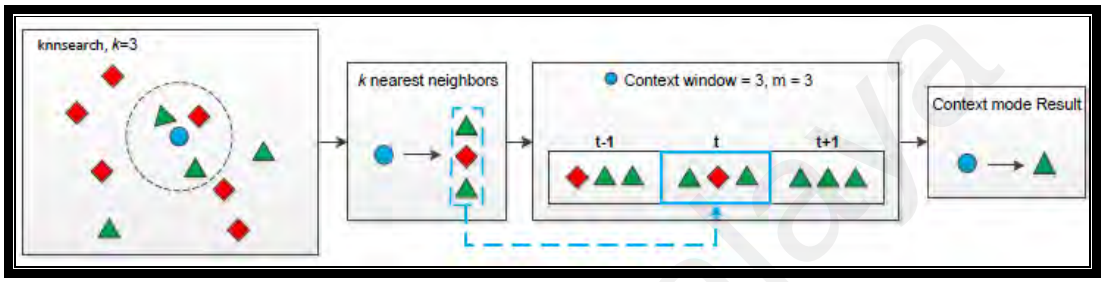


Figure 4.18: K-NN when k=3

Mean-shift clustering model (Derpanis, 2005): The algorithm in a mean shift clustering model is classified as nonparametric, in that it involves no previous knowledge related to the number of clusters but is not restricted only to the cluster shape. The major concept behind the mean clustering model is that it deals with the points within a d-dimensional space according to the function in the empirical density that involves the correlation between the dense areas of a given space along with the local maximum point of the distribution model. Figure 4.19 shows an example of a Mean-shift clustering model, and Equation (4.5) represents the Mean-shift clustering model.

Kernel function $K(x_i - X)$

$$\boldsymbol{m(x)} = \frac{\sum_{xi \in N(x)} \boldsymbol{K(x_i - X) x_i}}{\sum_{xi \in N(x)} \boldsymbol{K(x_i - X)}} \quad \textit{where } \boldsymbol{N(x)} \text{ is the neighborhood of } \boldsymbol{x}, \text{a set of points} \tag{4.5}$$

for which $\boldsymbol{K(x)} \neq \boldsymbol{0}$

Universiti Malaya

Let the data be a finite set S embedded in the n-dimensional Euclidean space, X. Let K be a flat kernel that is the characteristic function of λ in X.

$$K(x) = \begin{cases} 1 \ if \ \|x\| \leq \lambda \\ 0 \ if \ \|x\| > \lambda \end{cases}$$

In each iteration of the algorithm, $s \leftarrow m(s)$ is performed for all $s \in S$ simultaneously.

$$f(x) = \sum_i K(x - x_i) = \sum_i k\left(\frac{\|x - x_i\|^2}{h^2}\right)$$

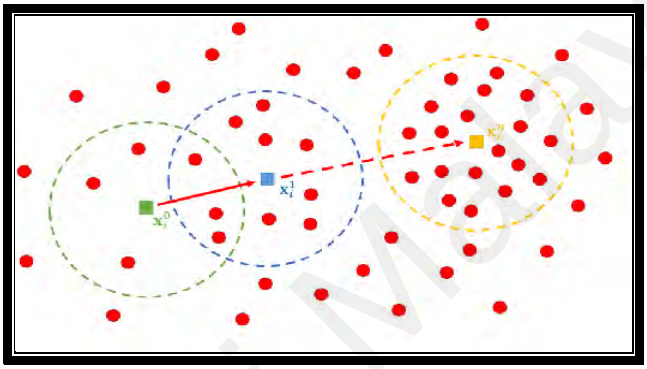


Figure 4.19: Mean shift algorithm procedure for a data point xi

Support Vector Machine (SVM) (Burges, 1998): This technique is classified as nonlinear, because it is characterized by revealing more results than other techniques. This method involves applying a nonlinear approach to identify entry information related to an elevated dimensional space. Moreover, by following a linear-based system to classify data, SVM provides better classification or regression outcomes. Figure 4.20 shows a sample SVM classifier, and Equation (4.6) represents the SVM classifier.

$$\left[\frac{1}{n}\sum_{i=1}^{n} \max(0, 1 - y_i\,(w.x_i - b))\right] + \lambda\,\|w\|^2$$

(4.6)

Universiti Malaya

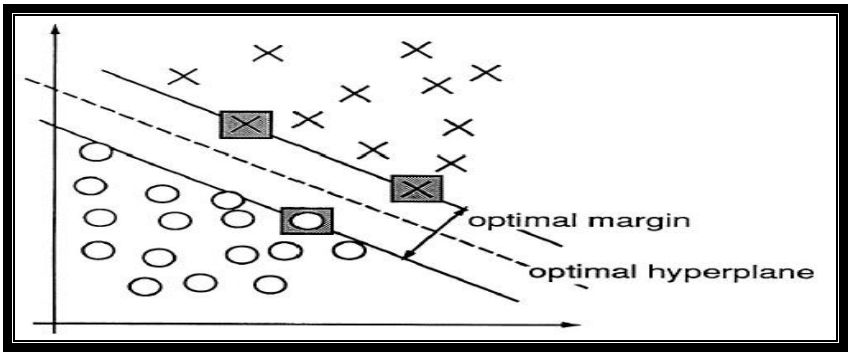


Figure 4.20: SVM example

Hidden Markov Model (HMM) (Ramage, 2007): HMM is classified as a statistical model that is utilized to combine data according to an ordered sequence of variables. This model is produced based on certain variables derived from a Markov chain, where the initial variables are considered to be buried. Equation (4.7) is a simple representation of an HMM model. Figure 4.21 shows the probabilistic parameters of a HMM.

$$\boldsymbol{P(S_{1:T}, Y_{1:T}) = P(S_1)P(Y_1|S_1)\ \prod_{t=2}^{T} P(S_t|S_{t-1})P(Y_t|S_t),} \quad \textbf{(4.7)}$$

$$\boldsymbol{P(O) = P(o_t, o_{t-1}, \ldots., o_1|A) = \prod_{t=1}^{T} P(o_t|o_{t-1}; A) = \prod_{t=1}^{T} A_{ot-1ot}\ \cdots}$$

By setting the Markov model parameters, the probability of any sequence of states, as illustrated in this equation, can be computed.

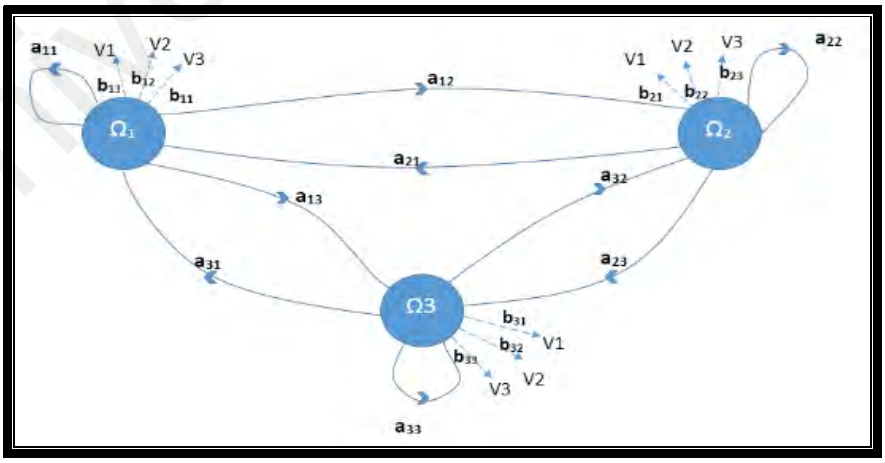


Figure 4.21: Representation of HMM with 3 hidden states and 3 output states

Dynamic Time Warping (DTW) (Senin, 2008): This model is utilized to identify the maximum meeting point of two signals. The algorithm in this model can be used to

Universiti Malaya

measure the length between two points released by signals and to measure their alignment along with the signals. The dynamic model employs these distances to identify the maximum distance it achieved, thereby identifying the easier and less expensive path. There are several DTW extensions, e.g. Derivative Dynamic Time Warping (DDTW), Piecewise Dynamic Time Warping (PDTW) and Weighted Dynamic Time Warping (WDTW). The distance measures between the joint positions' sequences are calculated using the Euclidean distance metric. Equation (4.8) is a simple DTW model representation and Figure 4.22 shows a DTW example.

$$\boldsymbol{D(p,q) = D(q,p) =}$$
$$\sqrt{(\boldsymbol{q_1 - p_1})^2 + (\boldsymbol{q_2 - p_2})^2 + \cdots + (\boldsymbol{q_n - p_n})^2} = \sqrt{\sum_{i=1}^{n}(\boldsymbol{q_i - p_i})^2} \quad \textbf{(4.8)}$$

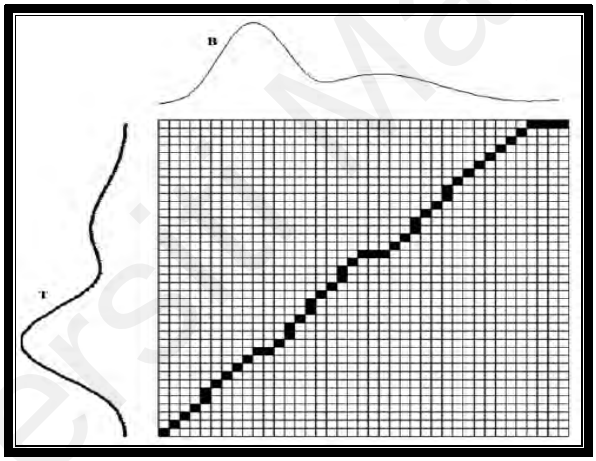


Figure 4.22: Aligning T and B, and visualizing optimal warping path in solid squares.

Time-delay neural network (TDNN) (Wohler & Anlauf, 1999): This model involves the use of specialized ANNs in a particular dataset in order to select the architecture that is more suitable for online networks. This provides an advantage to various real applications. In theory, this model is the result of various layers' expansion. This model depends on time delay that allows each neuron to perceive any previous actions revealed by the neurons. Equation (4.9) is a simple TDNN model representation and Figure 4.23 gives a TDNN model example.

Universiti Malaya

$$oj(t) = \begin{cases} 0 & if\ a_j(t) \leq 0 \\ a_j(t) & if\ a_j(t) < 1 \\ 1 & if\ a_j(t) \geq 1 \end{cases}$$

(4.9)

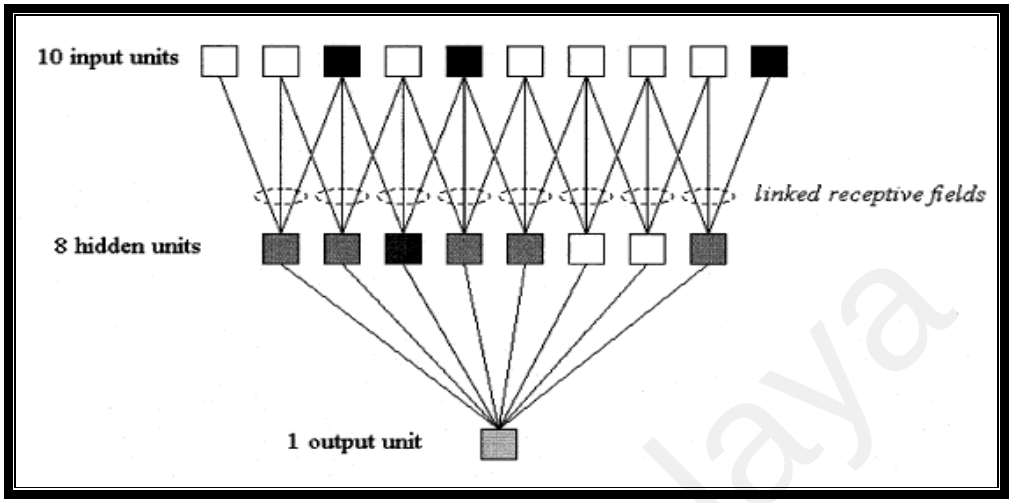


Figure 4.23: An example of one-layer time-delay neural network

Finite-state machine (FSM) (Holzmann, 1990): FSM is a system that can reveal a limited or large number of a possible set of states, taking into account that the ultimate status can be perceived but it is not practical. The FSM model can serve as a tool to identify and resolve certain issues. It can also be used according to certain approaches that can illustrate solutions for future developers and system controllers. Equation (4.10) represents the formal definitions of an FSM model and Figure 4.24 shows a sample FSM model.

In accordance with the general classification, the following formal definitions are given:

A deterministic finite state machine is a quintuple$(\sum, \boldsymbol{S}, \boldsymbol{s_0}, \boldsymbol{\delta}, \boldsymbol{F})$**,** where: **(2.10)**

$\sum$ is the input.

$\boldsymbol{S}$ is a finite, non-empty set of states.

$\boldsymbol{s_0}$ is an initial state, an element of $\boldsymbol{S}$.

$\boldsymbol{\delta}$ is the state-transition function: $\boldsymbol{\delta} : \boldsymbol{S} \times \sum \rightarrow \boldsymbol{S}$ ($\boldsymbol{\delta}$ would return a set of states).

$\boldsymbol{F}$ is the set of final states, a (possibly empty) subset of $\boldsymbol{S}$.

Universiti Malaya

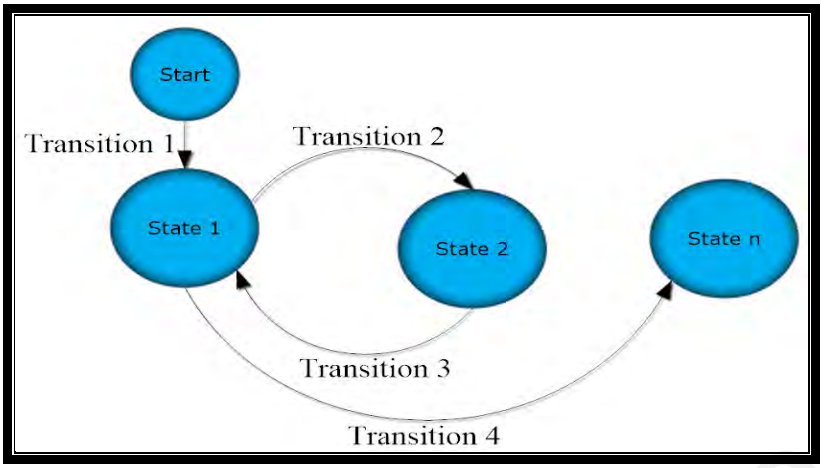


Figure 4.24: Finite-state machine example

Based on Figure 4.25, it appears the research community tends to work more on ASLR and hand gesture recognition than other scopes and applications, with 61% of research articles on ASLR and gesture recognition, 12.6% on desktop application, 8.0 % on VR application, 6.0% on gaming application, 5.0% on entertainment application and 8.0% of research articles on automatic number recognition.

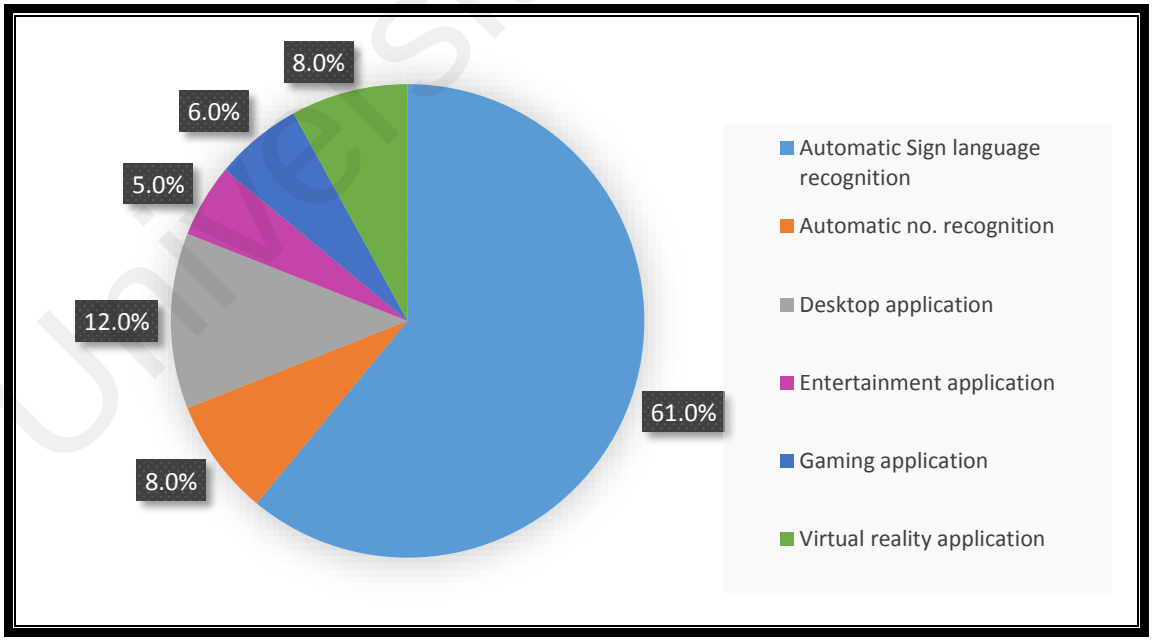


Figure 4.25: Statistical analysis of the main scopes and applications using gesture recognition

Universiti Malaya

Although research attempts and efforts in ASLR started several years ago, yet, it is still limited in many aspects. This area can still be perceived in its infancy. Based on our knowledge, no comprehensive frameworks have been developed which cover the present extensive scope. Research in this domain will certainly influence other domains that encompass HCI elements. In Chapter 2 (Literature Review), we summarized two main approaches that are used in translating SL into text, namely: 1) VBR approach, and 2) SBR approach. These approaches were discussed in this chapter with special emphasis on ArSL.

4.5.4 What are the application domains, commercial software, and devices that use gesture recognition for interaction in different environments?"

4.5.4.1 Application Domains

On account of the essential use of gesture recognition in various domains, this section presents a number of domains that involve gesture interactions.

Virtual Reality (VR): Virtual gestures and the increase in real-life applications have contributed to the rising demand for computer devices. The interactions in virtual applications involve the integration of gestures to facilitate realistic manipulations of virtual objects by using one's hands to display 3D interactions (Sharma, Huang, Pavlovic, Chu, & Schul, 1996) or 2D applications that support 3D interactions (Gandy, Starner, Auxier, & Ashbrook, 2000).

Robotics and Telepresence: Telepresence and tele-robotic systems are often applied in the field of space exploration and in conducting various research projects. Gestures that resemble human interactions are employed to monitor and interact with robots, and in the majority of cases, videos extracted from cameras are provided to the operator to monitor robot interactions (Goza, Ambrose, Diftler, & Spain, 2004). Gestures can be used to control robot hand and arm movements in order to identify real objects besides other movements of the robot.

Desktop and Tablet PC Applications: In desktop and tablet computing applications, gestures are used to provide another type of interaction that can be generated by using the

Universiti Malaya

mouse and keyboard (Stotts, Smith, & Gyllstrom, 2004). Many gestures that are involved in desktop computing applications require gesture demonstration through graphics or other applications that are based on pen gestures (Smith & Schraefel, 2004).

Sign Language: Using signs as a method of communication is considered an important part of human interaction. This is because sign language is considered to have a more structural base and is also more appropriate for test beds implemented in CV algorithms (Aujeszky & Eid, 2015). Moreover, such algorithms can be deemed a good way to facilitate communication among people with disabilities. Sign language for the deaf is a good example of encouraging the use of gestures and has received great attention in the literature (Dabre & Dholay, 2014) (Aujeszky & Eid, 2015).

Core applications refer to counting the core areas involved in computing applications. These may include information visualization, controlling and programming robots, desktop application, sign language, and VR, as illustrated in Table 4.7.

Table 4.7: Comparison of core computing applications

	Information Visualization	Control and program robots	Desktop Application	Sign Language	Virtual Reality
Principle	Interacting with data more naturally	Controlling and programming robots	Interacting with computer-generated environments more naturally	Automatic translation	Providing realism to man-made environments
Substitute	Mouse and keyboard	Mouse and keyboard	Mouse and keyboard	Camera-based devices, or DataGlove	DataGlove, etc.
Target	Scientific visualization, audio-visual presentations	Mobile robots, multi-DoF robots, programming by demonstration	Wearable computers	Communication systems for the disabled	Realistic manipulations of virtual objects

5.4.4.2 Commercial software and Devices

Hand gesture recognition techniques comprise an exciting research area that involves the implementation of various HCIs to computer-based devices. This section describes various products and software devices related to vision-based hand gesture recognition

Universiti Malaya

technology and that require interaction with a wide range of systems. These commercial devices developed so far are still in the primary stages of acceptance but can still be made robust according to user needs. A number of criteria are taken into account when designing and improving such software products in order to avoid technological difficulties that may impede their development. In developing these products, the support of affiliated associations is vital. The products need to be reviewed in terms of cost-effectiveness, efficiency, real-time application environments, and robustness in different real-life situations. Table 4.8 below compares various software products and devices that are involved in hand gesture recognition.

Universiti Malaya

Table 4.8: Comparison of different vision-based gesture recognition devices and software

	GestureTek ("GestureTek," 2008)	**Mgestyk** ("Mgestyk," 2009)	**Hand Gket** ("HandGKET," 2011)	**Omron** ("OMRON," 2012)	**Microsoft Kinect** ("Microsoft Kinect," 2012)	**Point Grab's** ("PointGrab's," n.d.)	**Soft Kinetic** ("SoftKinetic," 2012)
Year	2008	2009	2011	2012	2012	2012	2012
Windows PF	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Linux PF	No	No	No	Yes	No	Yes	Yes
Java Language	Yes	No	No	No	No	No	No
C++ Lang.	No	Yes	Yes	Yes	Yes	Yes	Yes
C# Language	No	Yes	Yes	Yes	Yes	No	Yes
ID - Camera	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Features - Provides Standard Gestures	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Features - Allows custom gestures	No	No	No	No	No	No	Yes
Technology	3D and 2D cameras for CV, face and body sensors, gesture recognition cameras in mobile devices and pointing frames.	Three-dimensional gesture recognition camera, software for hand gesture recognition	This middleware recognizes the user's hand gestures and generates keyboard or mouse events to control applications in the computer using CV techniques	Developed a new hand gesture recognition technology capable of simultaneously recognizing hand positions.	3D depth sensors for gesture recognition	Enables the integration of hand gestures with application; a gesture is a movement of the hands, fingers, arms, body, or face of an individual.	Real-time, three-dimensional gesture recognition
Interface	PC, smart TV, set-top Boxes	PC-based interface	PC-based interface	PC, TV	PC, smart TV, Xbox 360 video	PC, smart TV	PC, console, smart TV, set-top box
Product	Camera and software	Camera and software	Camera and software	Camera and software	3D depth sensors, RGB camera, software	Camera and software	Camera and software

Universiti Malaya

4.6 Summary and Discussion

This chapter identified the gaps in the ASLR and ArSLR domain, the representation of results and a discussion of the systematic literature review. In addition, a comprehensive literature investigation is conducted to summarise the research efforts on ArSLR approaches and ArSL signs, as illustrated in Chapter 2. On the basis of the aforementioned systematic literature review, the researcher outlined the main subject of the research, conducted the systematic literature review associated with the recent interest in the scientific research field and the gaps in research. In addition, the systematic literature review discovered that the vision- and sensor-based continuous ArSL approach, signer-independent and static backhand based on a smartphone are the issues that need to be addressed efficiently, especially in ArSLR.

The number of scientific publications and research conducted on ArSL is limited. That this topic has yet to be seriously addressed marks a research gap, which can be further considered by the research community. Meanwhile, the target language in the current study is ArSL.

The next chapter concentrates on the procedure used to create a comprehensive ArSL database.

Universiti Malaya

CHAPTER 5: ARABIC SIGN LANGUAGE DATABASE

5.1 Introduction

This chapter concentrates on the procedure used to create a comprehensive SL database for ArSL, which covers all the approaches and forms of signs for ArSLR development and evaluation based on SL experts' standard criteria and consensus. Two databases are created related to our framework, as discussed in Chapter 3. Figure 5.1 outlines the structure of this chapter.

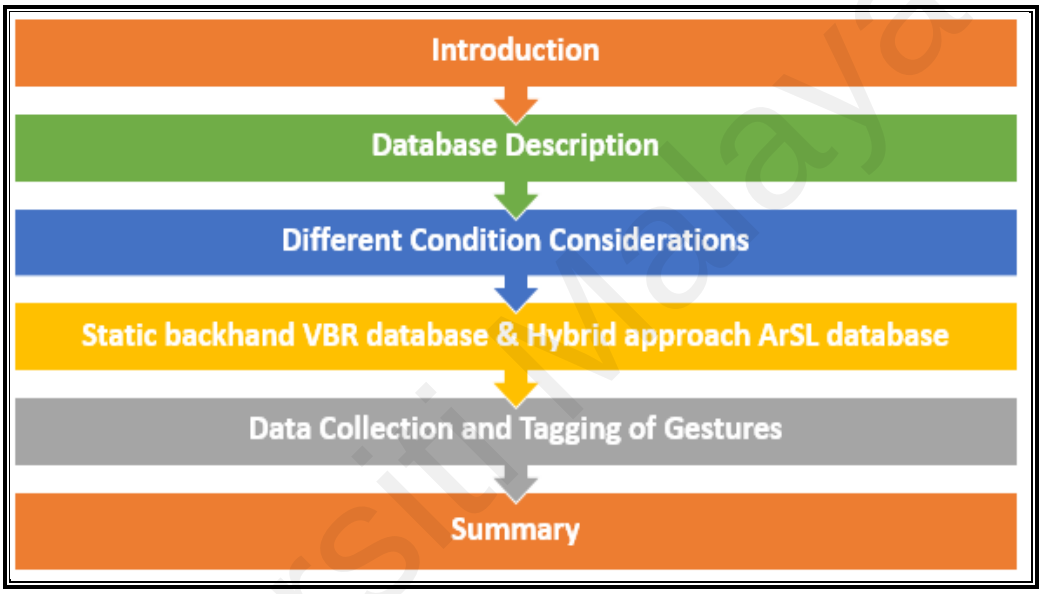


Figure 5.1: Outlines the structure of chapter 5

On the basis of our investigation, there has been no serious attention given to ArSLR given that there are no standard ArSL databases available for researchers in this field. Thus, in this work, a database is created based on ten signers (three male signers and seven female signers) who each performed 10 repetitions for each sign. The signers were SL expert practitioners and hearing- and speech-impaired individuals from the Prince Ali Bin Al-Hussein Club for the Deaf in Alzarqaa, Jordan; the Al-Amal Primary Mixed School for the Deaf in Alzarqaa, Jordan; and the Support Division for Students with Special Needs at the Deanship of Student Affairs, The University of Jordan in Amman, Jordan.

The remaining sections in this chapter discuss the database description and the list of the signs, isolated words and sentences constituting this database, and the descriptions of different condition considerations and database criteria. In addition, the data organisation,

Universiti Malaya

acquisition quality and devices are described. Thereafter, an overview of the data collection techniques and setup is presented. Finally, in this chapter, the data tagging and the final data format are discussed.

5.2 Database Description

The database has ArSL signs that have been created to evaluate our proposed frameworks in Chapter 3 and to serve as a bedrock to other frameworks in ArSLR. The database contents are shown in Table 5.1, Table 5.2, Table 5.3, Table 5.4 and Table 5.5, which are split into two main databases. The first database has a static backhand ArSL, such as Arabic letters and numbers, while the second database has signs in all ArSL forms and approaches with unrestricted sentence length and grammar.

A database is created under controlled lighting conditions and SL expert criteria, which can extend or add more signs. The signers start from silence, do the required gesture and end in silence. Additionally, the signers used the standard ArSL version accepted by the League of Arab States since February 2007.

In order to build an automatic ArSLR framework, a database is required for validation. Accordingly, 39 Arabic Alphabet characters (28 Arabic alphabets are represented by one static sign and the 11 remaining characters represent other orthographical representation of the Arabic alphabets), 21 Arabic numbers, 60 Arabic isolated words and 60 continuous ArSL sentences. Note that the sentences in Table 5.4 (continuous dataset 1) is provided by Assaleh et al. (2010), and the SL expert evaluated the sentences. This expert suggested the addition of Table 5.5 to cover the continuous form of scope.

5.3. Different Condition Considerations

5.3.1. Lighting and background conditions for vision-based dataset

All the VBR hand signs in our DB are shown with a white background and with direct light on the signer's hand. This will give the flexibility for the researchers to make any further processing on the images such as background subtraction, or any other processing as researchers can add their own background images to the dataset without any limitation on the clothing of the user.

Universiti Malaya

5.3.2. Data Organization

The signs are organized as follows:

1. **First Database:**

   This database has a mixed dataset, backhand Alphabet (Letters) and numbers.

2. **Second Database:**

   This database has five dataset; Alphabet (Letters), Numbers, Isolated-words, continuous dataset 1, and continuous dataset 2. Note that, the sequence of devices recordings (smart gloves and Kinect readings) make up a sign language word, and a sequence of words make up a sign language continuous sentence.

5.3.3. Signer Independence

To achieve the signer independence, 10 different signers with different ages, experience, and situation were selected with the ranging age from 15 to 45 years, ranging experience from 5 to 20, SL expert, hearing and speech-impaired signers.

Table 5.1: Arabic letters dataset

English Translation	Arabic Letters	#	English Translation	Arabic Letters	#
KKaf	ق	**.11**	Aleph	ا	**.1**
Kaf	ك	**.11**	Baa	ب	**.1**
Lam	ل	**.13**	Taa	ت	**.3**
Meem	م	**.14**	Thaa	ث	**.4**
Noon	ن	**.15**	Gym	ج	**.5**
Haa	ه	**.16**	Hhaa	ح	**.6**
Waaw	و	**.17**	Khaa	خ	**.7**
Yaa	ي	**.18**	Dal	د	**.8**
marbpta Tta	ة	**.19**	Thal	ذ	**.9**
hamza+Alff	أ	**.31**	Raa	ر	**.11**
Waaw+Hamza	ؤ	**.31**	Zay	ز	**.11**
Alf maksora	ئ	**.31**	Syn	س	**.11**
Hamza ala nabera	ئـ	**.33**	Shen	ش	**.13**
Hamza	ء	**.34**	Sad	ص	**.14**
Alff+hamza	إ	**.35**	Daad	ض	**.15**

Universiti Malaya

Alff+maada	آ	**.36**	Ttaa	ط	**.16**
Alff maksora	ى	**.37**	Thaa	ظ	**.17**
No		**.38**	Ayn	ع	**.18**
The	ال	**.39**	Ghayn	غ	**.19**
			Faa	ف	**.11**

Table 5.2: Arabic Numbers dataset

English Translation	Arabic Number	#	English Translation	Arabic Number	#
Twenty	عشرون	**.51**	Zero	صفر	**.41**
Thirty	ث ن	**.51**	One	واحد	**.41**
Forty	اربعون	**.53**	Two	اثنان	**.41**
Fifty	خمسون	**.54**	Three	ثلاثه	**.43**
Sixty	ستون	**.55**	Four	اربعه	**.44**
Seventy	سبعون	**.56**	Five	خمسه	**.45**
Eighty	ثمانون	**.57**	Six	سته	**.46**
Ninety	تسعون	**.58**	Seven	سبعه	**.47**
A hundred	مئه	**.59**	Eight	ثمانيه	**.48**
a thousand	الف	**.61**	Nine	تسعه	**.49**
			Ten	عشره	**.51**

Table 5.3: Arabic Isolated-Words dataset

English Translation	Arabic Isolated-word	#	English Translation	Arabic Isolated-word	#
Patient	مريض	**.91**	Happy	سعيد	**.61**
a headache	صداع	**.91**	Sad	حزين	**.61**
Medicine	دواء	**.93**	Important	مهم	**.63**
Injection	حقنة	**.94**	Angry	غاضب	**.64**
check up	فحص	**.95**	Beautiful	جميل	**.65**
Blood	دم	**.96**	Love	يحب	**.66**
Surgery	عمليه جراحيه	**.97**	Hate	يكره	**.67**
Food	طعام أكل	**.98**	Worry	قلق	**.68**
Water	ماء شرب	**.99**	Serious	جاد	**.69**
How?	كم	**.111**	New	جديد	**.71**
Where?	اين	**.111**	Neighbor	جار	**.71**
Why?	لماذا سبب	**.111**	Friend	صديق	**.71**
University	جامعه	**.113**	Gift	هديه	**.73**
Plane	طائره	**.114**	Welcome	اهلا	**.74**
Sister	أخت	**.115**	Home	بيت	**.75**
Television	تلفاز	**.116**	How are you?	كيف الحال	**.76**
Guest	ضيف	**.117**	Peace be upon you	السلام عليكم	**.77**
Enemy	عدو	**.118**	Problems	مشاكل	**.78**
please enter	تفضل	**.119**	Bus	حافلة	**.79**
Shame	عيب	**.111**	Taxi	تكسي	**.81**
Sleeps	ينام	**.111**	Gas station	محطة بنزين	**.81**
Wakes up	يستيقظ	**.111**	Wages	أجرة	**.81**

Universiti Malaya

Hears	عمسي	.113	Street	شارع	.83
Silence	تكسي	.114	Money	نقود	.84
Smell	مشي	.115	Profit	ربح	.85
Helps	دعاسي	.116	Bank	مصرف (بنك)	.86
Yesterday	امس	.117	Jordanian Dinar	دينار	.87
Goes	بهذ ي	.118	Free	مجانا	.88
Comes	يئي	.119	Hospital	مستشفى	.89
Relatives	قارب	.111	Doctor	طبيب	.91

Table 5.4: Arabic Continuous Dataset 1

English Translation	Arabic Sentence	
I went to the soccer club.	ذهبت الى نادي كرة القدم	.111
I love car racing.	أنا أحب سباق السيارات	.111
I bought an expensive ball.	اشتريت كرة ثمينة	.113
On Saturday I have a soccer match.	يوم السبت عندي مباراة كرة قدم	.114
There is a soccer field in the club.	في النادي ملعب كرة قدم	.115
There will be a bike racing tomorrow.	غدا سيكون هناك سباق دراجات	.116
I found a new ball in the field.	وجدت كرة في الملعب	.117
How old is your brother?	كم عمر أخيك ؟	.118
My mom had a baby girl today.	اليوم ولدت أمي بنتا	.119
My brother is still breast feeding.	أخي ما زال يرضع	.131
My grandfather is at our home.	جدي في بيتنا	.131
My kid bought an inexpensive ball.	اشترى ابني كرة رخيصة	.131
My sister read a book.	قرأت أختي كتابا	.133
My mother went to the market this morning.	ذهبت أمي إلى السوق في الصباح	.134
Is your brother home?	هل أخوك في البيت ؟	.135
My uncle’s house is big.	بيت عمي كبير	.136
In two month my brother will get married.	سيتزوج أخي بعد شهرين	.137
In two months my brother will get divorced.	سيطلق أخي بعد شهرين	.138
Where does your friend work?	أين يعمل صديقك ؟	.139
My brother plays basketball.	أخي يلعب كرة السلة	.141
I have two brothers.	عندي أخوان	.141
What is your father’s name?	ما اسم أبيك	.141
Yesterday my grandfather was sick.	كان جدي مريضا امس	.143
Yesterday my father died.	مات ابي امس	.144
I saw a beautiful girl.	رأيت بنتا جميلة	.145
My friend is tall.	صديقي طويل	.146
I do not eat close to bedtime.	أنا لا آكل قبل النوم	.147
I ate delicious food at the restaurant.	أكلت طعاما لذيذا في المطعم	.148
I like drinking water.	أنا أحب شرب الماء	.149
I like drinking milk in the evening.	أنا أحب شرب الحليب في المساء	.151
I like eating meat more than chicken.	أنا أحب أكل اللحم أكثر من الدجاج	.151
I ate cheese and drank juice.	أكلت جبنة وشربت عصيرا	.151
Next Sunday the price of milk will go up.	في الأحد القادم سيرتفع سعر الحليب	.153
Yesterday morning I ate olives.	أكلت زيتونا صباح امس	.154
I will buy a new car in a month.	سأشتري سيارة جديدة بعد شهر	.155
He washed for morning prayer.	توضأ لصلاة الصبح	.156
I went to Friday prayer at 10:00 o’clock.	ذهبت الى صلاة الجمعة الساعة العاشرة	.157
I saw a big house on TV.	شاهدت بيتا كبيرا على التلفاز	.158

Universiti Malaya

Yesterday I went to sleep at 10:00 o'clock.	ذهبت إلى النوم في الساعة العاشرة	**159.**
I went to work this morning by my car.	ذهبت إلى العمل في الصباح بسيارتي	**161.**

Table 5.5: Arabic Continuous Dataset 2

English Translation	Arabic Sentence	
The rain came down from the sky profusely	نزل المطر من السماء بغزارة	**161.**
The man went to the sea early	ذهب الرجل إلى البحر مبكرا	**161.**
We bought a new house	اشترينا بيتا جديدا	**163.**
I saw a cat rushing down the street	رأيت قطة مسرعة في الشارع	**164.**
I love animals that are not dangerous	أنا أحب الحيوانات غير الخطرة	**165.**
My brother rides horses skilfully	أخي يركب الخيل بمهارة	**166.**
My uncle is a ship captain at the sea	عمي قبطان سفينة في البحر	**167.**
My uncle is a policeman	خالي رجل أمن	**168.**
All of us went to the book fair	ذهبنا جميعا إلى معرض الكتاب	**169.**
There is an aquarium in our house	يوجد حوض سمك في بيتنا	**171.**
I saw the bird on the tree	رأيت العصفور على الشجرة	**171.**
I like helping poor people	أحب مساعدة الفقراء	**171.**
The manager came to our class	جاء المدير إلى صفنا	**173.**
Lions live in the forest	تعيش الأسود في الغابات	**174.**
I like the quiet of the night	أحب هدوء الليل	**175.**
My brother traveled to Malaysia	سافر أخي إلى ماليزيا	**176.**
I will go to the cinema with my friend	سأذهب إلى السينما مع صديقي	**177.**
My father worked as a judge in the court	يعمل أبي قاضيا في المحكمة	**178.**
Justice is important in life	العدل مهم في الحياة	**179.**
I will be a doctor someday	سأكون طبيبا يوما ما	**181.**

5.4. Static backhand VBR database

In order to model the static backhand ArSLR framework, one smartphone camera (made of Apple iPhone X, 12-megapixel resolution) was utilized to capture images of the signs made by the signers. The images saved in JPG format for analysis in the final phases of the database collection.

The image acquisition system is made of 30-cm square white acrylic with mounted LED light strips, to provide uniform intensity over all of the images. Once the signers performed the signs in the correct way, the signers will wait a moment, then the capturing process will start and the image will be stored into the computer. Figure 5.2 presents the image acquisition environment.

Universiti Malaya

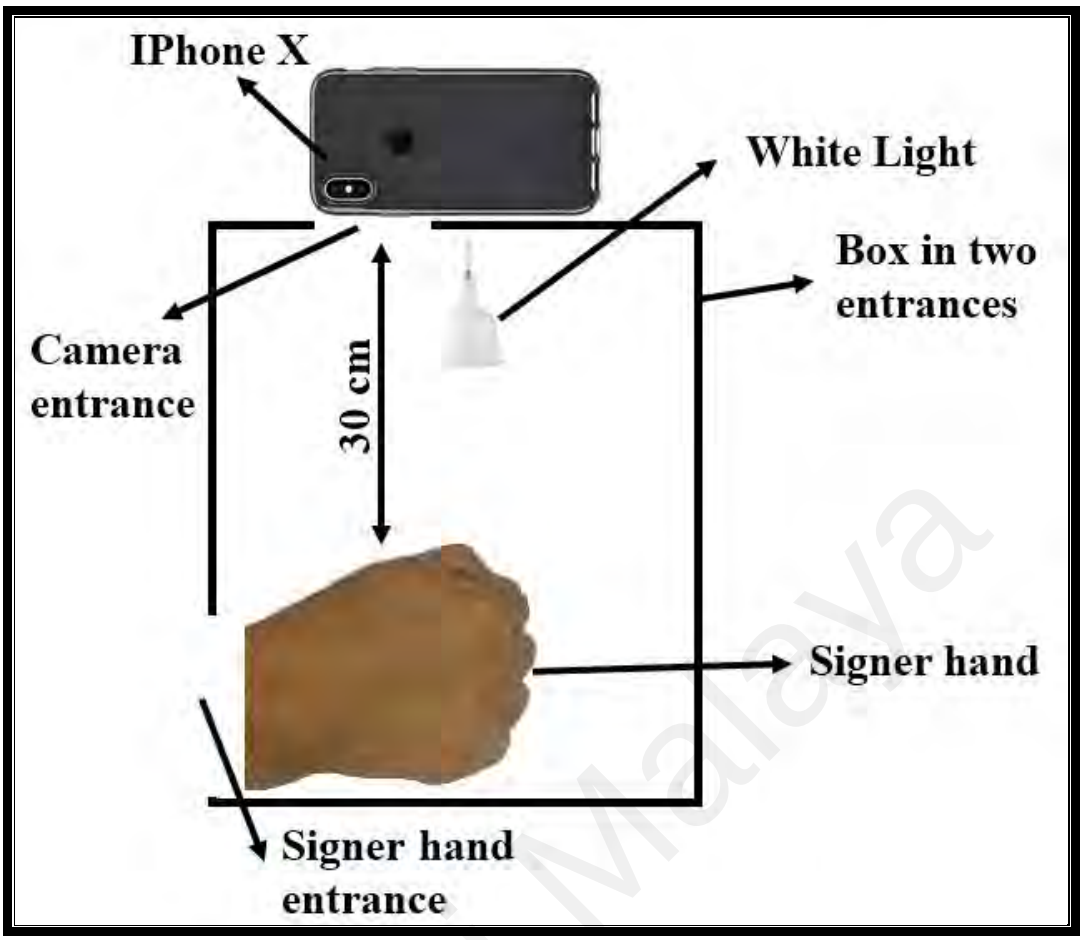


Figure 5.2: The image acquisition environment

5.5 Hybrid approach ArSL database

For the modelled hybrid ArSLR framework, smart data gloves and one Microsoft Kinect are used.

5.5.1. Smart Data Gloves

The smart gloves (Idris, 2012) are selected among other kinds of gloves for both personal and technical reasons.

First, cost-wise, this study supports the use of this local device, because it is cheaper than other smart gloves in the same category. Then, technically speaking, this device overcomes the Bluetooth communication problem that usually arises in SBR by reducing the movement limitation. Moreover, it has an efficient number of sensors with accurate

Universiti Malaya

readings. More details about data collection and tagging are described in Section 5.5 and the smart gloves package is presented in Figures 5.3 and 5.4.

Smart gloves have a Bluetooth adapter with an external battery. Therefore, a PC with a Bluetooth connection for data collection and communication is required in this study.

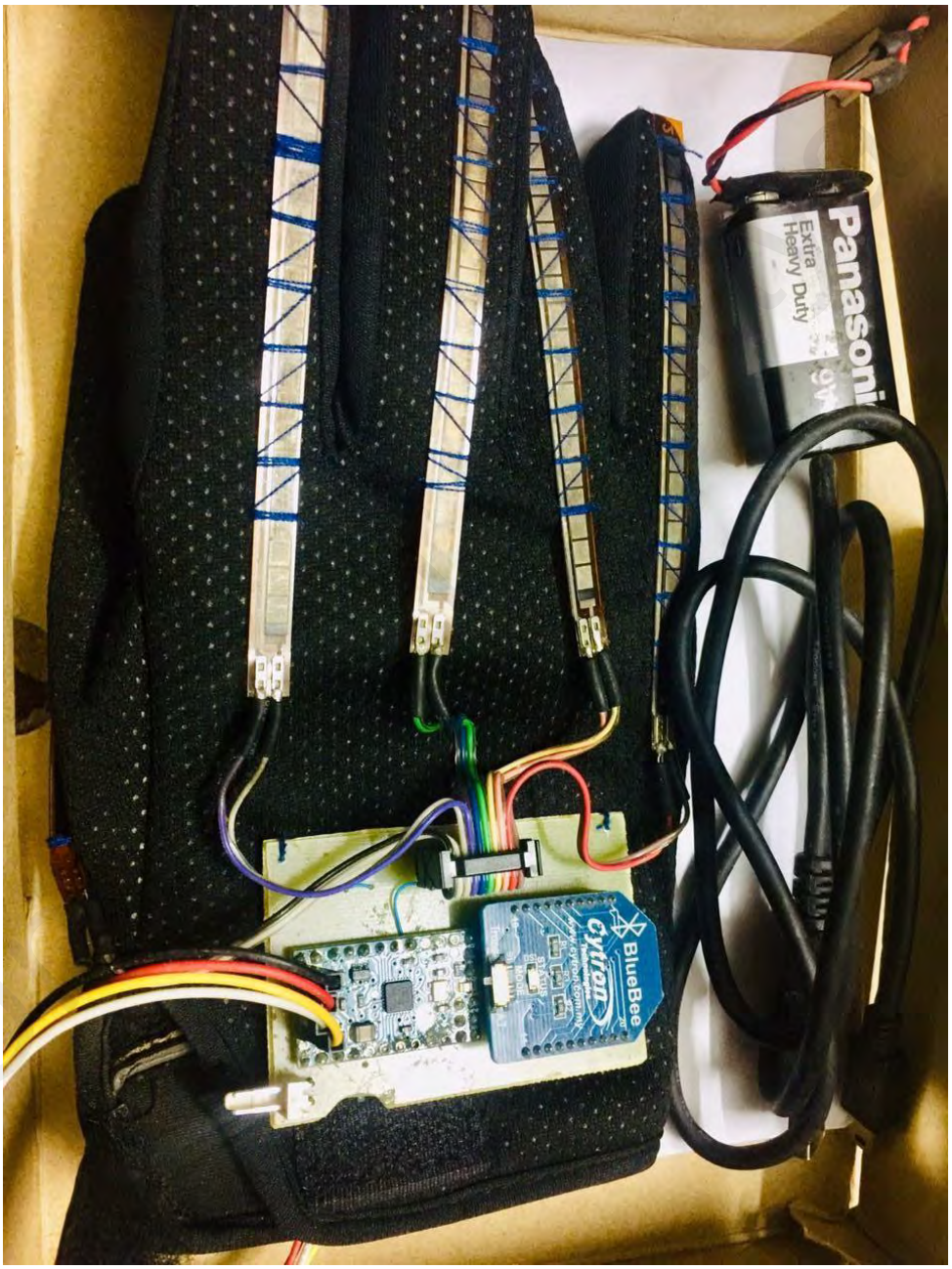


Figure 5.3: Smart Data Gloves Package (Idris, 2012)



Figure 5.4: Smart Data Gloves (Idris, 2012)

The Smart glove has one 3-axis accelerometer sensor and 5 embedded bend sensors (one per finger) as shown in Figure 5.5.

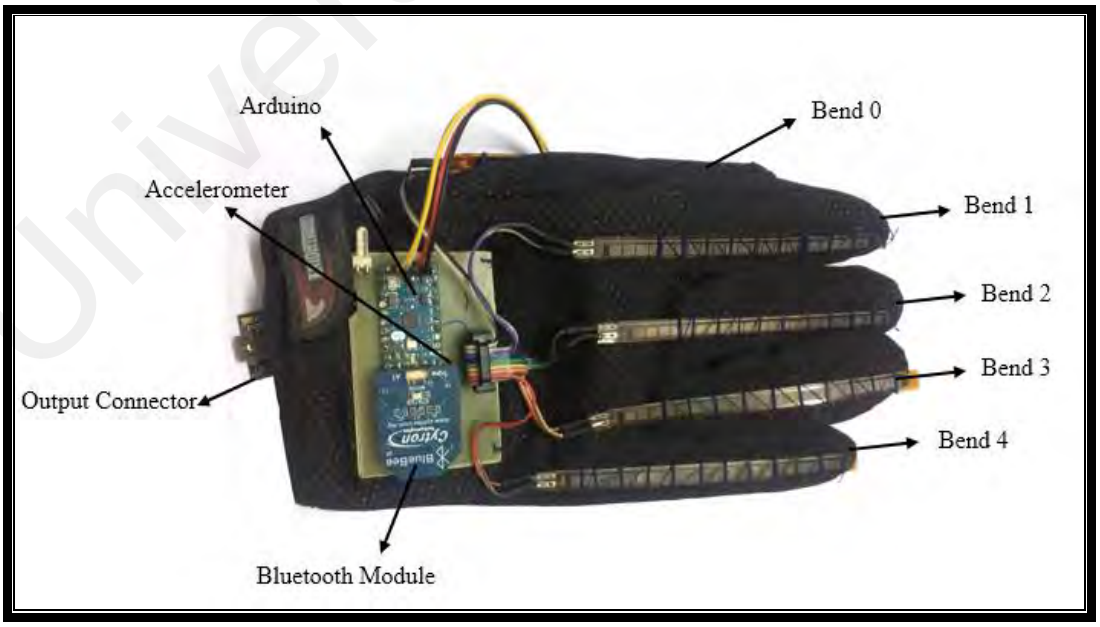


Figure 5.5: Smart data gloves Sensors

Hand movement measurements include both static and dynamic accelerations. The static acceleration provides pitch and roll rotational angles as shown in Figure 5.6, while the dynamic acceleration provides 3-axis that represents instantaneous hand movement as shown in Figure 5.7.

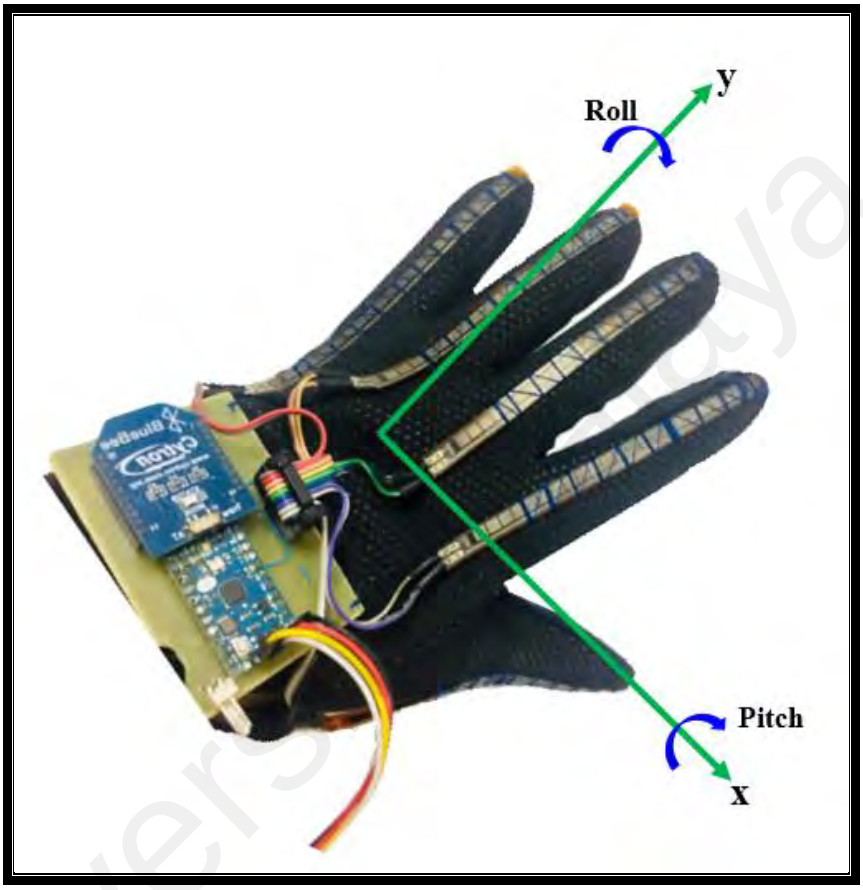


Figure 5.6: Pitch and Roll Angles

The bend percentages are recorded for 5-flexion measurements (thumb, index, middle, ring, and little fingers) in the range of 0.0% - 100.0%. Moreover, two rotational angles for pitch and roll are recorded in hand orientation range as -90˚ to +90˚. Finally, the hand instantaneous acceleration values for 3-axis (x, y, and z) are recorded in a range between -2g and +2g.

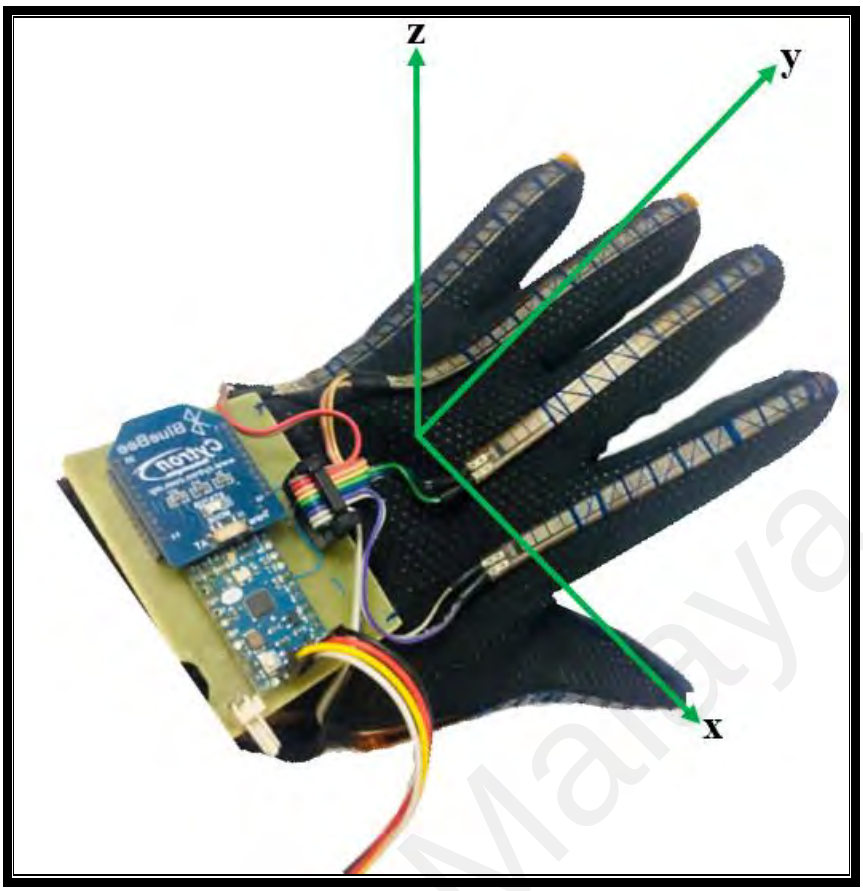


Figure 5.7: Axes (x, y and z) Reference.

Microsoft Excel Worksheet (.xls) is used to record the acquired feature data file for left and a right hand that each row containing 5-flexion measurements, 2-hand orientation angles, and 3-instantaneous acceleration measurements.

	Flexion Measurements					**Hand Orientation**		**Hand Acceleration**		
Left	Thumb	Index	Middle	Ring	Little	Roll	Pitch	Ax	Ay	Az
Right	Little	Ring	Middle	Index	Thumb	Roll	Pitch	Ax	Ay	Az

Universiti Malaya

5.5.2. Microsoft Kinect

Microsoft Kinect was originally released as a motion sensing input device by Microsoft for the Xbox 360 video game console. Later Microsoft released a version for Windows PCs which turned out as a device to be used for other purposes than developing games. Tang (2011) is proposed a new method for HGR using the RGB color model and depth data from Microsoft's Kinect sensor. For VBR database using Kinect, the recorded videos (color and depth) are converted to images based on a frame rate of 30 fps, then the SL expert selected the main images (one or two images) that covered the purpose sign.

5.6. Data Collection and Tagging of Gestures

5.6.1 Data Collection

Data collected from signers plays the most important role in a gesture detection and recognition framework whether it is the sample templates of gestures or information on how to recognize a gesture.

In sensor-based dynamic gestures, signs are composed of a stream of letters or words that are not physically separated. In vision-based, such a problem is less of a concern due to the fact that video images can be manually tagged using visual inspection with a high level of accuracy. A sign language expert is able to visually identify the boundaries of different words in a sentence. However, this is not the case for sensor-based signs. In such a case, the sensor readings cannot be examined visually for manual tagged. One of the solutions for manual tagging of smart gloves sensor readings is to use a camera in recording the signing. Once the signing is completed, the video recordings can be synchronized with the sensor readings to detect the boundaries of the signs.

To collect the data for gestures, we used Microsoft Kinect to extract features such as distance, elevation, curvature and palm area hand features of the signer. In addition, we used Smart Glove to extract features and collect data from the fingers such as flexion

Universiti Malaya

measurements, hand movements, orientation angles and instantaneous acceleration measurements of the hand signers. This involves the detection of human joints using the skeleton detection feature available in Microsoft Kinect and the flex sensor and accelerometer information for detection of finger states using Smart Glove.

5.6.2 Data Tagging

As one of the main step in supervised learning, tagging is required to set classes for glove readings and prepare them for the modeling stage. In ArSL framework, dynamic signs are composed of several gestures. Hence, boundaries must be located carefully to mark successive gestures. Therefore, a video segment was recorded for each repetition to carry out manual tagging. The following diagram in Figure 5.8 shows the main faces followed in collection and tagging data.

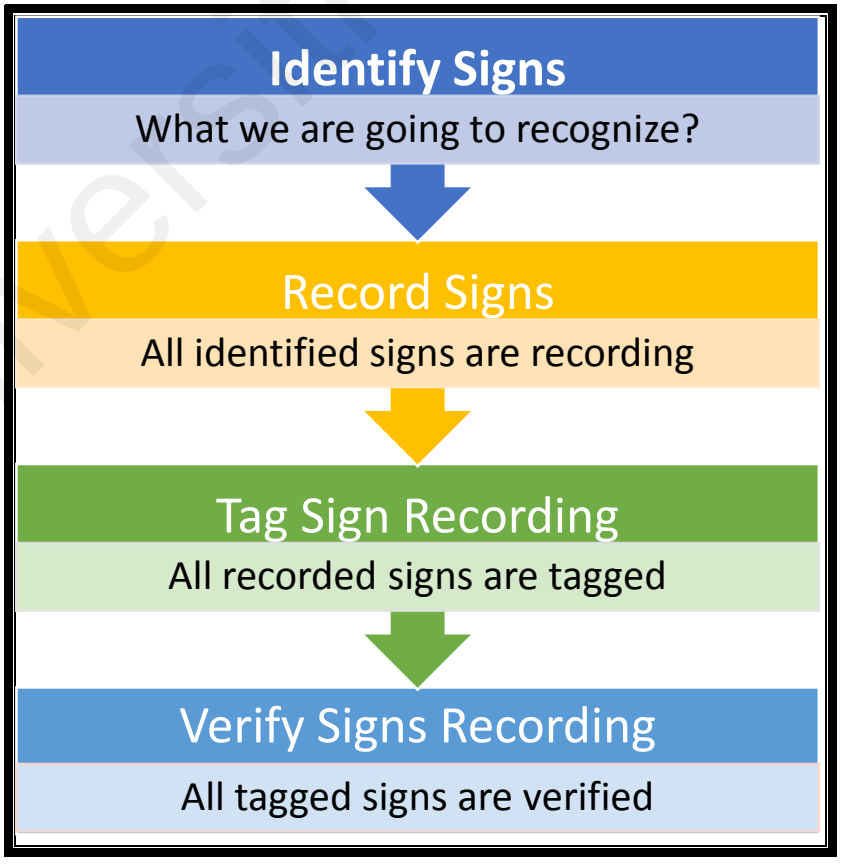


Figure 5.8: Data collection and tagging flow

Universiti Malaya

5.6.2.1 Microsoft Kinect Tagging

In order to collect the data, we applied the recommended procedure by Microsoft Kinect Developer Summit (2011).

- **Identifying Signs**

The initial step before data collection is to identify the signs that we are going to recognize. This thesis aimed at recognizing gestures specified in standard ArSL as illustrated in tables 5.1, 5.2, 5.3, 5.4, and 5.5 which can be found in section 5.1.

- **Record Signs**

After identification of the signs, we employed target users (SL expert, practitioners, and hearing and speech-impaired people) for the collection of signs data. All singers performed 10 repetitions for each sign.

- **Tag Sign Recording**

After signs have been recorded successfully, and SL expert signer observed all record. Then, the recorded videos are converted to images (color and depth images), and SL expert chooses the main images (one or two images) that covered the purpose sign in each gesture. Thereafter, all signs recorded are tagged which will represent the collection of coordinates for each sign.

- **Verify Sign Recording**

All tagged signs are verified to check whether the actual tags represent the correct collection of coordinates.

5.6.2.2 Smart Data Gloves Tagging

The Smart Glove will be used to identify the signs of gesture. Therefore, we need to identify all ArSL alphabet, numbers, words, and sentences.

Universiti Malaya

- **Identifying Signs**

The initial step before the collection of data is to identify the gestures that we are going to recognize. This thesis aimed at recognizing gestures specified in standard ArSL. Therefore the focus was only to recognize the standard ArSL as illustrated in tables 5.1, 5.2, 5.3, 5.4, and 5.5 which can be found in section 5.1. In smart gloves data collection, each signer performed 10 repetitions for each sign.

- **Record Signs**

We used to target users (such as SL expert, practitioners, and hearing and speech-impaired people) for the collection gesture data after the identification of the gestures. All singers performed 10 repetitions for each sign, isolated-word, and sentence.

- **Tag Sign Recording**

When signs are recorded successfully, SL expert observed all the record, and digital camera record a video for each signer repetition. For each sign record, 10 (.XLS) files were recorded for the left hand and 10 (.XLS) files for the right hand. Afterward, for each repetition, tagging was recorded separately for each hand based on their timestamps and the files integrated together to have a label associated with 20 features for each file. In addition, we synchronized the files by deleting unmatched tags because of the non-uniform sampling rate. Finally, we wrote a MATLAB code to save ten files with 20 features and tags in Comma-Separated Values (.CSV) files that were saved into a cell array of $feature\ vectors(FVs)$. Figure 5.9 is shows the process used in smart gloves data tagging.

- **Verify Sign Recording**

All tagged gloves signs are verified to check whether the actual tags represented the correct collection of sensor information.

Universiti Malaya

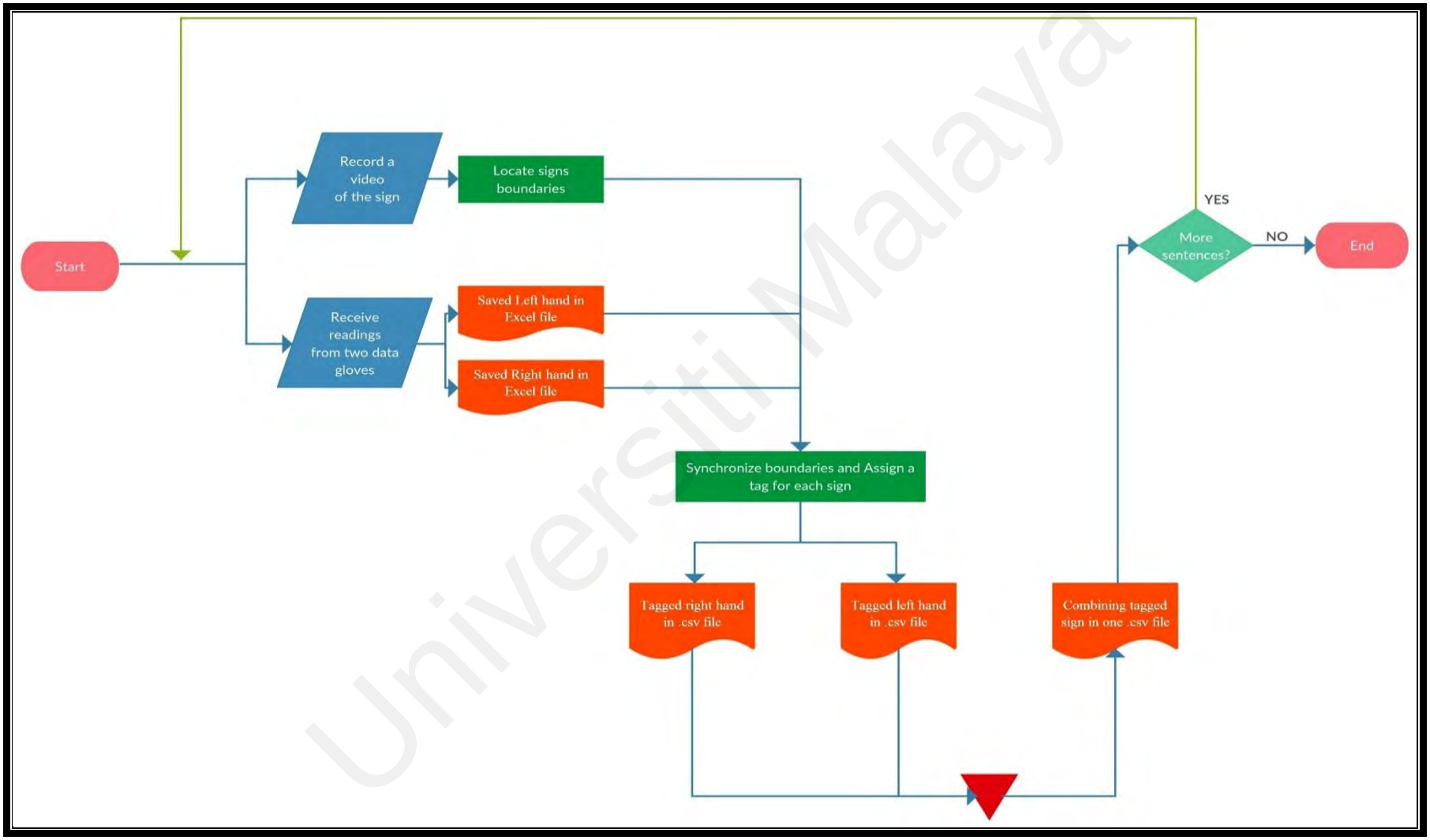


Figure 5.9: Data Tagging flowchart

Universiti Malaya

5.7 Summary

This chapter described the procedure used to create the ArSL database, which coversed all approaches and forms of signs. The database consists of two main databases related to our framework, which was discussed in chapter 3. These databases are created based on 10 signers (three male signers and seven female signers) who performed 10 repetitions for each sign. The signers are SL expert practitioners and hearing- and speech-impaired individuals. In addition, the list of signs used in this database and the data organisation were presented. Furthermore, the data acquisition quality and devices were presented together with the data collection techniques and setup. Finally, the data tagging and final data format are described. The design of the proposed framework shall be presented in the next chapter

Universiti Malaya

CHAPTER 6: FRAMEWORK DESIGN AND IMPLEMENTATION

6.1 Introduction

Chapters three and five explain the research methodology and the approaches used for database collection and labelling, respectively. In Chapter 6, two frameworks are modelled in order to recognise ArSL. In the first framework in Section 6.2, a usable static backhand and signer-independent ArSLR framework is modelled based on VBR approach, which translates signs of letters and numbers into their corresponding texts using a smartphone camera based on the multi-classification approach. The second framework depicted in Section 6.3 models a usable and hybrid continuous ArSL approach, the signer-independent and automatic ArSLR framework, which integrates the VBR approach using Microsoft Kinect and the SBR approach using smart data gloves.

6.2 Framework 1: Static backhand Arabic Sign Language Recognition

This framework have three main GR phases, which are integrated to recognize static backhand Arabic letters and numbers in the ArSL, as shown in Figure 6.1. The main purpose of the framework is to lower the errors inherent in the static backhand ArSLR.

Universiti Malaya

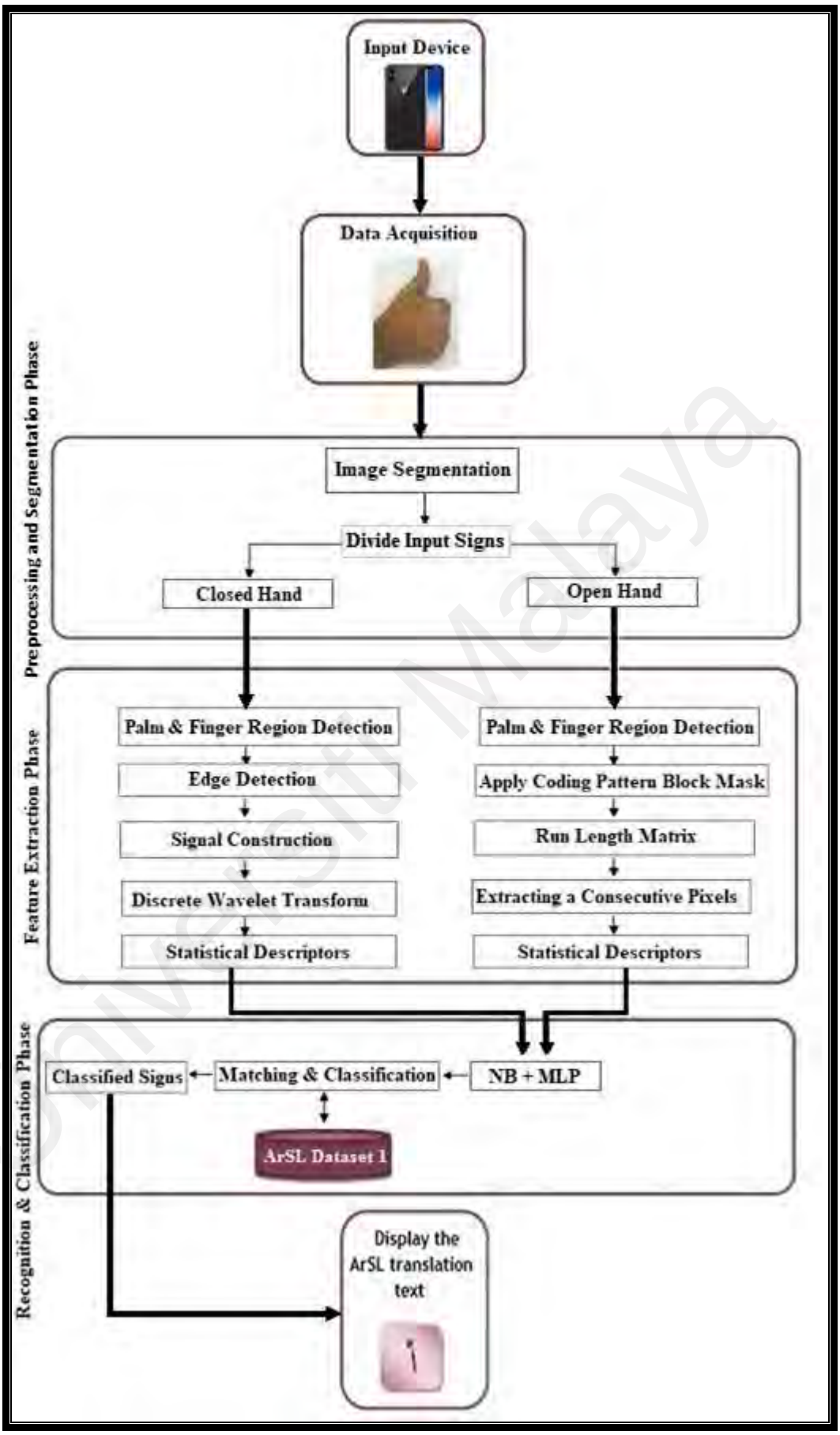


Figure 6.1: Proposed framework for static backhand ArSLR

Universiti Malaya

Figure 6.1 explains the entire framework, which contains three main phases: pre-processing and segmentation, feature extraction, and recognition and classification.

1- **Pre-Processing and Segmentation Phase:**

   A- Image segmentation represents the step of hand detection where the proposed framework obtains input signs dataset in RGB, then converted them through binarization into white and black signs dataset in order to identify hand localizations.

   B- Sign status sequence: Compared with open hand signs, no finger goes out from the palm within closed hand signs as they appear as circles. Thus, the algorithm of gesture position recognition is applied to divide the input signs, which are represented by images into open and closed hand sign categories, depending on the position of the hand (alphabet sign & number sign) and a circularity shape feature.

2- **Feature Extraction Phase:** Open hand and closed hand signs have various features. Therefore, the features extracted from the two sign categories vary, and these features are usually located in the upper hand region. Hence, the algorithm uses white and black images to determine the upper-hand area in the previous step and sets it as the Region of Interest (ROI).

3- **Recognition & Classification Phase:** In this phase, the NB and MLP classifier are utilized to recognition in open and closed hand sign categories.

6.2.1 Pre-Processing Phase

6.2.1.1 Image Segmentation and Detection

The image background is removed with a special tool during the front-hand and backhand removal pre-processing in order to reveal the image of a hand, which represents key recognition system data for ArSL. Then the system moves on to the image detection stage. Many good tools are commonly applied to remove image backgrounds (Phung,

Universiti Malaya

Bouzerdoum, & Chai, 2005), but in this thesis, the system input images use the same background for all sign images of numbers and letters. Subsequently, dataset images are in standby for extraction through the binarization process, where the backgrounds and foregrounds of letter and number sign images are extracted. A suitable binarization threshold value should be first identified in the binarization step. A binarization algorithm that relies on the iterative thresholding method is employed in this thesis as a basic and suitable way of identifying various codes as marks for foreground and background areas and to construct the binarization threshold value. Consequently, the foreground region is extracted from the original input image as a sign image. The instance of an image segmentation through iterative thresholding is shown in Figure 6.2.

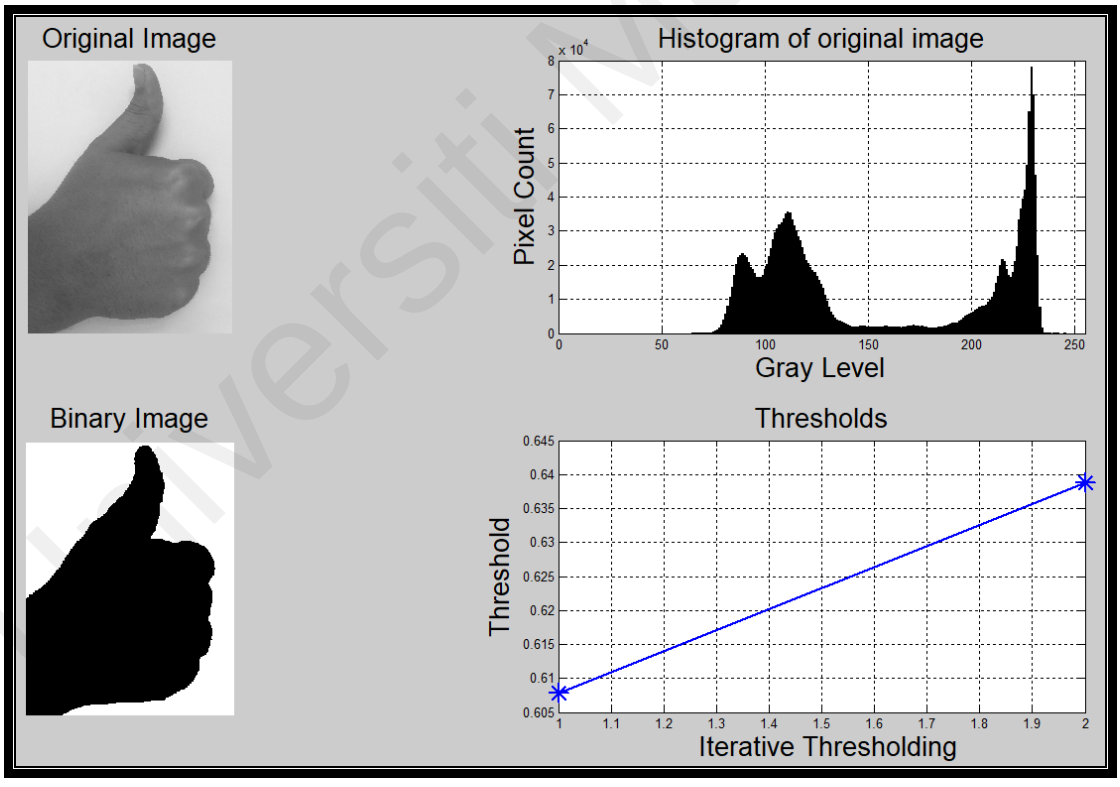


Figure 6.2: Image segmentation using an iteration Thresholding

6.2.1.2. Signs status categories:

The characteristics of some finger letters and numbers vary, so their patterns cannot be treated by one algorithm. On the other hand, the features of finger numbers and letters that

Universiti Malaya

are identical must be categorized in the same group. Therefore, this study split all finger numbers and letters into two main groups based on sign circularity and taking into account suitable tools to address the two categories.

Finger numbers and letters can be segmented into two core groups based on their appearance:

1. Closed hand signs.
2. Open hand signs.

Closed hand sign category: In this category, the sign shape appears as a circle and no finger leaves the forehand, as seen in Figure 6.3. Letters "ص, ض, ف, ق, هـ, و" and)0(صفر are members of the closed hand sign group.

Open hand sign category: In this category, the sign shape is less circular because it has a minimum of one finger that extends over the palm in both side and upward directions. Figure 6.4 illustrates the shapes of open hand signs. The letters in this group are "أ, ب, ت, ث, ج, ح, خ, د, ذ, ر, ز, س, ش, ط, ظ, ع, غ, ك, ل, م, ن, ي" and the numbers are)1(واحد,)2(اثنان,)3(ثلاثة,)4(اربعة,)5(خمسة,)6(ستة,)7(سبعة,)8(ثمانية,)9(تسعة")

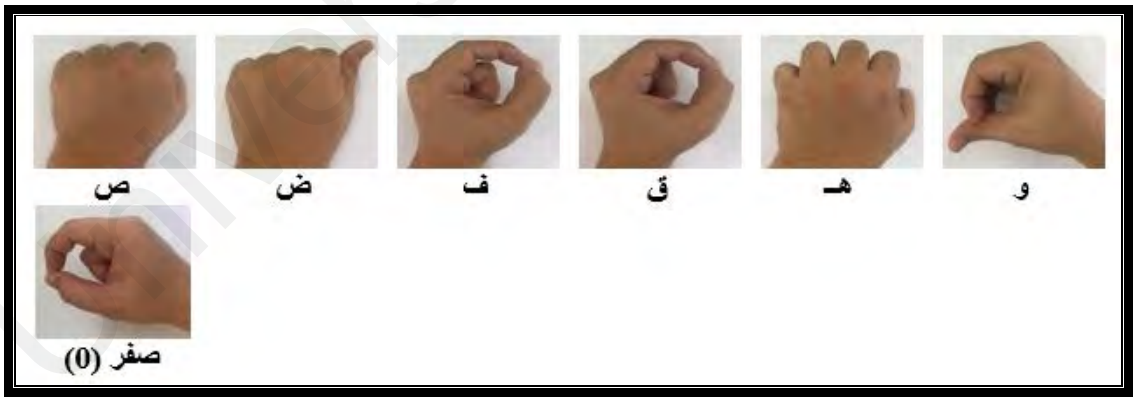


Figure 6.3: Closed hand static signs for ArSL letters and numbers

Universiti Malaya



Figure 6.4: Open hand static signs for ArSL letters and numbers

A- Hand Configuration Analysis

Suppose the proposed system generates a clear background hand image (right or left) as illustrated in Figure 6.5, where the system is required to identify the configuration of a finger input image comprising the finger state and finger parts based on the definition of sign language. The finger parts represent finger names that consist of the "pinky-ring-middle-index and the thumb" while the finger states denote the finger status like flexion or extension.

Universiti Malaya

The image of a hand can be described as a border shape identical to the palm, which is directly linked to five borders with rectangular shape denoted by fingers. Based on the proposed system, backhand images were selected to represent the hand images dataset.



Figure 6.5: Input image example.

The finger parts are naturally ordered from right to left as the "thumb-index-middle-ring and pinky" in the left-hand case and as illustrated in Figure 6.6. Moreover, the human hand is fundamentally composed of the palm and five fingers. The thumbs and fingers comprise of interconnected bones in which the thumb has two bones whereas the finger has three bones (Figure 6.6) (Erol, Bebis, Nicolescu, Boyle, & Twombly, 2007; Dipietro, Sabatini, & Dario, 2008).

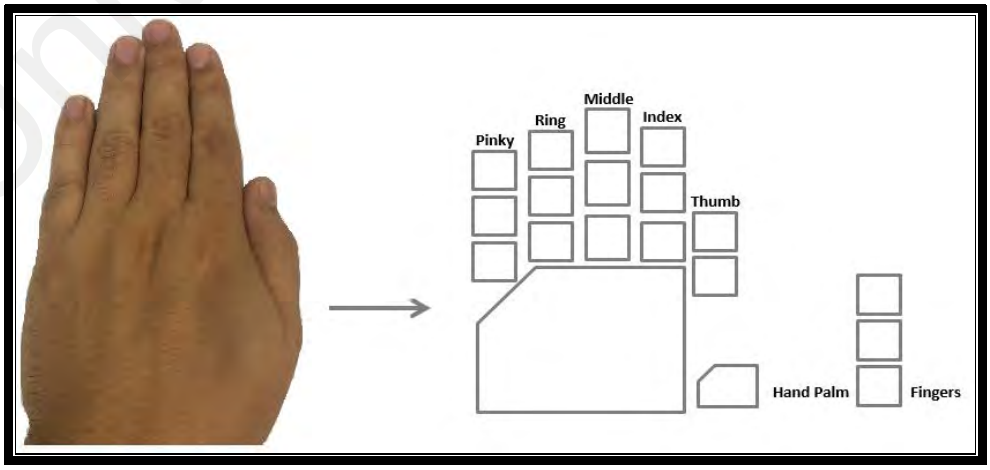


Figure 6.6: Hand configuration analysis.

Universiti Malaya

In nature, finger movement is flexible within two freedom degrees: movement to the back or front of the hand and spreading of the fingers from each other within a plane as described by the palm. As seen in Figure 6.7, a finger can exhibit in either extension or flexion state. Two or more bones can be seen in the extension state image, whereas only one bone can be seen in flexion state.

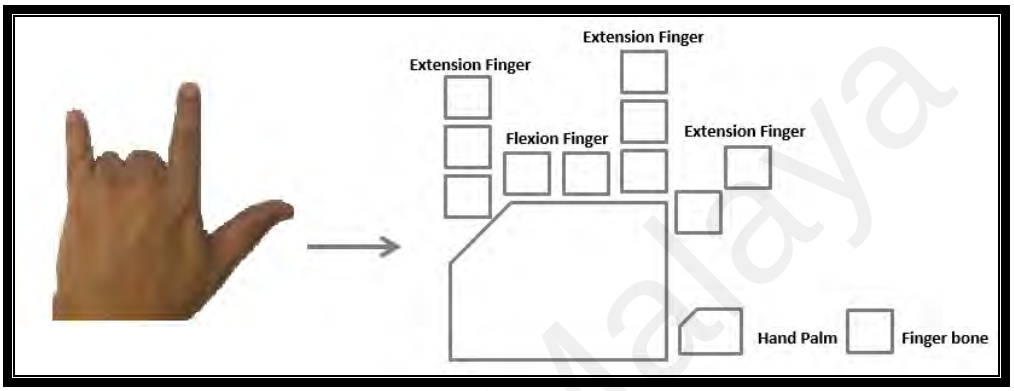


Figure 6.7: Left-hand Fingers state.

B- Open Hand Sign Category Description

For the open hand sign category, the fingers' positions exhibit in the upward trend. The fingers are fixed in their locations through the fourteen bones within the fingers as illustrated in Figure 6.6. The state of a finger (flexion or extension) can be determined by the fingertip, which represents the highest position of each finger. As shown in Figure 6.8, the sequence of a finger is used to determine its name and length.

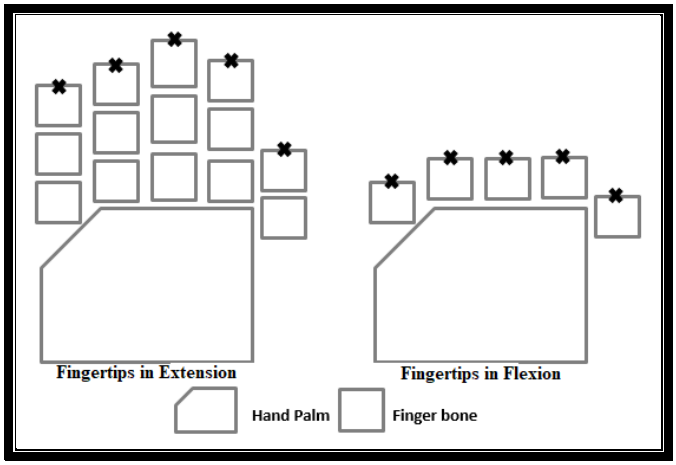


Figure 6.8: Fingertips in extension & flexion

Universiti Malaya

This provision implies that the five fingertips can be taken as a result of the quantization and sampling of continuous signals in the horizontal and vertical directions accordingly.

The signals are turned into digital values and serves as a representation of block regions as depicted in Figure 6.9. Although a block mask with $5x5$ dimensions encompasses all five fingertips. However, it expands into a horizontal direction and as a result causes digital errors. Furthermore, the block mask is separated into sub-blocks, each of which is assigned a quantized value in order to increase the robustness of the mask against the errors in digitization and the signs are differentiated through the quantized value of the distribution. Figure 6.9 represents the hand configuration analysis and the blocking mask with $10x10$ dimensions applied in this study.

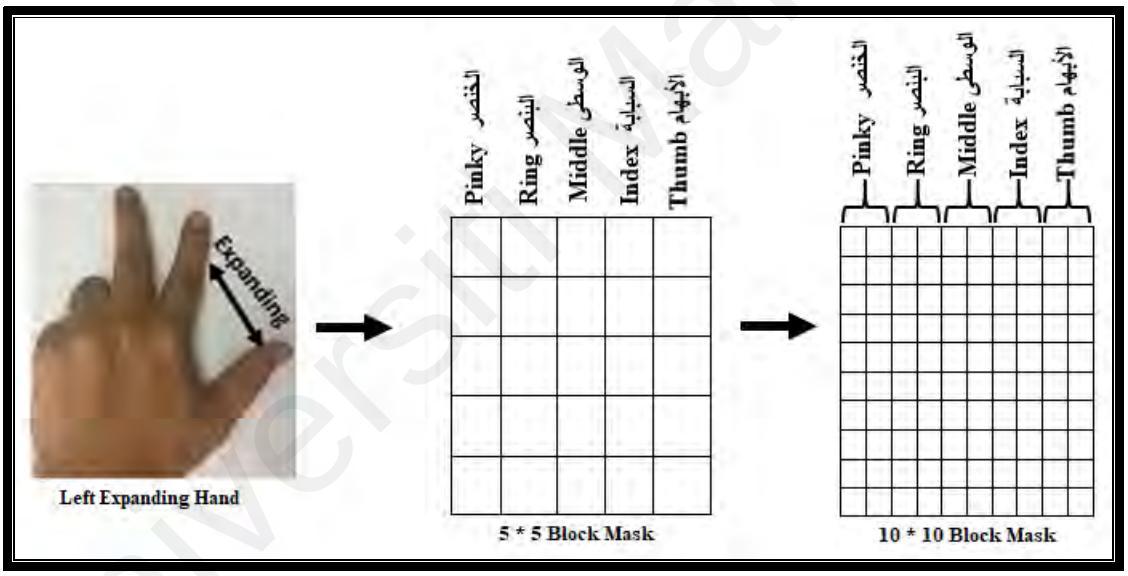


Figure 6.9: Hand configuration analysis

C- Close Hand Sign Category Description

The contour of the hand edges in the category of closed hand signs is deemed a circle, so the signs become similar to each other. The information in the upper areas of closed hand signs is more significant than in lower areas and as a result, the upper areas are utilized as ROI to recognize the signs in the closed hand category as depicted Figure 6.10. Various graph patterns are obtained from the 1D sign signals generated by the upper hand areas.

University of Malaya

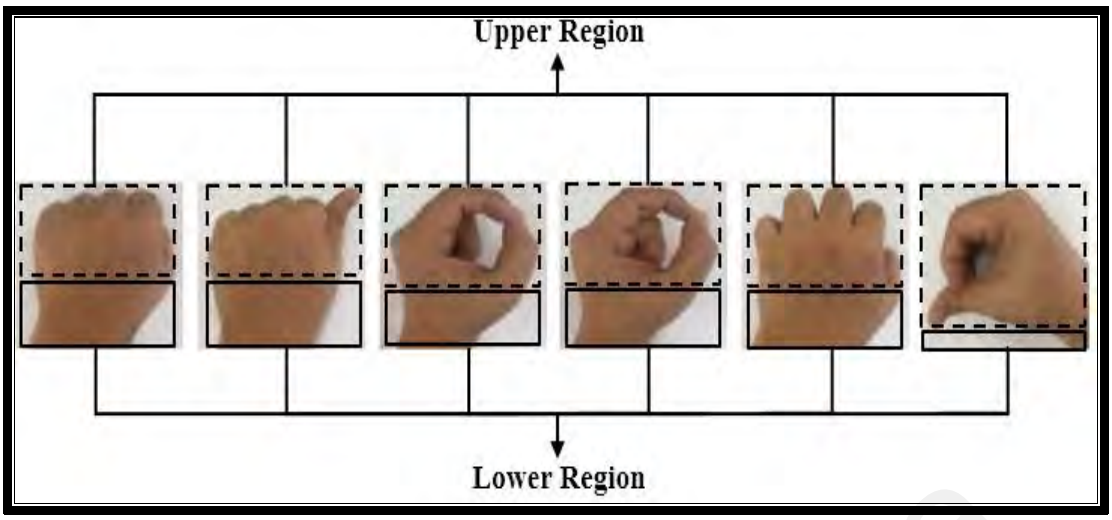


Figure 6.10: Region of interest of the close hand signs

6.2.1.3 Shape Factor and Axis Ratio Method

There is an approximately circular similarity of the hand edge contour in a closed hand sign, in comparison with the open hand signs according to the prior definition given in the section of gesture category.

As illustrated in Figures 6.11 and 6.12, the extended fingers are represented by dash lines. The output result of the shape factor, which denotes the circularity of number and letter sign samples indicates that the closed hand signals have high shape factor values of approximately 0.64-0.78, which refers to as high circularity. On the other hand, the shape factor values for the open hand category are very low, roughly 0.27-0.59. The threshold values can be used for classification in two classes due to the tolerance of circularity between open and closed hand signs.

Universiti Malaya

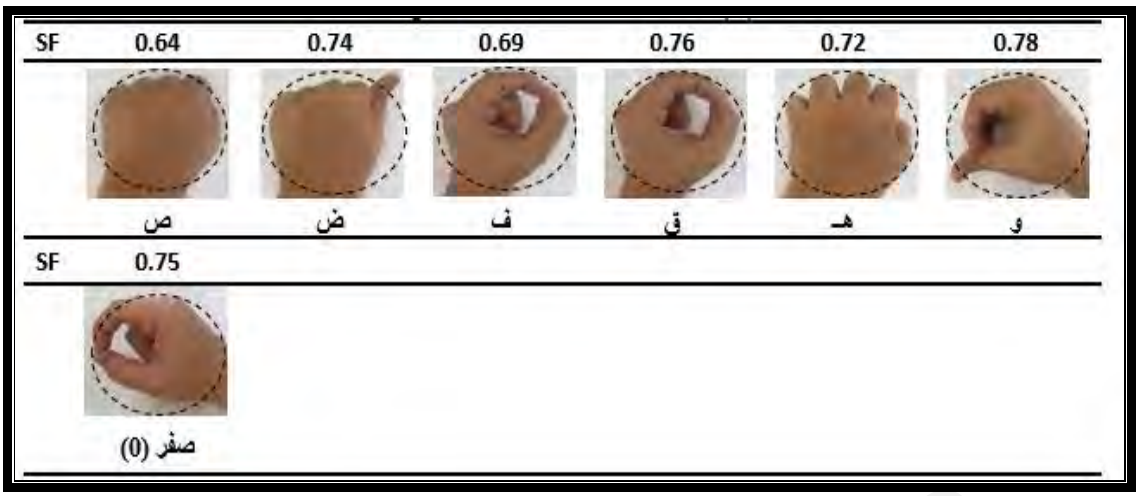


Figure 6.11: Circularity shape factor for close hand signs

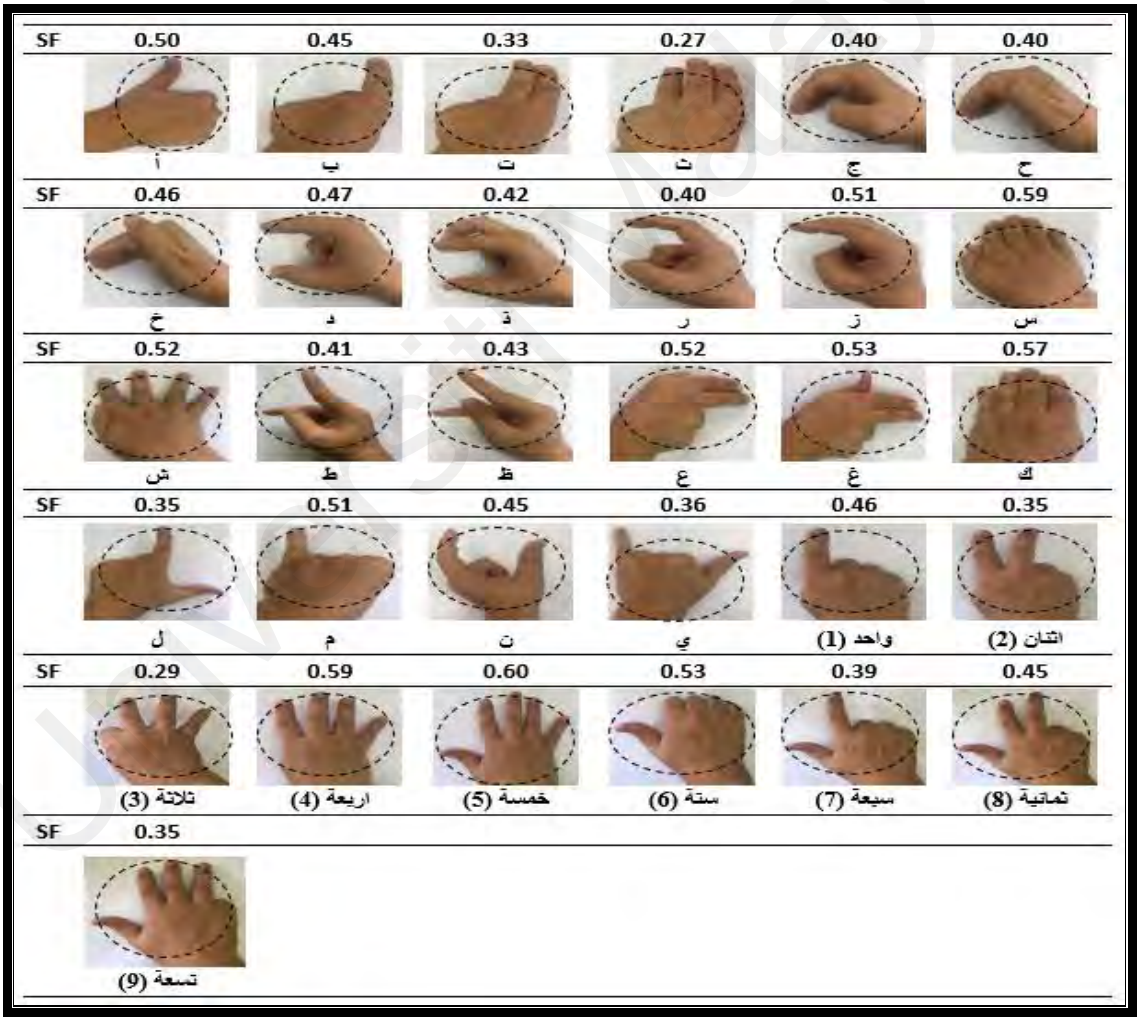


Figure 6.12: Circularity shape factor for open hand signs

Therefore, an appropriate algorithm should be determined to compute sign circularity and identify suitable value for threshold circularity for both closed and open hand

categories classification. Then the signs can be categorized into the closed or open hand based on the identified threshold. Hence, in this framework, the circularity threshold values for closed and open hand categories were identified beforehand. Subsequently, the circularity value of any new sign can be calculated and compared with the threshold. Furthermore, the axis ratio and shape factor are used here to identify the circularity (Rathod, Sapate, & Khatirkar, 2012). Besides, 3800 samples of signs comprising 3100 open signs and 700 closed signs were used in the experiment to determine the suitable threshold value for circularity. In this framework, the threshold value for the shape factor is 0.63 and for the axis ratio, it is 1.7.

The proposed flow chart of the sign category algorithm is depicted in Figure 6.13 below.

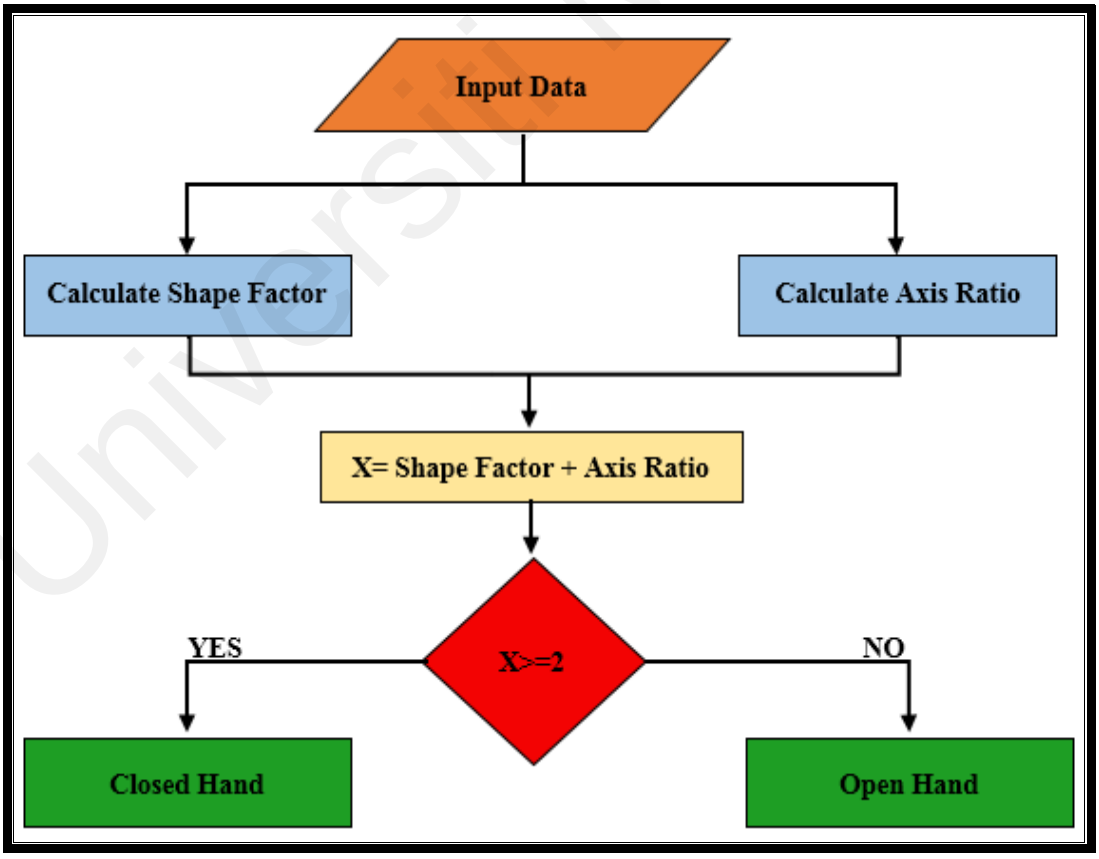


Figure 6.13: Flowchart for proposed sign category algorithm

Universiti Malaya

A- Shape Factor (SF)

Circularity represents a quantity without dimensions that scales the 'closeness' toward an ideal circle. It is also computed by dividing the area of the hand ratio of its perimeter, as described in equation (6.1). The shape factor of circularity (S) was presented in (Rathod et al., 2012).

$$\boldsymbol{Shap\ factor} = \frac{\boldsymbol{4\pi A}}{\boldsymbol{P^2}} \quad \textbf{(6.1)}$$

Where *P* denotes the perimeter of the hand and *A* denotes the area of the hand. *A* represents the entire number of pixels within the hand and it can be computed by applying equation (6.2) (Krishnaveni & Radha, 2011).

$$\boldsymbol{A} = \iint \boldsymbol{I(x,y)dxdy} \quad \textbf{(6.2)}$$

$\boldsymbol{Where,}$

$$\boldsymbol{I(x,y)} = \begin{cases} \mathbf{1}, & \boldsymbol{Hand\ pixel} \\ \mathbf{0}, & \boldsymbol{Background} \end{cases}$$

The hand perimeter (*P*) represents the numbers of pixels within the contour of a proportional hand.

In this framework, the shape factor value is denoted by one (1) if the SF value for an entered sign is higher than the threshold; else it's equal to zero (0). The threshold (*T*) value calculated from the experiment with **Algorithm 6.1** is 0.63. Figure 6.3 shows the letters involved in the category of closed hand signs, which are "ص, ض, ف, ق, هـ , و and صفر" (0).

Universiti Malaya

Algorithm 6.1 (Proposed algorithm to determined sign category threshold detection)

Require: Determined sign category threshold detection

$a = Close\ hand\ signs;$

$b = Open\ hand\ signs;$

$\%\%\ Set\ the\ circularity\%\%$

$SF(a) = circularity\ SF\ of\ the\ close\ hand\ signs;$

$SF(b) = circularity\ SF\ of\ the\ open\ hand\ signs;$

$\%\ \%Find\ the\ local\ maximum\ of\ each\ group\%\%$

$maximum(SF(a)) = local\ maximum\ of\ the\ close\ hand\ sign\ category;$

$maximum\big(SF(b)\big) = local\ maximum\ of\ the\ open\ hand\ sign\ category;$

$T = min(value)\ between\ maximum(SF(a))\ and\ maximum(SF(b));$

B- Axis Ratio (AR)

As mentioned in the shape factor section, closed hand signs are more similar to a circle than open hand signs. Any circle can be described by the shape that has an identical length as the minor and major axes. Hence, the major axis length of an ellipse is usually represented by the major axis, where the ellipse has normalized second focal moments identical to the area. Moreover, the minor axis length of the ellipse is represented by the minor axis as illustrated in both Figures 6.14 and 6.15. Then the division of the major axis magnitude or length of the minor axis length is defined by the axis ratio (AR).

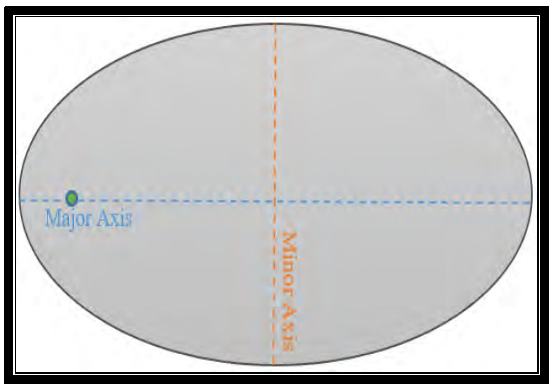


Figure 6.14: Ratio of the major axis to the minor axis.

Universiti Malaya

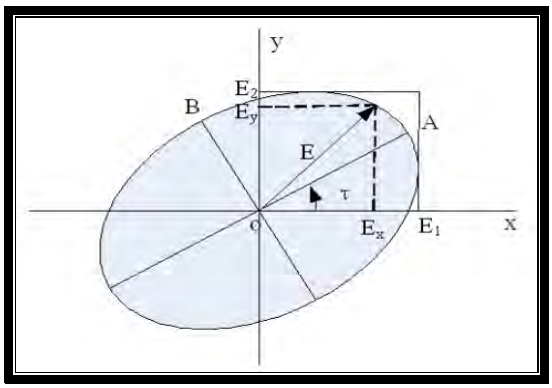


Figure 6.15: Polarization ellipse (Sun, Ou, Lu, & Tan, 2013)

$$\textbf{Axis Ratio} = \frac{\textbf{Half Major axis length}}{\textbf{Half Minor axis length}} \quad \mathbf{1} \leq \boldsymbol{Axis\ Ratio} \leq \infty \tag{6.3}$$

Therefore, the value of AR is equivalent to 1 if the value of AR for any input image is lower than the threshold value; else it becomes 0. In the current experiment, the threshold value obtained is equivalent to 1.7. Instances of the minor and major axes for closed and open hand signs are illustrated in Figure 6.16.

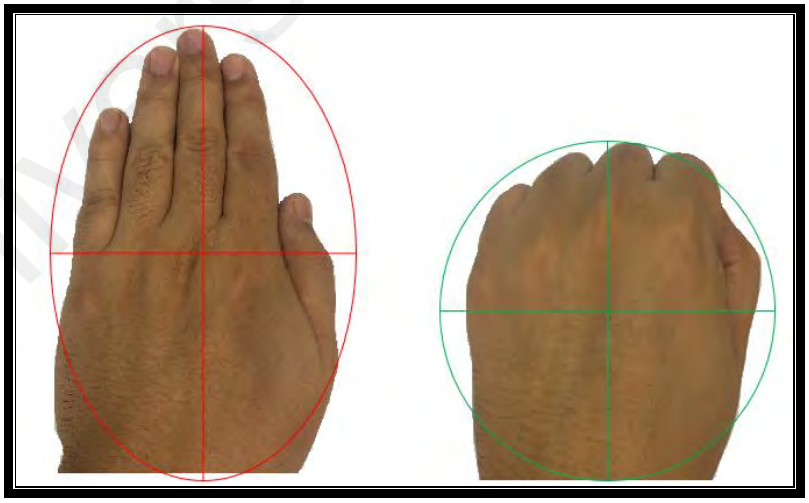


Figure 6.16: Major and minor axes of open and closed hand signs.

Universiti Malaya

6.2.2 FEATURE EXTRACTION PHASE

The features extracted from the closed and open hand sign categories vary due to the variations in the sign of both categories.

6.2.2.1 Feature Extraction of Closed Hand Signs

A closed hand sign within an image can be described by a hand with all fingers in flexion state and whose shape seems circular. As the closed hand sign category has seven signs with similar shapes (Pugeault & Bowden, 2011), recognition in this category becomes very difficult. Therefore, the signs from the closed hand category are recognized using the Discrete Wavelet Transform (DWT) alongside statistical descriptors. DWT was selected because the small alternations within a 2D image can be amplified within the edges of a 1D image and DWT has the ability to separate the small details between vague closed hand signs. Even though DWT has been used in other prior studies with an identical approach for the closed hand sign category, it is hereby used to represent hand edge bending in 1D rather than a 2D representation of images.

A- Area Detection for Closed Hand Sign Fingers

A closed hand sign comprises two portions: the finger area and the palm area (Figure 6.17). Because significant information for recognizing letters is only involved in the finger area, Algorithm 6.2 is proposed an open window in a finger area to effectively process hand image data based on the area of a finger. The system initially identifies the row involved in the middle point position. Then this position is set as the bottom window in the finger area. The window is filled such that they fit in the area of a finger on the other side. Thus, the right and left sides are located in the rightmost and leftmost columns in the hand area, while the upper side is taken from the first row in the hand area.

Universiti Malaya

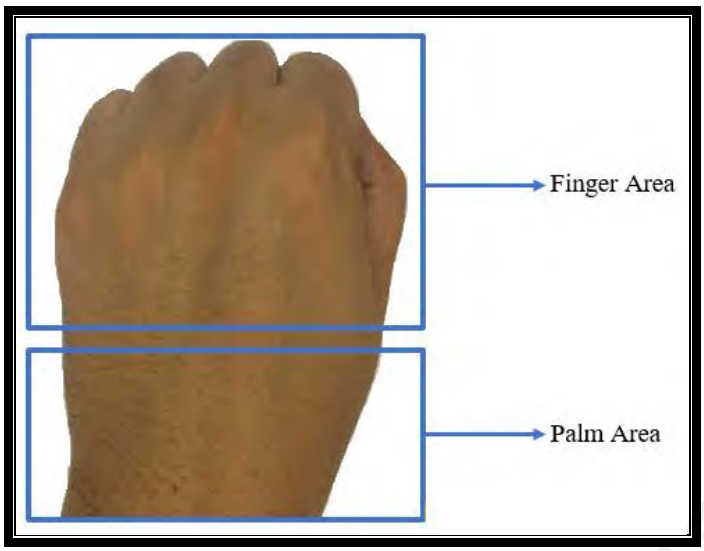


Figure 6.17: Finger area window.

Algorithm 6.2 Finger Region Detection

Require: Determined the region detection

$X = input\ sign;$

$\%\ locate\ the\ middle\ point\ of\ a\ hand\ region$

$(C_a, C_b) = middle\ point\ position\ of\ an\ input\ image\ (X);$

$\%\ Find\ all\ sides\ of\ the\ finger\ region\ window$

$bottom = row\ of\ the\ centroid\ (C_a);$

$top = row\ of\ the\ first\ hand\ pixel\ from\ the\ top;$

$left = column\ of\ the\ first\ hand\ pixel\ from\ the\ left;$

$right = column\ of\ the\ first\ hand\ pixel\ from\ the\ right;$

$Finger_region = a(top{:}bottom;\ left{:}right);$

B- Edge Detection

Forehand image position and finger shape information has been employed in prior studies to perform sign recognition. In this framework, closed hand signs are taken from backhand images based on the mobile device proposed, where the significant information is only available within the backhand finger contour. Several good algorithms for edge recognition can be used for this problem such as Canny edge detection, Laplacian of

Universiti Malaya

Gaussian (LoG), etc. (Bin & Samiei, 2012)(Kaur & Singh, 2016). The LoG detector is recommended for use in this study because it provides good outcomes without noise interference.

Marr (1982) proposed an LoG, whose operator identifies the best filter to detect edges using the image Signal-to-Noise Ratio (SNR). The LoG detector operates in two stages: a Gaussian function is used to form a low-pass smoothing filter; then the Laplacian operator is used as a high-pass filter based on the second derivative of zero used for detection of the edges (Bin & Samiei, 2012).

The Laplacian method of edge detection operates by finding the zeros that cross the image edges using the second derivative. Computing the image derivative can determine the location of the edge, which has a one-dimensional slope shape.

Let $f(x, y)$ be a LoG function of the image being represented by the second derivative, which is given by equation (6.4) (Kaur & Singh, 2016):

$$\boldsymbol{\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}} \quad \textbf{(6.4)}$$

For an image with double edges, the LoG detector first smooths the image and then calculates the Laplacian. In addition, the edge can be determined by identifying the zero that crosses double edges. Let the value of pixel intensity be $L(x, y)$, then the image Laplacian can be expressed by equation (6.5) (Kaur & Singh, 2016):

$$\boldsymbol{L(x, y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}} \quad \textbf{(6.5)}$$

Figure 6.18 represents a popular usage of the Laplacian filter for discrete approximations.

	1				1		
1	4	1		1	-4	1	
	1				1		
L_x						L_y	

Figure 6.18: Laplacian filter in X direction and Y directions.

Universiti Malaya

C- Closed Hand Signal Construction

Within the closed hand category, the 2D backhand scenes vary slightly from forehand scenes, which makes it very hard to recognize sign meaning. The signs of letters in forehand scenes can be differentiated by the positioning of the thumb (Pugeault & Bowden, 2011). In fact, it is clear that the alphabet sign can be distinguished from the front and not back side. For instance and as illustrated in Figure 6.19, prolonging the thumb for the sign of letter 'ض' will change it to be covered by new fingers for the sign of letter 'ص' (Figure 6.19). Accordingly, the thumb position is deemed important information for differentiation. In addition, the letter signs in 2D recognition for the closed hand category are more similar than those in the open hand category; thus they become difficult in differentiation. Furthermore, the signatures extracted from diverse points on the 2D hand contour are regarded as optimal 1D data for closed hand sign differentiation. The signature method has been proposed to transform the contour of a 2D hand into a signature with one dimension, which is considered as preliminary data for analysis. The Euclidean distance, which represents the distance between a centroid and every contour point, is used to generate the 1D signature signal (da Fontoura Costa & Cesar Jr, 2009). The points obtained from different contour sizes vary due to the variation in hand positions and sizes from different distances. Consequently, normalization becomes necessary since different hand contour size may be identified as another letter instead of the intended letter. Closed hand sign normalization is applied with this method to control the various hand edge sizes in order to make it identical. Lastly, an additional extraction process is used to generate the normalized signature signals for the closed hand sign category.

Universiti Malaya



Figure 6.19: Example of flexing and extending thumb

D- Discrete Wavelet Transform (DWT) With a Statistical Descriptor

Figure 6.20 illustrates how difficult it is to recognize closed hand signs in 2D edge contours. Thus, small variations within the 2D image can be enlarged by a 1D bending signal. Nevertheless, it is still very challenging to differentiate some Arabic letters (such as 'ف', 'ق'). Tools based on the frequency domain can be utilized to classify the small variations in 1D signature signals, which denote low-frequency elements. In this thesis, the DWT is chosen as a tool to detect small variations, as DWT is known as a strong tool for analyzing stationary and non-stationary signal elements. However, DWT overrides the Discrete Fourier Transform (DFT), which fails in analyzing the elements of non-stationary signals (Sifuzzaman, Islam, & Ali, 2009). In addition, DWT is also characterized by its ability to expose similar signal sides, after which it can separate the tiny details of ambiguous closed hand sign signals. The similarities of the closed hand sign shape and the Haar Wavelet Transform (HWT) has resulted in employing the HWT a mother wavelet.

Universiti Malaya

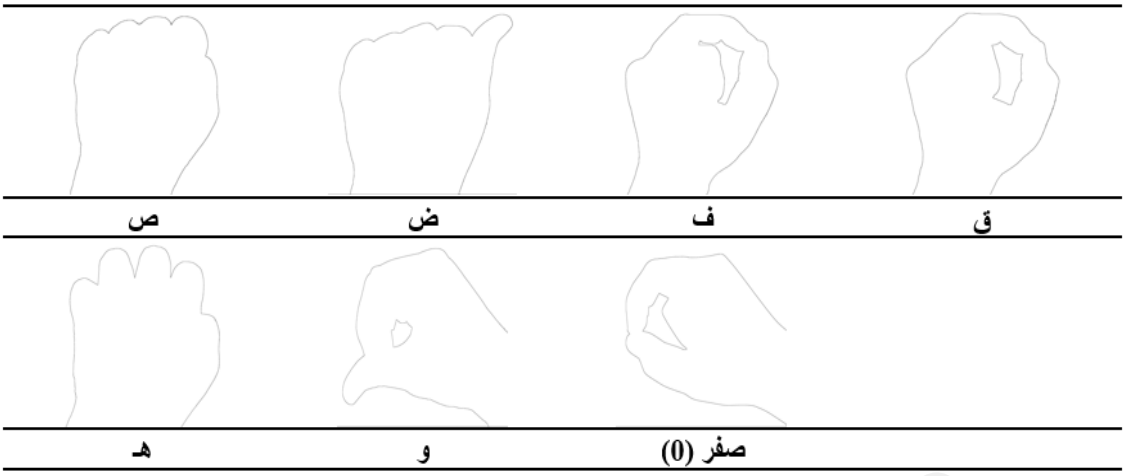


Figure 6.20: Edge detection for closed hand signs

In the further recognition stage of the recognition process, the feature vectors for closed hand signs are measured by applying several statistical measures such as skewness (β_1), kurtosis (β_2), mean (μ) and variance (σ^2).

Approximation coefficients are used to compute these statistical descriptors given in the following equations:

$$\textbf{Skewness } (\boldsymbol{\beta 1}) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x-\mu}{\sigma}\right)^3 \quad \textbf{(6.6)}$$

$$\textbf{Kurtosis } (\boldsymbol{\beta 2}) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x-\mu}{\sigma}\right)^4 \quad \textbf{(6.7)}$$

$$\textbf{Mean } (\mu) = \frac{1}{N}\sum_{i=1}^{N} X_i \quad \textbf{(6.8)}$$

$$\textbf{Variance } (\boldsymbol{\sigma 2}) = \frac{1}{N}\sum_{i=1}^{N}(x-\mu)^2 \quad \textbf{(6.9)}$$

6.2.2.2 Feature Extraction from Open Hand Signs

Signs of an open hand are represented by extending fingers that stand out from the hand's palm, but the closed hand signs do not have any fingers that stand out. As illustrated in Figure 6.21, extended fingers represent spaced or adjacent fingers and the spaces among fingers are restricted by the hands limitations (Lee & Cohen, 2004). In addition, three positions are available for extended fingers such as the palm right side, the

Universiti Malaya

palm left side and the topmost of the palm. The finger signs are named in this framework based on the finger positions. Thus, the signs of fingers situated on the right or left side of the palm are denoted by horizontal signs, while the signs of fingers situated at the topmost of the palm are denoted by vertical signs as depicted in Figure 6.22.

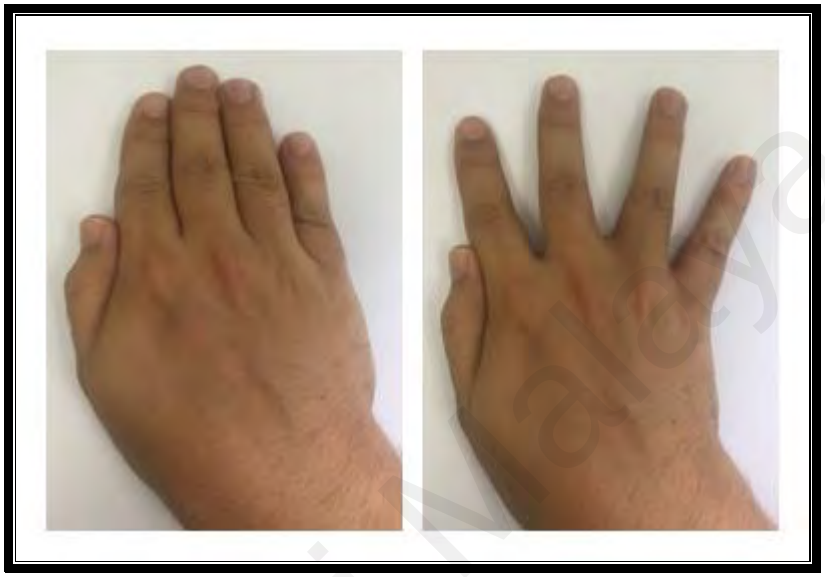


Figure 6.21: Adjacent and spaced fingers.

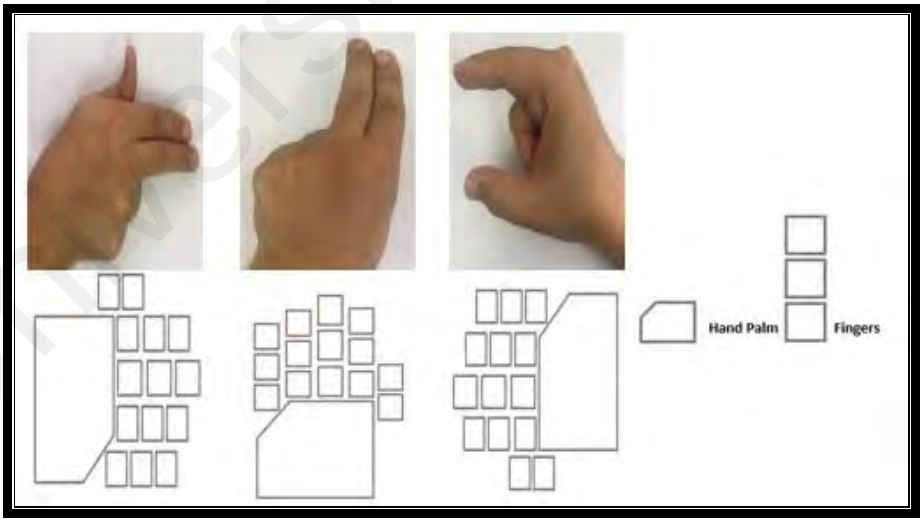


Figure 6.22: Hand finger directions.

Open hand signs are typically recognized using geometric computation and the curvature method (Malima, Özgür, & Çetin, 2006; Yin & Xie, 2007). Both methods involve similar basic steps to identify the number of prolonged fingers. Nevertheless,

knowledge of the number of prolonged fingers is not sufficient for differentiating open hand signs, because several signs have an identical number of prolonged fingers, such as letters ت, ع, ل, ن, and ي.

As shown in Figure 6.23, all these letters have two prolonged fingers, thus the names of the prolonged fingers and their numbers should be known for open hand signs differentiation.

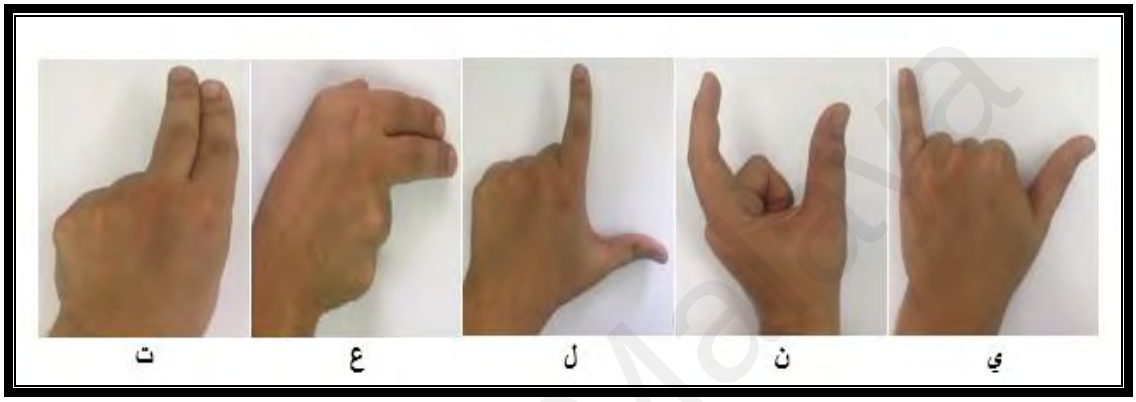


Figure 6.23: Open hand signs with two fingers out

The Run-length Co-occurrence Matrix (RCM) is one of the most common matrices utilized in gray-scale images that consists of different texture patterns. It has been used for instance in beef quality forecasting and wood categorization but has not been constructed for sign language recognition. Open hand signs of ArSL finger numbers and letters have various hand shapes but not texture patterns. Thus, hand shape patterns were created in the current study, which depends on the area level distribution according to how the patterns occupy the block mask. Moreover, 31 patterns for open hand signs were created. The handshape vision can be identified through statistical feature descriptors that are computed from the area level distributions selected in both vertical and horizontal directions.

Universiti Malaya

A- Area Detection for Open Hand Sign Fingers

The process described in Section (6.2.2.1.A) Area detection for closed hand sign fingers is also used to detect the regions of a finger. The row that consists of the centroid position is first detected by the system. The detected row is then used as a bottom for the fingers area window. The other sides of the window are located to fit with the area of a finger. The right and left sides are located on the rightmost and leftmost columns of the hand area, while the topmost side is taken from the first row within the hand area.

B- Construction of Area Level Coding Masks Pattern

The location and extended finger number are used here to detect open hand signs. Therefore, the 10 x10 block mask according to our proposed technique aforementioned in Section 6.2.1.2.B is utilized to detect the names and positions of the fingers in order to differentiate the signs in the open hand. Five different bins, 0, 1, 2, 3 and 4, represent the quantized areas of the hand.

- "If the block has no hand pixels, then 0 bin is assigned to represent the hand area".
- "If the hand area is less than 25.01% of the block area, then 1 bin is assigned to represent the hand area".
- "If the hand area ranges between 25.01% and 50.00% of the block area, then 2 bins are assigned to represent the hand area".
- "If the hand area ranges between 50.01% and 75.00% of the block area, then 3 bins are assigned to represent the hand area".
- "If the hand area is over 75.00% of the block area, 4 bins are assigned to represent the hand area".

Finger area segmentation is based on the hand structure model as well as the proposed 10×10 blocking mask as depicted in Figure 6.24.

Universiti Malaya

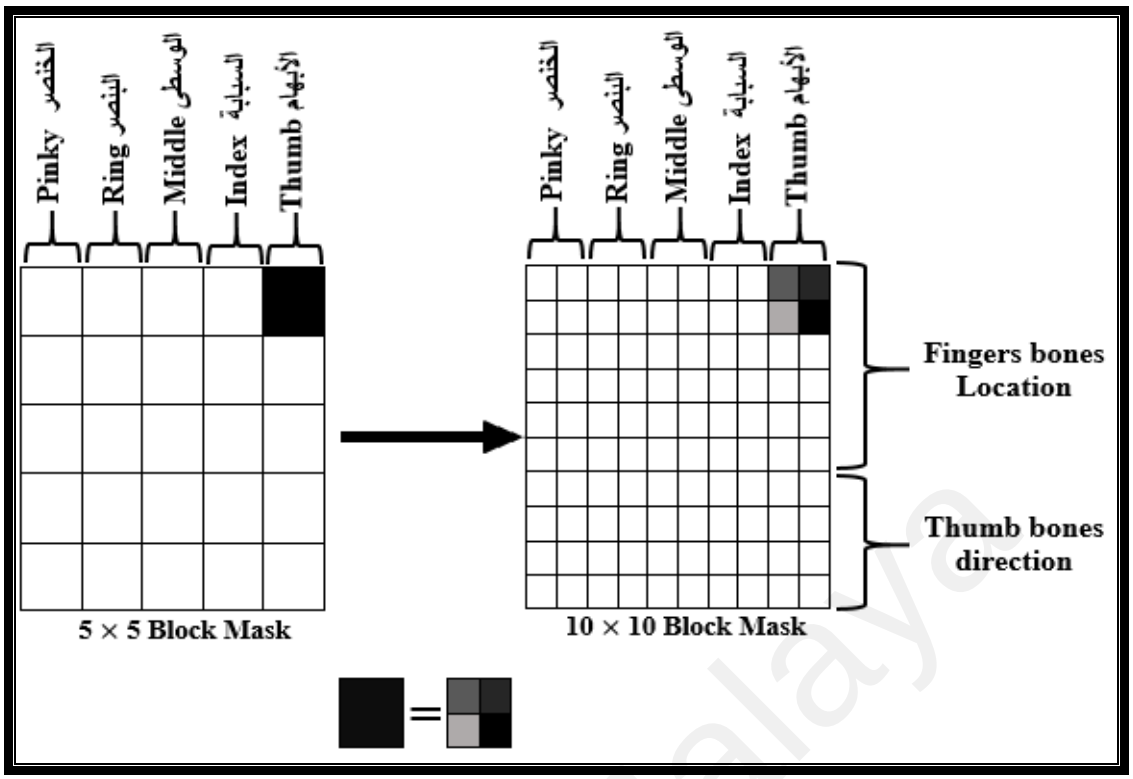


Figure 6.24: Finger locations and directions in 5*5 & 10*10 block masks

C- Statistical Descriptors and Run-Length Matrix (RLM)

RLM is employed in this work to extract the consecutive pixel group that attains the Quantized Area Levels (QAL) within the coding table. The RLM algorithm is an optimal tool for extracting textual features (Nailon, 2010). Figure 6.25 shows instances of open hand sign category coding patterns depending on the numbers occupying the blocks.

The coding table is illustrated in Figure 6.26, which shows the structure of the 10 x10 matrix besides the five QAL codes numbered from 0 to 4. An RLM of area level is composed of ten columns that denote the run numbers ($j = 1, 2, \ldots, 10$) and five main rows that denote five QAL ($i = 0, 1, \ldots, 4$). For instance, the element in the area level RLM for the $(4,5)th$ position denotes how often five sequential elements for a QAL $i = 4$ occurred. In this state, five sequential elements for a QAL $i = 4$ are repeated three times in the coding table, therefore the element in the $(4,5)th$ location is equal to 3.

Universiti Malaya

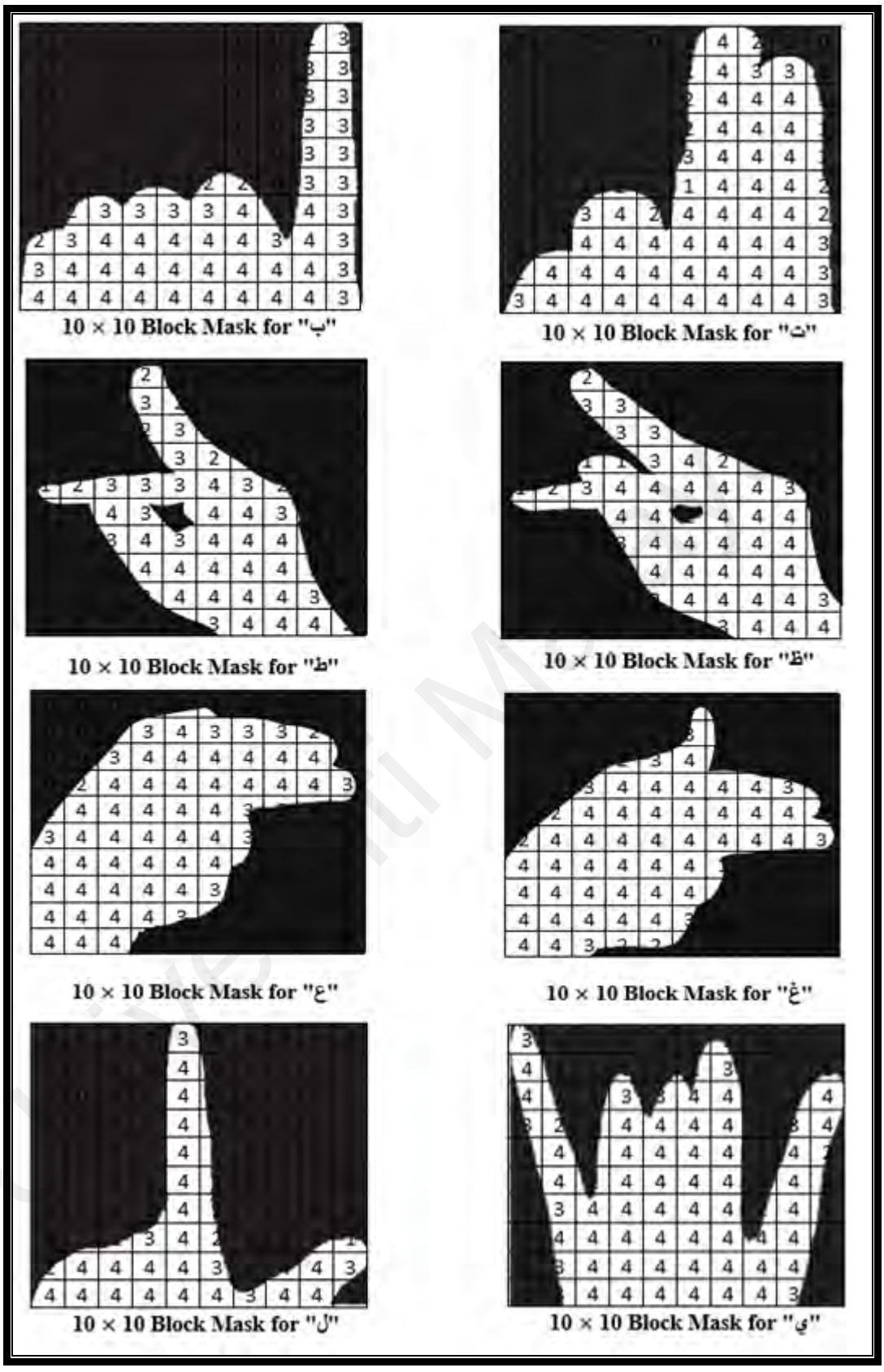


Figure 6.25: Examples of coding patterns based on occupied blocks.

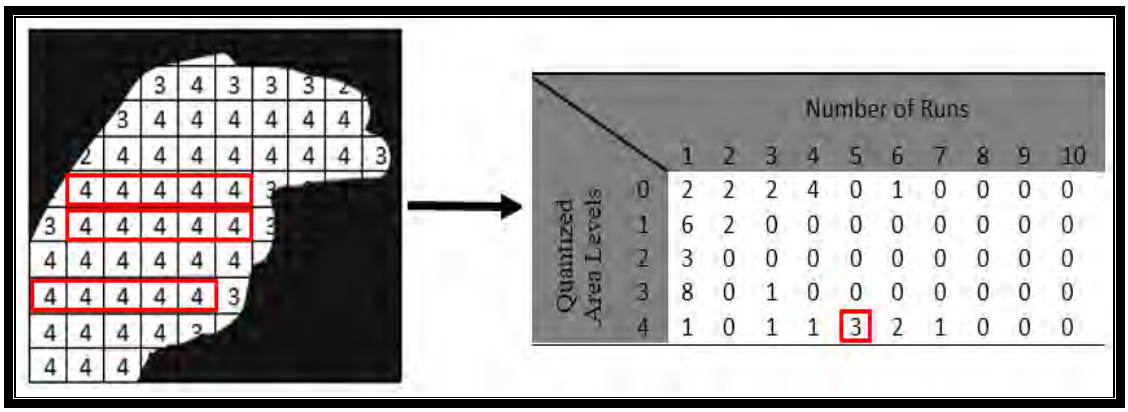


Figure 6.26: RLM computation.

The quantized area levels are distributed differently from sign to sign. Figure 6.27(A) represents a sign for adjacent fingers, which has a distribution that extends over the quantized area and is four larger than the spaced finger sign distribution in Figure 6.27(B). Various signs can have an identical number of extended fingers, like 'ب', 'ع' and 'ل'. Therefore, these letters can be recognized with this method. The long-run distribution of letter 'ل', which has an area level of 0, is more than that of letters 'ب' and 'ع'. Moreover, the long run distribution of letter 'ع', which has an area level of 4, is more than that of letters 'ب' and 'ل'. The area level and combined short run distribution of letter 'ب' are not the same as the letters 'ل' and 'ع'.

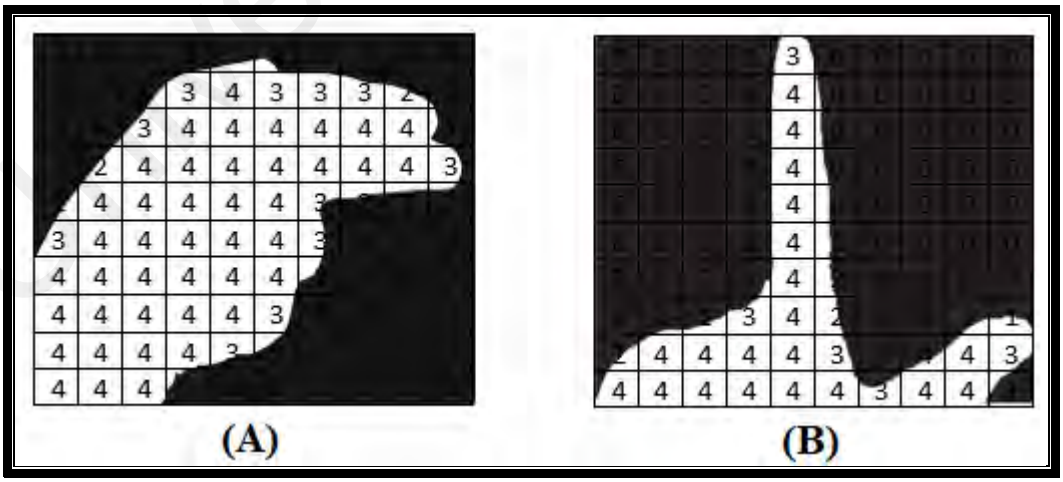


Figure 6.27: Area level distribution.

The proposed method extracts the features from RLM in the 90° and 0 directions using the similar statistical analysis tools applied in (Xu, Kurani, Furst, & Raicu,

2004)(Galloway, 1975). As shown below, 28 feature descriptors are available for every sign.

- ✓ Let $\boldsymbol{i}$ be the quantized area levels ($\boldsymbol{i} = \mathbf{0}, \mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}$)
- ✓ $\boldsymbol{j}$ be the run length ($\boldsymbol{j} = \mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \ldots, \mathbf{10}$)
- ✓ $\boldsymbol{p}(\boldsymbol{i}, \boldsymbol{j})$be the $(\boldsymbol{i}, \boldsymbol{j})\boldsymbol{th}$ element of the area − level run length matrix
- ✓ $\boldsymbol{M}$ total # of area levels
- ✓ $\boldsymbol{N}$ max. RLM
- ✓ $\boldsymbol{P}$ # of elements in the coding table.

- "Equations 6.10 and 6.11 present the short and long run parameter measures of the distributions of short and long runs only for area levels 0 and 4".

$$\textbf{Short run parameter} = \sum_{j=1}^{N} \frac{p(i,j)}{j^2} \quad \textbf{(6.10)}$$

$$\textbf{Long run parameter} = \sum_{j=1}^{N} p(i,j) \times j^2 \quad \textbf{(6.11)}$$

- "Equations 6.12 and 6.13 present the short and long run emphasis measures of the distributions of short and long runs in the coding table (Galloway, 1975)(Haralick, 1979)".

$$\textbf{Short run emphasis} = \frac{\left[\sum_{i=1}^{M} \sum_{j=1}^{N} \frac{p(i,j)}{j^2}\right]}{\sum_{i=1}^{M} \sum_{j=1}^{N} p(i,j)} \quad \textbf{(6.12)}$$

$$\textbf{Long run emphasis} = \frac{\left[\sum_{i=1}^{M} \sum_{j=1}^{N} p(i,j) \times j^2\right]}{\sum_{i=1}^{M} \sum_{j=1}^{N} p(i,j)} \quad \textbf{(6.13)}$$

- "Area level non-uniformity measures the similarity of area level values throughout the image. When the runs are equally distributed throughout the quantized area levels, this parameter has the lowest value (Galloway, 1975)(Haralick, 1979)".

$$\textbf{Area level non} - \textbf{uniformity} = \frac{\sum_{i=1}^{M} \left(\sum_{j=1}^{N} p(i,j)\right)^2}{\sum_{i=1}^{M} \sum_{j=1}^{N} p(i,j)} \quad \textbf{(5.14)}$$

Universiti Malaya

- "Run length non-uniformity measures the similarity of run lengths throughout the image. When the runs are equally distributed over all lengths, this parameter has the lowest value (Galloway, 1975)(Haralick, 1979)".

$$\textbf{Run length non} - \textbf{uniformity} = \frac{\sum_{i=1}^{N}\left(\sum_{j=1}^{M} p(i,j)\right)^2}{\sum_{i=1}^{M}\sum_{j=1}^{N} p(i,j)} \quad \textbf{(5.15)}$$

- "Equations 6.16 and 6.17 present the low and high area level run emphasis measures of the distribution of low area level values (Xu et al., 2004)".

$$\textbf{Low area} - \textbf{level run emphasis} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}\frac{p(i,j)}{i^2}}{\sum_{i=1}^{M}\sum_{j=1}^{N} p(i,j)} \quad \textbf{(6.16)}$$

$$\textbf{High area} - \textbf{level run emphasis} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} p(i,j) \times i^2}{\sum_{i=1}^{M}\sum_{j=1}^{N} p(i,j)} \quad \textbf{(6.17)}$$

- "Equations 6.18 and 6.19 present the short run low and long run high area level emphasis measures of the joint distribution of short runs and low area level values (Xu et al., 2004)".

$$\textbf{Short run low area level emphasis} = \frac{\left[\sum_{i=1}^{M}\sum_{j=1}^{N}\frac{p(i,j)\times i^2}{j^2}\right]}{\sum_{i=1}^{M}\sum_{j=1}^{N} p(i,j)} \quad \textbf{(6.18)}$$

$$\textbf{Long run high area level emphasis} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} p(i,j) \times i^2 \times j^2}{\sum_{i=1}^{M}\sum_{j=1}^{N} p(i,j)} \quad \textbf{(6.19)}$$

6.2.3 RECOGNITION AND CLASSIFICATION PHASE

The features discussed above are hereby employed on three multi-classifiers, namely the NBC and MLP, for the performance comparison purpose which is briefly described below.

Universiti Malaya

6.2.3.1. Naïve Bayes Classifier (NBC)

NBC is considered a strong technique that is based on machine learning. This classifier has been applied in different domains and applications owing to its accuracy, speed, and simplicity. The Bayesian classification approach entails measuring the trade-off among different classification decisions using cost and probability functions. The Bayes' probability rule was used to develop NBC (Duda et al., 2012), and this rule states:

$$\boldsymbol{P(Y|X_1, \dots, X_n) = \frac{P(X_1,\dots,X_n|Y)P(Y)}{P(X_1,\dots,X_n)}} \quad \textbf{(6.20)}$$

This represents the likelihood occurrence of acquiring $\boldsymbol{Y}$ in the presence of conditions $\boldsymbol{X_1}$ $\boldsymbol{to}$ $\boldsymbol{X_n}$ or the posterior likelihood of $\boldsymbol{Y}$ in the presence of the prior likelihood of $\boldsymbol{Y}$ and probability of conditions $\boldsymbol{X_1}$ $\boldsymbol{to}$ $\boldsymbol{X_n}$. Accordingly, this rule can be explained for the case of letters as a posterior likelihood of a letter in the presence of a prior likelihood of this letter and the probability coefficient. The highest value of computed probabilities is then used to make a decision in order to attain minimum classification error. Based on the rule of Bayes, NBC is expressed mathematically as:

$$\boldsymbol{P(X_1, \dots, X_n|Y) = \prod_{i=1}^{n} P(X_i|Y)} \quad \textbf{(6.21)}$$

The assumption of NBC that all features are independent and labeled as class Y is not true for all statuses. Equation (6.21) represents a simplification of equation (6.20), which reduces to multiplication. Principally, NBC converts the multivariate problem into a univariate problem and disregards the correlation possibilities between inputs (Islam, Wu, Ahmadi, & Sid-Ahmed, 2007). The diverse sign letters in the present case are processed as classes by the classifier. The most popular rule utilized by the NBC classifier is to choose the class with maximum a posterior probability (MAP). Then the rule selects a class $\boldsymbol{C_k}$ if:

$$\boldsymbol{P(C_k|x) > P(C_k|x), j \neq k} \quad \textbf{(6.22)}$$

All previous class likelihoods are equal, then equation (6.22) is simplified further as follows:

Universiti Malaya

$$P(\boldsymbol{C_k}|\boldsymbol{x}) = \frac{P(\boldsymbol{x}|\boldsymbol{C_k})\boldsymbol{P}(\boldsymbol{C_k})}{\boldsymbol{P}(\boldsymbol{x})} \quad \textbf{(6.23)}$$

Here, $P(\boldsymbol{C_k}|\boldsymbol{x})$ can be maximized by only maximizing $P(\boldsymbol{x}|\boldsymbol{C_k})$, where $\boldsymbol{P}(\boldsymbol{x})$ represents class independence. Then equation (6.24) is given:

$$P(\boldsymbol{C_k}|\boldsymbol{x}) > P(\boldsymbol{x}|\boldsymbol{C_k}), \boldsymbol{j} \neq \boldsymbol{k} \quad \textbf{(6.24)}$$

The implication is that C_k will be chosen if the condition in equation (6.24) is achieved. Two functions (prior and probability) are required for learning in the NBC case. Furthermore, NBC is characterized by its ability to classify the necessary parameters through a small quantity of training data samples (Duda et al., 2012).

6.2.3.2 Multi-Layer Perceptron Neural Networks (MLP)

Recently, ANNs have been widely employed because they are able to imitate natural intelligence by learning from the experience (Rumelhart, McClelland, & Group, 1987). ANNs create input-output mapping without deriving model equations in order to learn from the examples. Besides, different applications, such as automatic control, optimization, function approximation, forecasting control, pattern categorization, and others benefit from ANNs (Wasserman, 1993; Lippmann, 1987). Figure 6.28 shown structure of a MLP classifier (input layer, hidden layer and output layer).

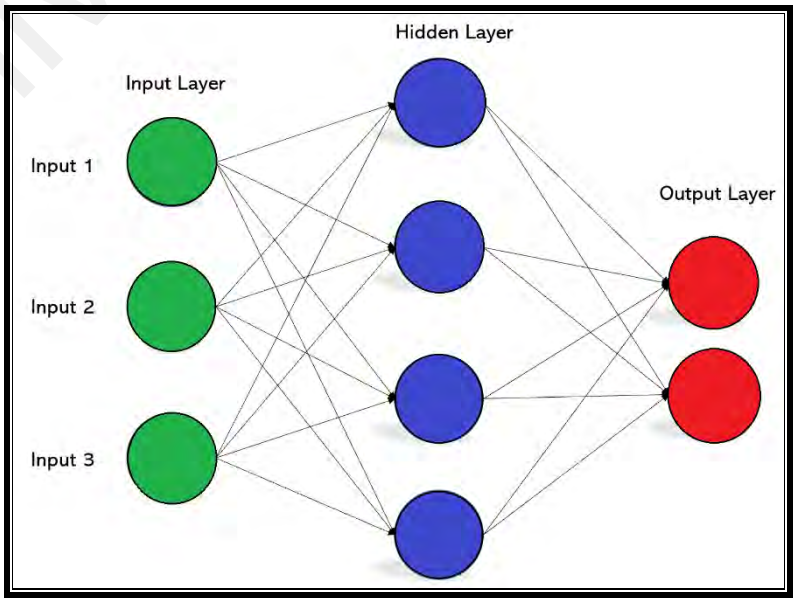


Figure 6.28: Structure of a MLP classifier

Universiti Malaya

The MLPNN architecture consists of 5 input layers, 2 hidden layers and 1 output layer (Figure 6.29).

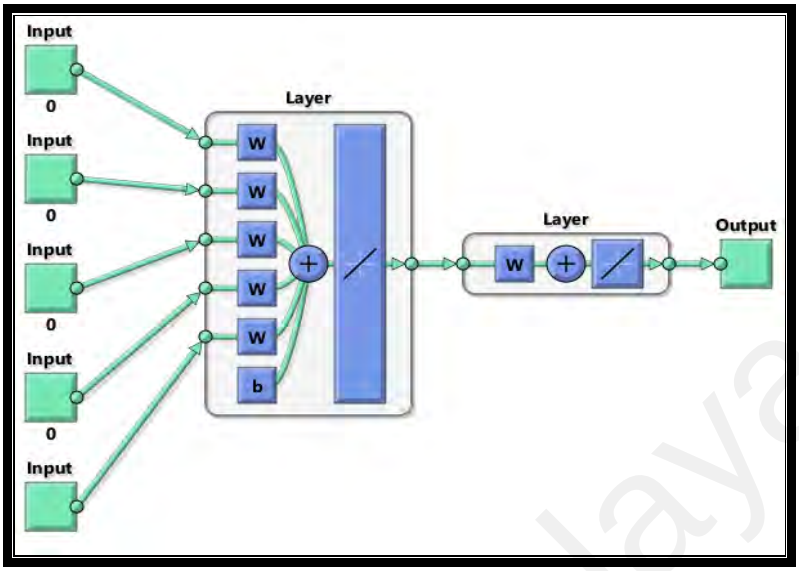


Figure 6.29: Example of MLP with five inputs

Neurons are similar simple units of processing that are interconnected to form an ANN. An adaptable weight factor is linked to every connection with a neuron. An interior activity level is generated through the sum of the input weighted factors for each neuron in the network.

$$\boldsymbol{a_i = \sum_{j=1}^{n} w_{ij}\, x_{ij} - w_{io}} \qquad \textbf{(6.25)}$$

Where x_{ij} is the jth input signal to the ith neuron, w_{io} is the threshold value linked to unit i and w_{ij} is the connection weight that extends between input j and neuron i. The threshold value is processed as a typical weight for an imposed input at -1. The neuron output yi is generated by passing the interior activity across the nonlinear function (φ). Then yi is given by:

$$\boldsymbol{y_i = \varphi(a_i)} \qquad \textbf{(6.26)}$$

The deserved network input/output is attained by modifying the connections' weights throughout the training process. On the other hand, the neurons in a Feed forward Network (FFN) with multilayers are arranged as layers without lateral or feedback connections. Typically, neuron layers regardless of the output layers are denoted by

Universiti Malaya

hidden layers. Input signal propagation over the network is based on the layer-by-layer principle, where the signal spreads in a straight-ahead direction. The back propagation (BP) algorithm (Rumelhart et al., 1987) is defined as a direct method of iterative training that is used for multilayer FFNs with nonlinear sigmoidal threshold units. The BP algorithm employs training data, including P input-output vector pairs that describe the problem. Furthermore, the mean square difference between the desired output and the actual output can be reduced by employing the Generalized Least-Mean-Square (GLMS) algorithm together with the BP algorithm (Lippmann, 1987). The minimized error function in the BP algorithm is denoted by the square difference average of the desired output and the obtained output from every neuron in the output layer. Equation 6.27 expresses the error function.

$$\boldsymbol{E = \frac{1}{P} \sum_p \sum_k (d_{pk} - o_{pk})^2} \quad \textbf{(6.27)}$$

Where $\mathbf{k}$ is the element index in the output vector, $\mathbf{P}$ is the training vector pair index, $\mathbf{o_{pk}}$ is the $\mathbf{k^{th}}$ output vector element when assuming the $\mathbf{P}$ pattern as the network input and $\mathbf{d_{pk}}$ is the $\mathbf{k^{th}}$ element of the desired pattern vector $\mathbf{p^{th}}$.

By reducing the cost function exemplified in equation (6.27), a new updated rule is obtained to modify the weights of connections linking the neurons. The expression for modifying the weight of the connection between neuron $\boldsymbol{i}$ in layer $\boldsymbol{m}$ and neuron $\boldsymbol{j}$ in layer $\boldsymbol{m+1}$ is defined as:

$$\boldsymbol{\Delta w_{ij} = \eta \delta_j o_i} \quad \textbf{(6.28)}$$

Where $\boldsymbol{i}$ is the unit index of a layer, $\boldsymbol{o_i}$ is the unit $\boldsymbol{i}$ output in the $\boldsymbol{m^t h}$ layer and $\boldsymbol{\delta_j}$ is the delta error term in the back propagation case from the $\boldsymbol{j^{th}}$ unit in layer $\boldsymbol{m+1}$. The expression for the delta error term $\boldsymbol{\delta_j}$ is:

$$\boldsymbol{\delta_j = [d_j - o_j] o_j [1 - o_j]} \quad \textbf{(6.29)}$$

Neuron *j* is an output,

Universiti Malaya

$$\delta_j = y_i \left[1 - y_j\right] \sum_k \delta_k w_{kj} \quad (6.30)$$

Neuron j is found in a hidden layer, while **k** is the neuron index in layer$(m + 2)$, which is ahead of neuron j's layer.

6.2.4. Multi-Classifier Combination

In this section, the input features that represent hand features are first extracted in the feature extraction phase. As seen in Figure 6.30, these features are then assembled to feed three various classifiers, each of which will obtain its own likelihood probability. Subsequently, all likelihood probabilities obtained are multiplied by equal weights and are pooled to produce the total probability $P(O|\lambda)$ of all input signs.

Throughout the classification, $P(O|\lambda)$ is calculated depending on the probabilities of the three classifiers $P(O|\lambda_k)$, $P(O|\lambda_a)$ and $P(O|\lambda_g)$. Although various weights can be set for different features, yet, identical weights are applied to all features extracted in this study. However, the sign with the maximum average likelihood probability is set as the recognized gesture. The difference between likelihood probabilities obtained by applying a multi-classifier with this method can increase or diminish the discriminatory power.

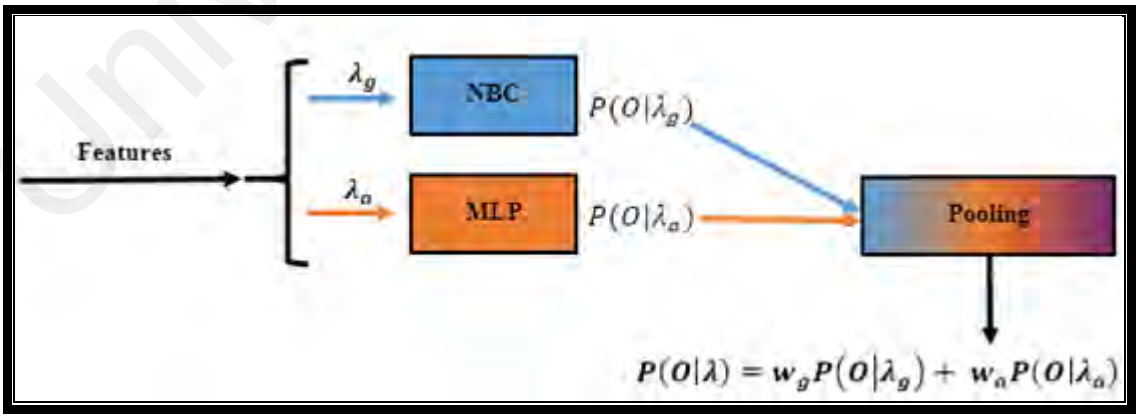


Figure 6.30: Framework of combining multi-classifiers.

Universiti Malaya

6.3 Framework 2: Hybrid Approach for ArSLR Based on Data Combining

This framework shows how the VBR and SBR approaches can be used to achieve accurate ArSLR made via the signer-independent framework. An ad-hoc solution for the joint calibration of the two devices is firstly presented. Then, a set of feature descriptors is introduced both for the smart gloves and for the Microsoft Kinect depth data. For the smart gloves, simple pre-processing and feature extraction techniques are applied to capture, extract and emphasise the temporal dependence of the data. For the Microsoft Kinect, the hand is initially extracted from the acquired data and divided into the palm and finger regions. Consequently, four different sets of feature descriptors are extracted, representing different clues, such as the distances of the fingertips from the centre of the hand and from the palm plane, the curvature of the hand contour and the geometry of the palm region. Finally, the proposed feature sets are fed to multi-class SVMs. Figure 6.31 presents the proposed hybrid automatic continuous ArSLR framework.

Universiti Malaya

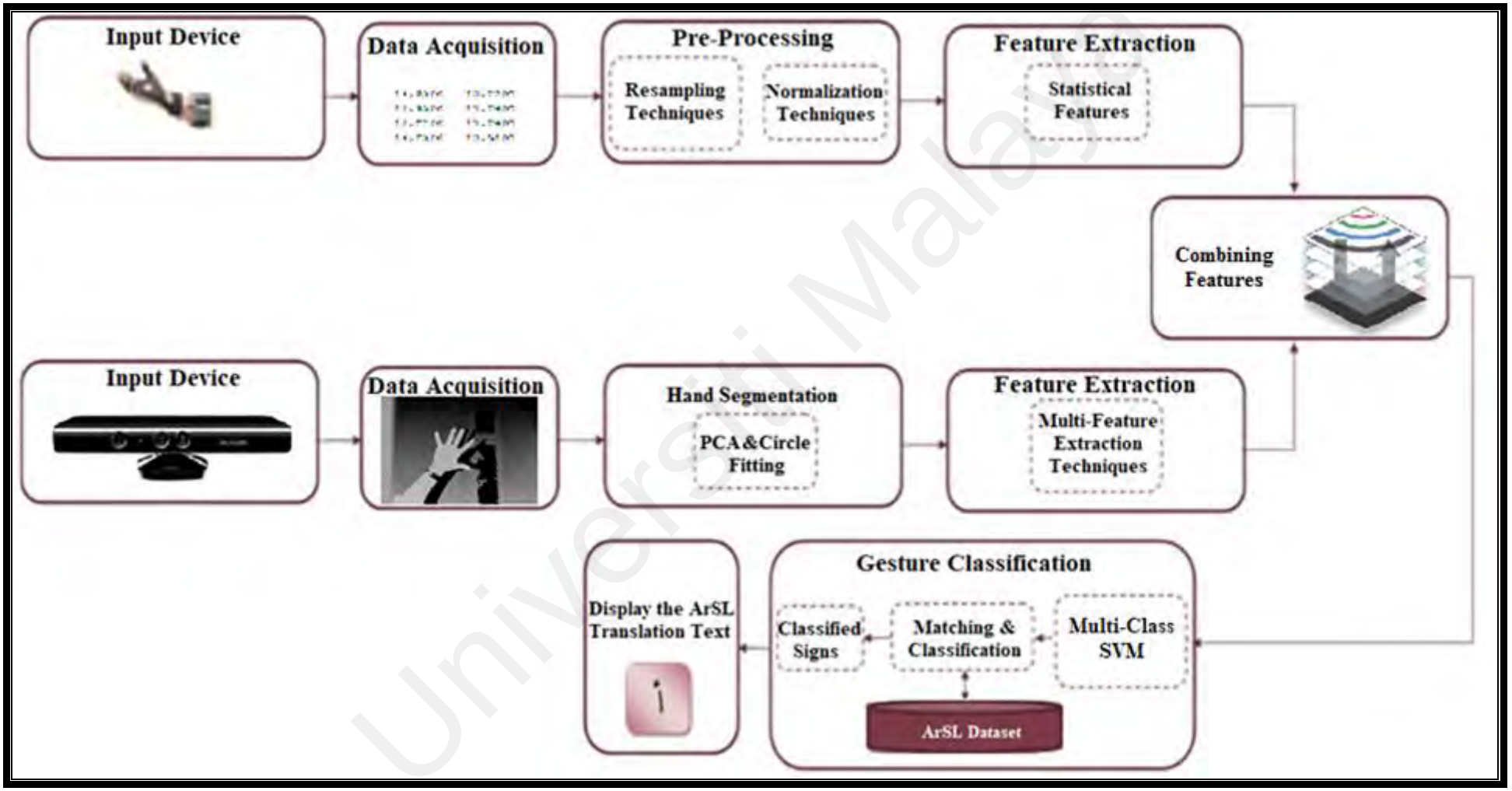


Figure 6.31: Main phases for proposed hybrid approach for automatic ArSLR framework 2

Universiti Malaya

6.3.1 Pre-processing and feature extraction phase from the smart glove

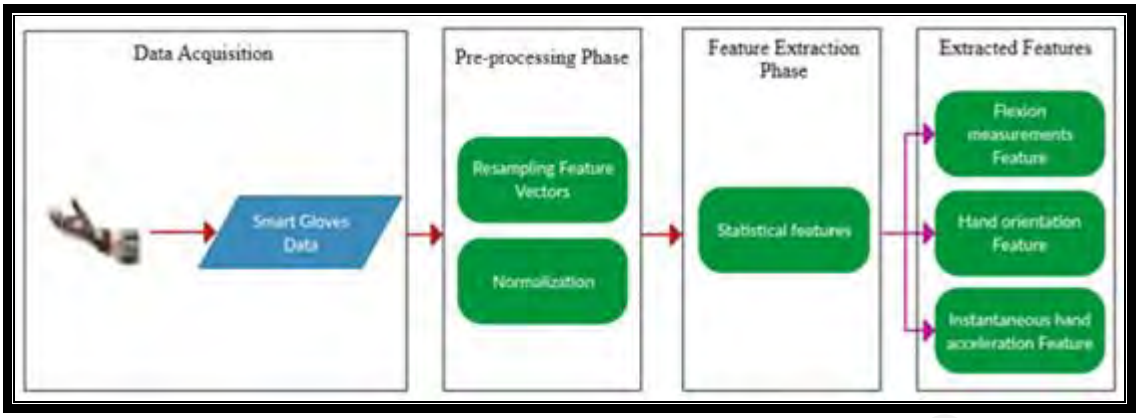


Figure 6.32: Architecture of the proposed smart data gloves for ArSLR

5.3.1.1 Pre-processing Phase

Pre-processing is an essential foundational stage in any data analysis task. The essence of this stage is to clean the dataset by removing the undesirable data or sequences that have has no value in the construction of the model. Various methods, such as resampling, normalisation and signal denoising, are employed to prepare the data for the next phase of processing.

A- Resampling Feature Vectors

The sampling rate of the collected data is about 30 readings per second, which appears to be in high correspondence with the velocity of the finger and hand movement. As a result, some of the feature vectors (FV) might not be useful for the analysis, thus becoming redundant. In order to avoid such a situation, resampling techniques were employed to eliminate the redundant feature vectors, thereby improving recognition time and dataset storage efficiency. The procedure for the resampling technique is described in Algorithm 6.3 by a factor Q (Tubaiz et al., 2015).

Universiti Malaya

Algorithm 6.3 Reducing the Number of Feature Vectors

Data: *Original feature vectors and their corresponding labels*
Result: *resampled feature vectors and their corresponding labels*
for all sign in the datasets do
for all repetitions do
if a set of feature vectors FVs has the same label L then
resample FVs by a factor Q
FVsnew $\Leftarrow$ *resampled FVs*
Labelsnew $\Leftarrow$ *L*
end if
end for
end for

Two resampling approaches were used with Q = 2 and 3, as described in Algorithm 6.3. The first approach used the 'resample' function in MATLAB to decimate the time-domain data by Q. The $\boldsymbol{resample}\ (\boldsymbol{data}, \boldsymbol{P}, \boldsymbol{Q})$ function was performed, such that the data were interpolated by a factor P and then decimated by a factor Q. In the second approach, Qth observation was maintained in the original data sequence, and the other observations were discarded. Thereafter, the factor Q reduces the length of the resulting feature vectors with their respective labels (Tubaiz et al., 2015).

Figures 6.33, 6.34 and 6.35 used 2 as the value of Q in both techniques to depict the resampling effect on the left-hand index flex sensor readings. To provide examples of the methods described in this chapter, sentences 5, 7 and 25 from Table 5.4 are randomly selected.

Universiti Malaya

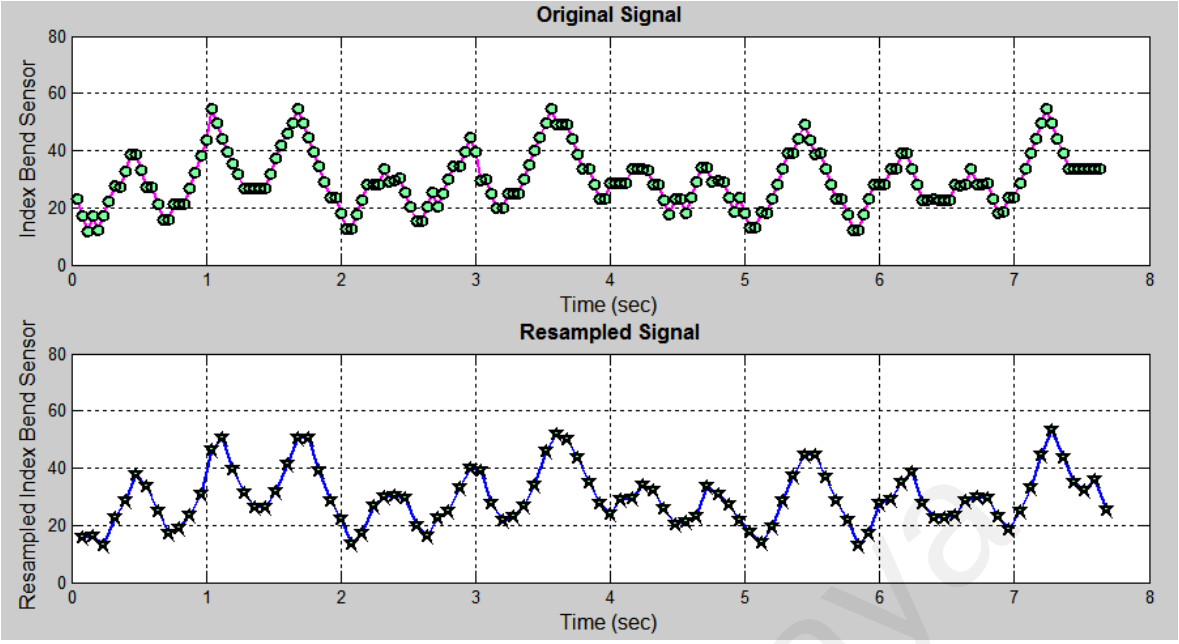


Figure 6.33: Original and resampling feature vectors of a signal for sentence 5

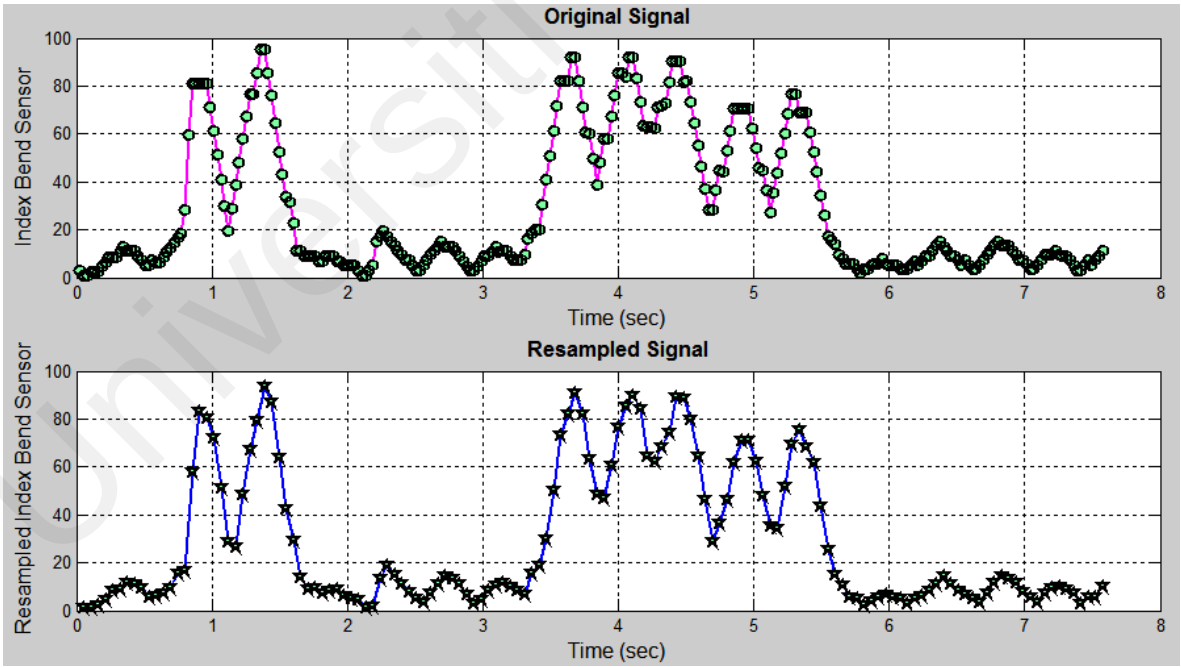


Figure 6.34: Original and resampling feature vectors of a signal for sentence 7

Universiti Malaya

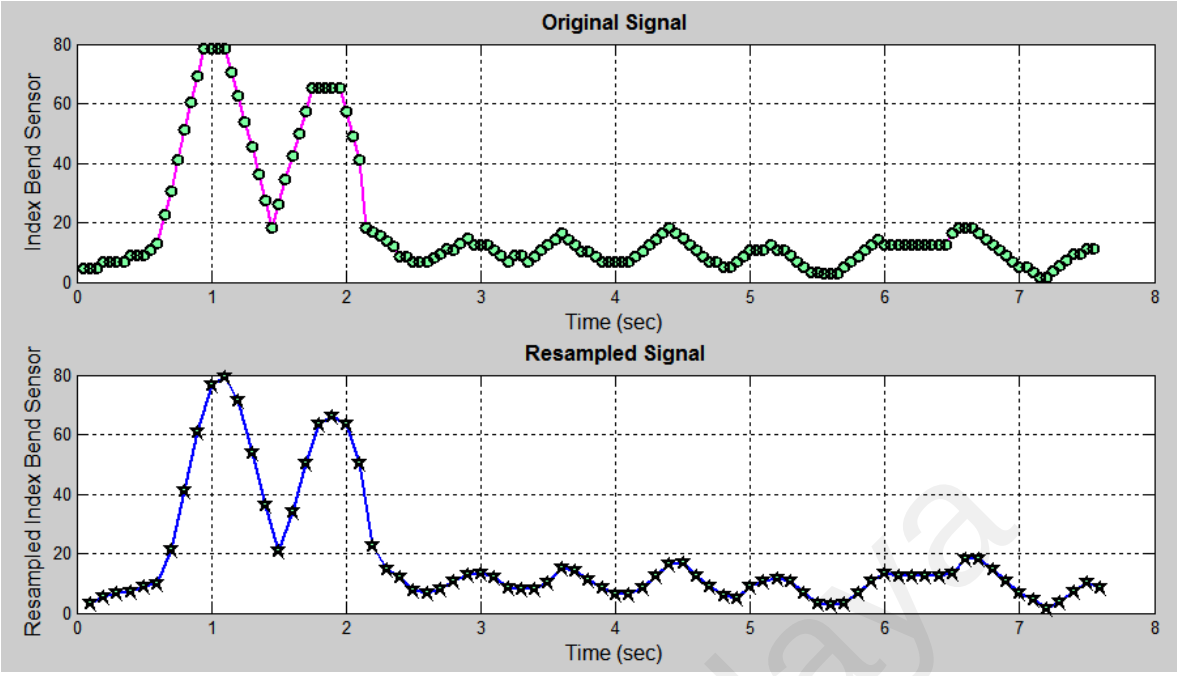


Figure 6.35: Original and resampling feature vectors of a signal for sentence 25

Results showed a slight difference between the two resampling approaches. Although the approaches produce nearly similar vector lengths and shapes, the values are slightly altered by the 'resample' function due to the low-pass filtering embedded in the 'resample' command.

B- Normalization:

Prior to further analysis, features from the smart data glove should be normalised to eliminate the varying scales and ensure a common range. For that reason, each feature in the dataset are standardised with the z-score, resulting in features with a zero mean and unit variance (Tubaiz et al., 2015).

Based on Equation 6.33, the value of the Z-score of data point x in a given set of data X with sample mean μ and standard deviation σ can be calculated as follows:

$$\boldsymbol{z = x - \frac{\mu}{\sigma}} \qquad \textbf{(6.33)}$$

where z is the value of z-scored that serves as the distance measure of x from μ in terms of σ.

Universiti Malaya

The methods maintain the properties of the original dataset shape, as shown in Figures 6.36, 6.37 and 6.38 for the index flex sensor readings of sentenced 5, 7 and 25, respectively, from Table 5.4.

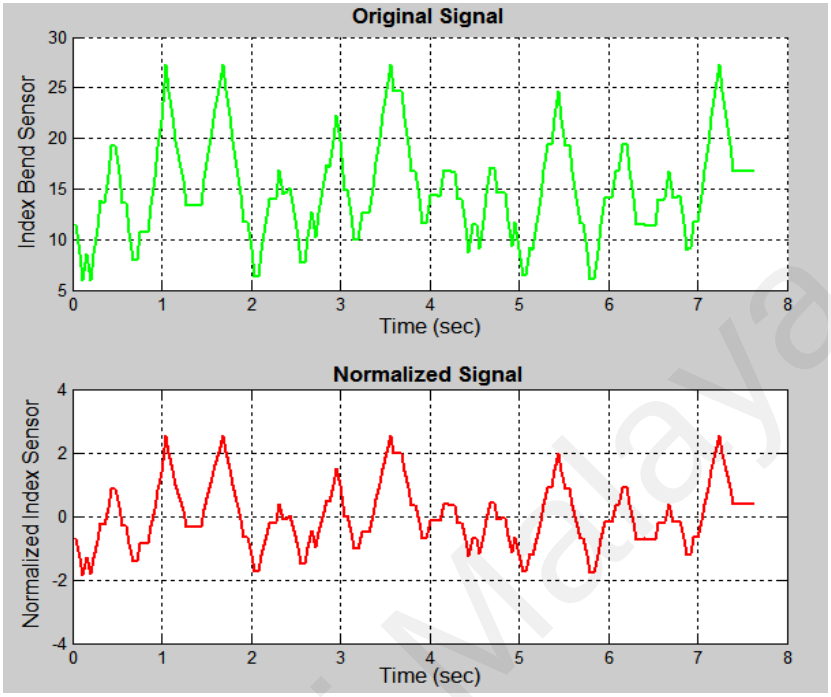


Figure 6.36: Normalization Effect for Sentence 5

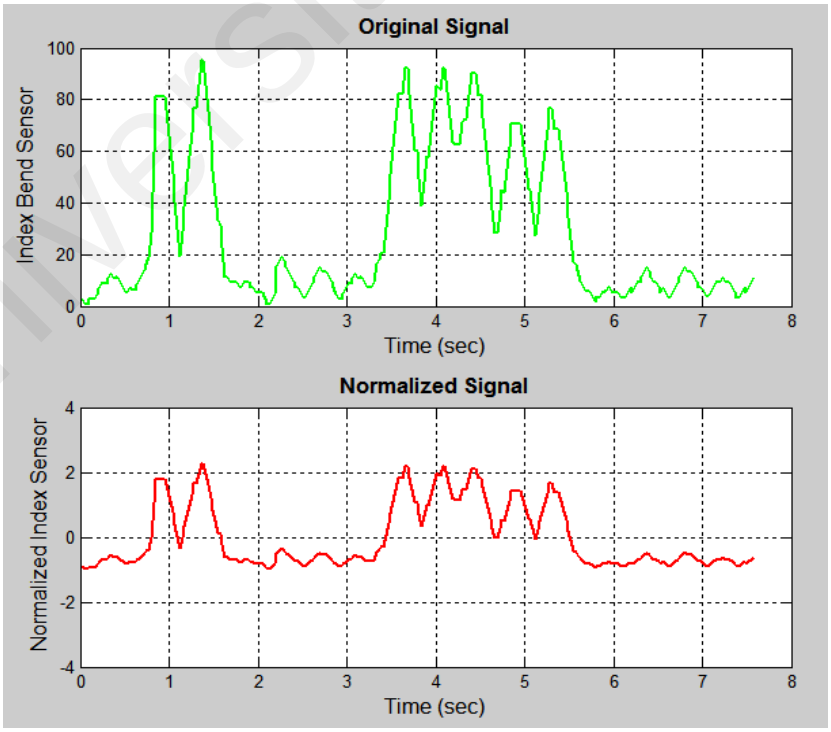


Figure 6.37: Normalization Effect for Sentence 7

Universiti Malaya

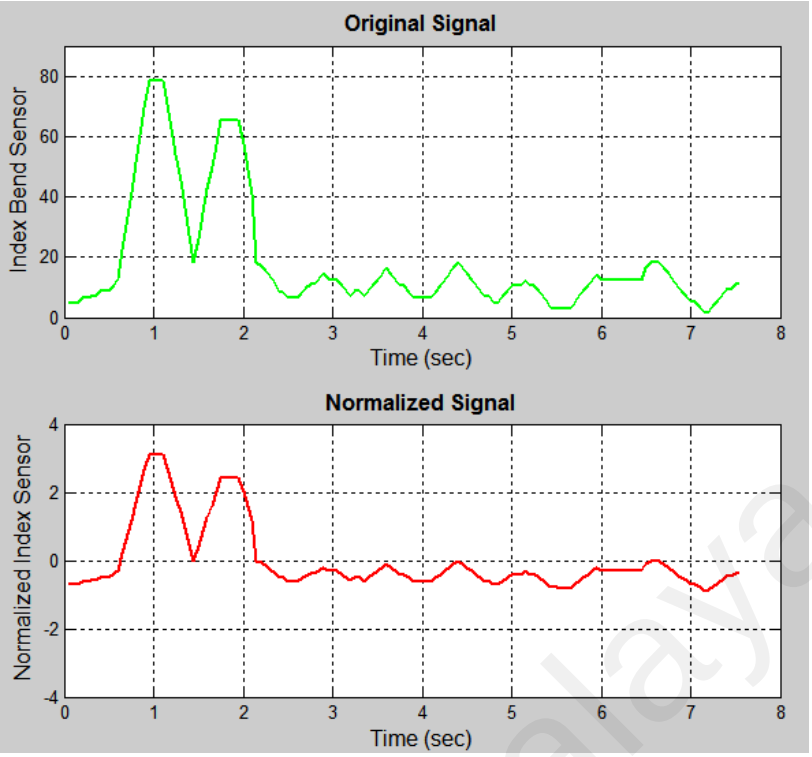


Figure 6.38: Normalization Effect for Sentence 25

The computation of the mean and standard deviation of the training set was carried out for the purpose of standardisation, as depicted in Equation 6.34. Thereafter, as indicated in Equation 6.35, μ_{train} and σ_{train} are saved from the training phase and then reused to standardise the testing set.

$$\boldsymbol{Z_{train} = X_{train} - \frac{\mu_{train}}{\sigma_{train}}} \quad \textbf{(6.34)}$$

$$\boldsymbol{Z_{test} = X_{test} - \frac{\mu_{train}}{\sigma_{train}}} \quad \textbf{(6.35)}$$

Where $\boldsymbol{z_{train}}$ is the z-scored training set that includes N standardized vectors for N features such that:

- $\boldsymbol{Z_{train} = \{Z_{f1}, Z_{f2}, Z_{f3}, \dots, Z_{fn}\}}$

  $\boldsymbol{Z_{fi} = \{z_1, z_2, z_3, \dots, z_t\}}$

  $\boldsymbol{i = \{1, 2, 3, \dots, N\}}$

  T Is the total number of feature vectors in the original set $\boldsymbol{X_{train}}$

- $\boldsymbol{X_{train} = \{X_{f1}, X_{f2}, X_{f3}, \dots, X_{fn}\}}$

  $\boldsymbol{X_{fi} = \{x_1, x, x_3, \dots, x_t\}}$

Universiti Malaya

- $\mu_{train} = \{\mu_{f1}, \mu_{f2}, \mu_{f3}, \dots, \mu_{fn}\}$
- $\sigma_{train} = \{\sigma_{f1}, \sigma_{f2}, \sigma_{f3}, \dots, \sigma_{fn}\}$

6.3.1.2 Feature Extraction from smart data gloves device

In the next step, the useful features are extracted from raw data to ensure precision and conciseness.

Compared with the vision-based systems, the sensor-based systems are advantageous because they often do not require computationally expensive methods to extract the features. In this thesis, original data obtained from a smart data glove (Idris, 2012) indicate the bending of each finger as well as hand orientation and acceleration. These raw features provide important but noisy information. Thus, statistical analysis approaches are applied independently to extract the required features and enhance system performance, as explained in the section below.

In this thesis, the following features computed from the smart glove data are analysed:

1- **Flexion measurements:** The thumb, index, middle, ring, and little fingers dimensions are expressed in a space of 0.0% - 100.0% as bend percentages.

2- **Hand orientation:** It represents the two spanning angles between -90˚ and +90˚ range of roll and pitch.

3- **Instantaneous hand acceleration:** It depicts x, y and Z axes consisting of the values between −2g and +2g. A total of 10 measurements or features is reported. Their ranges are summarized in Table 4.7.

	Flexion Measurements					**Hand Orientation**		**Hand Acceleration**		
Left	Thumb	Index	Middle	Ring	Little	Roll	Pitch	Ax	Ay	Az
Right	Little	Ring	Middle	Index	Thumb	Roll	Pitch	Ax	Ay	Az

Universiti Malaya

- **Extraction the relevant Features based on multi-techniques for smart gloves data**

A- Statistical Features

Mean and standard deviation are used to obtain the descriptive features from the dataset. These statistical measures are often combined to generate quantitative summaries of a sample set (Tubaiz et al., 2015).

The mean is used to measure the central tendency and find the central value of the feature vector sequence. The average of the entire population is indicated as μ or a subset of T observations $X = \{x_1, x_2, x_3, \dots, x_T\}$. The sample μ is an estimate of the average population (see Equation 6.36).

$$\boldsymbol{\mu = \frac{1}{T} \sum_{K=1}^{T} x_k} \quad \textbf{(6.36)}$$

Conversely, the statistical deviation is measured by the standard deviation (σ). Thus, the sample σ is an estimate of the population standard deviation (see Equation 6.37).

$$\boldsymbol{\sigma = \left(\frac{1}{T-1} \sum_{K=1}^{T} (xk - \mu)^2\right)^{1/2}} \quad \textbf{(6.37)}$$

The statistical measure is the representation of the mean and standard deviation of the orientation and hand movement for the SL dataset in a particular interval. Subsequently, the mean and standard deviation of separate feature for a sliding window technique is computed by the proposed feature extraction method. A series of observations is obtained on the basis of a predefined window size. Then, as shown in Equations 6.38 and 6.39, μ and s for each feature are respectively calculated as follows:

$$\boldsymbol{\mu_i = \frac{1}{w} \sum_{k=1-\frac{w-1}{2}}^{i+\frac{w-1}{2}} x_k} \quad \textbf{(6.38)}$$

$$\boldsymbol{\sigma_1 = \left(\frac{1}{w-1} \sum_{k=1-\frac{w-1}{2}}^{i+\frac{w-1}{2}} (xk - \mu)^2\right)^{1/2}} \quad \textbf{(6.39)}$$

where W represents the window size, and i represents the current feature contained in a set of N features, such that i is a set of natural number illustrates as $i = \{1, 2, 3, \dots, N\}$.

Universiti Malaya

As a result, the mean values $\mu_{FV} = \{\mu_1, \mu_2, \mu_3, \dots, \mu_N\}$ of N sample for individual feature vector (FV), N and the standard deviations $\sigma_{FV} = \{\sigma_1, \sigma_2, \sigma_3, \dots, \sigma_N\}$ are computed. However, either one or both of these feature vectors can be added to its original feature at the end (as seen in Equation 6.40) (Tubaiz et al., 2015).

$$\boldsymbol{FV_i} = [\boldsymbol{FV_i} \;\; \boldsymbol{\mu_{FV_i}} \;\; \boldsymbol{\sigma_{FV_i}}] \quad \textbf{(6.40)}$$

Thereafter, the feature vector of the window has dropped by FV and the initial step is reciprocated as explained in algorithm 6.4.

Algorithm 6.4 Window-Based Mean and Standard Deviation

Data: *Original feature vectors*
Result: *Feature vectors with window − based mean and standard deviation*
Define window size (w)
$StartInd = ceil\,(w/2)$;
$EndInd = floor\,(w/2)$;
for all *sign in the datasets* ***do***
  for all *repetitions* ***do***
    append $\frac{w-1}{2}$ *zero vectors to the beginning and end of the repetition*
    while $StartInd \leq i \leq (Size(FVs) - EndInd)$ ***do***
      Find feature − wise $\boldsymbol{\mu_{FV_i}}$ *and* $\boldsymbol{\sigma_{FV_i}}$ *in the range* $((i - EndInd) : (i + EndInd))$
      Concatenate $\boldsymbol{\mu_{FV_i}}$ *and* $\boldsymbol{\sigma_{FV_i}}$ *with the original features* $\boldsymbol{FV_i}$
    end while
    Delete the appended zero vectors
  end for
end for

The simple moving average in statistic represents the unweighted mean of an identical number of previous and future data. This statistical indicator provides a series of averages from sequential data subsets. This approach is treated as a filtering mechanism for low-pass as it reduces short-term fluctuations and preserves long-term trends; its output is a smooth form of the original signal. Figures 6.39, 6.40, 6.41, 6.42, 6.43 and 6.44 provide examples of the mean and standard deviations of a sliding window for a given input signal.

Universiti Malaya

Figures 6.39, 6.40, 6.41, 6.42, 6.43, 6.44 provide an example of the mean and standard deviation of a sliding window for a given input signal.

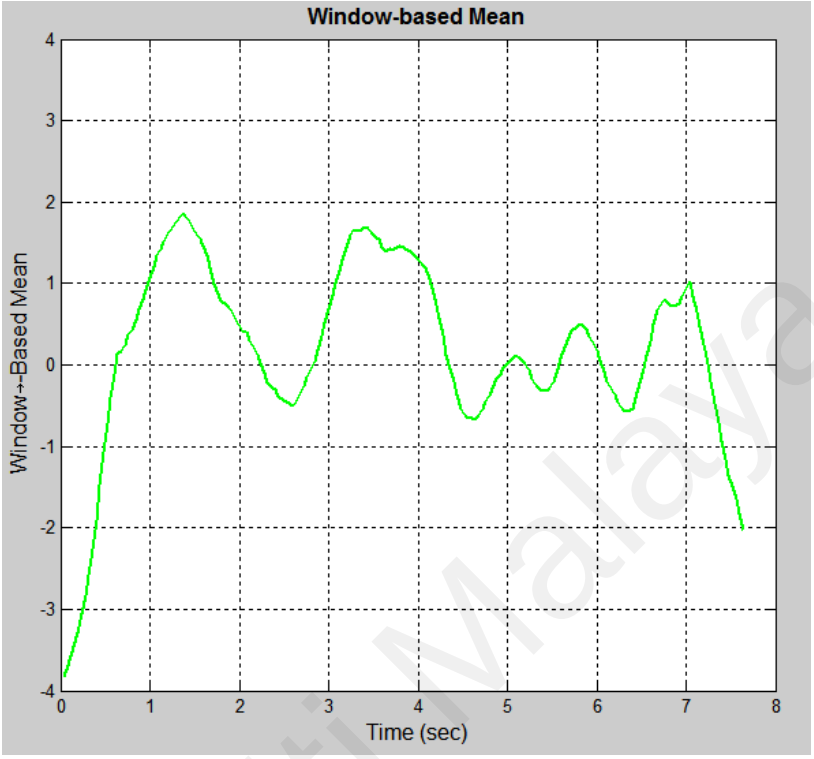


Figure 6.39: An Example of Statistical Mean for Sentence 5

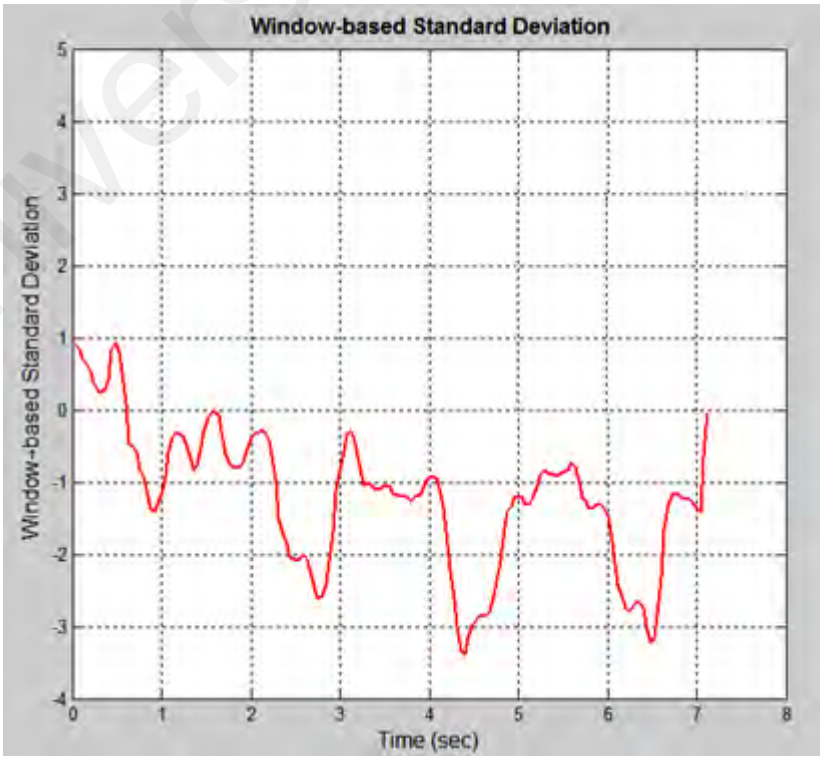


Figure 6.40: An Example of Standard Deviation for Sentence 5

Universiti Malaya

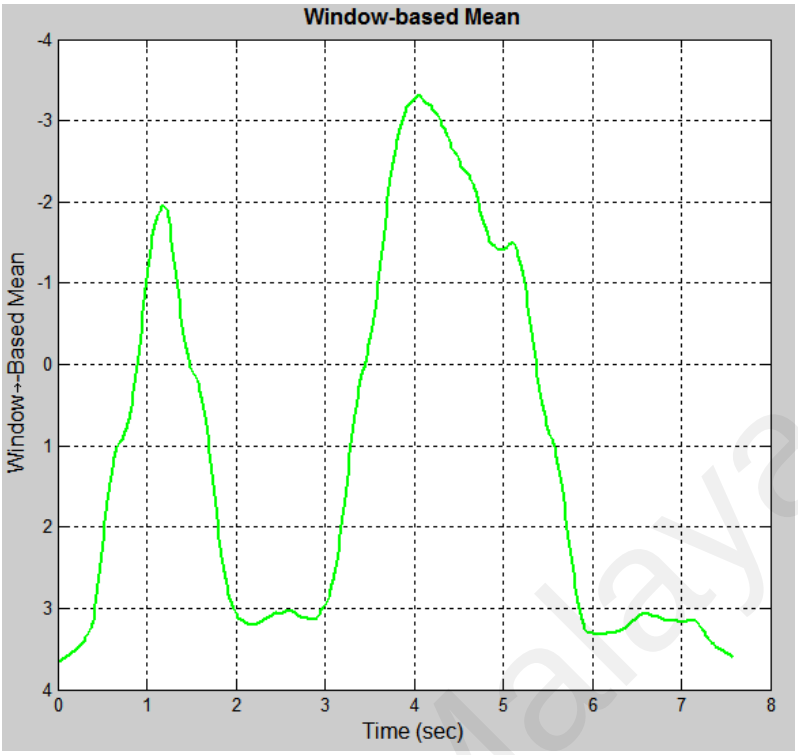


Figure 6.41: An Example of Statistical Mean for Sentence 7

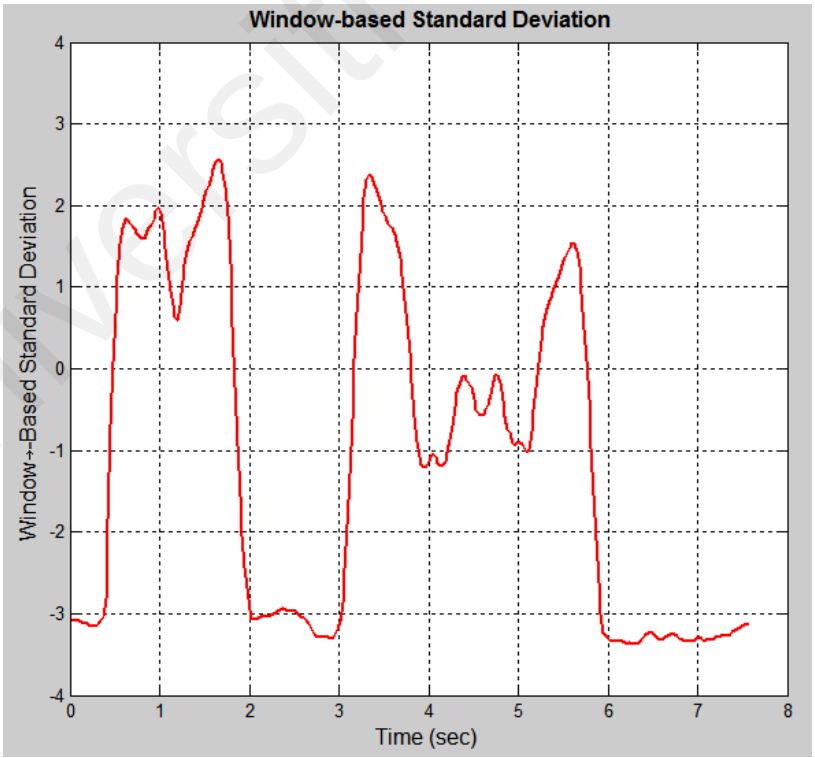


Figure 6.42: An Example of Standard Deviation for Sentence 7

Universiti Malaya

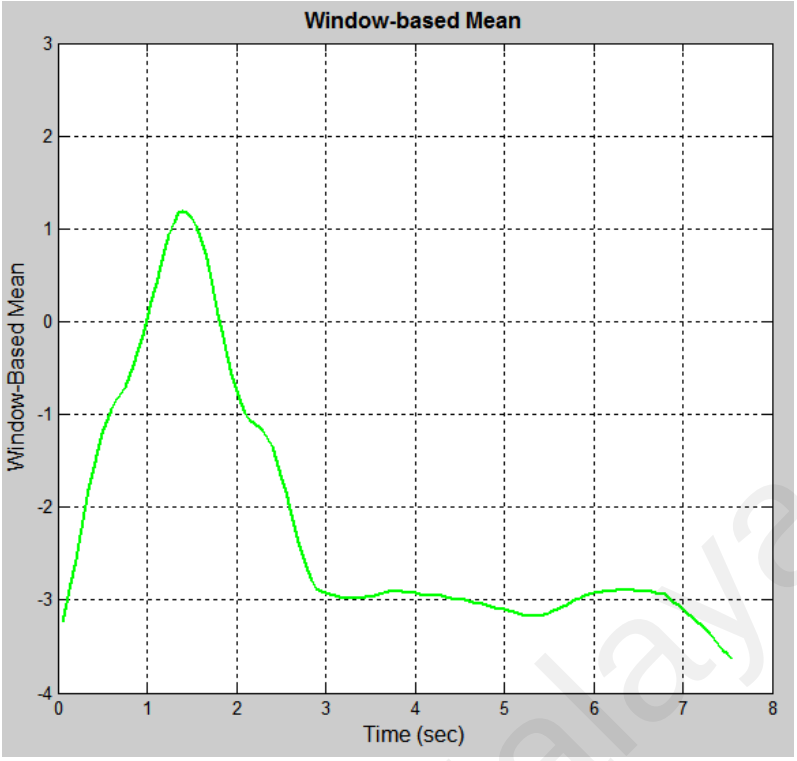


Figure 6.43: An Example of Statistical Mean for Sentence 25

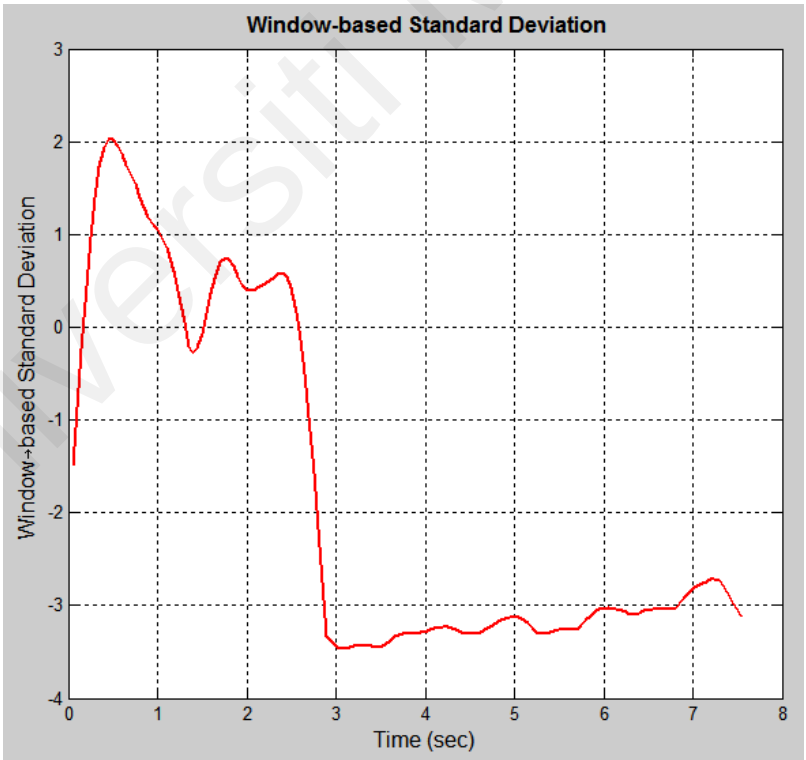


Figure 6.44: An Example of Standard Deviation for Sentence 25

Universiti Malaya

6.3.2 Hand Segmentation and Feature Extraction from the Microsoft Kinect:

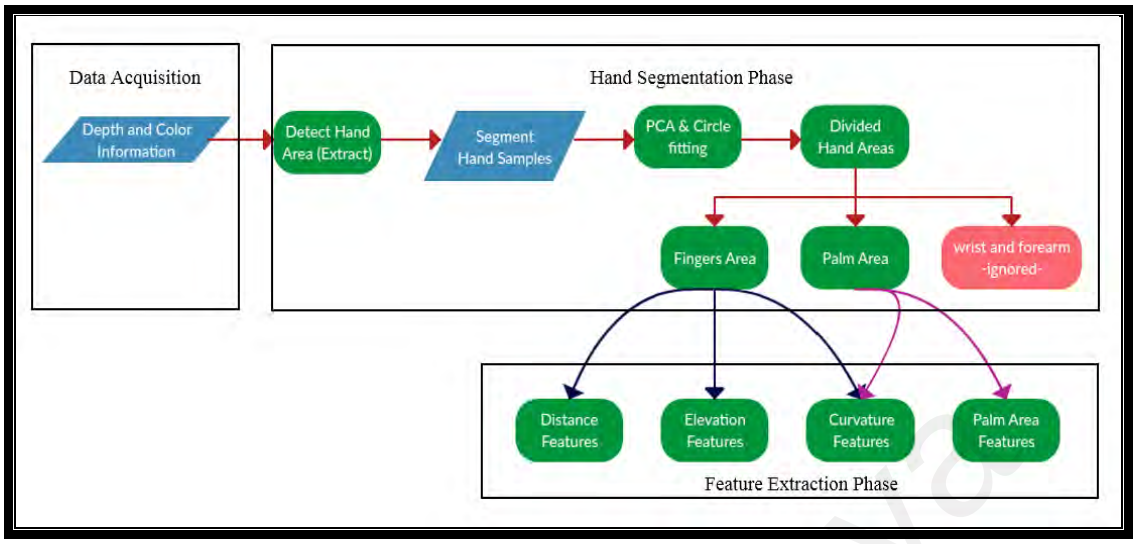


Figure 6.45: Architecture of the proposed Microsoft Kinect for ArSLR

6.3.2.1 Hand Segmentation phase:

In this phase, colour and depth information data are combined in order to obtain a potent SLR. Depth information can still produce the same result of recognition but not as robust as when it is combined with the colour information. Figure 6.46 shows the example for acquiring the colour image, whereas Figure 6.47 shows the example for acquiring the Microsoft Kinect depth image.



Figure 6.46: Acquired color image

Universiti Malaya



Figure 6.47: Acquired Microsoft Kinect depth image

The depth map is a point wherein the projection of the data generated by the Microsoft Kinect colour camera takes place and there exists a connection between the depth value and colour in each sample. The depth coordinate of the map $D(u, v)$ value obtained after the projection, which serves as the threshold value, is based on the colour information. In addition, the colour that is correlated to the samples are transformed to the CIELAB colour sample and, thereafter, the differences in the values between the initially obtained reference skin colour and the CIELAB colour space is determined when the colour samples that have a colour variation less than the threshold value are rejected.

The depth sample associated with the colour that is adaptable with the user skin colour, which can be part of the hand, face and any other uncovered part of the body, can only be retained in the first threshold. The next object closest to the Kinect outside the skin colour threshold is the hand region, among others.

Let the 3D point obtained via Kinect depth camera be represented with $\mathrm{X}_{u,v}$, which is also referred to as the back projection at point (u, v). The corresponding point $\mathrm{X}_{\min}$ is selected as the initial point for the hand detection process after performing the sample search on the threshold depth map with the minimum depth value $D_{\min}$. Our framework

Universiti Malaya

authenticates the availability of samples that have an identical value in the 5×5 region over X_{min}. This is done so as to avoid the selection at X_{min} as a hidden artifact due to the noise inherent in the measurement. However, the next nearest point can be selected for the iteration provided the cardinality of the threshold if is not satisfied.

Now, let the hand samples set be denoted by H. Provided that the points belonging to H has the same depth of X_{min} or greater in value T_{depth}, which relies on the size of the hand, then the representation of H is shown in Equation 5.45:

$$\boldsymbol{H = \{X_{u,v} | D(u,v) < D_{min} + T_{depth}\}} \quad \textbf{(6.45)}$$

A reference user's hand can be used to measure the T_{depth} value, but its constant empirical threshold value of 15 cm is useful in most cases while performing an experiment ($T_{depth} = 15\ cm$). An additional check H was carried out in order to eliminate the retained arm samples. In that check, each of the $X_{u,v} \in H$ that has 3D space distance from X_{min}, which exceeds the threshold T_{size}, is removed in which the size of the hand also depends upon (let the value of $T_{size} = 20\ cm$ for the experimental purpose). However, it should be noted that T_{depth} and T_{size} neither depends on sensor resolution nor hand position, rather it depends on the physical hand size.

The proposed algorithm can reliably carry out the segmentation of the hand samples and from the object sensor and other parts of the body. Figure 6.48 depicts the threshold depth map obtained using our proposed framework in which the closest sample is indicated in blue.

Universiti Malaya

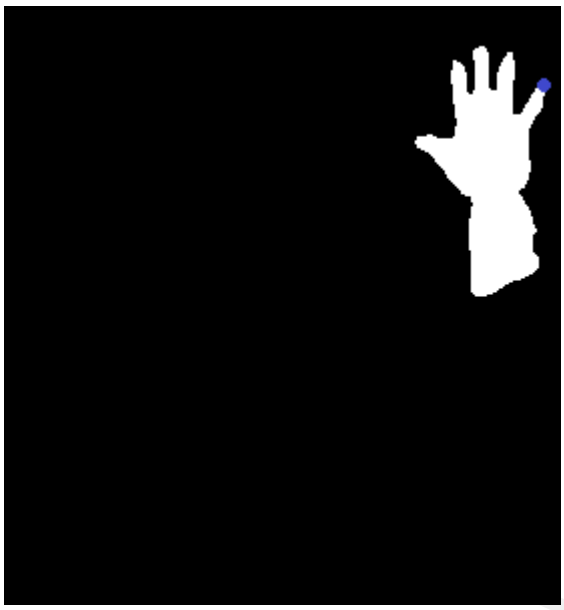


Figure 6.48: Extracted hand samples from Microsoft Kinect

The palm region detection is an essential step that is to be carried out as the first step in feature set extraction explained in Section 6.3.2.2. This is followed by building a 2D binary mask $B(u, v)$ on the lattice (u, v) in relation with the obtained depth map shown in Equation 6.46:

$$\boldsymbol{B} = \begin{cases} \mathbf{1}\ \boldsymbol{if}\ \boldsymbol{X_{u,v}} \in \boldsymbol{H} \\ \mathbf{0}\ \boldsymbol{otherwise} \end{cases} \tag{6.46}$$

i.e., non-zero entries of $B(u, v)$ for the indexes that correspond to the samples in H.

Finding the largest circle that is best suited to the palm region in $B(u, v)$ is one of the methods for our palm detection. In this regard, the starting point $\boldsymbol{C}$ of the circle fit algorithm must be determined first. However, the palm area in $\boldsymbol{B}$ is assumed to contain the maximum depth owing to the fact that the palm area is usually bigger compared to other areas, such as the wrist and fingers area.

A combination of 2D Gaussian kernel and a large standard deviation $\sigma = 150\,.\frac{1\,[m]}{D_{min}}$ is used to filter the 2D binary mask $B(u, v)$. However, it should be noted that the standard deviation value is rated in accordance with the smallest distance. This is done so that the equivalent distance value of the window measured in the metric unit from the Kinect can

Universiti Malaya

be obtained and to ensure the large support of the filter to acquire the density and thickness of the arm and hand areas. Then, the output of the Gaussian filter is contained in a blurred grayscale image $B(u, v)$ with values that are proportional to point's density, as presented in Figure 6.49 wherein the blue point has the maximum corresponding mask.

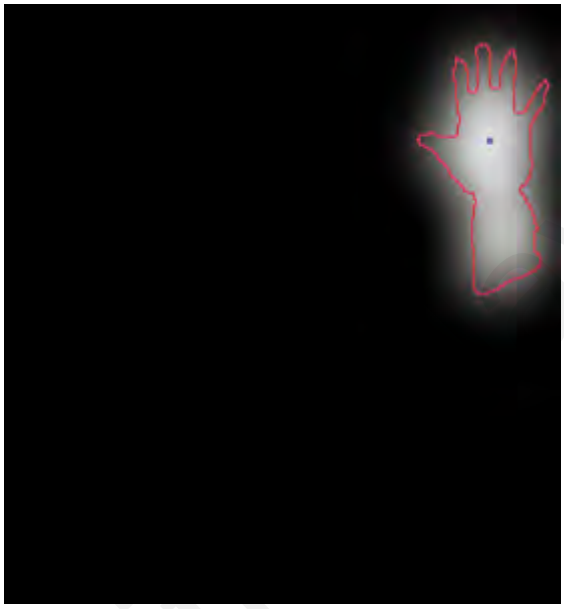


Figure 6.49: Samples of result of applying the Gaussian filter on the mask corresponding to H.

Assuming $\mathbf{C} = \mathbf{C_g}$, here $\mathbf{C_g}$ is the point of $B^f(u, v)$ with the maximum density and thickness value. However, $\mathbf{C_g}$ could be found elsewhere within the arm area but may not be close to the palm region in unfavorable cases where the depth point of the arm is greater than that of the hand.

Likewise, there could be many points with the highest density. In that case, further thresholding on $B^f(u, v)$ must be carried out. In order to overcome such a situation and produce a proper position for $\mathbf{C_g}$, the highest estimated density is denoted as $b_{max} = max_{u,v}\left(B^f(u, v)\right)$ and the threshold value, which is the value of $T_d = 0.9$, is denoted as $T_d \in [0,1]$, where $T_d \,.\, b_{max}$ tallies with the highest density of 90%. A recent 2D binary mask $B^T(u, v)$ is computed in Equation 6.47 as follows:

$$\boldsymbol{B^T(u,v)} = \begin{cases} \mathbf{1} & \boldsymbol{if\ B^f(u,v) \geq T_d \cdot b_{max}} \\ \mathbf{0} & \boldsymbol{otherwise} \end{cases} \quad \textbf{(6.47)}$$

where $B^T(u,v)$ consists of one or more blobs that represent possible candidates to contain $\mathbf{C_g}$. Each blob centroid is computed, which resulted in selecting the centroid with the closest blobs to the $\mathbf{C_g}$ to X_{min}, as described in the previous section.

The circle-fitting process consists of a circle together with the initial centre position $\mathbf{C} = \mathbf{C_g}$ and radius $r = 1\ [pxl]$, which is initially extended in $B(u, v)$, by incrementing the value of r until the 95% of the points inside it belong to H (the remaining 5% consists of the reserved errors that might be introduced from noise or the depth sensor of the artifacts). The position of $\mathbf{C}$ is moved the direction that maximises the density of the samples within the circle after computing the maximum value of the radius that satisfies the threshold. Thereafter, the radius r is incremented and the iteration continues until the maximum value of the circle is fitted within the palm-area.

Figure 6.50 shows that the circle is fitted in the hand along with the point $\mathbf{C_p}$ in blue color. The end of the position of $\mathbf{C}$, which is denoted by $\mathbf{C_f}$ correlates with the centre of the palm. Therefore, the 3D point $\mathbf{C_f}$, which is now referred to the hand centroid, is joined with the last radius $\mathbf{r_f}$ and has a vital role to play in the proposed algorithm. Figure 6.51 shows the regions subdivision in this order: palm (green), fingers (red) and wrist (purple).

Universiti Malaya

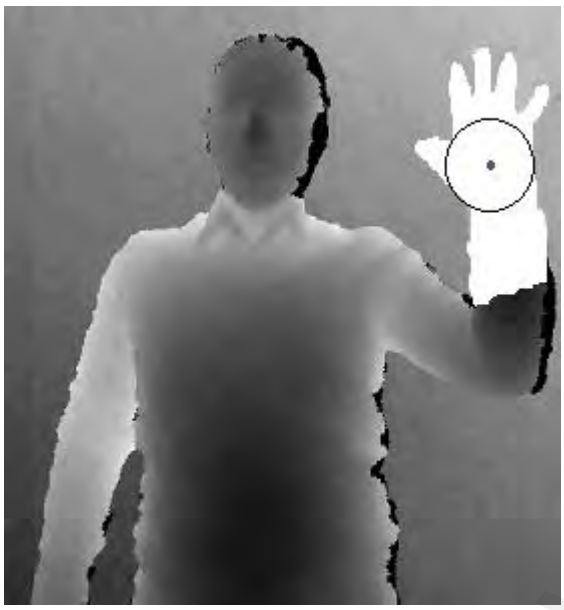


Figure 6.50: Circle fitted on the hand with the point $\mathbf{C_p}$ in blue



Figure 6.51: The regions hand subdivision

In some cases, the circle and palm areas do not properly correlate with each other due to the narrow and longer nature of the shape of the palm. Moreover, many of the obtained gestures are not parallel to the plane image, which causes the distortion of the circular shape by the angle projection. In order to eliminate this issue, an effective model in which the ellipse that is suitable to the palm region is introduced. Therefore, 12 regions that tally with various overlap angular directions is created, starting with $\mathbf{C_f}$ (50% of the

Universiti Malaya

superimposed angle in between the sectors is used) as indicated in Figure 6.52. Moreover, the point of the curved hand within the area nearest to the centre is chosen for each region.

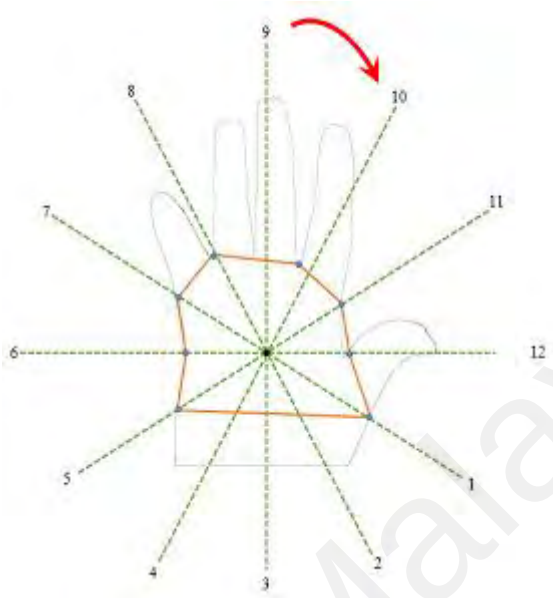


Figure 6.52: Angular directions applied for the computation of the ellipse

From this procedure, a polygon that is enclosed within the hand contour, which is approximate to the hand palm, is obtained. Notably, the reason for applying the partial overlay sector and the shortest distance within the sector is to ensure that the selection of the polygon corners is made at the basis of the fingers in which the sample is not a constituent of the polygon. Finally, the ellipse that is closest to the shape in the least-square value is evaluated by applying the same approach used in a past work (Fitzgibbon & Fisher, 1996). Figure 6.53 shows the suitability of the ellipse and the palm.

Universiti Malaya

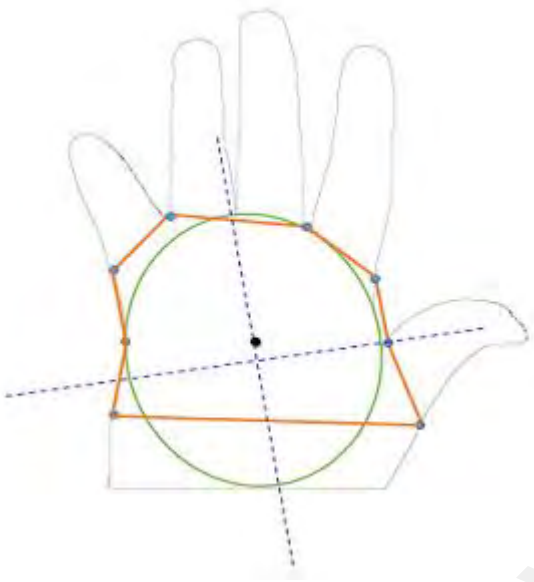


Figure 6.53: The fitting of the ellipse over the palm

Moreover, after the detection of all the available sample size, the random sample consensus (RANSAC) is employed to fit the 3D plane $\boldsymbol{\pi}$ on them. In addition, the PCA was used for the main axis extraction in the 3D points in *H* that correlate with the direction i_x of the vector while moving from the wrist down to the fingertips. Notably, the calculated direction obtained from this scenario is not accurate as it is based on the fingers' position on the performed gestures, thus providing a generic explanation of the hand orientation.

However, a 3D coordinate system that is focused on point $\mathbf{C_f}$ as initially illustrated can be constructed by projecting the axis $\boldsymbol{i_x}$ on a plane $\boldsymbol{\pi}$. For instance, let $\mathbf{i_x^{\pi}}$ denote the projection and $\mathbf{i_z^{\pi}}$ denote the position normal to the plane $\boldsymbol{\pi}$; then, the projection on the y-axis $\mathbf{i_y^{\pi}}$ can be computed by finding the cross-product of $\mathbf{i_z^{\pi}}$ and $\mathbf{i_x^{\pi}}$, which leads to the formation of a right-handed reference system $(\mathbf{i_x^{\pi}}, \mathbf{i_y^{\pi}}, \mathbf{i_z^{\pi}})$, with the points' coordinates denoted with $(\boldsymbol{x_{2D}}, \boldsymbol{y_{2D}}, \boldsymbol{z_{2D}})$.

It should also be noted that $\mathbf{C_f}$ could rest upon the folded finger across the hand palm but not necessarily on $\boldsymbol{\pi}$. $\mathbf{C_f}$ is projected to the $\boldsymbol{\pi}$ so that it can be allocated nearer to the actual centre of the hand, assuming that the correlated central point is denoted with $\mathbf{C_p}$.

Universiti Malaya

Figure 6.54 depicts the proposed 3D coordinate system. Now, all the needed information for the segmentation of $\boldsymbol{H}$ into three regions has been obtained as follows:

- P consisting of the points that match the palm of the hand (the ellipse or the circle holds the hand sample);
- W consisting of the points of H that rest upon the sub-space $\boldsymbol{x_{2D} \leq -r_f}$; and
- F consisting of the points of $H - P - W$ that are equivalent to the finger area.

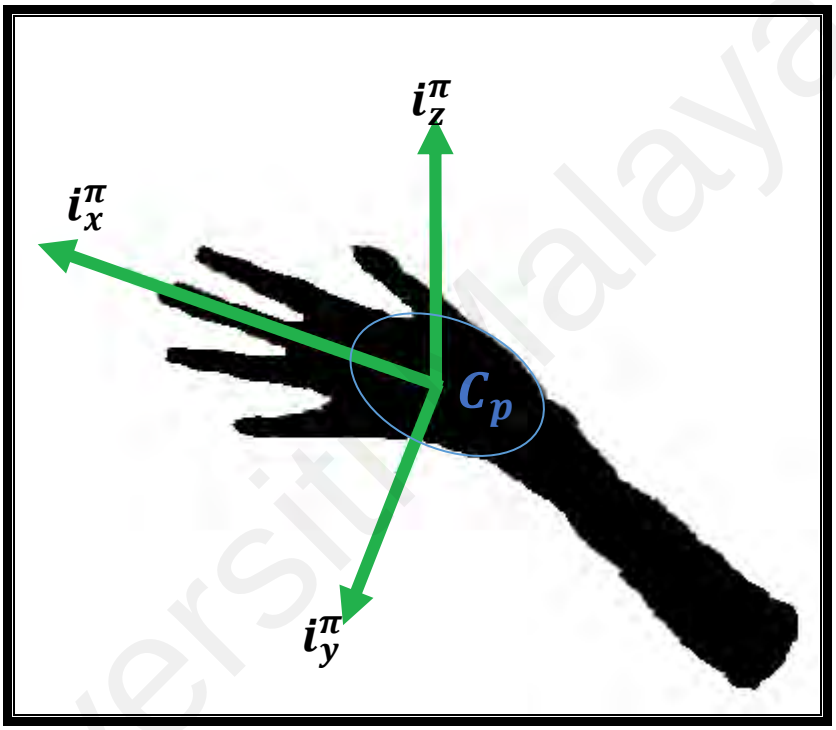


Figure 6.54: Reference system $\mathbf{i}_x^\pi, \mathbf{i}_y^\pi, \mathbf{i}_z^\pi$ computed on the basis of the estimated plane and of the PCA output, used for the features extraction

Finally, a pair of $H_e = (H - W) = (P + f)$ is estimated, which consists of the fingers, hand and palm point. The required information for feature extraction techniques in the proposed framework has been made available at this juncture.

University of Malaya

6.3.2.2 Extraction the Relevant Features based on Multiple techniques for Kinect Data

Depth data are used for calculating the four features of different types in this next procedure.

1- **Distance features:** The description of the three-dimensional (3D) Euclidean distance between the calculated middle of the palm and the fingertips.
2- **Elevation features:** The detailed description of the Euclidean distance between the suitable point of the palm sample and the fingertips, which could also be referred to as the finger height in relation to the palm.
3- **Curvature features:** The detailed description of the curve features within the finger and hand area.
4- **Palm area features**: The detailed description of the feature of the palm area curve, which is important for the estimation of the current position of the palm.

1- Distance features ($\mathbf{F^l}$)

A histogram construction representing the edge sample distance in F from the hand centroid C_P is the starting point of the feature set computation.

Let R_f represent the 3D radius r_f, which is back projected to the plane $\boldsymbol{\pi}$. Notably, R_f denotes the distance in between the C_f and ellipse edge when an effective fitting model is used with the ellipse whose value varies. In a given 3D point $X_i = X_{u,v} \in F$ found in the feature extracted from the finger, the standardised distance from the central point $d_{xi} = \|X_i - C_p\| - R_f,\ X_i \in F$ is calculated along with θ_{xi}, which is the angle between the axis i_x^{π} and the vector $X_i^{\pi} - C_p$ on the palm-plane π, where X_i^{π} is projections to X_i on π.

Therefore, $\boldsymbol{\theta}$ is quantised into separate sets of value θ_q using the similar procedure of quantisation Δ($\Delta = 2^{\circ}$ has been used in the current execution). Finally, the sector of the angle illustrated as $\boldsymbol{I(\theta_q) = \theta_q - \frac{\Delta}{2} < \theta \leq \theta_q + \frac{\Delta}{2}}$ then conforms to the value of

Universiti Malaya

each θ_q . Thus, the histogram $\boldsymbol{L(\theta)}$ is obtained by choosing the uttermost point within each of the $\boldsymbol{I(\theta_q)}$ sector.

$$\boldsymbol{L(\theta) = max_{I(\theta_q)}\, d_{xi}} \tag{6.48}$$

As depicted in Figures 6.55 and 6.56, a feature set of an angular region conforming to the finger, which is raised within an interval and a reference histogram $L_g^r(\theta)$ of the type, is created for each sign in the dataset for feature computation. In addition, the direction of the PCA main axes, as indicated in a previous section (6.3.2.1), is not accurate as it is affected by many issues, such as the maintained height of the wrist region and the total number of the fingers raised in the executed gesture hand detection. Therefore, the explicit correlation of the histogram in this scenario becomes impossible as there may not be a correct alignment of the obtained distance histogram with the hand gesture.

Figure 6.55 shows the finger edges F that represents the edge distances of the histogram with the equivalent feature regions. Meanwhile, Figure 6.56 presents the associated histogram $L(\theta)$ within the region that matches the different features $f_{g,j}^l$, in which the green stars are used to highlight the feature point.



Figure 6.55: Finger edges *F*

Universiti Malaya

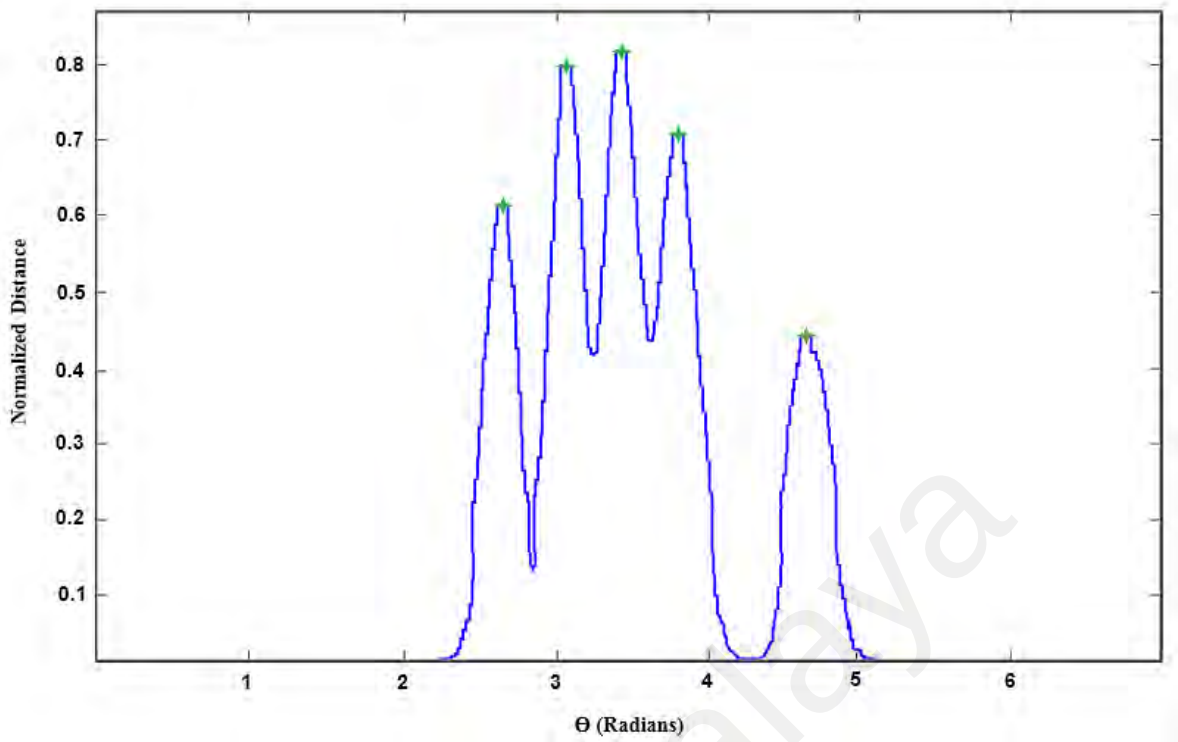


Figure 6.56: Samples of associated histogram $L(\theta)$ with the region corresponding to the various features.

For the purpose of comparing the implemented histogram gesture, the histogram gesture with each corresponding template is arranged together, and the argument that maximises the relation between the modified forms of the reference of each gesture histogram is also considered. The possibility of flipping the histogram is also considered owing to the fact that the hand can be positioned in such as a way that the dorsum or the palm faces the camera.

$$\Delta_g = \mathbf{arg\,max}_{\Delta}\ (\boldsymbol{p}\ (\boldsymbol{L}(\boldsymbol{\theta}), \boldsymbol{L}_g^r(\boldsymbol{\theta}+\Delta))) \tag{6.49}$$

$$\Delta_g^{rev} = \mathbf{arg\,max}_{\Delta}\ (\boldsymbol{p}\ (\boldsymbol{L}(-\boldsymbol{\theta}), \boldsymbol{L}_g^r(\boldsymbol{\theta}+\Delta)))$$

From the evaluation in Equations 6.49 and 6.50, $\boldsymbol{L}$ is used to represent the periodic function with period $\boldsymbol{2\theta}$,

where the value of the $a(.)$ and $b(.)$ relation between each other is represented by the $p(a(.), p(b(.))$ symbol.

Universiti Malaya

Then, the gesture of each reference and the obtained histogram can be coordinated by applying the translational shift Δ obtained from the evaluation. If the associated (reference) histogram and the histogram arranged next to the gesture are represented by $as\ L_g^r\ (\theta)$ and $L_g\ (\theta)$, respectively, then the $\Delta_g\ or\ \Delta_g^{rev}$ symbol can be used as the translational shift Δ to $L(\theta)$, which depends on the one that maximises the correlation.

$$\mathbf{L_g\ (\theta)} = \begin{cases} \mathbf{L(\theta - \Delta_g)\ if\ max_\Delta\ p\ \left(L(\theta), L_g^r\ (\theta + \Delta)\right)\ \geq max_\Delta\ p\ \left(L(-\theta), L_g^r\ (\theta + \Delta)\right)} \\ \mathbf{L(-\theta - \Delta_g^{rev})\ Otherwise} \end{cases} \tag{6.50}$$

Moreover, each gesture reference histogram could have different alignment Δ_g and regions that match different features of interest. This method is essentially applied to take care of the inefficient accuracy of the direction evaluated using PCA as applied in Section 6.3.2.1. Some samples of the implemented histograms for the different signs are shown in Figure 6.57.

If the various gestures represented by $\boldsymbol{G}$ are contained in the dataset, then the set feature F^l consists of separate finger values $j \in \{1, 2, 3, 4, 5\}$ in each gesture $g \in \{1, 2, 3, \dots, G\}$. The optimum value of the adjusted histogram in the angle area $I\left(\theta_{g,j}\right) = \theta_{g,j}^{min} < \theta < \theta_{g,j}^{max}$ is equivalent to the feature value $f_{g,j}^l$, which is related to the finger (j) within the gesture (g). j in gesture g (the details can be found in Figure 6.56), i.e:

$$\boldsymbol{f_{g,j}^l = \frac{maxL_g(\theta)}{L_{max}}} \tag{6.51}$$

The length L_{max} of the middle finger is used to normalise all features using the range of [0, 1] to scale all these features. This is done in order to show the difference in size of the hands of different peoples.

Universiti Malaya

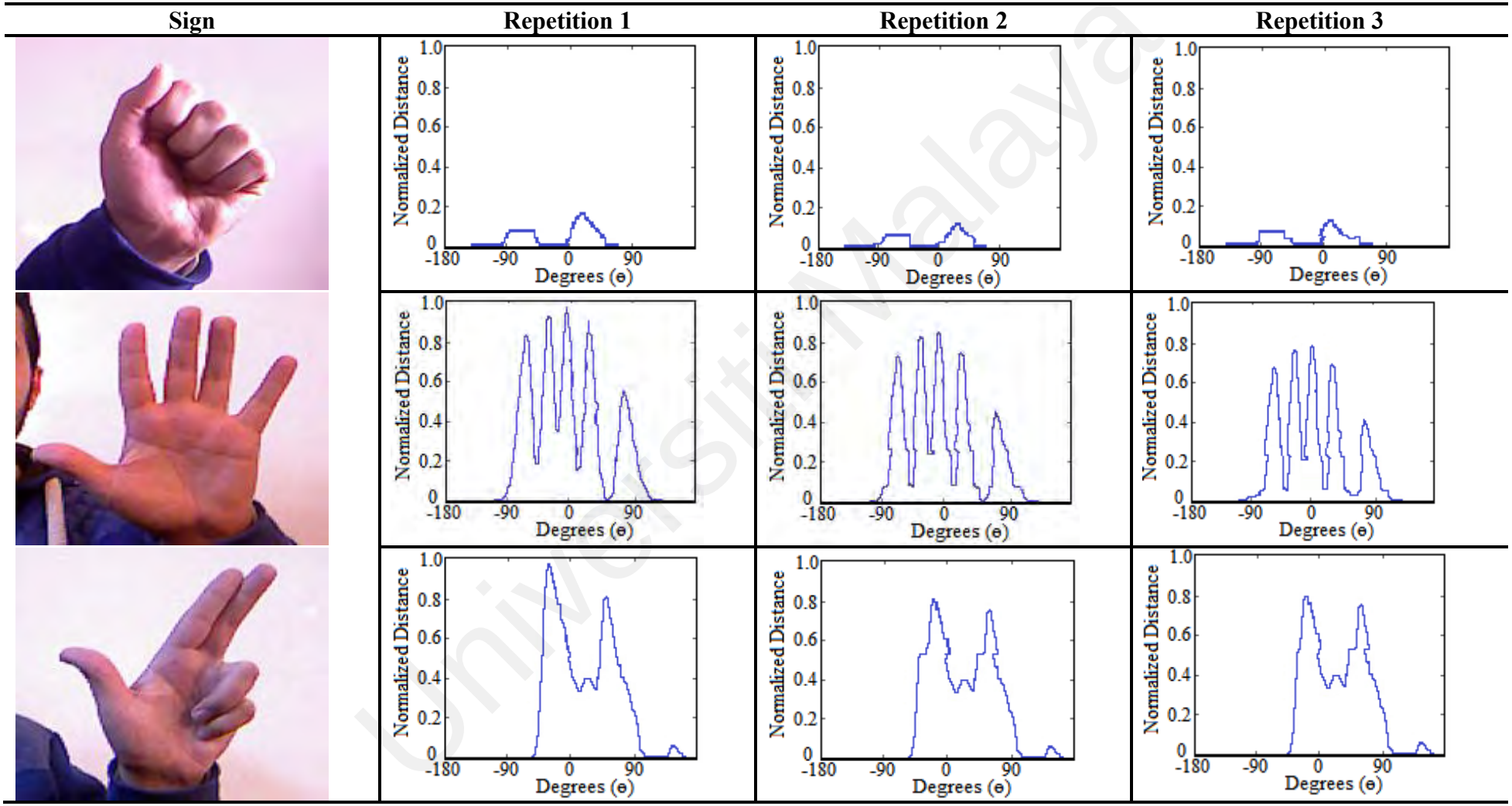


Figure 6.57: Samples of aligned distance histogram $L_g(\theta)$ for various signs.

Universiti Malaya

2- Elevation features (F^e)

The same approach used for the extraction of distance feature in the distance feature section is used for the construction of the elevation feature. The construction of this elevation starts by constructing a histogram that represents the separate sample distance X_j in F, which emanates from the plane **π** computed in Equation 6.52:

$$\boldsymbol{e_{x_j} = sgn\left(\left(X_j - X_j^{\pi}\right).i_y^{\pi}\right)\left|X_j - X_j^{\pi}\right|, X_j \in f} \quad \textbf{(6.52)}$$

where X_j^{π} is the projection of X_j on π. The e_{x_j} sign explains that X_j can be a member of any hemi-spaces defined by π, which means that X_j can be behind or in front of π.

The highest point of the absolute distance from the plane is chosen for a separate sector of the angular, which is equivalent to the θ_q value in the same way as illustrated for the feature distance produced by histogram $E(\theta)$:

$$\mathbf{E(\theta_q)} = \begin{cases} \mathbf{max\ e_{x_j}\ } \boldsymbol{if} \mathbf{\ \left|max\ e_{x_j}\right| > \left|min\ e_{x_j}\right|} \\ \mathbf{min\ \ e_{x_j}\ } \boldsymbol{otherwise} \end{cases} \quad \textbf{(6.53)}$$

Histogram $E(\theta_q)$ uses similar regions as those experimented in the distance features. The alignment information already calculated in the distance features section is used to carry out the alignment of the histogram $E(\theta)$, which matches the implemented gestures to different reference gestures in G. Here, histogram $E(\theta)$ is represented by $E^g(\theta)$, which is in alignment with the gesture template g^{th}. Then, the feature of the elevation can be calculated in accordance with the following:

$$\mathbf{f_{g,j}^e} = \begin{cases} \frac{\mathbf{1}}{\mathbf{L_{max}}}\ \mathbf{maxE^g(\theta)} & \mathbf{if\ \ |maxE^g(\theta)| > \ |minE^g(\theta)|} \\ \frac{\mathbf{1}}{\mathbf{L_{max}}}\ \mathbf{minE^g(\theta)} & \mathbf{otherwise} \end{cases} \quad \textbf{(6.54)}$$

Notably, the alignment calculated in the distance features section according to our framework is used to reduce the execution. This is because the feature distance is more dependable compared with the one computed from the elevation. In addition, the elevation feature vector F^e possesses a similar structure and number of elements as the distance features F^l.

Universiti Malaya

3- Curvature features (F^c)

The edges of the curved hand shape comprise the third proposed descriptor in this work. This approach relies on the constant integral according to past works (Manay et al., 2006; N. Kumar et al., 2012) and avoided differential operation for curvature description because the data acquired from the actual depth camera contain a great deal of noise.

The algorithm for the feature extractor accepts the binary mask $B(u, v)$ and the hand edge points H_e as the input and the subset of all the points $X_i \in H_c$, which are members of the hand contour of the boundary of H_e, denoted by $H_c = \delta H_e$. Given a set of S circular masks $M_s\ (X_i), s = 1,2,3, \dots, S$ of a radius r_s centred on each edge sample $X_i \in H_c$ in the experiments, the range value of 0.5 cm to 5 cm of the radius r_s is employed along with 25 masks.

Let $\mathrm{V(X_i, s)}$ stand for the curvature in $\mathrm{X_i}$, indicated as the samples number ration in H_c dropping from mask $M_s(\mathrm{X_i})$ over $M_s(\mathrm{X_i})$ size, i.e.:

$$\boldsymbol{V(X_i, s)} = \frac{\sum \mathbf{X_j} \in \mathbf{M_s}\,(\mathbf{X_i})\, \boldsymbol{B(X_j)}}{|\mathbf{M_s}\,(\mathbf{X_i})|} \qquad \textbf{(6.55)}$$

where $|\mathrm{M_s\,(X_i)}|$ expresses the cardinality of $\mathrm{M_s\,(X_i)}$. $B(\mathrm{X_j}) = B(u_j v_j)$ and $(u_j v_j)$ represent the 2D coordinates that correspond to $\mathrm{X_j}$. The value of $V(\mathrm{X_i}, s)$ is calculated for each sample $\mathrm{X_i} \in H_c$, and the value of radius $\mathrm{r}_s$is equivalent to the feature estimation level execution. The radius r_s is defined in metric units and then changed to the equivalent pixel size based on the distance interval between the hand and the device camera (Kinect). The descriptor in this way is invariant to the hand and Kinect distance interval.

In order to speed up the processing, the choice of substituting the circular mask with the simple square mask is added, after which the entire images are applied in order to facilitate the sample computation in the mask due to the constant rotation of the curvature

Universiti Malaya

masks. This method has been proven to be faster despite the rotation invariance and the technically unrecognised performance loss.

The $V(\mathrm{X_i}, s)$ value range of 0 to 1 represents the extremely convex and concave shape, which reveals the straight edge when the value of $V(\mathrm{X_i}, s) = 0.5$. The range of [0, 1] is quantised into N bins of similar-sized $b_1, b_2, b_3, \ldots, b_N$. The set $\mathrm{V}_{b,s}$ of the finger edge points $\mathrm{X_i} \in H_c$ with the equivalent value of $V(\mathrm{X_i}, s)$ dropping from the mask s to the bin b is indicated below.

$$\mathbf{V_{b,s}} \left\{ \boldsymbol{X_i} \left| \frac{(\boldsymbol{b-1})}{\boldsymbol{B}} < \boldsymbol{V(X_i, s)} \leq \frac{\boldsymbol{b}}{\boldsymbol{B}} \right. \right\} \tag{6.56}$$

A curvature feature is selected for each of the radius value s and bin b, represented by $f_{b,s}^{c}$, and the contour length $|H_c|$ is used to normalise the cardinality of the set $V(\mathrm{X_i}, s)$.

$$\boldsymbol{f_{b,s}^{c}} = \frac{|\mathbf{V_{b,s}}|}{|\boldsymbol{H_c}|} \tag{6.57}$$

The value of the curvature feature $f_{b,s}^{c}$ is taken from the interval value [0, 1], which represents the similar interval value divided by the features of the elevations and the distances. This is made possible with the help of normalisation. Finally, the curvature features $f_{b,s}^{c}$ within feature vector $\mathrm{F^c}$ with $B \times S$ entries are extracted followed by incrementing the index values $s = 1,2,3, \ldots, S$ and $b = 1,2,3, \ldots, N$. However, the graphical visualisation of complete curvature descriptor $\mathrm{F^c}$ is made possible by seeing each $f_{b,s}^{c}$ as the pixel value with the coordinates (b, s) in a grayscale image and reducing the size of $\mathrm{F^c}$ into S rows and N columns matrix. The illustrated example is shown in Figure 6.58.

Universiti Malaya

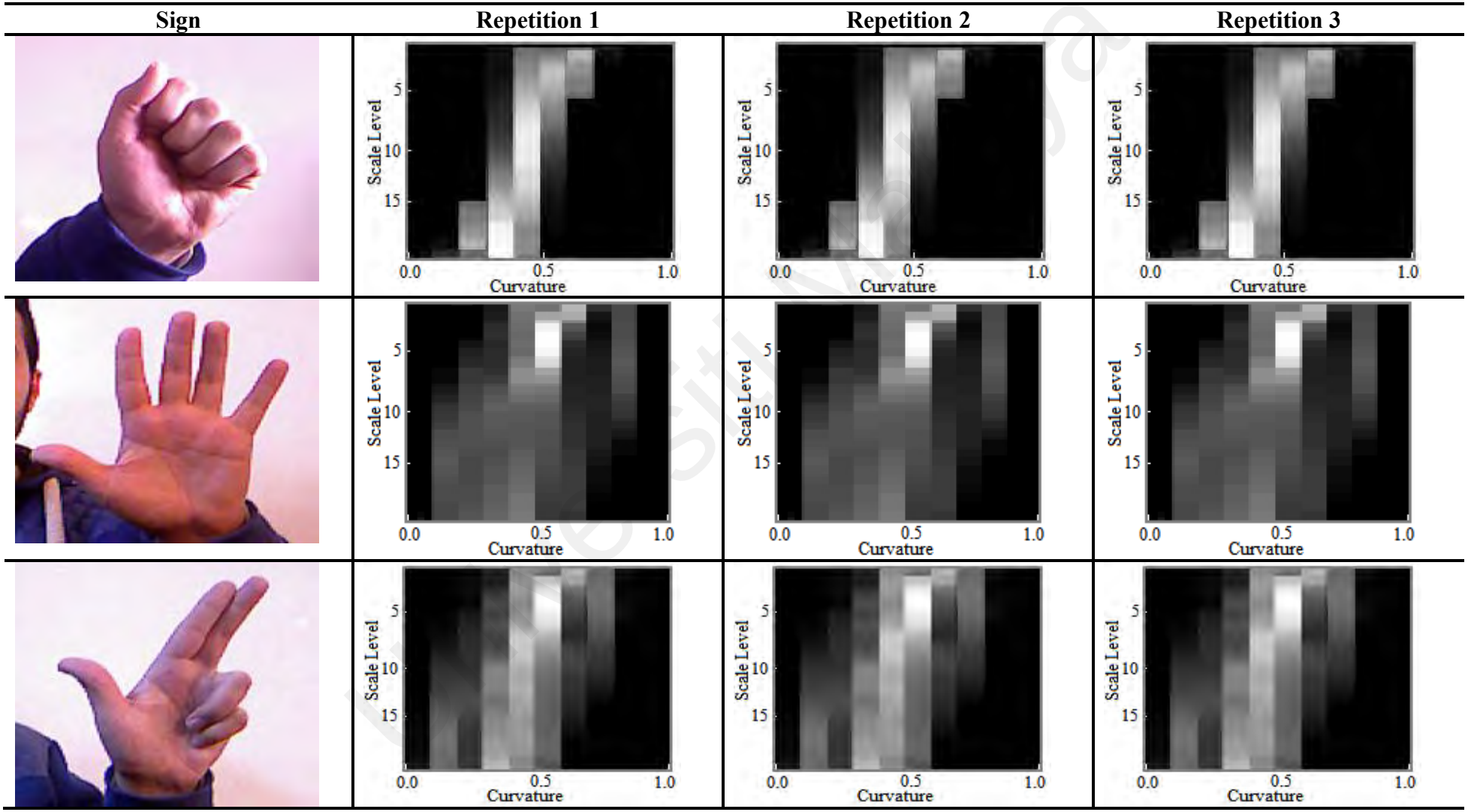


Figure 6.58: Samples of curvature descriptors for various signs.

4- Palm area features (F^a)

The final feature sets explain the sample displacement in the palm region P. The P variable might consist of the samples obtained from the finger, provided that there is a folding of the fingers across the palm area, but P is equivalent to palm area in an actual sense. The concept, as indicated in Figure 6.59, aims to split the area within into six distinct areas, as described across the plane π. Firstly, the two-part division was made on the ellipse or the circle describing the palm position, which consists of the upper and lower parts. The lower part is employed as an association to the position of the palm by fitting the 3D plane π_p to the region, whereas the upper half is made up of a division of five different areas, $namely, A_j,\ j = 1, 2, 3, 4, 5,$ which approximately agree with the regions closer to the separate fingers, as illustrated in Figure 6.59. This means that each region tallies to the affected area based on the finger position.



Figure 6.59: Example for hand sign that shown the regions closer to the separate fingers

Different area features are responsible for the shapes of the deformation in the matching region when moving or folding the connected finger. The association of P and the samples match the folded fingers over the palm, indicating how they are not occupied by the elevation or distance features but are rather used for the calculation of the palm

area feature. Here, the plane position area rigidly relies on the parameter that describes the palm area, the fingers' widths (a circle divided into an upper half portion which might be improved based on the hand of a definite user is utilised) on the position i_x^{π}, where the value of $\theta = 0$. Note that the centre C_f and radius r_f of the circle and axes of the ellipse that define the palm area and the hand segmentation have already been obtained in Section 6.3.2.1, leaving only the computation of the alignment of the θ directions element. In addition, the computed alignment information obtained in the feature distance section is used for the alignment of the regions format (increased by r_f, or extended based on the ellipse that consists of two axes) together with the direction of hand i_x^{π}. The same procedure used in the distance feature is employed to perform an adjustment in a separate gesture format so that the features on the area can be extracted for individual adjustment. Here, A_j^g, is used to represent the alignment with each gesture template, where g stands for the corresponding gesture.

Using the same procedure, the set of the points X_i in P corresponding to each of the regions A_j^g is calculated along with the distance between each sample X_i in A_j^g and π_p by using each area A_j^g. Then, the feature that corresponds to the area A_j^g is regarded as the average of the area samples distance A_j^g.

$$\mathbf{f}_{g,j}^{a} = \frac{\sum_{\mathbf{X_i} \in \mathbf{A}_j^g} \|\mathbf{X_i} - \mathbf{X}_i^{\pi}\|}{\left|\mathbf{A}_j^g\right|} \tag{5.58}$$

Then, palm area feature is selected as the matching feature of the area A_j^g. The whole area features are obtained from the vector F^a. This is computed by the feature area wherein one area represents separate fingers that correspond to the separate gesture available, by applying the similar principle of F^l and F^e. Finally, to ensure that the obtained value inside the $[0, 1]$ range is similar to other computed feature, the entrance of F^a is optimised.

Universiti Malaya

6.3.3 Combining Features for Recognition and Classification Phase

The proposed framework presented in Sections 6.3.1 and 6.3.2 has produced different feature vectors from two different devices: features from the smart data gloves device and those from the Microsoft Kinect device. These FVs are fed into the classifiers model for recognising the ArSL gestures. The manually extracted features obtained from the smart data gloves and the Microsoft Kinect are combined into one vector, which contains only the main descriptor features for every gesture.

Each of these vectors can explain some important proof concerning the performance of the gestures. A multi-class SVM is employed to perform classification for the recognition. Each of the extracted FVs can be either applied for the classification or combined with other descriptors for the classification. The multiple feature descriptor (from smart gloves and Microsoft Kinect) can be combined by performing basic concatenation on the chosen FV. The goal of this framework is to carry out sign classification into G classes, one for each sign in the designed database.

A multi-class SVM classifier is employed in the classification technique which depends on the *one-against-one* approach (Chang & Lin, 2011). In this classification approach, a set of sign expressions of binary SVM classifiers are employed for train and test each of the class against another. Each computed result is selected as a vote for a particular sign, and the sign with the highest number of vote for each sample in a test set is chosen, thus referring to the classification result. An indefinite Gaussian radial basis function (RBF) kernel and the parameter for the classification algorithm are selected to apply fusion matrix and cross-validation on the feature set for training and testing model the model. Figure 6.60 presents the cross-validation techniques.

Universiti Malaya

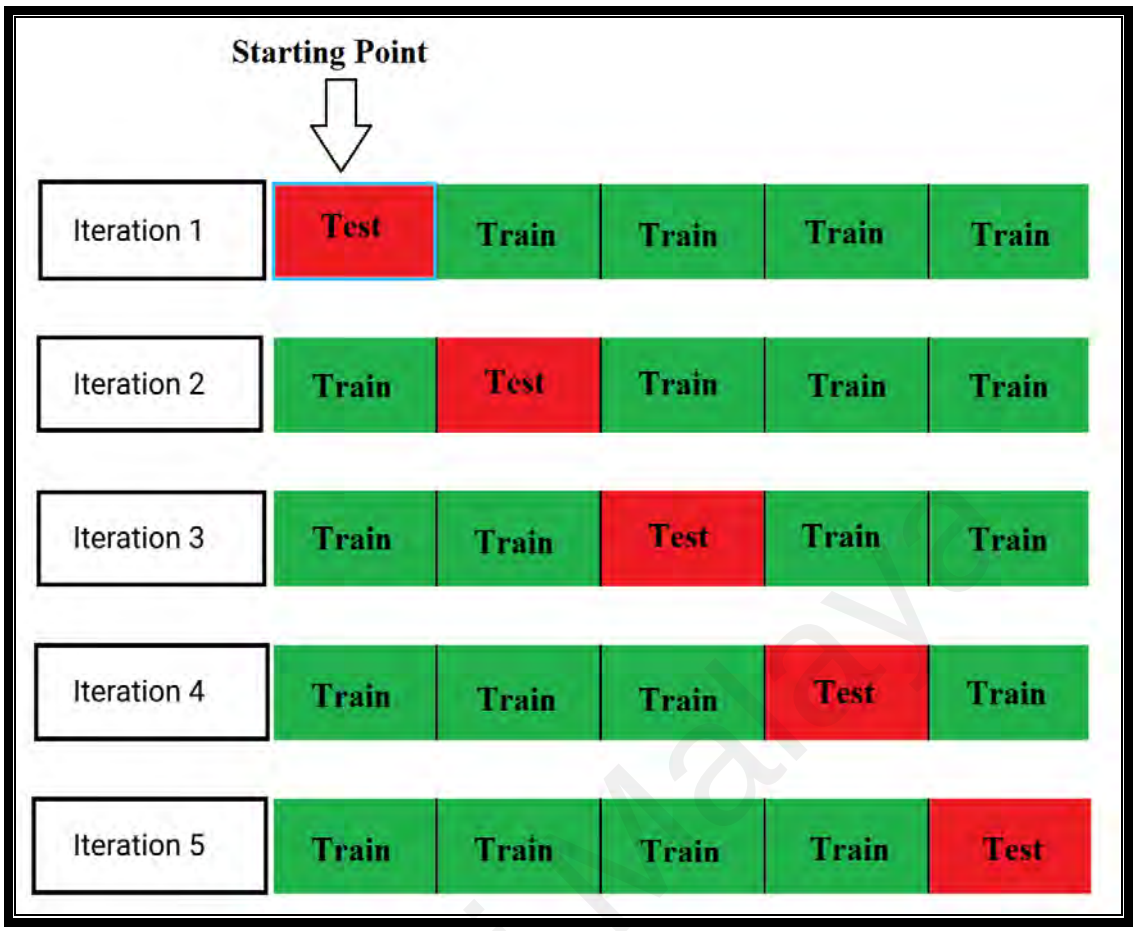


Figure 6.60: Cross-validation techniques

In a given training dataset consisting of M user ($M = 10$), the RBF kernel space of parameters (C, γ) is shared by a regular grid. However, the training dataset is grouped into two rations for each set of the parameter, which consists of $M - 1$ users for classification, and the rest is used for testing and validation. The iteration is carried out M times by updating the user feature for validation; thus, the parameter sets that produce the highest precision, in average, are chosen as the grid search output. Finally, all the M users of the training set are classified using SVM with the optimal parameter.

Universiti Malaya

6.4 Summary

This chapter presented the proposed frameworks for modelling and developing ArSLR. The first framework represents a usable static backhand and signer-independent approach for ArSLR of letters and numbers signs based on the VBR approach using a smartphone camera. The second framework was a usable hybrid VBR and SBR approach for signer-independent continuous ArSLR development using Microsoft Kinect and smart data gloves. The next chapter will evaluate and analyse the obtained results of each framework and explain the output of each phase of the framework.

Universiti Malaya

CHAPTER 7: EVALUATION RESULTS AND DISCUSSION

7.1 Introduction

Chapters three and six presented the detailed descriptions of the methodology, the phases and the method of each framework modelling. Meanwhile, the current chapter presents and discusses the obtained results of each framework and explains the output of each phase of the framework. In Section 7.2, the obtained results for framework 1 (the static backhand ArSLR framework) are presented. In Section 7.3, the obtained results for framework 2 (hybrid approach, signer-independent and automatic ArSLR framework based on VBR and SBR) are presented, whils Section 7.4 discusses the results and compares the proposed framework with the related works. Section 7.5 summarises the framework results.

7.2 Results of Framework 1: Static Backhand Arabic Sign Language Recognition

The main experimental setups carried out in each phase of this framework is presented in the next subsection. Other subsections discuss the revealed experimental classification and the obtained recognition results after comparing with other related works.

7.2.1. System Experimental Setups

In this section, the training and testing phases for framework 1 are presented. The classification algorithm is trained and evaluated by employing evaluation metrics for system performance testing. In addition, the experimental setups used for determining the optimal lighting test, the number of features, and the number of hidden layers for MLP for the closed and open hand are also presented.

Universiti Malaya

7.2.1.1. Training and Testing Phases

As discussed earlier in Chapter 4, the database used for framework 1 is generated from 10 signers by SL experts and practitioners (consisting of 3 male signers and 7 female signers with ages ranging from 15–45 years). Each signer performed 10 repetitions for each static backhand ArSL, including Arabic alphabet characters and Arabic numbers. A total of 38 static backhand Arabic characters represented by one static sign are recorded for each signer using a smartphone camera (Apple iPhone X with a 12MP resolution).

The training and testing phases are then conducted to train and test each of the classifiers. The training phase is mainly applied for learning in order to fit the target parameters of the classifier, including an input layer, an output layer, hidden layers, stop error and learning rate. This phase is a combination of several iterations wherein the error in each iteration that is equal to the difference between the expected response of the classifier and the actual one is computed and then used at each layer in order to set the weight of the input. The iterations continue until the computed error equals the predefined error value, which is called the target error. In each iteration, the computed error decreases until it reaches the target one.

In contrast, the testing phase is independent of the training set but has the same probability distribution. It is applied to evaluate the classifier performance using a new set of inputs. The database is divided into training and testing sets, which comprise 70% and 30% of the total data, respectively.

7.2.1.2. Lighting Condition Test

A lighting test is initially conducted during the recording of the dataset in order to determine the most accurate light condition in which to capture the hand signs. Thus, static signs are recorded by different users under both direct and corner light conditions, after which they are compared. As illustrated in Table 7.1, the signs recorded under direct

Universiti Malaya

light conditions ae more accurate than those recorded under corner light conditions. The recorded accuracies for direct and corner light conditions are 95.63% and 83.86%, respectively. Hence, all the imported vision-based hand signs from the recorded dataset to the framework worked under direct light with the user.

Table 7.1: Accuracy of lighting conditions

Light condition	Accuracy
(a) Direct light conditions	95.63
(b) Corner light conditions	83.86

7.2.1.3. Selection of the optimal number of hidden layers for MLP classifier

The MLP consists of the input, hidden and output layers. Each framework must have a specific number of hidden layers based on its target and requirements, such that the random selection of this number results in either over-fitting or under-fitting of the target problem. This means that the number of hidden layers is a model hyper-parameter that must be carefully determined. Thus, the experiment conducted in the current study is set to observe the most suitable number of hidden layers in MLP networks for both open and closed hand sign categories. The number of classes in the open hand sign group is equal to 18 whilst that of the extracted features is 30. The closed hand sign group consists of 14 features extracted from 1D signals and 9 classes.

The number of hidden layers in this experiment is changed from 1 to 6. As shown in Figure 7.1, the best-recorded numbers of hidden layers in the MLP of the closed and open hand signs are 4 and 3, respectively, with the corresponding accuracy rates of 87.10% and 88.90%.

Universiti Malaya

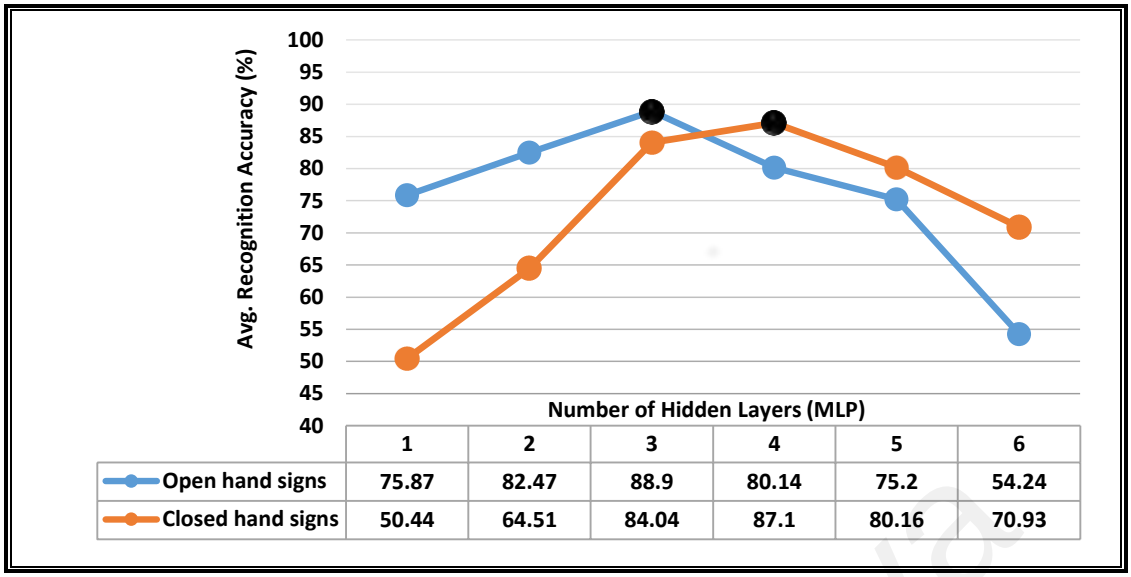


Figure 7.1: Optimal Number of Hidden Layers in MLP

7.2.1.4. Selection of the optimal number of features

All the imported signs are segmented into two classes (open hand and closed hand), and each class, in turn, is segmented into two sub-classes (letter signs and number signs). Each class consists of various features that differ from other extracted features. In machine learning, in order to enhance classification performance, the most effective features must be selected from the original set of features due to a large number of features extracted from the inputs. In other words, an optimal feature subset must be realised by ensuring that it describes the basic concept of the original set of features. The main benefit of this stage consists of the reduction in dimension, reduction in the time of executions or processing, enhancement of the classifier accuracy, reduction in the resource consumption and a decrease in the memory storage. The most revealed numbers of features for both closed and open hands are illustrated in Tables 7.2 and 7.3, respectively.

Universiti Malaya

Table 7.2: Features number for closed hand signs

Feature Number	Accuracy (%)
2	25.67
4	37.83
6	49.99
8	61.81
10	75.40
12	86.80
14	95.63
16	95.69 with long time in classification and recognition stage

Based on the changing number of features between 2 and 16, Table 7.2 shows that the system accuracy is directly related to the number of features until it reaches a specific threshold. When such a threshold value is exceeded, the system performance becomes worse. Hence, the best-recorded system accuracy for closed hand signs is attained when the number of the extracted features is equal to 14.

Table 7.3: Feature number for open hand signs

Feature Number	Accuracy (%)
5	43.15
10	56.82
15	67.54
20	76.41
25	87.19
30	95.63
35	95.69 with long time in classification and recognition stage

For the open hand case, the system accuracy is directly related to the number of features for a specific threshold number of feature attained. Exceeding such value, in turn, reduces the system speed. The number of the extracted feature is found to be 30 when the number of features is changed between 5 and 35.

Universiti Malaya

7.2.2 Performance Metrics of Static Backhand Closed Signs

The recorded evaluation matrices values for the Static Backhand closed signs are illustrated in Table 7.4.

Table 7.4: Evaluation matrices results for closed Hand signs

Signs	Sensitivity	Specificity	Precision	FPR	FNR	F1-score
ص	0.9400	0.9986	0.9495	0.0014	0.0600	0.9447
ض	0.9500	0.9986	0.9500	0.0014	0.0500	0.9500
ق	0.9500	0.9983	0.9406	0.0017	0.0500	0.9453
ف	0.9400	0.9986	0.9495	0.0014	0.0600	0.9447
ه-	0.9800	0.9994	0.9800	0.0006	0.0200	0.9800
و	0.9200	0.9994	0.9787	0.0006	0.0800	0.9485
zero	0.9800	0.9977	0.9245	0.0023	0.0200	0.9515

On the basis of the sensitivity results that reflect the recognition rate per class, the most correctly classified characters are the number zero (0) and the letter (ه-) due to their special shapes. For specificity, the result is as high as 99%, which means that the probability of any sign taken from other classes to be correctly recognised is about 0.99; in other words, the average recognition rate of the other classes (letters and numbers) is 99%. The highest precision value is recorded for the letter (ه-), making the proposed framework attain an accuracy rate of 98.00% in classifying such a letter. The opposite of specificity, sensitivity and precision values are the FPR and FNR, which stand for the errors in classification and recognition. Specifically, FPR and FNR represent errors in actual positives/negatives that are predicted positives/negatives. The minimum recorded errors in the classification whilst considering these FPR and FNR values are for the letter (ه-). The F1-score, in turn, is used to convey the balance among the sensitivity and precision. Hence, the best-recorded balance for the letter (ه-) is 98.00%.

Universiti Malaya

7.2.3 Performance Metrics of Static Backhand Open Sings

The recorded evaluation matrices values for open hand signs are illustrated in Table 7.5.

Table 7.5: Evaluation matrices results for open hand sign

Signs	Sensitivity	Specificity	Precision	FPR	FNR	F1-score
أ	0.9900	0.9994	0.9802	0.0006	0.0100	0.9851
ب	0.9600	0.9992	0.9697	0.0008	0.0400	0.9648
ت	0.9700	0.9989	0.9604	0.0011	0.0300	0.9652
ث	0.9500	0.9977	0.9223	0.0023	0.0500	0.9360
ج	0.9600	0.9983	0.9412	0.0017	0.0400	0.9505
ح	0.9300	0.9986	0.9490	0.0014	0.0700	0.9394
خ	0.9400	0.9983	0.9400	0.0017	0.0600	0.9400
د	0.9300	0.9983	0.9394	0.0017	0.0700	0.9347
ذ	0.9500	0.9983	0.9406	0.0017	0.0500	0.9453
ر	0.9100	0.9980	0.9286	0.0020	0.0900	0.9192
ز	0.9200	0.9983	0.9388	0.0017	0.0800	0.9293
س	0.9300	0.9969	0.8942	0.0031	0.0700	0.9118
ش	0.9500	0.9986	0.9500	0.0014	0.0500	0.9500
ط	0.9600	0.9986	0.9505	0.0014	0.0400	0.9552
ظ	0.9500	0.9986	0.9500	0.0014	0.0500	0.9500
ع	0.9600	0.9992	0.9697	0.0008	0.0400	0.9648
غ	0.9700	0.9992	0.9700	0.0008	0.0300	0.9700
ك	0.9300	0.9977	0.9208	0.0023	0.0700	0.9254
ل	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
م	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
ن	0.9900	0.9994	0.9802	0.0006	0.0100	0.9851
ي	0.9800	0.9997	0.9899	0.0003	0.0200	0.9849
1	0.9700	0.9994	0.9798	0.0006	0.0300	0.9798
2	0.9900	0.9994	0.9802	0.0006	0.0100	0.9851
3	0.9800	0.9992	0.9703	0.0008	0.0200	0.9751
4	0.9400	0.9986	0.9495	0.0014	0.0600	0.9447
5	0.9300	0.9997	0.9894	0.0003	0.0700	0.9588
6	0.9900	0.9997	0.9900	0.0003	0.0100	0.9900
7	0.9700	0.9994	0.9798	0.0006	0.0300	0.9749
8	0.9600	0.9992	0.9697	0.0008	0.0400	0.9648
9	0.9700	0.9994	0.9798	0.0006	0.0300	0.9749

The sensitivity results attained the highest value of 0.99, indicating that the highest probability of signs taken from class to be correctly recognised has a probability of 0.99. The most correctly classified characters are the numbers 2 and 6 and the letters ن & أ. For the specificity,)ل and م(achieved the highest value similar to the value recorded for all letters and numbers, which is equal to 0.99. This value means that the probability of

Universiti Malaya

recognising any number or letter sign is 99%. In other words, the average recognition rate is 99%. The precision, which stands for the proportion of predicted positives that are actually positive, is 100% for letters م ,ل. Thus, the system is completely correct in the classification of those letters.

The errors in classification are defined by the FPR (1-specificity) and FNR (1-sensitivity). Thus, zero error in classification is recorded for the letters (م ,ل). Considering the F1-score, the recorded balance (related to the lowest variation) amongst the sensitivity and precision attained the highest value for number 6.

Universiti Malaya

7.3. Results of Framework 2: Hybrid Approach for ArSLR Based on Data Combining

In this section, the system experimental setup and performance in each phase of the framework is presented. In addition, other subsections present the experimental result classification and the recognition outcomes, along with the inclusion of the comparison between the obtained framework results and those of the related works.

7.3.1. System Experimental Setups

This section describes the system experimental setup and then selects the optimal values of the parameters for pre-processing, feature extraction and input data devices used in the proposed framework. In addition, a description of the database utilised for training and testing is presented.

7.3.1.1. Training and Testing Phases

The training phase is applied in the learning to fit the classifier parameters, which mainly consists of the input layer, an output layer, hidden layers, stop error, learning rate and any other required parameters. For framework 1, this phase includes various iteration executions in order to obtain the minimum possible error, which is the difference between the expected response of the classifier and the actual one.

The testing phase uses the independent set on the training phase. However, both have the same probability distribution. The purpose of the testing phase is to assess the classifier performance using a set of inputs that are different from the one used in the training phase.

The database is divided into the ratios of 70% and 30% of the total datasets for the training and testing sets, respectively.

Universiti Malaya

7.3.1.2. Optimised Parameters for Pre-processing and Feature Extraction to Enhance System Performance Based on Smart Data Gloves Data

Two efficient techniques are applied to pre-process the signs before extracting features in order to enhance the collected data and to improve the feature vectors. This is done to ensure that the maximum possible recognition rate with the minimum possible recognition time is obtained. These techniques pass through the resampling and normalisation process, which is deployed to normalise the coordination of the system and the resampling and to decrease the number of feature vectors.

The output of the pre-processing experiment produced three types of feature vectors: the original feature vector without resampling, resampling factor with filtering using a ratio of 1:2 and resampling factor with filtering using a ratio of 1:3, as described in Section 5.3.1.1. Then, the feature extraction technique is applied on these types to extract the optimal resultant features used on the proposed framework.

In the feature extraction phase, the implementation of the results is based on many scenarios (as presented in Table 7.7) to identify the optimal parameters of the proposed framework. The result shows that the combination of Mean (**M**) and standard deviation (**SD**) features outperformed other techniques, especially when the normalisation and the resampling filtering are applied using a ratio of 1:2, which provided accurate results as presented in Table 7.6.

Universiti Malaya

Table 7.6: Optimized Parameters and Recognition Rates (%)

Type of FVs	Alphabet	Numbers	Isolated-words	Continuous Data 1	Continuous Data 2
Original FV with M	93.15%	94.53%	90.49%	91.65%	92.10%
Original FV with SD	92.90%	93.15%	88.84%	90.42%	89.70%
Original FV with M & SD	94.90%	95.16%	93.81%	94.65%	94.80%
Resampled (1:2) with M	94.00%	95.10%	91.40%	92.50%	93.16%
Resampled (1:2) with SD	93.70%	94.79%	90.16%	91.60%	91.95%
Resampled (1:2) with M & SD	**96.36%**	**97.29%**	**94.46%**	**97.12%**	**98.26%**
Resampled (1:3) with M	93.80%	94.90%	91.16%	92.80%	93.16%
Resampled (1:3) with SD	92.90%	93.50%	90.78%	91.50%	92.65%
Resampled (1:3) with M & SD	95.90%	96.85%	94.05%	96.73%	97.70%

7.3.1.3 Combining the hybrid approaches and selecting the optimal extracted features set

All the input data for framework 2 are recorded using smart data gloves and Microsoft Kinect. Four features are extracted by using the Microsoft Kinect device, namely, distance, elevation, curvature and palm area feature. Tables 7.7 to 7.11 show comparisons of the accuracy in using each single input device and the combination of all the input data classes.

Universiti Malaya

Table 7.7: Accuracy of letter signs in terms of extracted features set

Feature set	Accuracy
Smart Data Gloves	96.36
Smart Data Gloves & Distance features (Kinect)	97.12
Smart Data Gloves & Elevation features (Kinect)	96.90
Smart Data Gloves & Curvature features (Kinect)	97.81
Smart Data Gloves & Palm area features (Kinect)	96.56
Smart Data Gloves & Kinect (D+E+C+P)	99.38

Firstly, in the recognition of alphabet signs, the recognition accuracy obtained by using only smart data gloves as an input device is equal to 96.36%. When a combined approach (hybrid approach) consisting of smart data gloves with curvature features is applied using the curvature features as a single feature set, the best accuracy result of 97.81% is obtained. Moreover, the smart data gloves with distance feature set offered the second highest accuracy value of 97.12%. The elevation and palm area features set attained the accuracies of 96.90% and 96.56%, respectively. However, when these four feature sets are combined with the smart data gloves, an efficient accuracy of 99.38% is obtained.

Table 7.8: Accuracy of numbers signs in terms of extracted features type

Feature set	Accuracy
Smart Data Gloves	97.29
Smart Data Gloves & Distance features (Kinect)	98.30
Smart Data Gloves & Elevation features (Kinect)	97.80
Smart Data Gloves & Curvature features (Kinect)	98.56
Smart Data Gloves & Palm area features (Kinect)	97.60
Smart Data Gloves & Kinect (D+E+C+P)	99.42

Secondly, the number of signs recognition attained the recognition accuracy of 97.29% when only smart data gloves are used as an input device. When the combination of smart data gloves with curvature features is applied using the curvature features as a single feature set, the best accuracy result of 98.56% is obtained. In addition, the smart data gloves with distance feature set offered the second highest accuracy value of 98.30%, whereas the elevation and palm area feature set attained the accuracies of 97.80% and

Universiti Malaya

97.60%, respectively. Moreover, the combination of the four feature sets with smart data gloves yields an accuracy of 99.42%.

Table 7.9: Accuracy of isolated words in terms of extracted features type

Feature set	Accuracy
Smart Data Gloves	94.46
Smart Data Gloves & Distance features (Kinect)	95.97
Smart Data Gloves & Elevation features (Kinect)	95.16
Smart Data Gloves & Curvature features (Kinect)	96.64
Smart Data Gloves & Palm area features (Kinect)	94.90
Smart Data Gloves & Kinect (D+E+C+P)	99.15

Thirdly, the recognition accuracy of 94.46% is obtained by using smart data gloves as an input device for the recognition of isolated words. The combined approach (hybrid approach), which comprises mart data gloves with curvature features (Kinect), obtained the best accuracy of 96.64% when the curvature features are used as a single feature set. In addition, the second highest accuracy value of 95.97% is obtained by using the smart data gloves with the distance feature set, whereas the elevation and palm area feature sets attained the accuracies of 95.16% and 94.90%, respectively. However, the ensemble of the four feature sets with smart data gloves produced an efficient accuracy of 99.15%.

Table 7.10: Accuracy continuous dataset 1 in terms of extracted features type

Feature set	Accuracy
Smart Data Gloves	97.12
Smart Data Gloves & Distance features (Kinect)	97.93
Smart Data Gloves & Elevation features (Kinect)	97.67
Smart Data Gloves & Curvature features (Kinect)	98.10
Smart Data Gloves & Palm area features (Kinect)	97.43
Smart Data Gloves & Kinect (D+E+C+P)	99.25

Fourthly, the continuous dataset 1 recognition obtained a recognition accuracy of 97.12% using smart data gloves alone as an input device. When the curvature features are used as a single feature set by applying the hybrid approach consisting of smart data gloves with curvature features, the best accuracy result of 98.10% is obtained. Moreover, the smart data gloves with distance feature set offered the second highest accurate value

Universiti Malaya

of 97.93%. The elevation and palm area feature sets attained the accuracies of 97.67% and 97.43%, respectively, whereas the combination of the four feature sets with smart data gloves attained 99.25% accuracy.

Table 7.11: Accuracy of continuous dataset 2 in terms of extracted features type

Feature set	Accuracy
Smart Data Gloves	98.26
Smart Data Gloves & Distance features (Kinect)	98.70
Smart Data Gloves & Elevation features (Kinect)	98.54
Smart Data Gloves & Curvature features (Kinect)	98.97
Smart Data Gloves & Palm area features (Kinect)	98.40
Smart Data Gloves & Kinect (D+E+C+P)	99.55

Lastly, 98.26% recognition accuracy was obtained on the recognition of continuous dataset 2 using only smart data gloves as an input device. The best accuracy result of 98.97% is obtained by applying the combined approach (hybrid approach), which consists of smart data gloves with curvature features, using the curvature features as a single feature set. The elevation and palm area feature sets attained the accuracies of 98.54% and 98.40%, respectively, and the smart data gloves with distance feature set offered the second highest accuracy value of 98.70%. Moreover, an accuracy of 99.55% is obtained when the four feature sets are combined with the smart data gloves.

The summary of the recorded accuracy percentages related to each data type in terms of the extracted feature set is presented in Figure 7.2.

Universiti Malaya

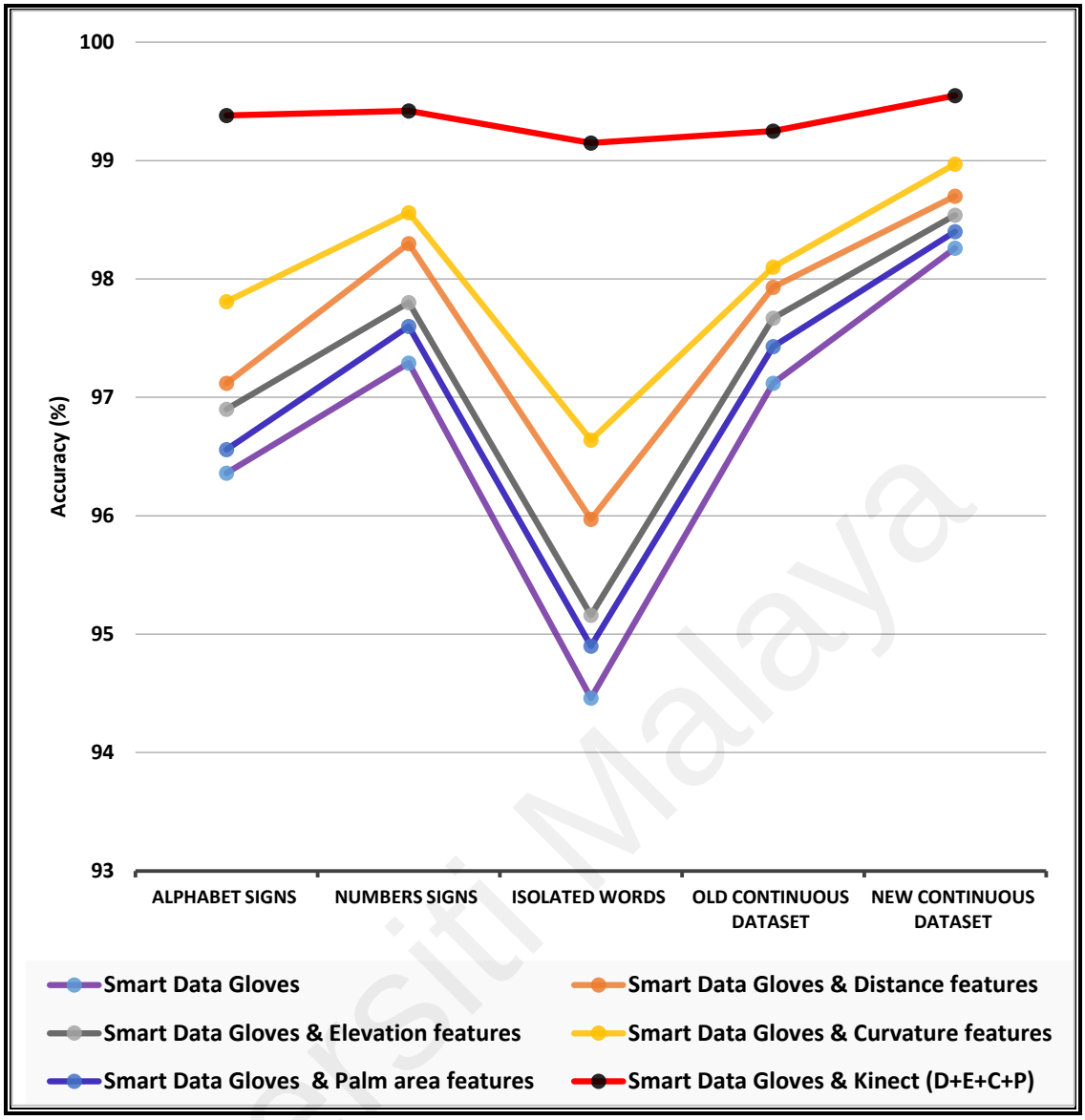


Figure 7.2 Recorded accuracy percentages related to each data type in terms of extracted features set

In conclusion, as shown in Tables 7.8 to 7.12, using the hybrid approach (vision and sensor-based) by combining feature sets extracted from input signs in the classification and recognition offers more accurate results than using any single device or feature type. Results also show that the curvature features achieved the best recognition results when combined with smart data gloves, which does not depend on the positions of the centroid and palm plane or the computation of the hand orientation. Therefore, the proposed approach performs better in complex configurations.

Universiti Malaya

7.3.2 Performance metrics results for hybrid Approach

The computed evaluation performance metrics for the five input datasets are shown in Tables 7.12 to 7.16, respectively.

Table 7.12: Computed evaluation performance metrics for Arabic letters

Signs	Sensitivity	Specificity	Precision	FPR	FNR	F1-score
G1	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G2	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G3	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G4	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G5	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G6	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G7	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G8	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G9	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G10	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G11	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G12	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G13	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G14	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G15	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G16	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G17	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G18	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G19	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G20	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G21	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G22	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G23	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G24	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G25	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G26	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G27	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G28	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G29	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G30	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G31	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G32	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G33	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G34	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G35	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G36	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G37	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G38	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G39	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000

Universiti Malaya

As shown in Table 7.12, a sensitivity value of 1.00 is recorded for most of the input Arabic letters signs. The highest value implies that the probability of a sign taken from the letters class to be correctly recognised is high (probability is 1). In other words, almost all the input signs are correctly recognised by each classifier. For the specificity value, the value of 1.00 is recorded for all signs. This implies that the probability of recognising any sign correctly on the proposed framework using the dataset is 1.00. The precision value of 1.00 means the percentage of truly positive signs among all signs is classified as positive (100%). Thus, it computes the proportion of all predicted positives that are actually positive.

The FPR and the FNR measure the classification errors, wherein those metrics stand for the approach error in classification. They are the proportions of the actual positives/negatives that are predicted as positives/negatives, respectively. The obtained values of those metrics are almost zero, indicating that there is no error in the classification of the positives with a slight error in the classification of the negatives. The F1-score, in turn, is deployed to convey the balances among the sensitivity and precision. Hence, the recorded balance for all gestures is between 99% and 100%. This proves the effectiveness of the system in recognising all the input alphabet gestures without any classification error.

Universiti Malaya

Table 7.13: Computed evaluation performance metrics for Arabic Numbers

Signs	Sensitivity	Specificity	Precision	FPR	FNR	F1-score
G40	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G41	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G42	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G43	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G44	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G45	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G46	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G47	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G48	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G49	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G50	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G51	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G52	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G53	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G54	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G55	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G56	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G57	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G58	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G59	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G60	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000

In Arabic number gestures for sensitivity, which reflects the recognition rate per class, results show that the value of 1.00 is recorded for almost all the input signs. Thus, the system recognised all the inputs for Arabic number signs. The specificity and precision value of 1.00 are recorded for all the input signs. Hence, the framework has a 100% probability in recognising and classifying any sign from the numbers dataset.

The approach error in classification is measured using the FPR (1-specificity) and the FNR (1-sensitivity). The FPR and FNR are the proportions of actual positives/negatives that are actually predicted positives/negatives. The F1-score is used to convey the balance among the sensitivity and precision. Hence, the recorded balance values for all the number of signs ranged between 0.98 and 1.00. In such balance, zero error and 100% classification percentage reveal the system's efficiency in classifying and recognising all the input number signs without any classification error.

Universiti Malaya

Table 7.14: Computed evaluation performance metrics for Arabic Isolated-Words

Signs	Sensitivity	Specificity	Precision	FPR	FNR	F1-score
G61	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G62	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G63	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
G64	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G65	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G66	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G67	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G68	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
G69	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G70	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G71	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G72	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G73	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G74	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G75	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G76	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G77	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G78	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G79	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G80	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G81	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G82	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G83	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G84	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G85	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G86	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G87	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G88	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G89	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G90	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G91	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G92	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G93	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G94	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G95	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G96	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G97	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G98	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
G99	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G100	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G101	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G102	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G103	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G104	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G105	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G106	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G107	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G108	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G109	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G110	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000

Universiti Malaya

G111	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G112	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G113	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G114	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
G115	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
G116	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G117	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G118	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G119	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G120	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000

For isolated-word signs, the average recorded values for sensitivity ranged between 0.97 and 1.00. Such sensitivity ranges revealed that the framework was able to recognise almost all the input of isolated-words signs. In addition, a percentage of 100% is recorded for both the specificity and precision metrics for all the input signs. However, these values prove that the framework can completely recognise and classify all the input signs from the isolated-word dataset.

The error in classification is measured using two metrics consisting of FPR (1-specificity) and the FNR (1- sensitivity) with the optimal values of 0%. The FPR and FNR are the proportions of actual positives/negatives that are actually predicted positives/negatives. The FPR value is 0% for all the inputs, and this reveals that there was no error in the classification of positives without any truly null significant sign. A slightly small error is recorded in the classification of negatives based on the FNR values.

The F1-score is utilised to convey the balance between sensitivity and precision. Hence, the recorded balance values for all the isolated-word signs ranged between 0.98 and 1.00. The achieved balance for the sensitivity and precision record of zero error percentage and 100% classification reveal that the framework can efficiently classify and recognise all the input isolated-word signs.

Universiti Malaya

Table 7.15: Computed evaluation performance metrics for Arabic continuous dataset 1

Signs	Sensitivity	Specificity	Precision	FPR	FNR	F1-score
G121	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G122	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G123	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G124	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G125	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G126	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G127	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G128	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G129	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G130	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G131	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G132	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G133	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G134	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G135	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G136	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G137	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G138	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
G139	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G140	0.9700	1.0000	1.0000	0.0000	0.0300	0.9848
G141	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G142	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G143	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G144	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G145	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G146	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G147	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G148	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G149	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G150	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G151	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G152	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G153	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G154	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G155	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G156	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G157	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G158	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G800	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G160	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000

For continuous dataset 1, a sensitivity percentage between 97% and 100% is recorded for the input Arabic old database sentences. This proves that the framework can recognise nearly all the input continuous dataset 1. The recorded percentages of the specificity and precision metrics for all the input gestures are both 100%. Hence, the framework completely recognised and classified all the input continuous dataset 1.

Universiti Malaya

As in previous data cases, the error in the classification is measured using two metrics which consist of FPR (1-specificity) and the FNR (1- sensitivity) with their optimal value of 0%. The FPR and FNR are the proportions of the actual positives/negatives that are actually predicted positives/negatives. The FPR value is 0% for all the inputs. Thus, there is no classification error of positives and no truly null significant signs. A little error between 1% and 3% is recorded in the classification of negatives based on the FNR values, which are 1- sensitivity.

The F1-score is utilised to convey the balance between sensitivity and precision. Hence, the recorded balance values of all the continuous dataset 1 ranged between 98% and 100%. It can be summarised that the framework is completely efficient in classifying and recognising all the input signs for continuous dataset 1.

Table 7.16: Computed evaluation performance metrics for Arabic continuous dataset 2

Signs	Sensitivity	Specificity	Precision	FPR	FNR	F1-score
G161	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G162	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G163	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G164	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G165	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G166	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G167	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G168	0.9800	1.0000	1.0000	0.0000	0.0200	0.9899
G169	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G170	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G171	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G172	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G173	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G174	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G175	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G176	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G177	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950
G178	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G179	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000
G180	0.9900	1.0000	1.0000	0.0000	0.0100	0.9950

For the continuous dataset 2, the recorded sensitivity percentages ranged between 98% and 100% indicating that nearly all the continuous dataset 2 are recorded efficiently.

Universiti Malaya

The percentage of both specificity and precision metrics for all the input signs is 100%. Thus, the framework is capable of recognising and classifying all the input signs for continuous dataset 2.

For the new database sentences, the error in classification is measured using two metrics that consist of FPR (1-specificity) and the FNR (1- sensitivity) with 0% as their optimal value. The FPR and FNR are the proportion of actual positives/negatives that are predicted positives/negatives. The FPR value is 0% for all the inputs; thus, there is no classification error of positives and no truly null significant signs. In contrast, a little error ranging between 1% and 2% is recorded in the classification of negatives based on the FNR values.

The F1-score is utilised to convey the balance between sensitivity and precision. Hence, the recorded balance values of all the continuous dataset 2 ranged between 99% and 100%. It can be summarised that the framework efficiently classified and recognised all the input gestures for continuous dataset 2.

Finally, all the datasets recorded almost 100% percentage for sensitivity, specificity and precision metrics. The FPR, which stands for the misclassified rate, recorded almost zero percentage for all five classes with slightly FNR percentages due to the misclassification of negatives. Moreover, the F1-score results prove that the proposed framework is efficient and capably balanced the recorded sensitivity and precision for all cases.

Universiti Malaya

7.4 Result Discussion

In this section, the results of the evaluation of the proposed ArSL framework are presented and discussed.

7.4.1. Experimental Results and Discussions for Framework 1

The proposed framework is evaluated on the basis of an experimental evaluation of its ability to recognise input data by matching them with their corresponding signs. In this framework, 38 static signs for backhand finger numbers and letters are deployed. In practice, some numbers and letters are clearly different, whilst others have identical hand shapes with various meanings. As a result, the framework may attain low recognition accuracy for ambiguous signs when it utilises the same approach for the recognition of all the signs. Consequently, sign grouping is utilised to solve the problem of ambiguity among signs by dividing identical signs into groups (closed and open hand sign groups). Therefore, the system can select a suitable approach to address the signs in every group.

Furthermore, three classifiers are applied in the classification phase of the framework (e.g. NB and MLP). The details of these classifiers are described in Chapter 6. The recorded accuracy and other basic evaluation metrics related to each classifier are illustrated in Section 7.4.1.1. In addition, the obtained results of both closed and open hand signs cases are presented and discussed.

7.4.1.1. Recorded Accuracy and Evaluation Metrics for Applied Classifiers

Each classifier is initially trained and tested using randomly selected training and testing sets by ensuring that each set has signs from all classes and all signers. The recognition accuracy of each classifier is then recorded, as shown in Table 7.17 and Figure 7.3. Next, the three classifiers are combined to form the ensemble classifier and its accuracy is recorded and compared with those of single classifiers.

Universiti Malaya

Table 7.17: Accuracy of classification techniques based on all dataset size

Classification Techniques	Accuracy (%)
NB	86.30
MLP	88.00
NB + MLP (Our Framework)	95.63

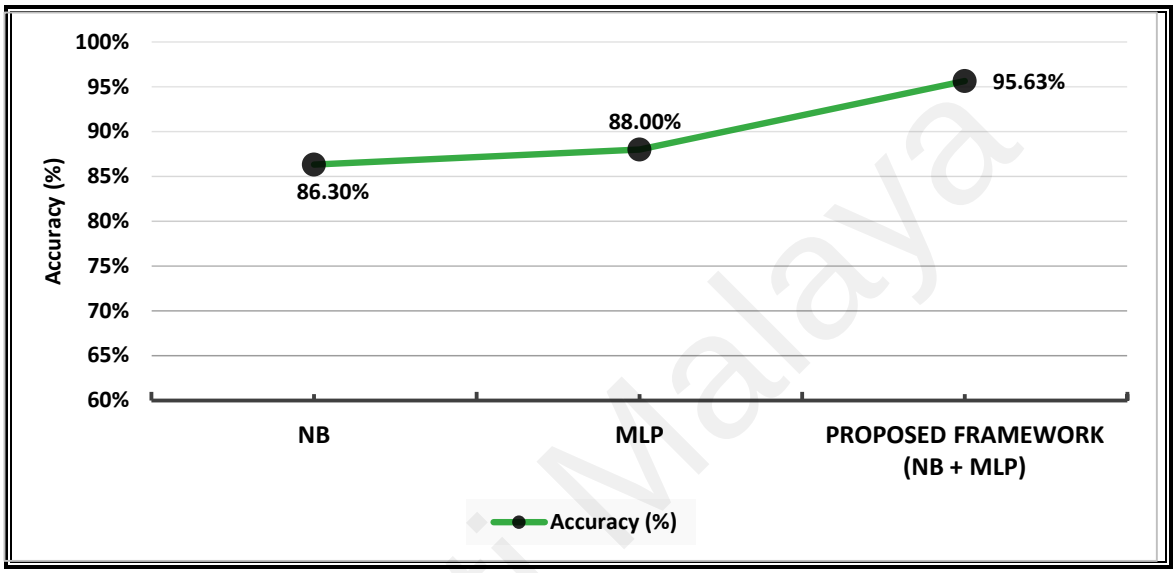


Figure 7.3: Accuracy of classification techniques

As shown in Table 7.17 and Figure 7.3, the MLP classifier recorded the highest classification accuracy (88%) compared with other classifiers. In contrast, the proposed framework (multiple classifiers) outperforms the three single classifiers as it combines the benefits inherent from each classifier, thus serving as a solution to their limitations. The recognition accuracy of our proposed framework is 95.63%.

7.4.1.2 Confusion Matrix of Closed hand signs

Table 7.18 and Figure 7.4 illustrates the confusion matrix of the closed hand signs for database 1.

Universiti Malaya

Table 7.18: Confusion matrix of closed hand signs

	ص	ض	ق	ف	هـ	و	Zero	No detection	No. of misclassified samples
ص	**94**	3			2			1	**6**
ض	5	**95**							**5**
ق			**95**	5					**5**
ف			6	**94**					**6**
هـ					**98**		2		**2**
و						**92**	6	2	**8**
Zero					1		**98**	1	**2**
Total									**34**

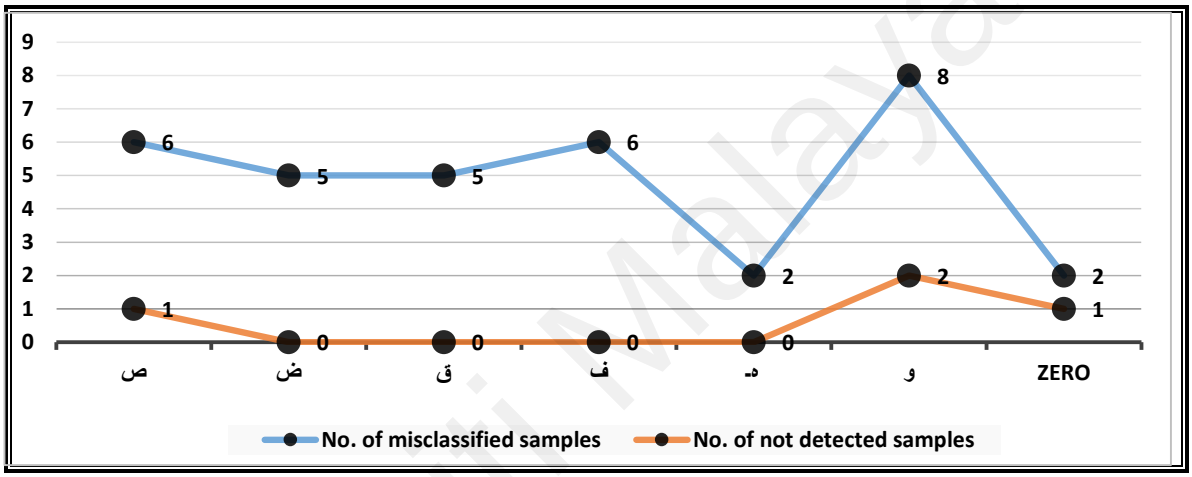


Figure 7.4: Recognition rates of closed hand signs

In the closed hand case, some signs of letter ص are wrongly recognised as ض and هـ. This is due to the sharing of similar hand and closed fingers. The same issue is found in letters (ص and ض), (ق and ف), (هـ , و) and the number zero. Only one of the **ص** and zero signs and two of the **و** signs are not detected. This means that the classifier could not exclusively recognise the sign due to the highest shared likelihood value among two or more signs.

7.4.1.3 Confusion matrix of open hand signs

Table 7.19 and Figure 7.5 illustrates the confusion matrices for open hand signs.

Universiti Malaya

Table 7.19: Confusion matrix of Open hand signs

	أ	ب	ت	ث	ج	ح	خ	د	ذ	ر	ز	س	ش	ط	ظ	ع	غ	ك	ل	م	ن	ي	1	2	3	4	5	6	7	8	9	No detection	No. of misclassified samples
أ	**99**																1																1
ب		**96**	2	1																												1	4
ت		1	**97**	2																													3
ث			2	**95**								1						1							1								5
ج					**96**	2	2																										4
ح					3	**93**	4																										7
خ					3	3	**94**																										6
د								**93**	3	2	2																						7
ذ								3	**95**	1																						1	5
ر								2	2	**91**	3			1	1																		9
ز								1	1	2	**92**			1	2																	1	8
س												**93**	2					3							2								7
ش												2	**95**													2					1		5
ط										1	1			**96**	2																		4
ظ										1				3	**95**																	1	5
ع																**96**	2															2	4
غ																3	**97**																3
ك				2								3						**93**								2							7
ل																			**98**										1			1	2
م	1																			**97**			2										3
ن																					**99**	1											1
ي																					2	**98**											2
1		1																					**97**	1								1	3
2																								**99**						1			1
3				2																					**98**								2
4												3	1					2								**94**							6
5												2	2					2								1	**93**						7
6																												**99**				1	1
7	1	1																										1	**97**				3
8																								1					1	**96**	1	1	4
9				1																							1			2	**97**		3
Total																																	**132**

Universiti Malaya

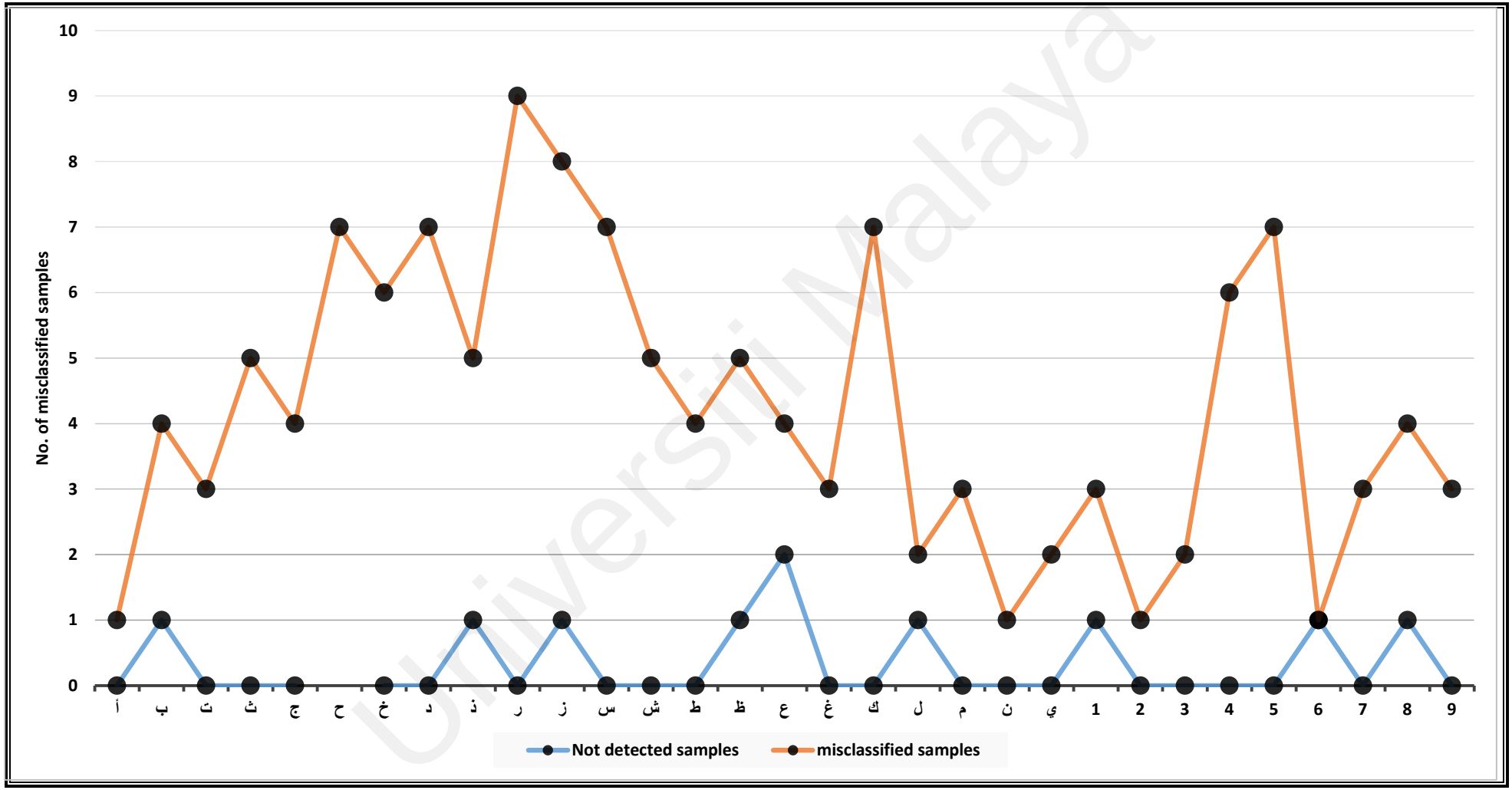


Figure 7.5: Recognition rate of open hand signs

Universiti Malaya

In the open hand case, the misclassification occur among أ and غ as both letters have a small part above the letter body. Another misclassification is recorded among ,ج( )ث,ب( )غ ,ع(,)ظ ,ط(,)ش ,س(,)ز ,ر(,)ذ ,د(dna)خ ,ح as each group shared the same hand and open fingers. The sign gesture 'ر' is incorrectly recognised as 'ز ,ذ ,د' because these are performed using one finger and the same hand. Similar signs are also observed in rows 10 and 11. In row 2 of the table, the sign gesture 'ز' is wrongly recognised as the signs for 'ر, ط', because both gestures shared similar open fingers. Likewise, in rows 4, 5, 6, 7 and 8, the signs are very similar.

A few number of signs related to the letters ل ,ز ,ذ ع , ظ,ب and the numbers 6 and 8 are not detected. This means that the classifier cannot exclusively recognise the sign due to the highest shared likelihood value among the two or more signs.

Next, the accuracy rate is calculated by the equation in Chapter 3, Section 3.5 (Evaluation Phase):

$$\textbf{Recognition rate} = \frac{\textbf{Number of correctly recognized signs}}{\textbf{Total number of signs}} \times 100\% \qquad \textbf{(3.1)}$$

The n**umber of total signs = # of signs × # of singers × # of repetitions**.

Number of total signs = 38 × 10 ×10 = **3800 sample signs**.

Based on Tables 7.18 and 7.19 and Figures 7.5 and 7.6,

Number of misclassified samples + Not detected sample = **166 simple signs,**

Number of correctly recognised = 3800-166= **3634 sample signs.**

All the input equation values are thus obtained.

Accuracy rate= (3634 / 3800) * 100%

Accuracy rate= 0.9563 * 100%

Accuracy rate= 95.63%

In conclusion, the results prove that the proposed system recorded high accuracy in classifying and recognising letters and numbers for both cases of the closed and open hand signs. The average classification for overall recognition accuracy is 95.63%. In

Universiti Malaya

contrast, the total number of misclassified samples for the closed and open hand signs are 34 and 132, respectively. This large difference in the number of misclassified samples is made possible as the number of open hand signs (31) is much larger than that of closed hand signs (7).

7.4.1.4 Error Analysis for open and closed hand signs cases

This section presents the details of the ArSL letters and numbers that are not always correctly recognised by the proposed framework. Low recognition rate may have occurred because the signers made certain hand gestures ineffectively or some signs are very similar. Some letter signs that are confused by the proposed framework are shown in Table 7.20.

Table 7.20: Misclassified letter signs by the proposed system

#NO.	Actual letter	Misclassified as	Error (%) Total number of times
1.	**ر**	د, ذ, ز	9
2.	**ز**	ر, ط	8
3.	**ح**	خ, ج	7
4.	**خمسة)5(**	س, ش, ك	7
5.	**د**	ذ , ر, ز	7
6.	**س**	ك, ش, 3(	7
7.	**ك**	س, ث, اربعة)4(	7
8.	**اربعة)4(**	س, ش, ك	6
9.	**خ**	ح, ج	6
10.	**ث**	ت , س , ك	5
11.	**ب**	ت	4

The error percentage column shows the total number of misclassifications that occur for each letter presented in the actual letter column. For the sign gesture 'ر', it is wrongly recognised as 'د, ذ, ز'. This is made possible as both sign gestures are performed using one finger. In row 2 of Table 7.20, the sign gesture 'ز' is incorrectly recognised as the signs for 'ر, ط' due to the sharing of a similar open finger and hand by both gestures. In rows 3 and 9, the signs for 'ح and خ' are wrongly recognised as the signs for 'خ, ج and ح, ج' respectively, because of the sharing of the similar hand and open finger by both gestures. Such similarities of signs are also found in rows 4 to 8, 10 and 11. The highest

University Malaya

number of misclassification time is attained by the sign gesture 'ر', which has high similarities with other letters, whilst the lower one is attained by the sign gesture 'ب', which shares the same shape (sign shape) with the sign gesture 'ت'.

7.4.1.5 Comparison with related works

In this section, a comparison of the proposed framework with some related works reviewed in Chapter 2 is presented.

7.4.1.5.1 Comparison with ArSL frameworks

Table 7.21 below shows a comparison of the proposed framework and some of the vision-based related works reviewed in Chapter 2.

Table 7.21: Comparison among our framework and other ArSL frameworks

Ref.	Number of signs	Features Classification	Signer Dependency	Recognition Rate (%)
Our Framework	**38**	**NBC & MLP**	**Signer-Independent**	**95.63%**
(Samir Elons et al., 2013)	30	PCNN	Signer-Dependent	90%
(Maraqa et al., 2008)	28	Elman network algorithms	Signer-Dependent	89.66%
(Assaleh et al., 2005)	30	polynomial classifier	Signer-Dependent	93.41%
(Mohandes, 2001)	28	SVM	Signer-Dependent	87%

It is obvious from Table 7.21 that the highest number of signs are used for the training and testing of the proposed framework, which includes 38 signs divided into 28 Arabic letters (represented by one static sign and the other 10 Arabic number). Meanwhile, the number of signs used in other works range from 28 to 30. Concerning the classification phase, the proposed framework outperforms the baseline in using the multiple classifiers

University Malaya

for the sign classification compared to other works that used single classifiers. For example, Elons et al. (2013) used the PCNN classifier, Maraqa et al. (2008) used the Elman network algorithms, Assaleh et al. (2005) used polynomial classifiers and Mohandes (2001) used SVM. Another difference between the proposed framework and the related works is that the former is signer-independent in which 10 different signers with different ages, experiences and situations are selected, whereas other works are signer-dependent. This is considered a critical contribution to the proposed framework. The proposed framework records the highest recognition accuracy of 95.63%.

7.4.1.5.2 Comparison with Other Languages

Table 7.22 presents the comparison of the performance of the proposed framework with other related works applied to non-Arabic languages.

Table 7.22: Comparison among our framework and related works on other languages

Ref.	Scope	Classification	Accuracy (%)
Our Framework	**ArSL**	**Multi-classifier**	**95.63**
(Abhishek et al., 2016)	ASL	Finger Posture Recognition	92.00
(Pan et al., 2016)	ASL	SVM	94.00
(Huong et al., 2015)	VSL	Template matching by Mahalanobis distance	91.50
(Funasaka et al., 2015)	ASL	Decision tree	82.71
(Jasim et al., 2014)	Bangla SL	KNN	88.60
(Hartanto et al., 2014)	Indonesian SL	Euclidian distance & Template matching	63.00
(Rajathi et al., 2013)	TSL	Proximal SVM	91.00

Those other works presented for recognition include the following: American Sign Language (ASL) by Abhishek (2016) and Pan (2016), Vietnamese Sign Language (VSL)

Universiti Malaya

by Huong et al. (2015), ASL by Funasaka et al. (2015), Bangla Sign Language (Bangla SL) by Jasim et al. (2014), Indonesian Sign Language (Indonesian SL) by Hartanto et al. (2014) and Taiwan Sign Language (TSL) by Rajathi et al. (2013). Among these works, the highest recorded accuracy of 95.63% is attained by the proposed framework. This is as a result of the use of multiple feature extractors and multiple classifiers in order to enhance the recognition rate.

7.4.2 Experimental Results and Discussions for Framework 2

7.4.2.1 Confusion Matrices for Different Input Datasets

In order to obtain a more accurate experimental analysis, the confusion matrix related to the recognition of each type of dataset is presented in Tables 7.23 to 7.27. Such matrices show that the recognition of some gestures is not always optimal for all the data types. The related confusion matrix for each data type is listed in Tables 7.23 to 7.27.

Universiti Malaya

Table 7.23: Confusion matrix for Arabic letters

	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	G16	G17	G18	G19	G20	**No detection**	**No. of misclassified samples**
G1	**99**																				**1**	**1**
G2		**99**																			**1**	**1**
G3			**100**																			**0**
G4				**100**																		**0**
G5					**99**																**1**	**1**
G6						**99**															**1**	**1**
G7							**98**														**2**	**2**
G8								**100**														**0**
G9									**99**												**1**	**1**
G10										**100**												**0**
G11											**99**										**1**	**1**
G12												**99**									**1**	**1**
G13													**99**								**1**	**1**
G14														**98**							**2**	**2**
G15															**99**						**1**	**1**
G16																**99**					**1**	**1**
G17																	**98**				**2**	**2**
G18																		**100**				**0**
G19																			**100**			**0**
G20																				**100**		**0**

Universiti Malaya

Cont.

	G21	G22	G23	G24	G25	G26	G27	G28	G29	G30	G31	G32	G33	G34	G35	G36	G37	G38	G39	No detection	No. of misclassified samples
G21	**100**																				**0**
G22		**99**																			**1**
G23			**99**																		**1**
G24				**100**																	**0**
G25					**100**																**0**
G26						**100**															**0**
G27							**100**														**0**
G28								**100**													**0**
G29									**100**												**0**
G30										**98**										**2**	**2**
G31											**99**									**1**	**1**
G32												**100**									**0**
G33													**99**							**1**	**1**
G34														**100**							**0**
G35															**99**						**1**
G36																**99**					**1**
G37																	**100**				**0**
G38																		**100**			**0**
G39																			**100**		**0**
Total																					**24**

Universiti Malaya

The confusion matrix for Arabic alphabet characters illustrated in Table 7.23 shows that there is no confusion among the alphabet gestures. All the recorded errors during the recognition are due to ‘no detection cases’, which represent the inability of the classifier to uniquely identify some gestures due to the high shared likelihood value among two or more gestures. Such gestures are G1, G2, G5, G6, G7, G9, G11-G17, G30, G31 and G33. The total number of misclassified samples is 24 out of 3900 samples; thus, the recognition accuracy is 99.38% with 0.6% error in recognition.

Universiti Malaya

Table 7.24: Confusion matrix for Arabic Numbers

	G40	G41	G42	G43	G44	G45	G46	G47	G48	G49	G50	G51	G52	G53	G54	G55	G56	G57	G58	G59	G60	No detection	No. of misclassified samples
G40	**99**																					**1**	**1**
G41		**99**																				**1**	**1**
G42			**98**																			**2**	**2**
G43				**99**																		**1**	**1**
G44					**99**																	**1**	**1**
G45						**100**																	**0**
G46							**99**															**1**	**1**
G47								**100**															**0**
G48									**99**													**1**	**1**
G49										**100**													**0**
G50											**100**												**0**
G51												**100**											**0**
G52													**100**										**0**
G53														**100**									**0**
G54															**99**							**1**	**1**
G55																**100**							**0**
G56																	**100**						**0**
G57																		**99**				**1**	**1**
G58																			**100**				**0**
G59																				**98**		**2**	**2**
G60																					**100**		**0**
Total																							**12**

Universiti Malaya

Table 7.24 shows the confusion matrix that relates to the Arabic number characters. As can be clearly seen, there is no confusion among number gestures. However, 12 of the input samples are misclassified due to ‘no detection errors’, in which the classifier could not identify some gestures uniquely because they share the same highest likelihood value among two or more gestures. Such gestures are G40-G44, G46, G48, G54, G57 and G59. The total number of misclassified samples is 12 out of 2100 samples with the recognition accuracy of 99.42% with 0.5% error in recognition.

Universiti Malaya

Table 7.25: Confusion matrix for Arabic Isolated-Words

	G61	G62	G63	G64	G65	G66	G67	G68	G69	G70	G71	G72	G73	G74	G75	G76	G77	G78	G79	G80	No detection	No. of misclassified samples
G61	**98**																				**2**	**2**
G62		**98**																			**2**	**2**
G63			**97**																		**1**	**1**
G64				**99**																		**0**
G65					**100**																	**0**
G66						**100**																**0**
G67							**98**														**2**	**2**
G68								**97**													**3**	**3**
G69									**99**												**1**	**1**
G70										**100**												**0**
G71											**100**											**0**
G72												**100**										**0**
G73													**98**								**2**	**2**
G74														**99**							**2**	**2**
G75															**98**						**2**	**2**
G76																**98**					**2**	**2**
G77																	**100**					**0**
G78																		**100**				**0**
G79																			**100**			**0**
G80																				**98**	**2**	**2**

Universiti Malaya

Cont.

	G81	G82	G83	G84	G85	G86	G87	G88	G89	G90	G91	G92	G93	G94	G95	G96	G97	G98	G99	G100	No detection	No. of misclassified samples
G81	**100**																					**0**
G82		**100**																				**0**
G83			**100**																			**0**
G84				**100**																		**0**
G85					**98**																**2**	**2**
G86						**98**															**2**	**2**
G87							**100**															**0**
G88								**99**													**1**	**1**
G89									**99**												**1**	**1**
G90										**100**												**0**
G91											**98**										**2**	**2**
G92												**100**										**0**
G93													**100**									**0**
G94														**99**							**1**	**1**
G95															**100**							**0**
G96																**100**						**0**
G97																	**100**					**0**
G98																		**97**			**3**	**3**
G99																			**99**		**1**	**1**
G100																				**100**		**0**

Universiti Malaya

Cont.

	G101	G102	G103	G104	G105	G106	G107	G108	G109	G110	G111	G112	G113	G114	G115	G116	G117	G118	G119	G120	No detection	No. of misclassified samples
G101	**99**																				**1**	**1**
G102		**100**																				**0**
G103			**100**																			**0**
G104				**100**																		**0**
G105					**98**																**2**	**2**
G106						**99**															**1**	**1**
G107							**100**															**0**
G108								**100**														**0**
G109									**99**												**1**	**1**
G110										**100**												**0**
G111											**99**										**1**	**1**
G112												**98**									**2**	**2**
G113													**100**									**0**
G114														**97**							**3**	**3**
G115															**97**						**3**	**3**
G116																**99**					**1**	**1**
G117																	**100**					**0**
G118																		**99**			**1**	**1**
G119																			**99**		**1**	**1**
G120																				**100**		**0**
Total																						**51**

Universiti Malaya

Table 7.25 illustrates the confusion matrix related to Arabic isolated-word characters. As can be seen, there is no confusion among the isolated-word gestures. The misclassification error is due to the inability of the classifier to identify some gestures uniquely because they share the same highest likelihood values among two or more gestures. Those gestures are G85, G86, G88, G89, G91, G94, G98, G99, G101, G105, G106 as well as G109, G111, G112, G114-116, G118 and G119. The total number of misclassified samples is 51 out of 4,000 samples. Thus, the recognition accuracy of 98.72% with 1.257% recognition error is attained.

Universiti Malaya

Table 7.261: Confusion matrix for Arabic continuous dataset 1

	G121	G122	G123	G124	G125	G126	G127	G128	G129	G130	G131	G132	G133	G134	G135	G136	G137	G138	G139	G140	No detection	No. of misclassified samples
G121	**99**																				**1**	**1**
G122		**98**																			**2**	**2**
G123			**100**																			**0**
G124				**98**																	**2**	**2**
G125					**100**																	**0**
G126						**99**															**1**	**1**
G127							**100**															**0**
G128								**99**													**1**	**1**
G129									**100**													**0**
G130										**99**											**1**	**1**
G131											**98**										**2**	**2**
G132												**99**										**1**
G133													**100**									**0**
G134														**99**							**1**	**1**
G135															**100**							**0**
G136																**100**						**0**
G137																	**100**					**0**
G138																		**97**			**3**	**3**
G139																			**100**			**0**
G140																				**97**	**3**	**3**

Universiti Malaya

Cont.

	G141	G142	G143	G144	G145	G146	G147	G148	G149	G140	G151	G152	G153	G154	G155	G156	G157	G158	G159	G160	No detection	No. of misclassified samples
G141	**99**																				**1**	**1**
G142		**98**																			**2**	**2**
G143			**100**																			**0**
G144				**100**																		**0**
G145					**98**																**2**	**2**
G146						**100**																**0**
G147							**100**															**0**
G148								**99**													**1**	**1**
G149									**98**												**2**	**2**
G150										**100**												**0**
G151											**100**											**0**
G152												**100**										**0**
G153													**99**								**1**	**1**
G154														**100**								**0**
G155															**98**						**2**	**2**
G156																**100**						**0**
G157																	**100**					**0**
G158																		**100**				**0**
G159																			**99**		**1**	**1**
G160																				**100**		**0**
Total																						**30**

Universiti Malaya

Table 7.26 shows the confusion matrix related to the Arabic continuous dataset 1. As can be seen, there is no confusion in the continuous dataset 1. The misclassification error related to those gestures is due to the inability of the classifier to recognise some gestures uniquely because they share the same highest likelihood values among two or more gestures. Those gestures are G121, G122, G124, G126, G128, G130, G131, G134, G138, G140, G141, G142, G145, G145, G148, G149, G153, G155 and G159. The total number of misclassified samples is 30 out of 4000 samples; thus, the recognition accuracy is 99.25% with 0.75% error in recognition.

Universiti Malaya

Table 7.27: Confusion matrix for Arabic continuous dataset 2

	G161	G162	G163	G164	G165	G166	G157	G168	G169	G170	G171	G172	G173	G174	G175	G176	G177	G178	G179	G180	No detection	No. of misclassified samples
G161	**100**																					**0**
G162		**100**																				**0**
G163			**100**																			**0**
G164				**98**																	**2**	**2**
G165					**100**																	**0**
G166						**99**															**1**	**1**
G167							**100**															**0**
G168								**98**													**2**	**2**
G169									**100**													**0**
G170										**100**												**0**
G171											**100**											**0**
G172												**99**									**1**	**1**
G173													**100**									**0**
G174														**100**								**0**
G175															**99**						**1**	**1**
G176																**100**						**0**
G177																	**99**				**1**	**1**
G178																		**100**				**0**
G179																			**100**			**0**
G180																				**99**	**1**	**1**
Total																						**9**

Universiti Malaya

The confusion matrix related to the Arabic continuous dataset 2 is illustrated in Table 7.27. As can be seen, there is no confusion in the continuous dataset 2. The misclassification error related to those gestures is due to the inability of the classifier to recognise some gestures uniquely because they share the same highest likelihood values among two or more gestures. Those gestures are G164, G166, G168, G172, G175 and G180. The total number of misclassified samples is 9 out of 2,000 samples; thus, the recognition accuracy is 99.55% with 0.45% error in recognition.

Universiti Malaya

7.4.2.2 Comparison with Previous works

This section conducts a comparison of the current work and some of the most related works published recently in terms of the type of sign, classification, tools, signer dependency and recognition accuracy.

7.4.2.2.1 Comparison with other dynamic ArSL Framework

Tables 7.28 and 7.29 show comparisons of the proposed framework with some ArSL-related works.

Table 7.28: Comparison between Isolated words framework and other ArSL frameworks

Source	Type of Sign form	Classification	Tools	Signer Dependency	Recognition Rate (%)
Our Framework	Isolated words	SVM	Smart Gloves & Kinect	10 Signer-Independent	99.15%
(Amin et al., 2016)	Isolated words	HMM	Microsoft Kinect	Signer-Independent	92.5%
(Elons et al., 2013)		Hybrid PCNN	-	Signer-Independent	96%
(Mohandes et al., 2012)		HMM	-	Signer-Independent	95%
(Albelwi et al., 2012)		KNN	LifeCam VX-5500	-	90.55%
(Assaleh et al., 2012)		KNN	DG5-VHand data glove	Signer-Independent	92.5%
(Ritchings et al., 2012)		-	DataGlove	4 Signer-Independent	93%
(Shanableh et al., 2007)		HMM	Digital Camera	Signer-Independent	95%

Universiti Malaya

Table 7.29: Comparison between Continuous signs framework and other ArSL frameworks

Source	Type of Sign form	Classification	Tools	Signer Dependency	Recognition Rate (%)
Our Framework	Continuous Dataset 1	SVM	Smart Gloves & Kinect	10 Signer-Independent	99.25%
	Continuous Dataset 2				99.55%
(Tubaiz et al., 2015)	Continuous Dataset	MKNN	DG5-VHand gloves	Signer-Dependent	98.9%
(Tolba et al., 2013)		PCNN	Digital Camera	-	70%
(Assaleh et al., 2010)		HMM	Digital Camera	Signer-Dependent	73.3%
(Tuffaha et al., 2015)		Modified the polynomial classifier to suitable with sequential data	DG5-VHand gloves	-	85%

For the isolated-word signs, the proposed framework records the highest recognition accuracy of 99.15% compared to other related works presented by Amin et al. (2016), Elons et al. (2013), Mohandes et al. (2012), Albelwi et al. (2012), Assaleh et al. (2012), Ritchings et al. (2012) and Shanableh et al. (2007).

For continuous signs, the proposed framework records the highest recognition accuracy of 99.25% and 99.55% compared to other related works presented by Tubaiz et al. (2015), Tolba et al. (2013), Assaleh et al. (2010) and Tuffaha et al. (2015).

This is due to the combination of two input devices (vision and sensor-based approaches) compared to a single device, which has a critical effect on recognition accuracy. Moreover, the input data are recorded for the proposed framework by 10 signers compared to 1 to 3 signers used in other works. This is considered a critical difference due to the ability of the framework to recognise gestures performed by 10 signers with different ages, experiences and situations. Another notable difference is in the use of both Microsoft Kinect and smart data gloves as the input devices, which have had obvious effects on the recognition accuracy.

Universiti Malaya

The results in Tables 7.28 and 7.29 prove the efficiency and accuracy of the proposed framework, respectively, which records the highest recognition accuracy rate by applying the largest number of signs as the input data.

7.4.2.2.2 Comparison with Other Languages

Table 7.30 illustrates the comparison of the performance of the proposed framework with other related works carried out in non-Arabic languages.

Table 7.30: Comparison among our framework and related works on other languages

Source	SL	Type of Sign form	Tools	Signer Dependency	Recognition Rate (%)
Our Framework	ArSL	Isolated words	Smart Gloves & Kinect	10 Signer-Independent	99.15
Our Framework	ArSL	Continuous Dataset 1	Smart Gloves & Kinect	10 Signer-Independent	99.25
		Continuous Dataset 2			99.55
(Baranwal et al., 2017)	ISL	Isolated words	Sony handycam	Signer-Dependent	85.00
(Fagiani et al., 2015)	LIS	Isolated words	Digital Camera	10 Signer-Independent	48.06
(Cooper et al., 2012)	DGS	Isolated words	Kinect	15 Signer-Independent	76.00
(Zaki et al., 2011)	ASL	Isolated words	3 Cameras	Signer-Dependent	89.10
(Koller et al., 2015)	DGS	Continuous	Digital Camera	9 Signer-Independent	47.00
(Bukhari et al., 2015)	ASL	Continuous	Data Gloves	Signer-Independent	92.00
(Kong et al., 2014)	CSL	Continuous	Cyber Gloves	Signer-Dependent	86.60
(Chai et al., 2013)	CSL	Continuous	Kinect	Signer-Dependent	96.32

Universiti Malaya

Here, the proposed framework is compared with the recognition systems proposed to recognise Indian sign language (ISL) by Baranwal et al. (2017), Italian Sign Language (LIS) by Fagiani et al. (2015), German Sign Language (DGS) by Cooper et al. (2012), American Sign Language (ASL) by Zaki et al. (2011), DGS by Koller et al. (2015), ASL by Bukhari et al. (2015), Chinese Sign Language (CSL) Kong et al. (2014) and CSL by Chai et al. (2013). The results show that the highest accuracy is recorded for the proposed framework. Kong et al. (2014) used multiple classifiers in their approach with a single input device (CyberGloves). In contrast, both smart data gloves and Kinect were used as the input devices in the current work and an ensemble of SVM classifiers is used in the classification stage. Thus, these two combinations proved the highest recorded accuracy for the proposed framework.

7.5 Summary and Discussion

This chapter presented and discussed the obtained results of both framework 1 (the static backhand ArSL recognition) and framework 2 (Hybrid Approach for ArSLR Based on Data Combination). The main results of this chapter could be summarised as follows:

1- For framework 1, the static backhand signs related to Arabic letters and numbers were recorded using a smartphone camera, where all the input data were segmented to detect the hand localisation and then categorised into open and closed hand signs. The conducted experimental setups for framework 1 revealed that the best lighting condition was the direct one, which offers more accurate recorded signs. Thus, all the imported hand signs to the framework could be made through a direct light on the user.

2- The overall recognition accuracies of the NB and MLP classifiers are 86.30% and 88.00%, respectively. Furthermore, the previous classifiers (NB and MLP) were combined into one classifier to perform the recognition and classification. The

Universiti Malaya

overall recognition accuracy result proved that the multi-classifier system outperformed the three single classifiers, as it combined their benefits and could avoid their problems, thus a recorded accuracy of 95.63% was attained for the proposed framework. All those values proved the efficiency of the system in recognising signs and defining misclassified ones; they also indicated that the multi-classifier was better than the single-classifier in terms of accuracy.

3- For framework 2, the static and dynamic hand gestures related to Arabic alphabets, numbers, isolated words and continuous sentences are recorded using a hybrid approach (smart data gloves and Microsoft Kinect). All the input signs are segmented into five classes: letters, numbers and isolated words, continuous dataset 1 and continuous dataset 2. In the recognition and classification phase, multi-class SVM classifiers were used.

4- The overall recognition accuracy for the combined input devices on the framework, when compared with a single device, was tested for all data types. The results revealed that such combination offers extremely higher recognition accuracies than by using single devices separately. The recorded accuracies for the combined input devices are 99.38%, 99.42%, 99.15%, 99.25% and 99.55% for the alphabets, numbers, isolated words, continuous dataset 1 and continuous dataset 2 classes, respectively. These values proved the efficiency of the system in recognising signs and defining misclassified ones. In turn, this finding clearly indicates that the hybrid approaches were better than single approaches in terms of accuracy.

Universiti Malaya

CHAPTER 8: CONCLUSION AND FUTURE WORKS

8.1 Overview

This chapter summarises the presented research work, research contributions and future work recommendations. Figure 8.1 outlines the structure of this chapter.

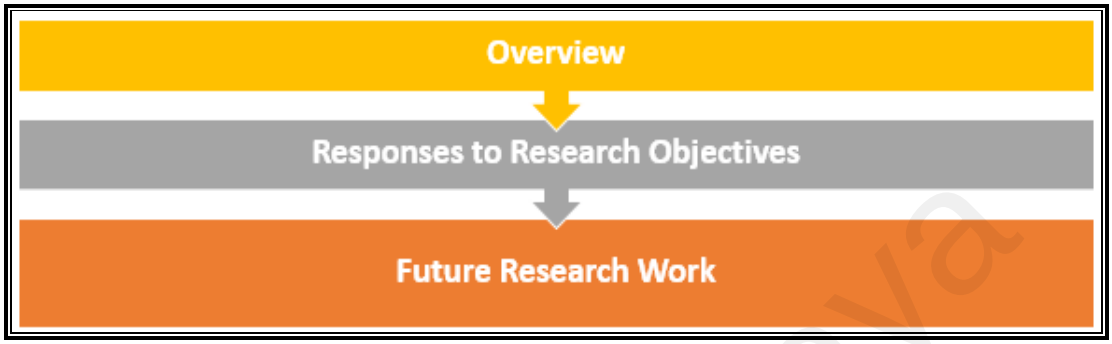


Figure 8.1: Outlines the structure of chapter 8

This thesis aimed to design and create a new ArSL database, which covers all approaches and forms based on the standard criteria and consensus of SL experts. In addition, this thesis proposed two frameworks for modelling and developing ArSLR. The first framework represents a usable static backhand and signer-independent approach for ArSLR of letters and numbers signs based on the VBR approach using a smartphone camera. The second framework represents a usable hybrid VBR and SBR approach for signer-independent continuous ArSLR development and evaluation using Microsoft Kinect and smart data gloves. To achieve the research objectives, a systematic literature review has been conducted to create research taxonomies and identify the research gaps on ArSLR approaches and ArSL forms of signs.

For the first framework, an ASLR system for ArSL is developed to better facilitate communication amongst people with normal hearing and those with hearing impairments. Optimal recognition results cannot be attained with just a single algorithm due to the varying sign shapes. In the proposed framework, the input signs to the sign language recognition system are first divided into open and closed hand signs. Consequently, the sign language recognition system identified suitable approaches to recognise open and closed hand signs. The DWT is integrated with 1D signature signals in closed hand signs

Universiti Malaya

to extract features that slightly vary from one another, which improved the recognition accuracy for ambiguous signs. In addition, open hand signs are differentiated through the distribution of quantised area levels, which are generated from the RLM. Statistical analysis approaches are employed to compute the feature descriptors. In our proposed framework, the multi-classification approach is implemented in the recognition and classification phase. The experiments confirmed the ability of the proposed framework to deal with signer-independence wherein no gloves are required. The presented framework also has the ability to recognise static backhand ArSL letters and numbers with a recognition accuracy of 95.63%.

The second framework shows how to jointly exploit the VBR and SBR approaches for the accuracy of ArSLR obtained via a signer-independent system. Firstly, a dedicated solution is presented for the combined calibration of the two different devices. Then, a set of novel feature descriptors is presented both for the smart gloves and for Microsoft Kinect. Finally, the proposed feature sets are fed to multi-class SVMs. Experimental results show that the obtained accuracies for the datasets of Arabic alphabets, Arabic numbers, isolated words and continuous sentences of this ArSL database are 99.38%, 99.42%, 99.15% and 99.55%, respectively. In conclusion, the created database and the proposed frameworks are able to increase the performance of the ArSLR.

8.2 Responses to Research Objectives

This section describes the findings, research methodology and deliverables of the defined research objectives. This thesis set out to develop a vision- and sensor-based signer-independent framework for ArSLR and to create a comprehensive ArSL database that covers all approaches and forms of signs. The overall research process is divided into three main sections: literature review, development and evaluation.

Universiti Malaya

Research Objective 1: To create research taxonomies based on a comprehensive literature investigation that summarize the research efforts on ArSLR approaches and ArSL forms of signs.

In order to achieve research objective 1, this thesis defined the following question:

Research Question 1: What are the main taxonomies of ArSLR approaches and ArSL forms of signs?

RQ1.1: What are the existing solutions and enabling technologies for ASLR?"

RQ1.2: What are the available ASLR taxonomies and representation techniques used in VBR and SBR?

RQ 1.3: What are the main VBR and SBR techniques and the algorithms used for recognition specifically?

RQ 1.4: What are the application domains, commercial software, and devices that use ASLR for interaction in different environments?

To achieve the first objective as described in Chapter 1 (Section 1.4), a systematic literature review and a comprehensive review are conducted (Chapter 4) as the first step. The systematic literature review is carried out by using the guidelines given out in a past work (Kitchenham et al., 2010). To conduct a systematic literature review, seven electronic databases are used: IEEE Xplore, Springer Link, Science Direct, ACM, Emerald Insight, Wiley and Google Scholar. Studies are selected by applying inclusion, exclusion and quality assessment criteria. After analysing 100 studies, the research questions are answered by conducting a comprehensive literature review on ASLR and ArSL pertaining to the related works. This is done to identify open issues, research gaps, related theory, dominating techniques and suitable tools and devices as mentioned in Section 2.6, 2.7, 2.8, 2.9, 2.10 and 4.5.

Universiti Malaya

Research Objective 2: To create a comprehensive SL database for ArSL that covers vision and sensor-based approaches and forms of signs for ArSLR development and evaluation based on SL experts standard criteria and consensus.

In order to achieve research objective 2, this thesis defined the following question:

Research Question 2: What are the standard criteria used by SL experts for creating ArSL database?

To achieve the second objective stated in Chapter 1 (Section 1.4), a comprehensive SL database for ArSL is created, as described in Chapter 5. The created database is based on 10 signers (3 male signers and 7 female signers), the signers performed 10 repetitions for each sign. The signers are selected from the SL expert practitioners, hearing and speech-impaired peoples.

The database content has been shown in Chapter 5 (Section 5.2) (Tables 5.1, 5.2, 5.3, 5.4 and 5.5), which was split into two main databases. The first database has a static backhand ArSL, such as Arabic letters and numbers, in which one smartphone camera (Apple iPhone X, 12MP resolution) is utilised to capture images of the signs made by the signers (Chapter 5, Section 5.4). The second database has a sign in all ArSL forms and approaches, which used smart data gloves and Microsoft Kinect (Section 5.5). The created database has 39 Arabic alphabet characters, 21 Arabic numbers, 60 Arabic isolated words and 60 continuous ArSL sentences. To achieve this objective, the data are tagged by applying a general recommended procedure provided by Microsoft (Section 5.6.2), which consists of four main steps, namely, identification of signs, recording of signs, tagging of sign recording and verification of the recorded signs.

Universiti Malaya

Research Objective 3: To model a static backhand signer-independent ArSLR framework based on VBR approach that translates signs of letters and numbers into their corresponding text using a smartphone camera.

In order to achieve research objective 3, this thesis defined the following question:

Research Question 3: How to model a static backhand and signer-independent ArSLR framework for ArSL letters and numbers?

To achieve the third objective as described in Chapter 1 (Section 1.4), a static backhand is designed and modelled based on a signer-independent ArSLR framework for ArSL letters and numbers (Chapter 6, Section 6.2 and Figure 6.1). The framework is modelled initially to collect static backhand sign data from SL experts and practitioners as described in Chapter 3 (Section 3.4.1) and Chapter 5. A smartphone camera is used to collect data for this framework as recommended based on the previous research and the SL experts.

The proposed framework has three main gesture recognition phases, which are integrated to recognise static backhand ArSL (Section 6.2). The proposed framework obtains input signs dataset in RGB, then converts them through binarisation into white and black. Thereafter, the input signs are divided and then represented by images as an open and closed hand sign categories. Each category has various feature extraction techniques. In the classification phase, multiple-classifier algorithms are applied, as mentioned in Chapter 3 (Section 3.4.2) and Chapter 6 (Section 6.2).

Universiti Malaya

Research Objective 4: To model a hybrid approach, signer-independent continuous ArSLR framework that integrates the VBR approach using Microsoft Kinect and the SBR approach using Smart Data Gloves.

In order to achieve research objective 4, this thesis defined the following question:

Research Question 4: How to model a hybrid approach for signer-independent continuous ArSLR framework?

To achieve the fourth objective as described in Chapter 1 (Section 1.4), a hybrid approach is designed and modelled based on a signer-independent ArSLR framework for ArSL letters, numbers, isolated words and continuous sentences (Chapter 6, Section 6.3 and Figure 6.31). The framework is modelled initially to collect the signs from hybrid approaches (vision and sensor-based devices). The database sign is created by SL experts and practitioners (Chapter 3, Section 3.4.1 and Chapter 5). For the sensor-based approach, smart data gloves are used, whilst for the vision-based approach, Microsoft Kinect is used.

The proposed framework shows how to jointly exploit the two approaches to have accurate ArSLR made via a signer-independent system. For the smart gloves, simple pre-processing and feature extraction techniques are applied, as mentioned in Chapter 3 (Section 3.4.2) and Chapter 6 (Section 6.3.1). For the Microsoft Kinect, the hand is firstly extracted from the acquired data and divided into palm and finger regions. Then, four main features are extracted, namely, distances, elevation, curvature and palm area region. Finally, the proposed feature sets (from smart gloves and Kinect) are fed into multi-class SVMs in Chapter 3 (Section 3.4.2) and Chapter 6 (Sections 6.3.2 and 6.3.3), respectively.

Universiti Malaya

Research Objective 5: To evaluate the performance of the modeled frameworks in terms of accuracy, and compare the performance with the related works.

In order to achieve research objective 5, this thesis defined the following question:

Research Question 5: What are the performance measures used to evaluate and compare ArSLR research works?

To achieve the fifth objective as described in Chapter 1 (Section 1.4), the performance measures used are identified to evaluate and compare ArSLR research works, as mentioned in Chapter 3 (Section 3.5). The evaluation is performed by the experimental result and the results are analysed and discussed in Chapter 7.

In this thesis, an evaluation measure was devised by using recognition rate, sensitivity, specificity, precision, false positive rate, false negative rate and F1-score. In addition, the proposed framework is evaluated based on features, devices and special conditions depending on pre-processing techniques, etc. Moreover, the proposed framework is compared with other related works on ArSL and other languages, as mentioned in Chapter 7 wherein the details of the evaluation are presented. The main results of the evaluation are discussed in Sections 7.2 and Section 7.3, and the comparisons of our frameworks and other works are presented in Sections 7.2.3 and 7.3.3.

Universiti Malaya

8.3 Future Work

1- More works can be done to improve or extend the database content by including a number for signers, adding more datasets and expanding its words.

2- Currently, the proposed ArSL framework has limited grammatical rules. Thus, more grammatical rules pertaining to the ArSL framework can be incorporated to enhance the framework's performance and accuracy.

3- Researchers could focus more on grammatical interpretation and incorporation of grammatical rules in the context of SLR framework.

4- An automatic tagging technique must be developed as it can reduce the tagging time. This is crucial when using a large database.

5- GR frameworks that are applicable in a real-life situations must be designed and modelled using such technologies as speech or text translation systems for sign language recognition frameworks. Such frameworks are indispensable and in high demand.

Universiti Malaya

REFERENCES

Abdel-Fattah, M. A. (2005). Arabic Sign Language: A Perspective. *Journal of Deaf Studies and Deaf Education*, *10*(2), 212–221. http://doi.org/10.1093/deafed/eni007

Abhishek, K. S., Qubeley, L. C. K., & Ho, D. (2016). Glove-based hand gesture recognition sign language translator using capacitive touch sensor. In *2016 IEEE International Conference on Electron Devices and Solid-State Circuits (EDSSC)* (pp. 334–337). http://doi.org/10.1109/EDSSC.2016.7785276

Abhishek, M., & Divakar, H. (2017). A Video Based Sign Language Recognition System. *International Journal of Engineering Science and Computing*, *7*(9), 1–3.

Abreu, J. G., Teixeira, J. M., Figueiredo, L. S., & Teichrieb, V. (2016). Evaluating sign language recognition using the myo armband. In *Virtual and Augmented Reality (SVR), 2016 XVIII Symposium on* (pp. 64–70). IEEE.

Abushariah, M. A. M. (2017). TAMEEM V1. 0: speakers and text independent Arabic automatic continuous speech recognizer. *International Journal of Speech Technology*, *20*(2), 261–280.

Abushariah, M. A. M., Ainon, R. N., Zainuddin, R., Elshafei, M., & Khalifa, O. O. (2010). Natural speaker-independent Arabic speech recognition system based on Hidden Markov Models using Sphinx tools. In *Computer and Communication Engineering (ICCCE), 2010 International Conference on* (pp. 1–6). IEEE.

Abushariah, M. A. M., Ainon, R. N., Zainuddin, R., Elshafei, M., & Khalifa, O. O. (2012). Phonetically rich and balanced text and speech corpora for Arabic language. *Language Resources and Evaluation*, *46*(4), 601–634.

Aha, D. W., & Bankert, R. L. (1996). A comparative evaluation of sequential feature selection algorithms. In *Learning from data* (pp. 199–206). Springer.

Ahmed, A. A., & Aly, S. (2014). Appearance-based Arabic Sign Language recognition using Hidden Markov Models. In *2014 International Conference on Engineering and Technology (ICET)* (pp. 1–6). IEEE. http://doi.org/10.1109/ICEngTechnol.2014.7016804

Akmeliawati, R., Dadgostar, F., Demidenko, S., Gamage, N., Kuang, Y. C., Messom, C., … SenGupta, G. (2009). Towards real-time sign language analysis via markerless gesture tracking. In *2009 IEEE Instrumentation and Measurement Technology Conference* (pp. 1200–1204). IEEE.

Albelwi, N. R., & Alginahi, Y. M. (2012). Real-Time Arabic Sign Language (ArSL) Recognition. In *International Conference on Communications and Information Technology (ICCIT)* (pp. 497–501). Tunisia.

Alghamdi, M., Elshafei, M., & Al-Muhtaseb, H. (2007). Arabic broadcast news transcription system. *International Journal of Speech Technology*, *10*(4), 183–195.

Al-Jarrah, O., & Al-Omari, F. A. (2007). Improving Gesture Recognition in The Arabic Sign Language Using Texture Analysis. *Applied Artificial Intelligence*, *21*(1), 11–33. http://doi.org/10.1080/08839510600938524

Al-Jarrah, O., & Halawani, A. (2001). Recognition of gestures in Arabic sign language using neuro-fuzzy systems. *Artificial Intelligence*, *133*(1-2), 117–138. http://doi.org/10.1016/S0004-3702(01)00141-2

AlKhateeb, J. H., Ren, J., Jiang, J., & Al-Muhtaseb, H. (2011). Offline handwritten Arabic cursive text recognition using Hidden Markov Models and re-ranking. *Pattern Recognition Letters*, *32*(8), 1081–1088.

Almohimeed, A., Wald, M., & Damper, R. I. (2011). Arabic Text to Arabic Sign Language Translation System for the Deaf and Hearing-Impaired Community. In *Proceedings of the Second Workshop on Speech and Language Processing for Assistive Technologies* (pp. 101–109). Association for Computational Linguistics. Retrieved from http://dl.acm.org/citation.cfm?id=2140499.2140515

Al-Rousan, M., Al-Jarrah, O., & Al-Hammouri, M. (2010). Recognition of dynamic gestures in arabic sign language using two stages hierarchical scheme. *International Journal of Knowledge-Based and Intelligent Engineering Systems*, *14*(3), 139–152. http://doi.org/10.3233/KES-2010-0197

AL-Rousan, M., Assaleh, K., & Tala'a, A. (2009). Video-based signer-independent Arabic sign language recognition using hidden Markov models. *Applied Soft Computing*, *9*(3), 990–999. http://doi.org/10.1016/j.asoc.2009.01.002

Al-Shamayleh, A. S., Ahmad, R., Abushariah, M. A. M., Alam, K. A., & Jomhari, N. (2018). A systematic literature review on vision based gesture recognition techniques. *Multimedia Tools and Applications*, *77*(21), 1–64. http://doi.org/10.1007/s11042-018-5971-z

Althobaiti, H., & Lu, C. (2017). A survey on Arabic optical character recognition and an isolated handwritten Arabic character recognition algorithm using encoded freeman chain code. In *Information Sciences and Systems (CISS), 2017 51st Annual Conference on* (pp. 1–6). IEEE.

Altun, O., & Albayrak, S. (2011). Turkish fingerspelling recognition system using Generalized Hough Transform, interest regions, and local descriptors. *Pattern Recognition Letters*, *32*(13), 1626–1632. http://doi.org/10.1016/j.patrec.2011.06.010

Amin, A., & Hong, Y. (2007). Sign Language Finger Alphabet Recognition from Gabor-PCA Representation of Hand Gestures. In *2007 International Conference on Machine Learning and Cybernetics* (Vol. 4, pp. 2218–2223). IEEE. http://doi.org/10.1109/ICMLC.2007.4370514

Amin, O., Said, H., Samy, A., & Mohammed, H. K. (2016). HMM based automatic Arabic sign language translator using Kinect. In *2015 Tenth International Conference on Computer Engineering & Systems (ICCES)* (pp. 389–392). http://doi.org/10.1109/ICCES.2015.7393081

Antonakos, E., Roussos, A., & Zafeiriou, S. (2015). A survey on mouth modeling and analysis for Sign Language recognition. *2015 11th IEEE International Conference and Workshops on Automatic Face and Gesture Recognition (FG)*, 1–7. http://doi.org/10.1109/FG.2015.7163162

Aran, O. (2008). Vision Based sign language recognition: Modeling and recognizing isolated signes with manual and non manual components. *PhD Thesis*. Retrieved from http://www.idiap.ch/~oaran/paublications/aran08phdthesis.pdf\nhttp://www.idiap.ch/~oaran/publications/aran08phdthesis.pdf

Assaleh, K., & Al-Rousan, M. (2005). Recognition of Arabic Sign Language Alphabet Using Polynomial Classifiers. *EURASIP Journal on Advances in Signal Processing*, *2005*(13), 2136–2145. http://doi.org/10.1155/ASP.2005.2136

Assaleh, K., Shanableh, T., Fanaswala, M., Amin, F., & Bajaj, H. (2010). Continuous Arabic Sign Language Recognition in User Dependent Mode. *Journal of Intelligent Learning Systems and Applications*, *02*(01), 19–27. http://doi.org/10.4236/jilsa.2010.21003

Assaleh, K., Shanableh, T., Fanaswala, M., Bajaj, H., & Amin, F. (2008). Vision-based system for continuous arabic sign language recognition in user dependent mode. *Proceeding of the 5th International Symposium on Mechatronics and Its Applications, ISMA 2008*, 25–29. http://doi.org/10.1109/ISMA.2008.4648841

Assaleh, K., Shanableh, T., & Zourob, M. (2012). Low Complexity Classification System for Glove-Based Arabic Sign Language Recognition. In *Proc. 19th International Conference. Neural Information Processing* (Vol. 7665, pp. 262–268). Berlin, Heidelberg: Springer Berlin Heidelberg. http://doi.org/10.1007/978-3-642-34487-9

Attia, M., & Somers, H. (2008). *Handling Arabic morphological and syntactic ambiguity*

Universiti Malaya

within the LFG framework with a view to machine translation (Vol. 279). University of Manchester Manchester.

Aujeszky, T., & Eid, M. (2015). A gesture recogintion architecture for Arabic sign language communication system. *Multimedia Tools and Applications*, *75*(14), 8493–8511. http://doi.org/10.1007/s11042-015-2767-2

Aujeszky, T., & Eid, M. (2016). A gesture recogintion architecture for Arabic sign language communication system. *Multimedia Tools and Applications*, *75*(14), 8493–8511. http://doi.org/10.1007/s11042-015-2767-2

Avraam, M. (2014). Static gesture recognition combining graph and appearance features. *International Journal of Advanced Research in Artificial Intelligence (IJARAI)*, *3*(2), 1–4.

Badi, H. (2016). A Survey on Recent Vision-Based Gesture Recognition. *Intelligent Industrial Systems*, *2*(2), 179–191. http://doi.org/10.1007/s40903-016-0046-9

Bairagi, A. G., & Kapse, Y. . (2018). Survey on Sign language to Speech Conversion. *International Journal of Innovative Research in Computer and Communication Engineering*, *6*(1), 267–274. http://doi.org/10.15680/IJIRCCE.2017.0601017

Baranwal, N., & Nandi, G. C. (2017). An efficient gesture based humanoid learning using wavelet descriptor and MFCC techniques. *International Journal of Machine Learning and Cybernetics*, *8*(4), 1369–1388.

Bastos, I. L. O., Angelo, M. F., & Loula, A. C. (2015). Recognition of static gestures applied to Brazilian sign language (Libras). In *Graphics, Patterns and Images (SIBGRAPI), 2015 28th SIBGRAPI Conference on* (pp. 305–312). IEEE.

Baxter, J. (2000). A model of inductive bias learning. *Journal of Artificial Intelligence Research*, *12*(1), 149–198. Retrieved from http://dl.acm.org/citation.cfm?id=1622248.1622254

Bay, H., Tuytelaars, T., & Van Gool, L. (2006). Surf: Speeded up robust features. In *European conference on computer vision* (pp. 404–417). Springer.

Bhuyan, M. K., Ghoah, D., & Bora, P. K. (2006). A Framework for Hand Gesture Recognition with Applications to Sign Language. In *2006 Annual IEEE India Conference* (pp. 1–6). http://doi.org/10.1109/INDCON.2006.302823

Bilal, S., Akmeliawati, R., Shafie, A. A., & Salami, M. J. E. (2011). Hidden Markov model for human to computer interaction: a study on human hand gesture recognition. *Artificial Intelligence Review*, *40*(4), 495–516. http://doi.org/10.1007/s10462-011-9292-0

Bin, L., & Samiei, M. (2012). Comparison for Image Edge Detection Algorithms. *IOSR Journal of Computer Engineering*, *2*(6), 1–4. Retrieved from www.iosrjournals.org

Binh, N. D., & Ejima, T. (2005). Hand gesture recognition using fuzzy neural network. In *Proc. ICGST Conf. Graphics, Vision and Image Proces (Cairo, 2005)* (pp. 1–6).

Birdal, A., & Hassanpour, R. (2008). Region Based Hand Gesture Recognition. In *16th international conference in central Europe on computer graphics, visualization and computer vision* (pp. 1–8). Iason.Zcu.Cz.

Boulay, B. (2007). *Human posture recognition for behavior understanding*. Universite de Nice-Sophia Antipolis.

Bourke, A. K., O'Brien, J. V, & Lyons, G. M. (2007). Evaluation of a threshold-based tri-axial accelerometer fall detection algorithm. *Gait & Posture*, *26*(2), 194–9. http://doi.org/10.1016/j.gaitpost.2006.09.012

Braddock, D., Rizzolo, M. C., Thompson, M., & Bell, R. (2004). Emerging technologies and cognitive disability. *Journal of Special Education Technology*, *19*(4), 49–56.

Breiman, L. (2001). Random Forests. *Machine Learning*, *45*(1), 5–32. http://doi.org/10.1023/A:1010933404324

Bretzner, L., Laptev, I., & Lindeberg, T. (2002). Hand gesture recognition using multi-scale colour features, hierarchical models and particle filtering. In *Proceedings of*

Fifth IEEE International Conference on Automatic Face Gesture Recognition (pp. 405–410). IEEE. http://doi.org/10.1109/AFGR.2002.1004190

Bukhari, J., Rehman, M., Malik, S. I., Kamboh, A. M., & Salman, A. (2015). American Sign Language Translation through Sensory Glove; SignSpeak. *International Journal Science and Technology*, *8*(1), 131–142. http://doi.org/10.14257/ijunesst.2015.8.1.12

Burges, C. J. C. (1998). A Tutorial on Support Vector Machines for Pattern Recognition. *Data Mining and Knowledge Discovery*, *2*(2), 121–167. http://doi.org/10.1023/A:1009715923555

Cadoz, C., & Wanderley, M. M. (2000). Gesture-music. Trends in gestural control of music, 12, 101.

Chai, X., Li, G., Lin, Y., Xu, Z., Tang, Y., Chen, X., & Zhou, M. (2013). Sign language recognition and translation with kinect. In *IEEE Conf. on AFGR* (Vol. 655).

Chanda, P., Auephanwiriyakul, S., & Theera-Umpon, N. (2012). Thai sign language translation system using upright speed-up robust feature and c-means clustering. In *Fuzzy Systems (FUZZ-IEEE), 2012 IEEE International Conference on* (pp. 1–6). IEEE.

Chang, C. C., Chen, J. J., Tai, W. K., & Han, C. C. (2006). New approach for static gesture recognition. *Journal of Information Science and Engineering*, *22*(5), 1047–1057. Retrieved from http://www.scopus.com/inward/record.url?eid=2-s2.0-33749567169&partnerID=40&md5=f206e540cbd59c5462a9a6e462e7eb3e

Chang, C.-C., & Lin, C.-J. (2011). LIBSVM: a library for support vector machines. *ACM Transactions on Intelligent Systems and Technology (TIST)*, *2*(3), 27.

Chaudhary, A., Raheja, J. L., Das, K., & Raheja, S. (2011). A survey on hand gesture recognition in context of soft computing. In *Communications in Computer and Information Science* (Vol. 133 CCIS, pp. 46–55). Springer Berlin Heidelberg. http://doi.org/10.1007/978-3-642-17881-8_5

Chen, D., Li, G., Sun, Y., Kong, J., Jiang, G., Tang, H., … Liu, H. (2017). An interactive image segmentation method in hand gesture recognition. *Sensors*, *17*(2), 253.

Chen, Y.-W., & Lin, C.-J. (2006). Combining SVMs with various feature selection strategies. In *Feature extraction* (pp. 315–324). Springer.

Cheok, M. J., Omar, Z., & Jaward, M. H. (2019). A review of hand gesture and sign language recognition techniques. *International Journal of Machine Learning and Cybernetics*, *10*(1), 131–153. http://doi.org/10.1007/s13042-017-0705-5

Choi, H.-R., & Kim, T. (2017). Combined Dynamic Time Warping with Multiple Sensors for 3D Gesture Recognition. *Sensors*, *17*(8), 1–15.

Chuan, C.-H., Regina, E., & Guardino, C. (2014). American sign language recognition using leap motion sensor. In *Machine Learning and Applications (ICMLA), 2014 13th International Conference on* (pp. 541–544). IEEE.

Cooper, H., Holt, B., & Bowden, R. (2011). Sign Language Recognition. In *Hilton A., Krüger V., Sigal L. (eds) Visual Analysis of Humans* (pp. 539–562). Springer, London. http://doi.org/10.1007/978-0-85729-997-0_27

Cooper, H., Ong, E.-J., Pugeault, N., & Bowden, R. (2012). Sign language recognition using sub-units. *Journal of Machine Learning Research*, *13*(Jul), 2205–2231.

Cote, M., Payeur, P., & Comeau, G. (2006). Comparative Study of Adaptive Segmentation Techniques for Gesture Analysis in Unconstrained Environments. In *Proceedings of the 2006 IEEE International Workshop on Imagining Systems and Techniques (IST 2006)* (pp. 28–33). IEEE. http://doi.org/10.1109/IST.2006.1650770

da Fontoura Costa, L., & Cesar Jr, R. M. (2009). *Shape classification and analysis: theory and practice*. CRC Press, Inc.

Dabre, K., & Dholay, S. (2014). Machine learning model for sign language interpretation using webcam images. In *2014 International Conference on Circuits, Systems,*

Communication and Information Technology Applications, CSCITA 2014 (pp. 317–321). http://doi.org/10.1109/CSCITA.2014.6839279

Delac, K., Grgic, M., & Grgic, S. (2005). Independent comparative study of PCA, ICA, and LDA on the FERET data set. *International Journal of Imaging Systems and Technology*, *15*(5), 252–260.

Derpanis, K. G. (2005). Mean Shift Clustering, 1–3. http://doi.org/10.1117/12.792998

Dipietro, L., Sabatini, A. M., & Dario, P. (2008). A Survey of Glove-Based Systems and Their Applications. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, *38*(4), 461–482. http://doi.org/10.1109/TSMCC.2008.923862

Domino, G., & Domino, M. L. (2006). *Psychological testing: An introduction*. Cambridge University Press.

Duda, R. O., Hart, P. E., & Stork, D. G. (2012). *Pattern classification*. John Wiley & Sons.

Easterbrook, S., Singer, J., Storey, M.-A., & Damian, D. (2008). Selecting empirical methods for software engineering research. In *Guide to advanced empirical software engineering* (pp. 285–311). Springer.

El-Bendary, N., Zawbaa, H. M., Daoud, M. S., Hassanien, A. E., & Nakamatsu, K. (2010a). ArSLAT: Arabic Sign Language Alphabets Translator. In *2010 International Conference on Computer Information Systems and Industrial Management Applications, CISIM 2010* (pp. 590–595). http://doi.org/10.1109/CISIM.2010.5643519

El-Bendary, N., Zawbaa, H. M., Daoud, M. S., Hassanien, A. E., & Nakamatsu, K. (2010b). ArSLAT: Arabic Sign Language Alphabets Translator. In *2010 International Conference on Computer Information Systems and Industrial Management Applications (CISIM)* (pp. 590–595). http://doi.org/10.1109/CISIM.2010.5643519

El-Bendary, N., Zawbaa, H. M., Daoud, M. S., Hassanien, A. E., & Nakamatsu, K. (2010c). ArSLAT: Arabic Sign Language Alphabets Translator. *2010 International Conference on Computer Information Systems and Industrial Management Applications, CISIM 2010*, *3*, 590–595. http://doi.org/10.1109/CISIM.2010.5643519

Elmahdy, M., Gruhn, R., Minker, W., & Abdennadher, S. (2009). Survey on common Arabic language forms from a speech recognition point of view. In *proceeding of International conference on Acoustics (NAG-DAGA)* (pp. 63–66).

Elons, A. S., Abull-Ela, M., & Tolba, M. F. (2013). A proposed PCNN features quality optimization technique for pose-invariant 3D Arabic sign language recognition. *Applied Soft Computing Journal*, *13*(4), 1646–1660. http://doi.org/10.1016/j.asoc.2012.11.036

Erol, A., Bebis, G., Nicolescu, M., Boyle, R. D., & Twombly, X. (2007). Vision-based hand pose estimation: A review. *Computer Vision and Image Understanding*, *108*(1), 52–73. http://doi.org/https://doi.org/10.1016/j.cviu.2006.10.012

Fagiani, M., Principi, E., Squartini, S., & Piazza, F. (2015). Signer independent isolated Italian sign recognition based on hidden Markov models. *Pattern Analysis and Applications*, *18*(2), 385–402.

Fang, G., & Gao, W. (2002). A SRN/HMM system for signer-independent continuous sign language recognition. In *Automatic Face and Gesture Recognition, 2002. Proceedings. Fifth IEEE International Conference on* (pp. 312–317). IEEE.

Fares, F., Mashagba, A., Fares, E., Mashagba, A., & Nassar, M. O. (2013). Automatic isolated-word Arabic sign language recognition system based on time delay neural networks: New improvements. *Journal of Theoretical and Applied Information Technology*, *57*(11), 42–47.

Universiti Malaya

Farghaly, A., & Shaalan, K. (2009). Arabic natural language processing: Challenges and solutions. *ACM Transactions on Asian Language Information Processing (TALIP)*, *8*(4), 14.

Fitzgibbon, A. W., & Fisher, R. B. (1996). A buyer's guide to conic fitting. *DAI Research Paper*.

Freeman, W., & Roth, M. (1995). Orientation histograms for hand gesture recognition. *International Workshop on Automatic Face and Gesture Recognition*, *12*, 296–301. http://doi.org/10.1.1.6.618

Funasaka, M., Ishikawa, Y., Takata, M., & Joe, K. (2015). Sign language recognition using leap motion controller. In *Proceedings of the International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA)* (p. 263). The Steering Committee of The World Congress in Computer Science, Computer Engineering and Applied Computing (WorldComp).

Galloway, M. M. (1975). Texture analysis using gray level run lengths. *Computer Graphics and Image Processing*, *4*(2), 172–179. http://doi.org/https://doi.org/10.1016/S0146-664X(75)80008-6

Gandy, M., Starner, T., Auxier, J., & Ashbrook, D. (2000). The Gesture Pendant: A Self-illuminating, Wearable, Infrared Computer Vision System for Home Automation Control and Medical Monitoring, 87. Retrieved from http://dl.acm.org/citation.cfm?id=851037.856538

GestureTek. (2008). Retrieved from http://www.gesturetek.com/

Goza, S. M., Ambrose, R. O., Diftler, M. A., & Spain, I. M. (2004). Telepresence control of the NASA/DARPA robonaut on a mobility platform. In *Proceedings of the 2004 conference on Human factors in computing systems - CHI '04* (pp. 623–629). New York, New York, USA: ACM Press. http://doi.org/10.1145/985692.985771

Gujrati, A., Singh, K., Soral, K. L., & Ambikapathy, M. (2013). Hand-talk Gloves with Flex Sensor: A Review. *International Journal of Engineering Science Invention*, 43–46.

Gupta, B., Shukla, P., & Mittal, A. (2016). K-nearest correlated neighbor classification for Indian sign language gesture recognition using feature fusion. In *Computer Communication and Informatics (ICCCI), 2016 International Conference on* (pp. 1–5). IEEE.

Gupta, S., Jaafar, J., & Ahmad, W. F. W. (2012). Static Hand Gesture Recognition Using Local Gabor Filter. *Procedia Engineering*, *41*, 827–832. http://doi.org/10.1016/j.proeng.2012.07.250

Habash, N. Y. (2010). *Introduction to Arabic Natural Language Processing*. *Synthesis Lectures on Human Language Technologies* (Vol. 3). Morgan and Claypool Publishers, USA. http://doi.org/10.2200/S00277ED1V01Y201008HLT010

Halawani, S. M. (2008). Arabic sign language translation system on mobile devices. *International Journal of Computer Science and Network Security (IJCSNS), 8*(1), 251–256. Retrieved from http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.129.2205&rep=rep1&type=pdf

Halawani, S. M., Daman, D., Kari, S., & Ahmad, A. R. (2013). An Avatar Based Translation System from Arabic Speech to Arabic Sign Language for Deaf People. *IJCSNS International Journal of Computer Science and Network Security*, *13*(12), 43–52.

HandGKET. (2011). Retrieved from https://sites.google.com/site/kinectapps/kinect

Haralick, R. M. (1979). Statistical and structural approaches to texture. *Proceedings of the IEEE*, *67*(5), 786–804.

Hartanto, R., Susanto, A., & Santosa, P. I. (2014). Real Time Static Hand Gesture Recognition System Prototype for Indonesian Sign Language.

Universiti Malaya

Hasan, M. M., & Mishra, P. K. (2012). Hand Gesture Modeling and Recognition using Geometric Features : A Review. *Canadian Journal on Image Processing and Computer Vision*, *3*(1), 12–26.

Hearing Impaired Definition. (2012). Retrieved from Retrieved 4 16, 2014, from Hearing Loss Association of North Carolina, a State Association of HLA of America: http://www.nchearingloss.org/hi.htm

Hemayed, E. E., & Hassanien, A. S. (2011). Edge-based recognizer for Arabic sign language alphabet (ArS2V-Arabic sign to voice). In *Computer Engineering Conference (ICENCO). IEEE* (pp. 121–127). http://doi.org/10.1109/ICENCO.2010.5720438

Holzmann, G. J. (1990). *Design and validation of computer protocols*. New Jersey: Prentice-Hall, Inc. Retrieved from http://cdn.worldcolleges.info/sites/default/files/x20v_1991.pdf

Hrúz, M., Trojanová, J., & Železný, M. (2012). Local Binary Pattern based features for sign language recognition. *Pattern Recognition and Image Analysis*, *22*(4), 519–526.

Hu, M., Shen, F., & Zhao, J. (2014). Hidden Markov models based dynamic hand gesture recognition with incremental learning method. In *2014 International Joint Conference on Neural Networks (IJCNN)* (pp. 3108–3115). http://doi.org/10.1109/IJCNN.2014.6889632

Huang, D.-Y., Hu, W.-C., & Chang, S.-H. (2009). Vision-Based Hand Gesture Recognition Using PCA+Gabor Filters and SVM. In *2009 Fifth International Conference on Intelligent Information Hiding and Multimedia Signal Processing* (pp. 1–4). IEEE. http://doi.org/10.1109/IIH-MSP.2009.96

Huong, T. N. T., Huu, T. V, Xuan, T. L., & Van, S. V. (2015). Static hand gesture recognition for vietnamese sign language (VSL) using principle components analysis. In *2015 International Conference on Communications, Management and Telecommunications (ComManTel)* (pp. 138–141). http://doi.org/10.1109/ComManTel.2015.7394275

Idris, M. H. (2012). Smart Glove for hearing impaired people. Johor: University Teknologi Malaysia. Malaysia.

Ionescu, B., Coquin, D., Lambert, P., & Buzuloiu, V. (2005). Dynamic Hand Gesture Recognition Using the Skeleton of the Hand. *EURASIP Journal on Advances in Signal Processing*, *2005*(13), 2101–2109. http://doi.org/10.1155/ASP.2005.2101

Islam, M. J., Wu, Q. M. J., Ahmadi, M., & Sid-Ahmed, M. A. (2007). Investigating the performance of naive-bayes classifiers and k-nearest neighbor classifiers. In *Convergence Information Technology, 2007. International Conference on* (pp. 1541–1546). IEEE.

Jasim, M., & Hasanuzzaman, M. (2014). Sign language interpretation using linear discriminant analysis and local binary patterns. In *Informatics, Electronics & Vision (ICIEV), 2014 International Conference on* (pp. 1–5). IEEE.

Jemina, N., Rachel, M., & Goswell, D. (2010). *Sign language interpreting : theory and practice in Australia and New Zealand* (2nd ed). Annandale, N.S.W. : Federation Press. Retrieved from http://trove.nla.gov.au/work/4010506?selectedversion=NBD45079847

Ju, S. X., Black, M. J., Minneman, S., & Kimber, D. (1997). Analysis of gesture and action in technical talks for video indexing. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition* (pp. 595–601). San Juan: IEEE Comput. Soc. http://doi.org/10.1109/CVPR.1997.609386

Just, A. (2006). *Two-Handed Gestures for Human-Computer Interaction*. Research report IDIAP 06-73, EPFL.

Kaaniche, M. B. (2009). GESTURE RECOGNITION FROM VIDEO SEQUENCES, 129.

Universiti Malaya

Kang, S., Chung, K., Rim, K.-W., & Lee, J.-H. (2012). Development of Real-Time Gesture Recognition System Using Visual Interaction. In *Proceedings of the International Conference on IT Convergence and Security 2011* (pp. 295–306). http://doi.org/10.1007/978-94-007-2911-7

Kanungo, T., Mount, D. M., Netanyahu, N. S., Piatko, C. D., Silverman, R., & Wu, A. Y. (2002). An efficient k-means clustering algorithm: analysis and implementation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *24*(7), 881–892. http://doi.org/10.1109/TPAMI.2002.1017616

Karam, M. (2006). A framework for research and design of gesture-based human computer interactions. *Science*, (November), 198. Retrieved from http://users.ecs.soton.ac.uk/amrk03r/Thesis.pdf

Karami, A., Zanj, B., & Sarkaleh, A. K. (2011). Persian sign language (PSL) recognition using wavelet transform and neural networks. *Expert Systems with Applications*, *38*(3), 2661–2667. http://doi.org/10.1016/j.eswa.2010.08.056

Kaur, S., & Singh, I. (2016). Comparison between Edge Detection Techniques. *International Journal of Computer Applications (0975*, *145*(15), 15–18.

Kausar, S., & Javed, M. Y. (2011). A Survey on Sign Language Recognition. In *2011 Frontiers of Information Technology* (pp. 95–98). IEEE. http://doi.org/10.1109/FIT.2011.25

Kim, J. S., Jang, W., & Bien, Z. (1996). A dynamic gesture recognition system for the Korean sign language (KSL). *IEEE Transactions on Systems, Man, and Cybernetics, Part B*, *26*(2), 354–359. http://doi.org/10.1109/3477.485888

Kirchhoff, K., Bilmes, J., Das, S., Duta, N., Egan, M., Ji, G., … Noamany, M. (2003). Novel approaches to Arabic speech recognition: report from the 2002 Johns-Hopkins summer workshop. In *2003 IEEE International Conference on Acoustics, Speech, and Signal Processing, 2003. Proceedings. (ICASSP '03)* (pp. 344–347). IEEE. http://doi.org/10.1109/icassp.2003.1198788

Kitchenham, B., & Charters, S. (2007). *Guidelines for performing Systematic Literature reviews in Software Engineering*. *Version 2.3*. Retrieved from http://scholar.google.com/scholar?hl=en&btnG=Search&q=intitle:Guidelines+for+performing+Systematic+Literature+Reviews+in+Software+Engineering#0\nhttp://www.dur.ac.uk/ebse/resources/Systematic-reviews-5-8.pdf

Kitchenham, B., Pretorius, R., Budgen, D., Brereton, O. P., Turner, M., Niazi, M., & Linkman, S. (2010). Systematic literature reviews in software engineering-A tertiary study. *Information and Software Technology*, *52*(8), 792–805. http://doi.org/10.1016/j.infsof.2010.03.006

Kjeldsen, F. C. M. (1997). Visual interpretation of hand gestures as a practical interface modality. Columbia University.

Koller, O., Forster, J., & Ney, H. (2015). Continuous sign language recognition: Towards large vocabulary statistical recognition systems handling multiple signers. *Computer Vision and Image Understanding*, *141*, 108–125. http://doi.org/10.1016/j.cviu.2015.09.013

Kong, W. W., & Ranganath, S. (2014). Towards subject independent continuous sign language recognition: A segment and merge approach. *Pattern Recognition*, *47*(3), 1294–1308.

Krishnaveni, M., & Radha, V. (2011). A Generalized Classification Approach for Indian Sign Language Datasets Using Instance Based Classifiers. In D. Nagamalai, E. Renault, & M. Dhanuskodi (Eds.), *Communications in Computer and Information Science* (pp. 414–422). Berlin, Heidelberg: Springer Berlin Heidelberg.

Kumar, G., & Bhatia, P. K. (2014). A detailed review of feature extraction in image processing systems. In *Advanced Computing & Communication Technologies (ACCT), 2014 Fourth International Conference on* (pp. 5–12). IEEE.

Universiti Malaya

Kumar, N., Belhumeur, P. N., Biswas, A., Jacobs, D. W., Kress, W. J., Lopez, I. C., & Soares, J. V. B. (2012). Leafsnap: A computer vision system for automatic plant species identification. In *Computer vision–ECCV 2012* (pp. 502–516). Springer.

Kurdyumov, R., Ho, P., & Ng, J. (2011). Sign language classification using webcam images. *Google Scholar*.

Kuznetsova, A., Leal-Taixe, L., & Rosenhahn, B. (2013). Real-Time Sign Language Recognition Using a Consumer Depth Camera. In *2013 IEEE International Conference on Computer Vision Workshops* (pp. 83–90). http://doi.org/10.1109/ICCVW.2013.18

Lafferty, J., McCallum, A., & Pereira, F. C. N. (2001). *Conditional random fields: Probabilistic models for segmenting and labeling sequence data*. *ICML '01 Proceedings of the Eighteenth International Conference on Machine Learning*. Retrieved from http://repository.upenn.edu/cis_papers/159/\nhttp://dl.acm.org/citation.cfm?id=655813

Lamar, M. V, Bhuiyan, M. S., & Iwata, A. (2000). T-CombNET-A Neural Network Dedicated to Hand Gesture Recognition. In *International Workshop on Biologically Motivated Computer Vision* (pp. 613–622). Springer.

Lantz, V. (2011). A Framework for Hand Gesture Recognition Based on Accelerometer and EMG Sensors. *IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans*, *41*(6), 1064–1076. http://doi.org/10.1109/TSMCA.2011.2116004

Lee, S. U., & Cohen, I. (2004). 3D hand reconstruction from a monocular view. In *Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004.* (Vol. 3, pp. 310–313 Vol.3).

Li, F., & Wechsler, H. (2005). Open set recognition using transduction. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *27*(11), 1686 – 1697. Retrieved from http://www.freepatentsonline.com/7492943.html\nhttps://www.google.com/patents/US7492943

Liang, R.-H., & Ouhyoung, M. (1998). A real-time continuous gesture recognition system for sign language. In *Automatic Face and Gesture Recognition, 1998. Proceedings. Third IEEE International Conference on* (pp. 558–567). IEEE.

Lichtenauer, J. F., Hendriks, E. A., & Reinders, M. J. T. (2008). Sign language recognition by combining statistical DTW and independent classification. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, *30*(11), 2040–2046.

Licsár, A., & Szirányi, T. (2002). Hand-Gesture Based Film Restoration. *Pris*, 95–103.

Lippmann, R. (1987). An introduction to computing with neural nets. *IEEE Assp Magazine*, *4*(2), 4–22.

Little, J. J. (2006). Simultaneous Tracking and Action Recognition using the PCA-HOG Descriptor. In *The 3rd Canadian Conference on Computer and Robot Vision (CRV'06)* (pp. 6–6). IEEE. http://doi.org/10.1109/CRV.2006.66

Lowe, D. G. (2004). Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, *60*(2), 91–110.

Lu, Z., Chen, X., Li, Q., Zhang, X., & Zhou, P. (2014). A hand gesture recognition framework and wearable gesture-based interaction prototype for mobile devices. *IEEE Transactions on Human-Machine Systems*, *44*(2), 293–299.

Luo, Q., Kong, X., Zeng, G., & Fan, J. (2010). Human action detection via boosted local motion histograms. *Machine Vision and Applications*, *21*(3), 377–389. http://doi.org/10.1007/s00138-008-0168-5

M. Al-Rousan and M. Hussain. (2001). Automatic recognition of Arabic sign language finger spelling. *International Journal of Computers and Their Applications - IJCA-*

Universiti Malaya

Special Issue on Fuzzy Systems, *8*(2), 80–88.

Malima, A. K., Özgür, E., & Çetin, M. (2006). A fast algorithm for vision-based hand gesture recognition for robot control. In *2006 IEEE 14th Signal Processing and Communications Applications* (pp. 1–4). IEEE (Institute of Electrical and Electronics Engineers). http://doi.org/10.1109/SIU.2006.1659822

Manay, S., Cremers, D., Hong, B. W., Yezzi, A., & Soatto, S. (2006). Integral invariants for shape matching. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *28*(10), 1602–1618.

Maraqa, M., & Abu-Zaiter, R. (2008). Recognition of Arabic Sign Language (ArSL) using recurrent neural networks. *1st International Conference on the Applications of Digital Information and Web Technologies, ICADIWT 2008*, *2012*(February), 478–481. http://doi.org/10.1109/ICADIWT.2008.4664396

Marin, G., Dominio, F., & Zanuttigh, P. (2014). Hand gesture recognition with leap motion and kinect devices. In *Image Processing (ICIP), 2014 IEEE International Conference on* (pp. 1565–1569). IEEE.

Meng, M. Q., & Liu, P. X. (2003). Visual gesture recognition for human-machine interface of robot teleoperation. In *Proceedings 2003 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2003) (Cat. No.03CH37453)* (Vol. 2, pp. 1560–1565). IEEE. http://doi.org/10.1109/IROS.2003.1248866

Mgestyk. (2009). Retrieved from http://www.mgestyk.com/

Microsoft Kinect. (2012). Retrieved from http://www.microsoft.com/en-us/kinectforwindows/

Mitra, S., Member, S., Acharya, T., & Member, S. (2007). Gesture Recognition : A Survey. *IEEE Transactions on Systems, Man and Cybernetics Part C: Applications and Reviews*, *37*(3), 311–324. http://doi.org/10.1109/TSMCC.2007.893280.

Mohamed Elhadj, Y. O., Zemirli, Z., & Ayyadi, K. (2013). Development of a bilingual parallel corpus of Arabic and Saudi Sign Language: Part I. In *Advances in Intelligent Systems and Computing* (pp. 285–295). http://doi.org/10.1007/978-3-642-32063-7_30

Mohandes, M. (2001). Arabic sign language recognition. In *Presented at the Int. Conf. Imag. Sci., Syst. Technol., Las Vegas, NV, USA* (Vol. 1, pp. 753–759).

Mohandes, M. A. (2013). Recognition of Two-Handed Arabic Signs Using the CyberGlove. *Arabian Journal for Science and Engineering*, *38*(3), 669–677. http://doi.org/10.1007/s13369-012-0378-z

Mohandes, M., A-Buraiky, S., Halawani, T., & Al-Baiyat, S. (2004). Automation of the arabic sign language recognition. In *2004 International Conference on Information and Communication Technologies: From Theory to Applications* (pp. 479–480). IEEE. http://doi.org/10.1109/ICTTA.2004.1307840

Mohandes, M., Aliyu, S., & Deriche, M. (2014). Arabic sign language recognition using the leap motion controller. *2014 IEEE 23rd International Symposium on Industrial Electronics (ISIE)*, 960–965. http://doi.org/10.1109/ISIE.2014.6864742

Mohandes, M., & Deriche, M. (2013). Arabic sign language recognition by decisions fusion using Dempster-Shafer theory of evidence. In *Computing, Communications and IT Applications Conference (ComComAp)* (pp. 90–94). IEEE. http://doi.org/10.1109/ComComAp.2013.6533615

Mohandes, M., Deriche, M., Johar, U., & Ilyas, S. (2012). A signer-independent Arabic Sign Language recognition system using face detection, geometric features, and a Hidden Markov Model. *Computers & Electrical Engineering*, *38*(2), 422–433. http://doi.org/10.1016/j.compeleceng.2011.10.013

Molina, J., Escudero-Viñolo, M., Signoriello, A., Pardàs, M., Ferrán, C., Bescós, J., … Martínez, J. M. (2013). Real-time user independent hand gesture recognition from time-of-flight camera video using static and dynamic models. *Machine Vision and*

Universiti Malaya

Applications, *24*(1), 187–204. http://doi.org/10.1007/s00138-011-0364-6

Murthy, G., & Jadon, R. (2009). A review of vision based hand gestures recognition. *International Journal of Information Technology and Knowledge Management*, *2*(2), 405–410. Retrieved from http://www.researchgate.net/publication/228659202_A_REVIEW_OF_VISION_BASED_HAND_GESTURES_RECOGNITION/file/79e415058417b1de00.pdf

Myers, B. A. (1998). A brief history of human-computer interaction technology. *ACM Interactions*, *5*(2), 44–54. http://doi.org/10.1145/274430.274436

Nailon, W. H. (2010). Texture analysis methods for medical image characterisation. In *Biomedical imaging* (Youxin Mao). InTech. http://doi.org/10.5772/8912

Naoum, R., Owaied, H., & Joudeh, S. (2012). Development of a new Arabic sign language recognition using k-nearest neighbor algorithm. *Journal of Emerging Trends in Computing and Information Sciences*, *3*(8), 1173–1178. Retrieved from http://isite.persiangig.com/Files/KNN/Arabic_Naoum.pdf

Nehaniv, C. L., Dautenhahn, K., Kubacki, J., Haegele, M., Parlitz, C., & Alami, R. (2005). A methodological approach relating the classification of gesture to identification of human intent in the context of human-robot interaction. In *Robot and Human Interactive Communication, 2005. ROMAN 2005. IEEE International Workshop on* (pp. 371–377). IEEE.

Noury, N., Barralon, P., Virone, G., Boissy, P., Hamel, M., & Rumeau, P. (2003). A smart sensor based on rules and its evaluation in daily routines. *Proceedings of the 25th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (IEEE Cat. No.03CH37439)*, *4*(fig 1), 3286–3289. http://doi.org/10.1109/IEMBS.2003.1280846

OMRON. (2012). Retrieved from http://www.omron.com/

Pan, T.-Y., Lo, L.-Y., Yeh, C.-W., Li, J.-W., Liu, H.-T., & Hu, M.-C. (2016). Real-time sign language recognition in complex background scene based on a hierarchical clustering classification method. In *Multimedia Big Data (BigMM), 2016 IEEE Second International Conference on* (pp. 64–67). IEEE.

Pansare, J. R., Dhumal, H., Babar, S., Sonawale, K., & Sarode, A. (2013). Real time static hand gesture recognition system in complex background that uses number system of Indian sign language. *International Journal of Advanced Research in Computer Engineering & Technology (IJARCET)*, *2*(3), pp–1086.

Pansare, J. R., Gawande, S. H., & Ingle, M. (2012). Real-time static hand gesture recognition for American Sign Language (ASL) in complex background. *Journal of Signal and Information Processing*, *3*(3), 364.

Parvini, F., & Shahabi, C. (2007). An algorithmic approach for static and dynamic gesture recognition utilising mechanical and biomechanical characteristics. *International Journal of Bioinformatics Research and Applications*, *3*(1), 4–23. http://doi.org/10.1504/IJBRA.2007.011832

Pavlovic, V. I., Sharma, R., & Huang, T. S. (1997). Visual interpretation of hand gestures for human-computer interaction: a review. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *19*(7), 677–695. http://doi.org/10.1109/34.598226

Phung, S. L., Bouzerdoum, A., & Chai, D. (2005). Skin segmentation using color pixel classification: analysis and comparison. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *27*(1), 148–154. http://doi.org/10.1109/TPAMI.2005.17

Pisharady, P. K., & Saerbeck, M. (2015). Recent methods and databases in vision-based hand gesture recognition: A review. *Computer Vision and Image Understanding*, *141*, 152–165. http://doi.org/10.1016/j.cviu.2015.08.004

PointGrab's. (n.d.). Retrieved from http://www.pointgrab.com/

Pugeault, N., & Bowden, R. (2011). Spelling it out: Real-time ASL fingerspelling recognition. In *2011 IEEE International Conference on Computer Vision Workshops*

Universiti Malaya

(ICCV Workshops) (pp. 1114–1119). http://doi.org/10.1109/ICCVW.2011.6130290
Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, *77*(2), 257–286.
Raees, M., & Ullah, S. (2014). Continuous Number Signs Recognition. In *2014 12th International Conference on Frontiers of Information Technology* (pp. 274–279). http://doi.org/10.1109/FIT.2014.58
Rajathi, P., & Jothilakshmi, S. (2013). A static Tamil sign language recognition system. *International Journal of Advanced Research in Computer and Communication Engineering*, *2*(4), 1885–1891.
Ramage, D. (2007). *Hidden Markov models fundamentals*. *CS229 Section Notes*. Retrieved from http://see.stanford.edu/materials/aimlcs229/cs229-hmm.pdf
Ranganath, S., Ghosh, D., & Kevin, N. Y. Y. (2004). Trajectory modeling in gesture recognition using cybergloves and magnetic trackers. *Proceedings of the 2004 IEEE Region 10 Conference TENCON*, *1*, 571–574. http://doi.org/10.1109/TENCON.2004.1414484
Rathod, A., Sapate, S. G., & Khatirkar, R. K. (2012). Shape factor analysis of abrasive particles used in slurry abrasion testing. *International Journal of Mechanical and Industrial Engineering*, *2*(4), 35–39.
RaySarkar, A., Sanyal, G., & Majumder, S. (2013). Hand Gesture Recognition Systems: A Survey. *International Journal of Computer Applications*, *71*(15), 25–37. http://doi.org/10.5120/12435-9123
Rekha, J., Bhattacharya, J., & Majumder, S. (2011a). Hand gesture recognition for sign language: A new hybrid approach. In *The 2011 International Conference on Image Processing, Computer Vision, and Pattern Recognition (IPCV)* (pp. 80–86).
Rekha, J., Bhattacharya, J., & Majumder, S. (2011b). Shape, texture and local movement hand gesture features for indian sign language recognition. In *Trendz in Information Sciences and Computing (TISC), 2011 3rd International Conference on* (pp. 30–35). IEEE.
Ren, Z., Yuan, J., Meng, J., & Zhang, Z. (2013). Robust part-based hand gesture recognition using kinect sensor. *IEEE Transactions on Multimedia*, *15*(5), 1110–1120. http://doi.org/10.1109/TMM.2013.2246148
Ren, Z., Yuan, J., & Zhang, Z. (2011). Robust hand gesture recognition based on finger-earth mover's distance with a commodity depth camera. In *Proceedings of the 19th ACM international conference on Multimedia* (pp. 1093–1096). ACM.
Ritchings, T., Khadragi, A., & Saeb, M. (2012). An Intelligent Computer-Based System for Sign Language Tutoring. *Assistive Technology*, *24*(4), 299–308. http://doi.org/10.1080/10400435.2012.680662
Rodríguez-Serrano, J. A., & Perronnin, F. (2009). Handwritten word-spotting using hidden Markov models and universal vocabularies. *Pattern Recognition*, *42*(9), 2106–2116.
Rokade, R., Doye, D., & Kokare, M. (2009). Hand gesture recognition by thinning method. In *Digital Image Processing, 2009 International Conference on* (pp. 284–287). IEEE.
Rumelhart, D. E., McClelland, J. L., & Group, P. D. P. R. (1987). *Parallel distributed processing* (Vol. 1). MIT press Cambridge, MA.
Rybach, D. (2006). *Appearance-Based Features for Automatic Continuous Sign Language Recognition*. Diploma Thesis, RWTH AACHEN UNIVERSITY, Germany.
Saffiong, K. (2018). Chapter 7: Human Computer Interaction. In *World of Computing* (pp. 147–154).
Samir Elons, A., Abull-ela, M., & Tolba, M. F. (2013). Neutralizing lighting non-homogeneity and background size in PCNN image signature for Arabic Sign

Universiti Malaya

Language recognition. *Neural Computing and Applications*, *22*(SUPPL.1), 47–53. http://doi.org/10.1007/s00521-012-0818-4

Schlömer, T., Poppinga, B., Henze, N., & Boll, S. (2008). Gesture recognition with a Wii controller. In *Proceedings of the 2nd international conference on Tangible and embedded interaction - TEI '08* (p. 11). New York, New York, USA: ACM Press. http://doi.org/10.1145/1347390.1347395

Schum, R. (2004). Psychological Assessment of Children with Additional Disabilities. *The Volta Review*, *104*(4), 237–255.

Senin, P. (2008). Dynamic Time Warping Algorithm Review. In *Information and Computer Science Department University of Hawaii at Manoa Honolulu* (pp. 1–23). USA 885. http://doi.org/10.1109/IEMBS.2007.4353810

Sethi, A., Hemanth, S., Kumar, K., Bhaskara Rao, N., & Krishnan, R. (2012). SignPro-An Application Suite for Deaf and Dumb. *International Journal of Computer Science Engineering & Technology*, *2*(5).

Shanableh, T., & Assaleh, K. (2007). Video-based feature extraction techniques for isolated Arabic sign language recognition. In *9th International Symposium on Signal Processing and Its Applications (ISSPA)* (pp. 1–5). IEEE. http://doi.org/10.1109/ISSPA.2007.4555408

Shanableh, T., & Assaleh, K. (2011). User-independent recognition of Arabic sign language for facilitating communication with the deaf community. *Digital Signal Processing: A Review Journal*, *21*(4), 535–542. http://doi.org/10.1016/j.dsp.2011.01.015

Sharma, R., Huang, T. S., Pavlovic, V. I., Chu, S., & Schul, K. (1996). Speech/gesture interface to a visual computing environment for molecular biologists. In *Proceedings of 13th International Conference on Pattern Recognition* (Vol. 3, pp. 964–968 vol.3). IEEE. http://doi.org/10.1109/ICPR.1996.547311

Shet, V. D., Prasad, V. S. N., Elgammal, A. M., Yacoob, Y., & Davis, L. S. (2004). Multi-Cue Exemplar-Based Nonparametric Model for Gesture Recognition. In *ICVGIP* (pp. 656–662).

Sifuzzaman, M., Islam, M. R., & Ali, M. Z. (2009). Application of wavelet transform and its advantages compared to Fourier transform.

Smith, G. M., & Schraefel, m. c. (2004). The radial scroll tool: scrolling support for stylus- or touch-based document navigation. In *Proceedings of the 17th annual ACM symposium on User interface software and technology - UIST '04* (p. 53). New York, New York, USA: ACM Press. http://doi.org/10.1145/1029632.1029641

Smreen, S. (2004). Linguistic Culture of Deaf. In *In: International Workshop on "Deaf Communication", Tunisia (in Arabic)*. Tunisia: In Arabic.

SoftKinetic. (2012). Retrieved December 2, 2015, from http://www.softkinetic.com/support/Forum/aft/746

Soudi, A., Neumann, G., & Van den Bosch, A. (2007). Arabic computational morphology: knowledge-based and empirical methods. In *Arabic Computational Morphology* (pp. 3–14). Springer.

Starner, T., & Pentland, A. S. (1996). *Real-Time American Sign Language Recognition Hidden Markov Models from Video Using*. *AAAI Technical Report FS-96-05*.

Stotts, D., Smith, J. M., & Gyllstrom, K. (2004). FaceSpace. In *Proceedings of the fifteenth ACM conference on Hypertext & hypermedia - HYPERTEXT '04* (p. 48). New York, New York, USA: ACM Press. http://doi.org/10.1145/1012807.1012827

Sun, L., Ou, G., Lu, Y., & Tan, S. (2013). Axial ratio bandwidth of a circularly polarized microstrip antenna. In *Advancement in Microstrip Antennas with Recent Applications*. InTech.

Suriya, M., Sathyapriya, N., Srinithi, M., & Yesodha, V. (2016). Survey on real time sign language recognition system: an LDA approach. In *International conference on*

Universiti Malaya

exploration and innovations in engineering and technology, ICEIET (pp. 219–225).
Swapna, B., Pravin, F., & Dharaskar Rajiv, V. (2011). Hand gesture recognition system for numbers using thresholding. In *Communications in Computer and Information Science* (Vol. 250 CCIS, pp. 782–786). http://doi.org/10.1007/978-3-642-25734-6_139
Symeonidis, K. (1996). Hand Gesture Recognition using neural networks. *Neural Networks*, 1–68. Retrieved from http://personal.ee.surrey.ac.uk/Personal/T.Windeatt/msc_projects/symeonidis/Final Project.pdf
Tang, M. (2011). Recognizing hand gestures with microsoft's kinect. *Stanford University, Department of Electrical Engineering*.
The Arabic Dictionary of Gestures for the Deaf. (2007). Retrieved from http://www.menasy.com/arab Dictionary for the deaf 2.pdf
Thirumuruganathan, S. (2010). *A detailed introduction to K-nearest neighbor (KNN) algorithm*. Retrieved from https://saravananthirumuruganathan.wordpress.com/2010/05/17/a-detailed-introduction-to-k-nearest-neighbor-knn-algorithm/
Tolba, M. F., Samir, A., & Aboul-Ela, M. (2013). Arabic sign language continuous sentences recognition using PCNN and graph matching. *Neural Computing and Applications*, *23*(3), 999–1010. http://doi.org/10.1007/s00521-012-1024-0
Tolba, M., Samir, A., & Abul-Ela, M. (2012). A proposed graph matching technique for Arabic sign language continuous sentences recognition. *2012 8th Int Conf Informatics Syst (INFOS)*, 14–20. Retrieved from http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=6236612
Triesch, J., & von der Malsburg, C. (1996). Robust classification of hand postures against complex backgrounds. In *Proceedings of the Second International Conference on Automatic Face and Gesture Recognition* (pp. 170–175). IEEE Comput. Soc. Press. http://doi.org/10.1109/AFGR.1996.557260
Tubaiz, N., Shanableh, T., & Assaleh, K. (2015). Glove-Based Continuous Arabic Sign Language Recognition in User-Dependent Mode. *IEEE Transactions on Human-Machine Systems*, *45*(4), 526–533. http://doi.org/10.1109/THMS.2015.2406692
Tuffaha, M., Shanableh, T., & Assaleh, K. (2015). Novel Feature Extraction and Classification Technique for Sensor-Based Continuous Arabic Sign Language Recognition. In *Neural Information Processing: 22nd International Conference, ICONIP, Part IV* (pp. 290–299). http://doi.org/10.1007/978-3-319-26561-2
Vogler, C., & Metaxas, D. (1998). ASL recognition based on a coupling between HMMs and 3D motion analysis. *Sixth International Conference on Computer Vision (IEEE Cat. No.98CH36271)*, (January). http://doi.org/10.1109/ICCV.1998.710744
von Agris, U., Zieren, J., Canzler, U., Bauer, B., & Kraiss, K. F. (2008). Recent developments in visual sign language recognition. *Universal Access in the Information Society*, *6*(4), 323–362. http://doi.org/10.1007/s10209-007-0104-x
Wachs, J. P., Kölsch, M., Stern, H., & Edan, Y. (2011). Vision-based hand-gesture applications. *Communications of the ACM*, *54*(2), 60. http://doi.org/10.1145/1897816.1897838
Wasserman, P. D. (1993). *Advanced methods in neural computing*. John Wiley & Sons, Inc.
Webel, S., Keil, J., & Zoellner, M. (2008). Multi-touch gestural interaction in X3D using hidden Markov models. In *Proceedings of the 2008 ACM symposium on Virtual reality software and technology - VRST '08* (p. 263). New York, New York, USA: ACM Press. http://doi.org/10.1145/1450579.1450644
Wei, S., Chen, X., Yang, X., Cao, S., & Zhang, X. (2016). A component-based vocabulary-extensible sign language gesture recognition framework. *Sensors*, *16*(4),

Universiti Malaya

1–16.
Winnemöller, H. (1999). Practical gesture recognition for controlling virtual environments. *Project for Bachelor of Science (Honours) of Rhodes University*.
Wohler, C., & Anlauf, J. K. (1999). An adaptable time-delay neural-network algorithm for image sequence analysis. *IEEE Transactions on Neural Networks / a Publication of the IEEE Neural Networks Council*, *10*(6), 1531–6. http://doi.org/10.1109/72.809100
Wu, Y., & Huang, T. S. (1999). Vision-based gesture recognition: A review. In *International Gesture Workshop* (pp. 103–115). Springer.
Wu, Y., & Huang, T. S. (2001). Hand modeling, analysis and recognition. *IEEE Signal Processing Magazine*, *18*(3), 51–60.
Xu, D.-H., Kurani, A. S., Furst, J. D., & Raicu, D. S. (2004). Run-length encoding for volumetric texture. *Heart*, *27*(25).
Yin, X., & Xie, M. (2007). Finger identification and hand posture recognition for human–robot interaction. *Image and Vision Computing*, *25*(8), 1291–1300.
Youssif, A. A. ., Aboutabl, A. E., & Ali, H. H. (2011). Arabic Sign Language (ArSL) Recognition System Using HMM. *International Journal of Advanced Computer Science and Applications (IJACSA)*, *2*(11), 45–51. Retrieved from http://ijacsa.thesai.org/
Yun, L., & Peng, Z. (2009). An automatic hand gesture recognition system based on Viola-Jones method and SVMs. *2nd International Workshop on Computer Science and Engineering, WCSE 2009*, *2*(0), 72–76. http://doi.org/10.1109/WCSE.2009.769
Zaki, M. M., & Shaheen, S. I. (2011). Sign language recognition using a combination of new vision based features. *Pattern Recognition Letters*, *32*(4), 572–577. http://doi.org/10.1016/j.patrec.2010.11.013
Zaman Khan, R. (2012). Hand Gesture Recognition: A Literature Review. *International Journal of Artificial Intelligence & Applications*, *3*(4), 161–174. http://doi.org/10.5121/ijaia.2012.3412
Zhang, X., Chen, X., Li, Y., Lantz, V., Wang, K., & Yang, J. (2011). A framework for hand gesture recognition based on accelerometer and EMG sensors. *IEEE Transactions on Systems, Man, and Cybernetics-Part A: Systems and Humans*, *41*(6), 1064–1076.

Universiti Malaya